REVERSE ENGINEERING
of a
Dyson Bladeless Fan



Design for Manufacturability

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EXECUTIVE SUMMARY

The reverse engineering of the Dyson Bladeless Fan was undertaken to understand its innovative design, optimize its assembly, and propose improvements for cost and manufacturing efficiency. This project began by disassembling the Dyson fan to inspect its internal components and model them using SolidWorks software. The detailed teardown allowed us to examine the fan's unique bladeless airflow system, which relies on an internal impeller and an air multiplier ring to create a smooth, uninterrupted air stream. This insight provided a solid foundation for subsequent analyses of the product's manufacturability, material selection, and design optimization.

After completing the teardown and modeling, we conducted a Design for Assembly (DFA) analysis to assess the assembly process and identify opportunities for part reduction. This analysis, alongside an economic study, revealed several inefficiencies in the current design, highlighting areas where manufacturing costs could be reduced without compromising the fan's performance. A thorough material selection analysis followed, where each part was evaluated for its strength, weight, price, and environmental impact, ensuring that the chosen materials would optimize both performance and sustainability. For example, plastic components were selected for their light weight and moldability, ideal for the fan's complex shapes.

In addition to material selection, a comprehensive manufacturing process selection analysis was performed to identify the most suitable production techniques for each component. The current manufacturing process for most components is injection molding and wire drawing; for the plastic components, rotational molding could be considered for larger, more complex parts to reduce material waste. For metal parts, die casting could offer higher precision for aluminum components. These assessments are aimed at streamlining production and ensuring consistent product quality. Finally, we proposed four fundamental design changes to improve the product's assembly and reduce manufacturing complexity: (1) eliminating three wire holders, (2) removing one of the swivel base supports, (3) removing the sensor hatch and its housing by attaching the sensor directly to the base, and (4) eliminating the LCD/PCB holder by adding projections to the swivel base plate. These changes significantly reduce part count and simplify assembly, leading to a more cost-effective and efficient production process.

This reverse engineering project gave us valuable insights into the Dyson bladeless fan's design and manufacturing processes. This resulted in optimized assembly, material use, and cost savings without compromising the fan's performance.

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PROJECT DESCRIPTION AND OBJECTIVES

The reverse engineering of the Dyson Bladeless Fan was initiated to understand its innovative design and optimize its manufacturability and cost efficiency. Dyson's bladeless fan represents a breakthrough in fan technology, utilizing an air multiplier technology that eliminates visible blades, creating a smooth airflow and enhancing safety and aesthetic appeal. Despite its high-performance features, the fan consists of 23 initially assumed parts, primarily plastic components, and presents design simplification and cost reduction opportunities. The objectives of this project were to perform a detailed teardown analysis, model the components, and evaluate the product's design for assembly, material selection, and manufacturing efficiency. The final goal was to propose design modifications that would simplify assembly, reduce costs, and improve product performance while maintaining the fan's functionality.

The project focused on the teardown and inspection of the Dyson fan's components, which revealed that 33 parts are made from PC/ABS plastic, one from ABS, two from POM, while small springs are made of stainless steel and the mesh panel of the filter is made of aluminum. All plastic components are produced through injection molding, while the metal parts use forging and wire drawing processes. Constraints included maintaining the fan's signature bladeless design and airflow technology while optimizing material use and assembly efficiency. Cost implications were considered throughout the project, aiming to reduce material and manufacturing costs while maintaining durability and product quality.

Objective Technical Specifications:

1. **Component Modeling:**
    - Dimensional tolerance between ±0.01 to ±0.05 inch for all parts in SolidWorks models.
2. **Material Selection:**
    - 33 parts in PC/ABS, 1 in ABS, 2 in POM, stainless steel for springs, and aluminum for the mesh panel.
    - Materials must meet ASTM standards for durability and performance.
3. **Manufacturing Process:**
    - Injection molding for plastic parts.
    - Forging and wire drawing for metal parts, with tolerances of ±0.05 inch.
    - Evaluate rotational molding, die casting, and laser cutting as alternatives.
4. **Part Reduction and DFA:**
    - Reduce part count by 10-15%.
    - Merge base plate with the base and consolidate wire holders.

5. **Assembly Efficiency:**
    - Reduce assembly time by 20%.
    - Consolidate wire holders and streamline base assembly with a 15% reduction in parts.
6. **Design Modifications:**
    - Secure single wire holder with a tolerance of ±0.05 inch.
    - Merge base and base plate.
    - Attach the sensor to hatch with a tolerance of ±0.05 inch.
    - Integrate PCB projections with a tolerance of ±0.01 inch.
7. **Cost Efficiency:**
    - Reduce material and production costs by 10-15%.
    - Achieve a 20% reduction in assembly time while maintaining product quality.

GAANT CHART

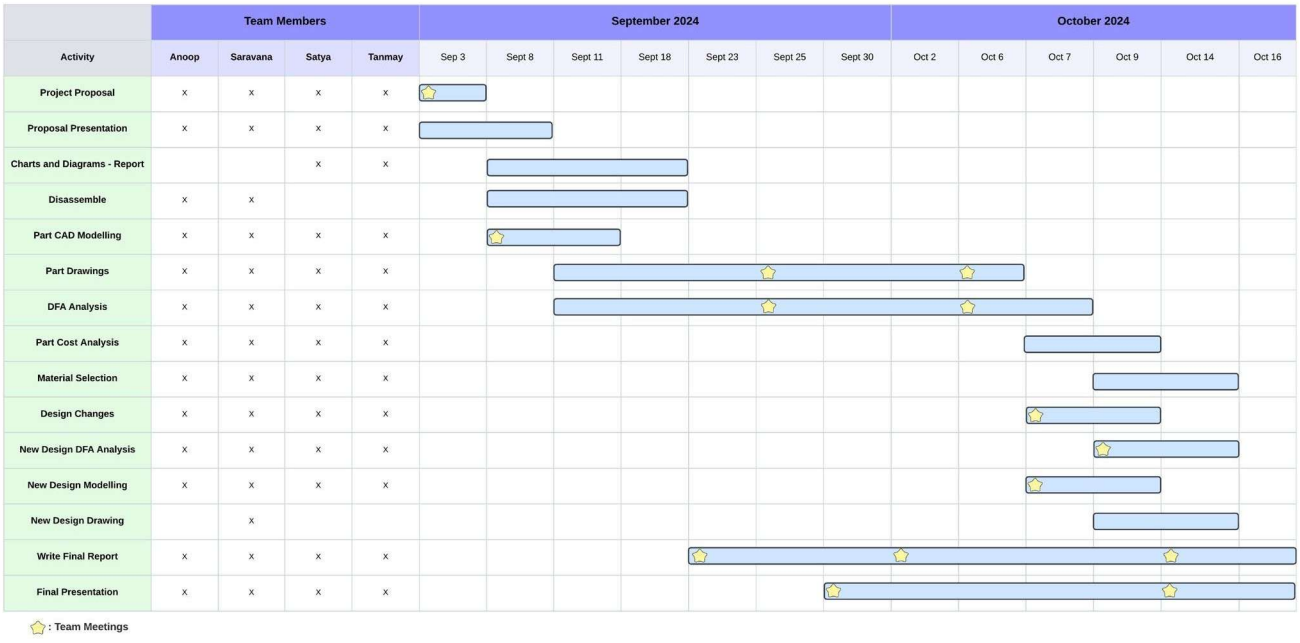


Figure 1: GAANT CHART

The GAANT chart in Figure 1 outlines the project tasks, how they were split among the group, and the deadlines for each task. The end goal was to complete the project. The milestones were various assignments, with the project report and presentation being the final milestones.

BLACK BOX DIAGRAM

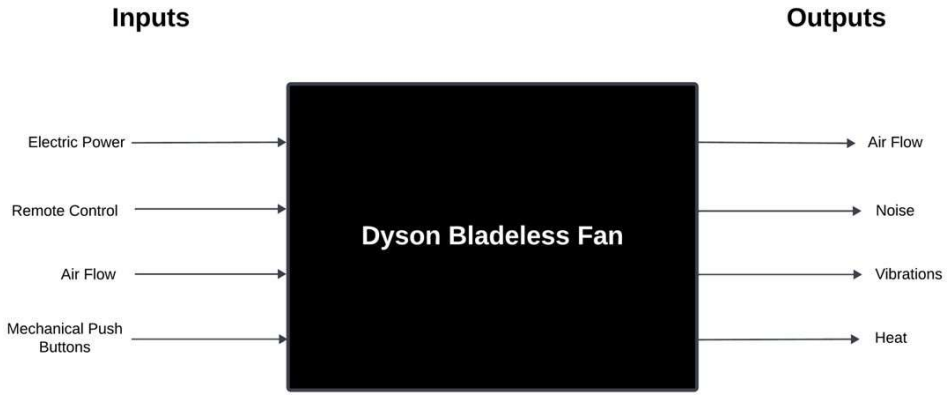


Figure 2: Black Box Diagram

The black box diagram in Figure 2 is the first step in reverse engineering for our chosen product. This was done at the beginning of the project to evaluate the product. Electric power is the primary source; either a mechanical button is pushed, or the same action is required via the remote control to switch on the fan. Airflow is also an input in the bladeless fan. The outputs are flow, noise, vibrations, and heat from the fan. In a black box diagram, the subsystems are not mentioned, and the operation of the fan can be misunderstood. The next step was disassembling the fan and then going through the different internal parts to understand their function. The glass box diagram below shows the same.

GLASS BOX DIAGRAM

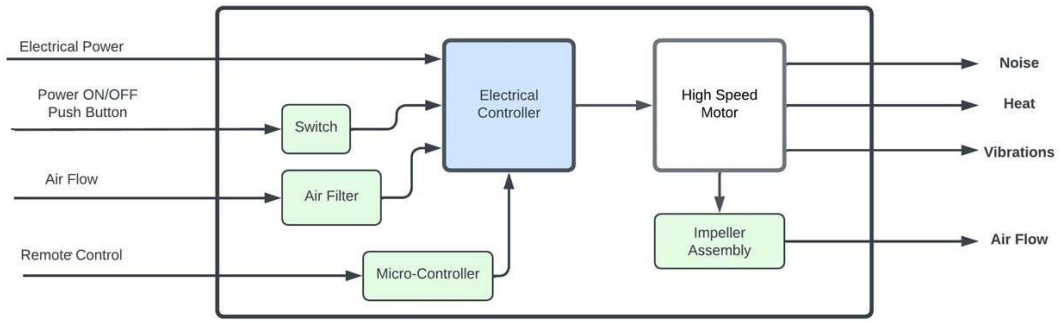


Figure 3: Glass Box Diagram

The glass box diagram in Figure 3 contains the internal parts in internal assemblies. These internal parts were disassembled carefully, and each was documented with their respective function, position internally, and complexity. The microcontroller regulates the high-speed motor, which drives the impeller assembly that generates airflow. The motor, in turn, connects to various

electrical and mechanical components that result in three essential by-products: noise, heat, and vibrations. The primary output of the system is airflow produced by the impeller. In contrast, the secondary outputs are the undesired effects of noise, heat, and vibrations caused by the motor and related components. This diagram effectively maps the flow of the fan system's inputs, control mechanisms, and outputs.

FISHBONE DIAGRAM

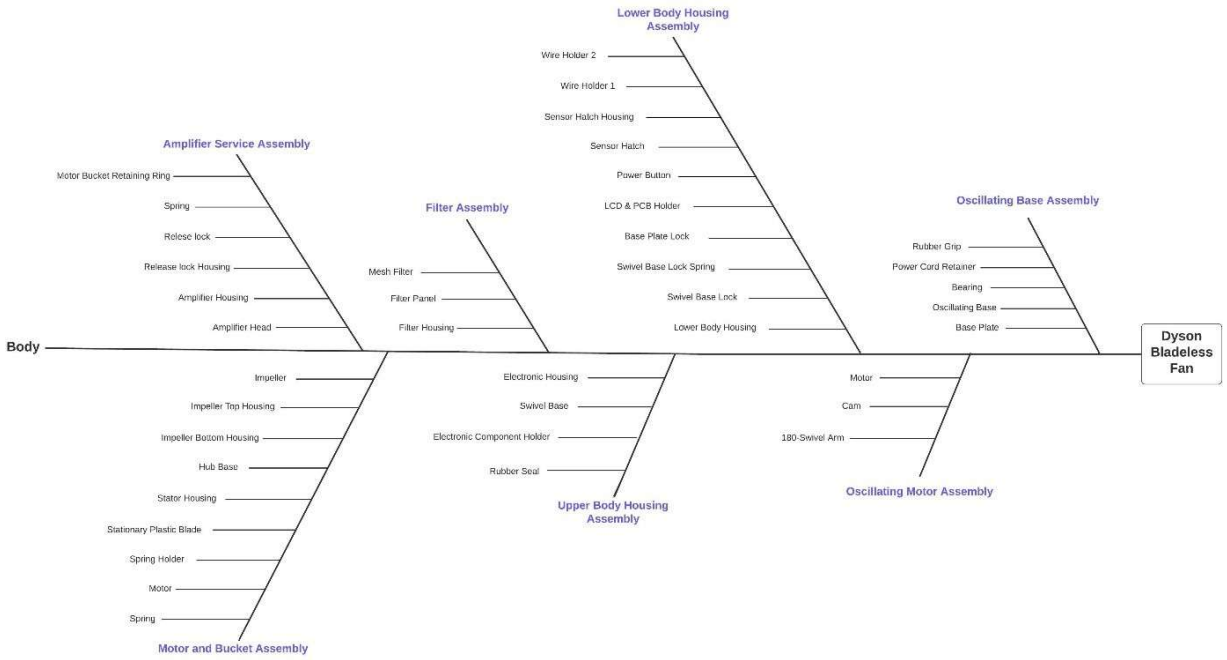


Figure 4: Fishbone Diagram

The fishbone diagram in Figure 4 is a comprehensive chart of all the different subassemblies in our fan. We have seven major subassemblies. Each of the lines in the subassembly indicates an independent part. They are listed below with their functions:

Amplifier Service Assembly:

The Amplifier Service Assembly controls and amplifies the airflow generated by the motor and impeller. The motor bucket retaining ring, spring, release lock, and amplifier housing are vital components. These parts work together to secure and maintain the alignment of the motor bucket and amplifier components. The amplifier head is critical in directing and shaping the airflow, contributing to the bladeless fan's smooth, consistent air output. The retention system ensures that the motor and amplifier remain securely in place while service-accessible.

Filter Assembly:

The Filter Assembly comprises the mesh filter, filter panel, and filter housing. This assembly is primarily responsible for maintaining air quality by filtering out dust and particulates before the

air is propelled through the system. The mesh filter is the primary filtration element, while the filter panel and housing securely hold it. This assembly ensures the air being circulated is clean, contributing to the fan's overall efficiency and user comfort.

Motor and Bucket Assembly:

The Motor and Bucket Assembly is the core of the fan's operation, housing the motor and impeller components that generate airflow. Essential parts include the impeller (top and bottom housing), hub base, stator housing, and stationary plastic blade. The motor drives the impeller, responsible for drawing in air and directing it into the amplifier for circulation. This assembly includes springs and other securing components to ensure smooth, vibration-free operation. This is the heart of the Dyson fan, where electrical energy is converted into mechanical motion to move air.

Lower Body Housing Assembly:

The Lower Body Housing Assembly comprises components like wire holders, the sensor hatch, the power button, the LCD and PCB holder, and the base plate lock. These components manage the fan's control systems and house the electronics responsible for its operation. The swivel base lock and spring stabilize the fan's rotational movement. This assembly is critical for user interaction, containing controls and electronic components that govern fan speed and operation.

Upper Body Housing Assembly:

The Upper Body Housing Assembly houses electronic components and supports the structural integrity of the fan. It includes electronic housing, a swivel base, and an electronic component holder, which keeps all the critical electronic systems securely in place. This assembly is essential for housing the fan's electrical systems and ensuring that the electronic and mechanical components work together efficiently, contributing to the fan’s overall durability and functionality.

Oscillating Base Assembly:

The Oscillating Base Assembly is responsible for the fan's movement, allowing it to rotate and distribute air evenly throughout a room. The components of the rubber grip, power cord retainer, bearing, oscillating base, and base plate are vital. The assembly’s bearing system enables smooth, continuous oscillation, while the rubber grip ensures stability during operation. This assembly enhances the fan's ability to circulate air across a wider area by enabling base movement.

Oscillating Motor Assembly:

The Oscillating Motor Assembly drives the fan's oscillating motion. It includes the motor, cam, and 180-degree swivel arm. These components allow the fan to pivot smoothly and efficiently, enabling it to distribute air across a room. The cam mechanism converts the motor’s rotational motion into a back-and-forth oscillation, allowing broad coverage without manual adjustment. This assembly plays a vital role in the functionality and versatility of the Dyson bladeless fan.

PATENT SEARCH

The patent **US8308445B2** relates to a bladeless fan assembly designed to create an air current using a nozzle that leverages the **Coanda effect**. This bladeless fan design improves traditional fan systems by eliminating visible blades, thus enhancing safety, reducing noise, and providing more even air distribution. The nozzle includes an internal passage for airflow, a mouth that directs the air over the Coanda surface, and an outlet to emit the air. This design allows the air to flow smoothly across the surface and entrain additional air from the surrounding environment. This leads to amplified airflow without turbulence typically associated with bladed fans.

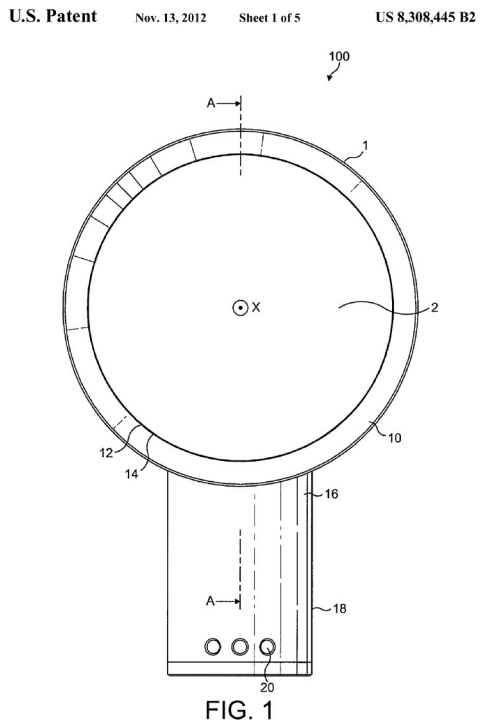


Figure 5: Patent Image-Amplifier Assembly

A key innovation of this fan is its ability to generate a more uniform air current, providing better cooling at a distance compared to conventional fans. The Coanda effect ensures that the air adheres closely to the curved surface of the nozzle, allowing for efficient entrainment and amplification of air. This enhances airflow and provides quieter operation, making it suitable for domestic environments like homes and offices.

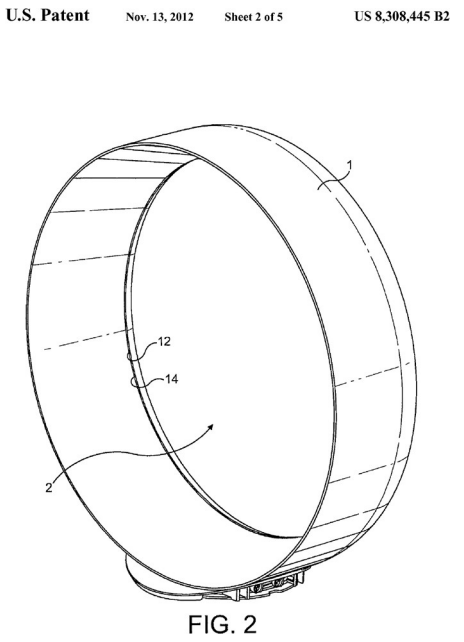


Figure 6: Patent Image-Amplifier Head

The fan's nozzle is designed flexibly, offering various shapes, including annular or circular, to optimize air distribution over a broader area. The design also allows for versatility in mounting, such as on a desk, wall, or ceiling. Additionally, the bladeless design opens the central part of the fan, which can be used for additional features like lighting or displays, further increasing the product's utility and appeal.

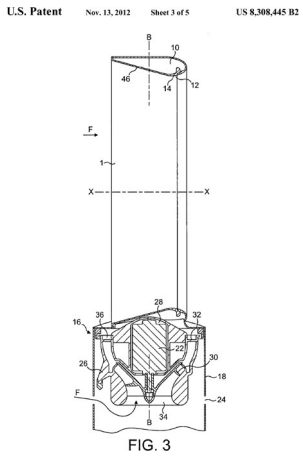


Figure 7: Patent Image-Airfoil, Impeller

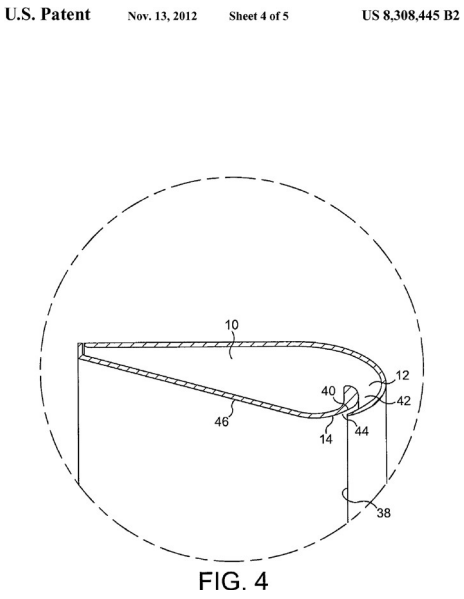


Figure 8: Patent Image-Airfoil

The patent covers multiple innovative aspects of the fan, including using a **DC brushless motor** to drive the airflow, ensuring efficient and reliable performance. The overall design reduces the number of moving parts, thus minimizing wear and tear and reducing manufacturing complexity. This makes the bladeless fan an attractive, modern alternative to traditional fans, combining aesthetics with functional enhancements.

DESIGN DOCUMENTATION

PARTS AND COMPONENTS DISASSEMBLY

After all the parts had been disassembled, a preliminary bill of materials was created. The table below shows the bill of materials and indicates whether the part is off the shelf.

Table 1: Preliminary Bill of Materials

SL.NO	Part Number	Description	Material	Quantity	OTS
1.	BF001-01	Amplifier head	PC/ABS	1	-
2.	BF001-02	Amplifier housing	PC/ABS	1	-
3.	BF001-03	Release lock Housing	PC/ABS	1	-
4.	BF001-04	Release lock	PC/ABS	1	-
5.	BF001-05	Spring	Stainless steel	1	YES
6.	BF002	Motor Bucket retaining ring	PC/ABS	1	-
7.	BF003-01	Impeller	PC/ABS	1	-
8.	BF003-02	Impeller top housing	PC/ABS	1	-
9.	BF003-03	Impeller bottom housing	PC/ABS	1	-
10.	BF003-04	Hub base	PC/ABS	1	-
11.	BF003-05	Stator housing	PC/ABS	1	-
12.	BF003-06	Stationary plastic blade	PC/ABS	1	-
13.	BF003-07	Spring Holder	PC/ABS	1	-
14.	BF003-08	Motor	-	1	YES
15.	BF003-09	Spring	Stainless steel	1	YES
16.	BF004-01	Filter Housing	ABS	1	-
17.	BF004-02	Filter Panel	Al	2	YES
18.	BF004-03	Mesh-Filter	Borosilicate	1	YES
19.	BF005-01	Electronic housing	PC/ABS	1	-
20.	BF005-02	Electronics Component Holder	PC/ABS	1	-
21.	BF005-03	Electronic housing	PC/ABS	1	-
22.	BF005-04	Rubber Seal	Rubber	1	YES
23.	BF006-01	Lower body housing	PC/ABS	1	-
24.	BF006-02	Swivel base lock	PC	1	-
25.	BF006-03	Swivel base lock spring	Stainless Steel	1	YES
26.	BF006-04	Base plate lock	POM	1	-
27.	BF006-05	LCD & PCB holder	PC/ABS	1	-
28.	BF006-06	Power button	PC/ABS	1	-
29.	BF006-07	Sensor hatch	PC/ABS	1	-
30.	BF006-08	Sensor hatch housing	PC/ABS	1	-
31.	BF006-09	Wire holder 1	PC/ABS	6	-
32.	BF006-10	Wire holder 2	POM	1	-
33.	BF007-01	Motor	-	1	YES
34.	BF007-02	Cam	PC/ABS	1	-
35.	BF007-03	180-swivel arm	Al	1	-

SL.NO	Part Number	Description	Material	Quantity	OTS
36.	BF008-01	Base plate	PC/ABS	1	-
37.	BF008-02	Oscillating base	PC/ABS	1	-
38.	BF008-03	Bearing	Stainless Steel	1	-
39.	BF008-04	Power cord retainer	PC/ABS	1	-
40.	BF008-05	Rubber grip	Silicon	1	-

We have 38 unique components, 46 total, and eight **Off-The-Shelf** components.

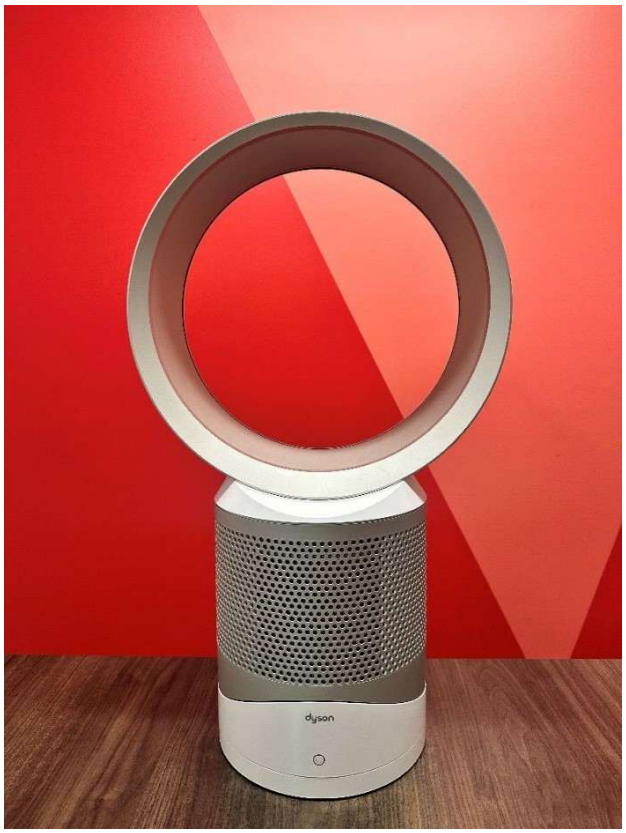


Figure 9: Dyson Fan

The disassembly procedure began with removing the parts from the top and moving them down. Our product, by design, follows a top-down approach. The procedure had around ten steps, beginning with removing the amplifier assembly, the filter assembly, the swivel base assembly, and then the base plate assembly. Once all the parts were disassembled, their dimensions were taken, and CAD modeling began. The images below are of the parts disassembled.



Figure 10: Teardown Image

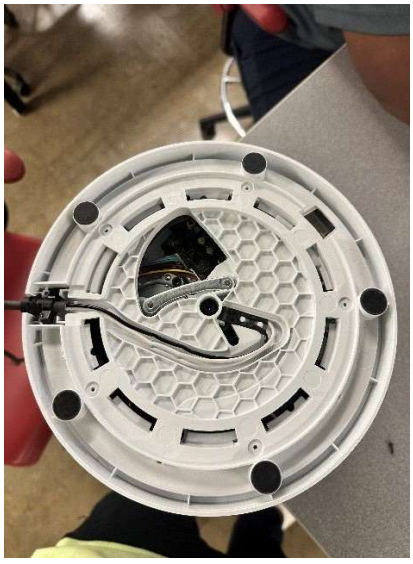


Figure 11: Oscillating Base Plate



Figure 12: Electronics Housing



Figure 13: LCD/PCB Holder

The parts were carefully disassembled, and all the electronics housing positions were documented. We also simultaneously documented all the parts and their locations and measured all mating surfaces carefully to ensure that the dimensions and positions matched when designing them on SolidWorks.



Figure 14: Filter

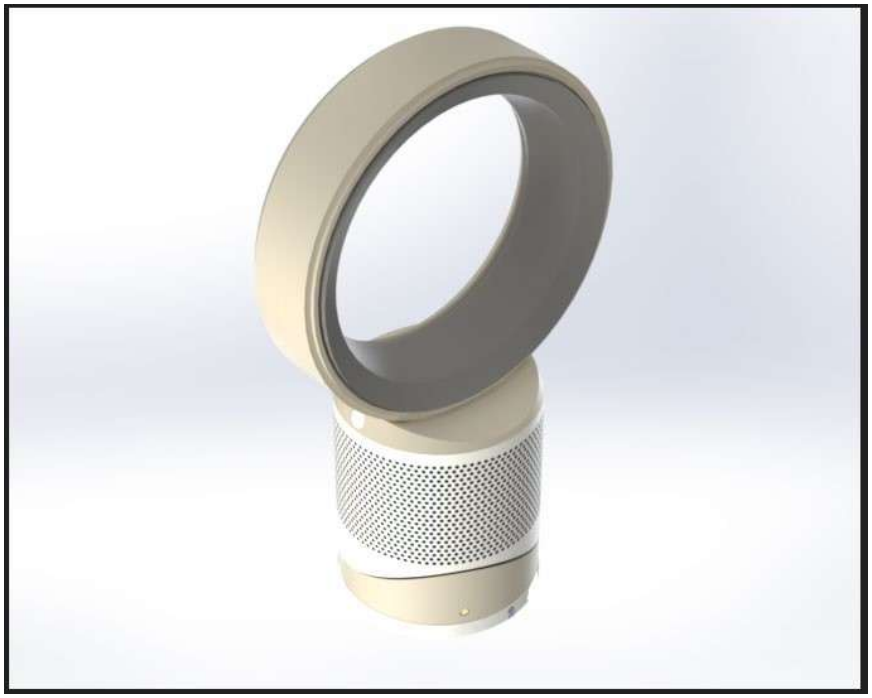


Figure 15:3D Render of Fan

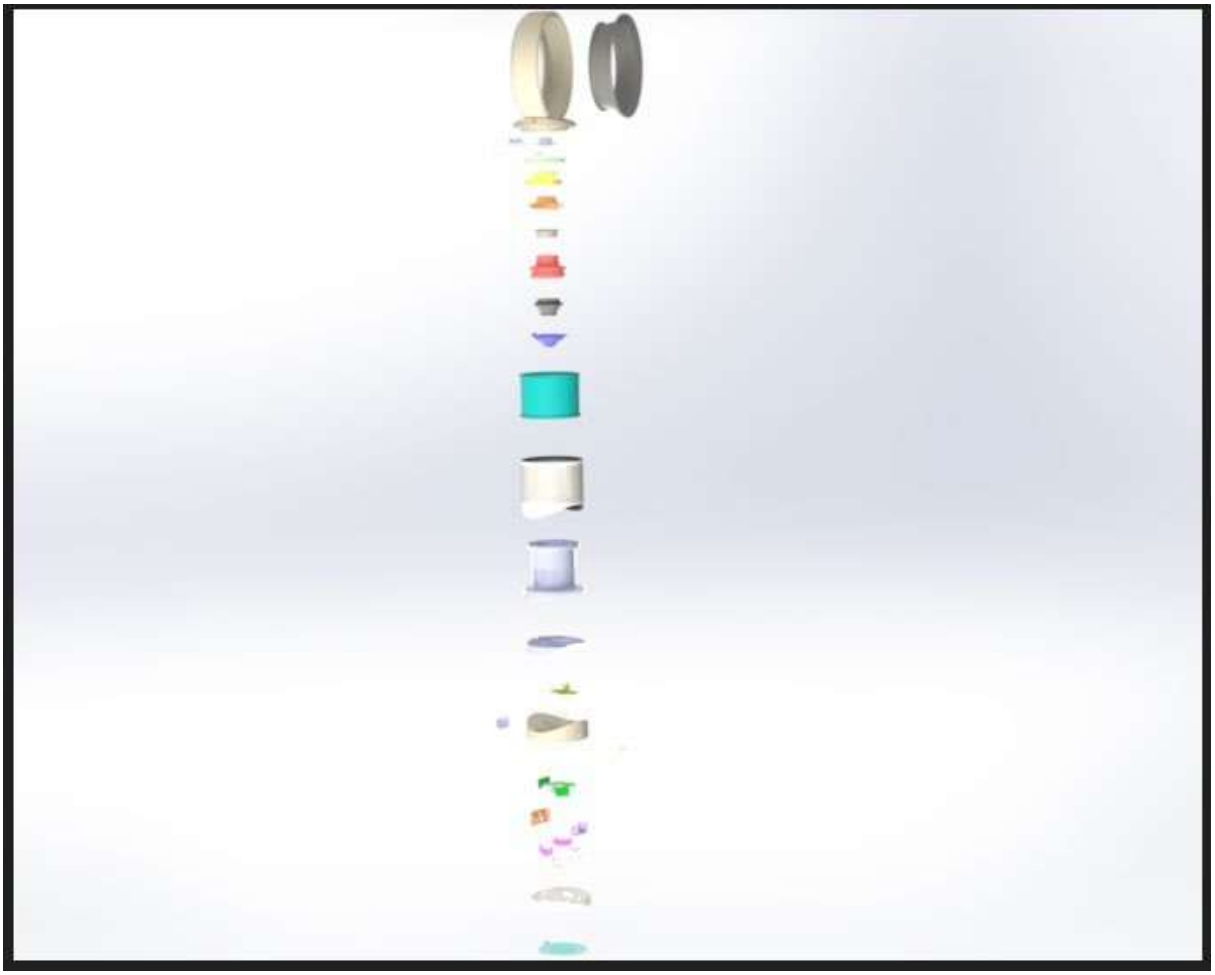


Figure 16: Exploded View

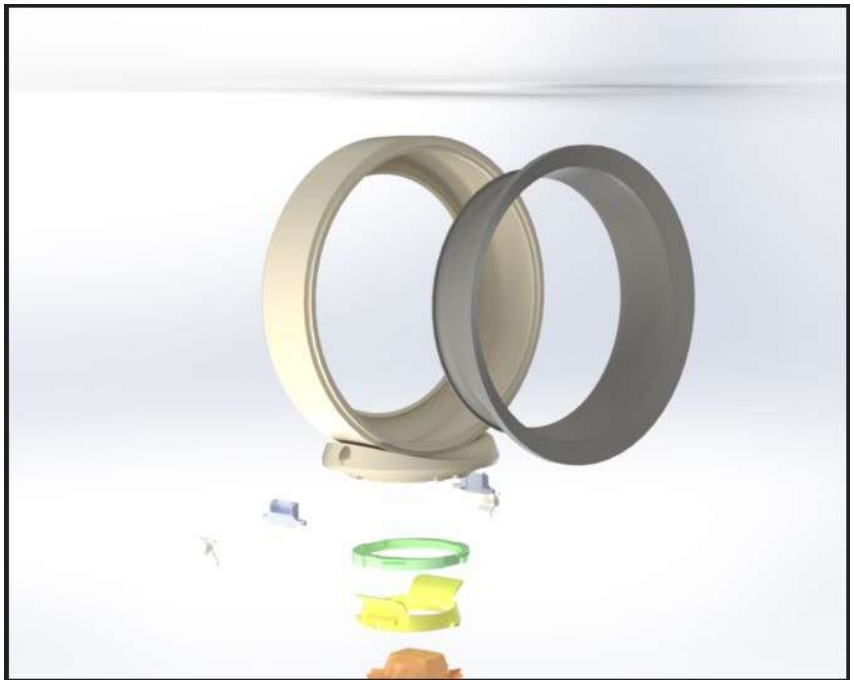


Figure 17: Amplifier Head 3D Render Exploded View

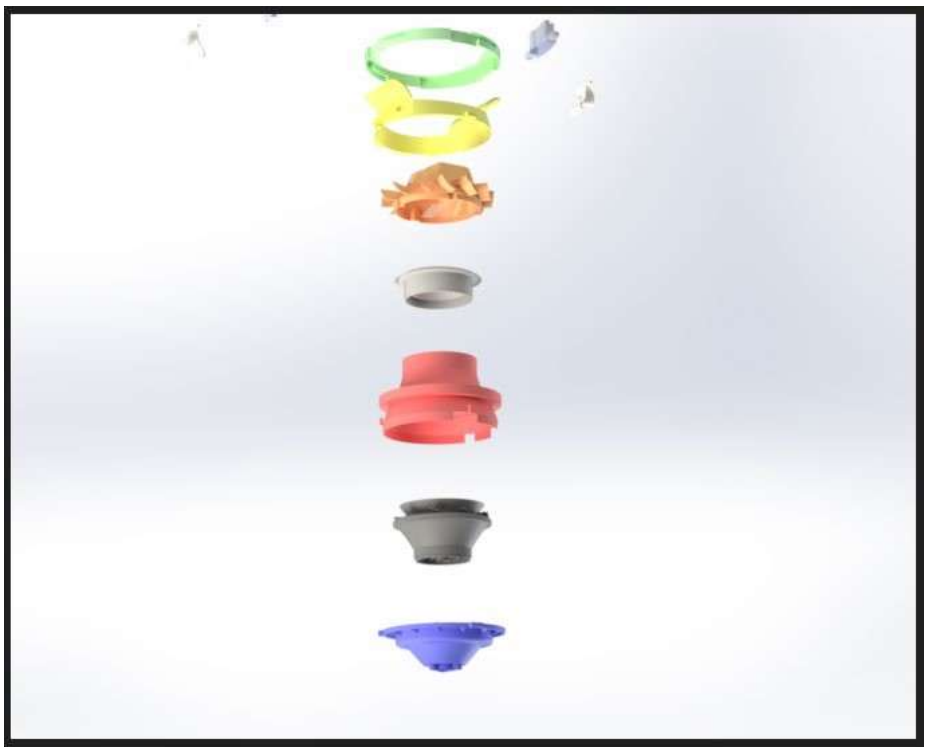


Figure 18: Upper Body Housing Assembly Exploded View

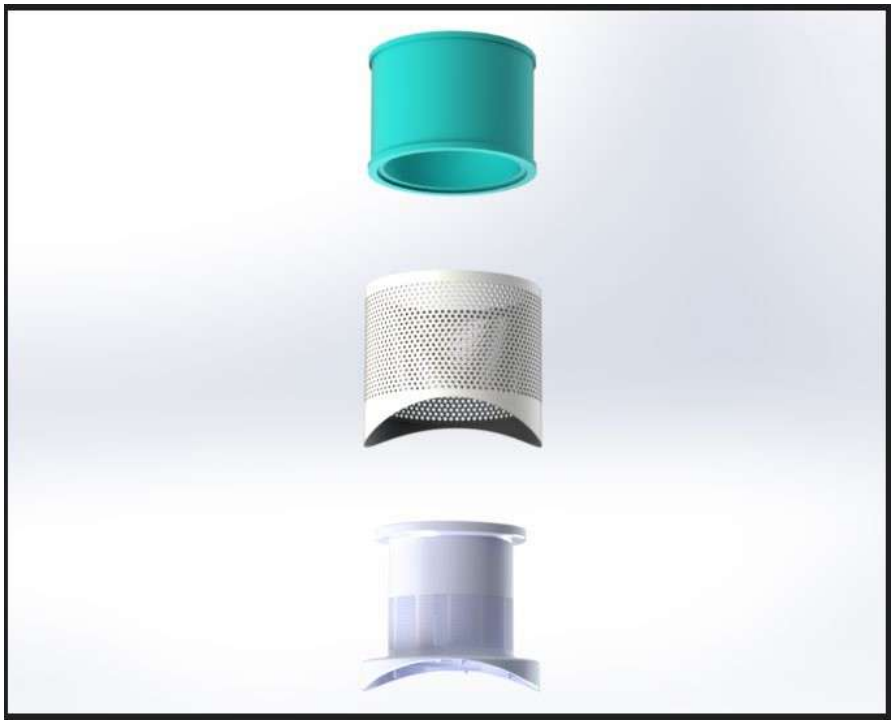


Figure 19: Filter Assembly Exploded View

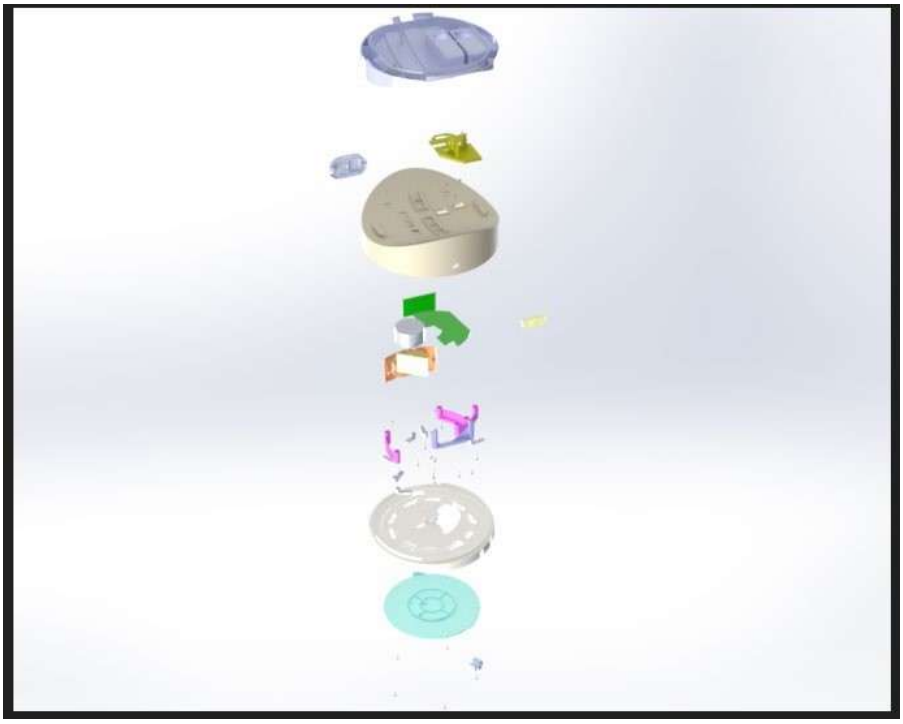


Figure 20; Lower Body Housing Assembly- Exploded View

MISCELLANEOUS PARTS TECHNICAL SPECIFICATIONS

Dyson Filter Specifications:

1. Technical Name = 360 Glass HEPA Filter
2. HEPA Filter made up of borosilicate microfiber pleated a thousand times.
3. Rubber seal forms a vacuum, sealing the filter within the machine.
4. Filter Life Specification = 12Months
5. Filter dimensions: (H) 6.2in (159 mm) (D) 8.4in (214 mm) Filter weight: 9oz (0.26kg)

Dyson Impeller Motor Specifications:

The main motor in the **Dyson DP01** (a brushless DC motor) operates at variable speeds depending on the fan setting. While Dyson does not typically disclose the exact **RPM (revolutions per minute)** for its fans, it's possible to estimate the range based on the fan's behavior.

Typically, brushless DC motors in air purifiers and fans like the Dyson DP01 can spin **up to 9,000-10,000 RPM** at their highest speed. Here's an approximate breakdown:

- **Low-speed settings:** The RPM may be in the range of **1,000-2,000 RPM**.
- **High-speed settings:** The RPM may peak between **9,000-10,000 RPM**.
- Voltage: 220-240Volts

These speeds allow the fan to generate strong airflow while remaining energy-efficient and quiet compared to traditional motor designs.

The **Dyson DP01** is an air purifier and fan combination unit in the Dyson Pure Cool series. It uses a brushless DC motor, which is standard in many of Dyson's fans and air purifiers due to its efficiency and quiet operation.

Key Motor Specifications of Dyson DP01:

- **Motor Type:** Brushless DC (Direct Current) motor
- **Power Output:** The power consumption is generally around 56 watts (in fan mode at the highest speed)
- **Speed Settings:** The DP01 features 10-speed settings, allowing for variable airflow.
- **Rotation:** It can oscillate up to 70 degrees to cover a wide area.

The brushless DC motor is known for its long life and energy efficiency, contributing to the DP01's effective air purification and cooling.

Dyson Oscillating Motor Specification:

The oscillation mechanism in the **Dyson DP01** is powered by a separate motor responsible for the fan's side-to-side movement. This motor is typically a **stepper or** low-power motor designed for precision and smooth oscillation.

Key Oscillation Motor Specifications of Dyson DP01:

- **Voltage:** 12-24Volts Dc
- **Motor Type:** Likely a stepper motor (used in most Dyson oscillating fans for precise control).
- **Oscillation Range:** 70 degrees, allowing the fan to sweep air across a wide area.
- **Control:** The oscillation is electronically controlled and can be turned on or off using the remote or app.
- **Power Consumption:** The oscillation motor consumes minimal power, likely only a few watts, as its primary function is to rotate the unit slowly and consistently.

Stepper motors are typically used in such applications because they offer fine control over movement, ensuring smooth and quiet oscillation.

Dyson Foam Seal:

The **foam seal** in the Dyson DP01 air purifier/fan is a crucial component for maintaining airtight integrity and noise reduction, though Dyson doesn't publicly list specific technical details about it. However, based on typical foam seals used in similar appliances, here are likely specifications and characteristics:

Likely Specifications of the Foam Seal in Dyson DP01:

1. **Material:**
   - **Closed-cell foam** or **polyurethane foam** is commonly used for such seals because it offers excellent resistance to air and water leaks and durability over time.
2. **Thickness:**
   - Typically ranges from **2 to 5 mm** depending on the area it's applied to, but the exact thickness would vary based on the gap it needs to seal.
3. **Density:**
   - Foam density is generally **30-50 kg/m³** for appliance seals, balancing between being compressible enough for a tight fit and firm enough to maintain a seal.

4. **Temperature Resistance:**

- The foam seal likely has a heat tolerance of around **-40°C to 80°C** (or higher), allowing it to handle the motor's heat and ambient temperature fluctuations without degrading.

5. **Functionality:**

- **Airtight Sealing:** Prevents air leakage around the filter and airflow pathways.
- **Sound Insulation:** Reduces noise from vibrations within the unit.
- **Dust Containment:** Helps prevent dust from escaping around filter edges.

6. **Longevity:**

- Foam seals are typically designed for **5-10 years** of use but may degrade over time due to heat, pressure, or exposure to contaminants.

Dyson Rubber AMP Seal:

- **AMP Seal**, which is a type of connector typically used in industrial and automotive applications to ensure waterproof or dustproof sealing,
- **Airflow Seals:** Dyson uses rubber or silicone gaskets and precision-engineered seals to ensure no air leaks from the filtration or fan system, maintaining high efficiency in air purification and airflow.
- **Electrical Seals:** Dyson products often have water-resistant seals, though specific AMP connectors for external connections aren't a typical feature.

DFA ANALYSIS

DFA Analysis Worksheet

Assembly Name: Dyson Bladeless Fan- DP-01 **Team:** Group 6 **Date:** 30-09-2024

If the answer is Yes to any of the metrics or questions enter a 1. If the answer is No then enter 0. Each cell must have a number.

Part		Complexity		Functional Analysis / Redesign Opportunity				Error Proofing		Handling			Insertion				Secondary Operations				
Part Number	Part Name	Number of Parts (N_p)	Number of Interfaces (N_i)	Theoretical Minimum Part	Part Can Be Standardized (if not already standard)	Cost (Low/Medium/High)	Practical Minimum Part	Assemble Wrong Part/ Omit Part	Assemble Part Wrong Way Around	Tangle, Nest, or Stick Together	Flexible, Fragile, Sharp or Slippery	Pliers, Tweezers, or Magnifying Glass Needed	Difficult to Align/ Locate	Holding Down Required	Resistance to Insertion	Obstructed Access/ Visibility	Re-orient Workpiece	Screw, Drill, Twist, Rivet, Bend, or Crimp	Weld, Solder, or Glue	Paint, Lube, Heat, Apply Liquid or Gas	Test, Measure or Adjust
BF001	Amplifier Service assembly																				
BF001-01	Amplifier head	1	1	1	0	M	1	0	0	1	0	0	0	0	0	0	0	0	1	0	0
BF001-02	Amplifier housing	1	6	1	0	M	1	0	0	1	0	0	0	0	0	0	0	0	1	0	0
BF001-03	Release lock Housing	2	3	0	0	L	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
BF001-04	Release lock	2	4	0	0	L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
BF001-05	Spring	2	2	0	0	L	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0
BF002	Motor Bucket retaining ring	1	3	1	0	L	1	0	0	1	0	0	1	0	1	0	1	1	0	0	0
BF003	Motor & Bucket assembly																				
BF003-01	Impeller	1	3	1	1	M	1	0	1	1	1	0	1	0	0	1	0	0	0	0	1
BF003-02	Impeller top housing	1	3	1	0	M	1	0	0	0	0	0	1	0	0	1	0	1	0	0	0
BF003-03	Impeller bottom housing	1	8	1	0	M	1	0	1	0	0	0	1	0	0	1	0	1	0	0	0
BF003-04	Hub base	1	1	1	0	L	1	0	1	1	0	0	0	0	0	1	0	0	1	0	0
BF003-05	Stator housing	1	3	1	0	L	1	0	0	1	0	0	1	0	0	1	0	1	0	0	0
BF003-06	Stationary plastic blade	1	3	1	1	M	1	0	0	1	1	0	0	0	0	1	0	1	0	0	0
BF003-07	Spring Holder	3	6	0	0	L	0	1	1	0	0	0	0	0	0	1	0	0	0	0	0
BF003-08	Motor	1	2	1	0	H	1	0	0	1	0	0	1	0	0	0	0	0	0	0	0
BF003-09	Spring	3	6	0	0	L	0	1	1	0	1	0	1	0	1	0	0	0	0	0	0
BF004	Filter assembly																				
BF004-01	Filter Housing	1	5	1	0	M	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
BF004-02	Filter Panel	2	7	1	0	L	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0
BF004-03	Mesh-Filter	1	2	1	1	M	1	1	0	1	1	0	0	0	0	0	0	0	1	0	0
BF005	Upper body housing assembly																				
BF005-01	Electronic housing	1	7	1	0	M	1	0	0	1	0	0	0	0	0	0	0	1	1	0	0
BF005-02	Swivel base	1	5	1	0	M	1	0	0	0	0	0	0	0	1	0	1	0	0	0	0
BF005-03	Electronics Component Holder	1	1	0	0	L	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0
BF005-04	Rubber Seal	1	2	1	1	L	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0
BF006	Lower body housing assembly																				
BF006-01	Lower body housing	1	14	1	0	M	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0
BF006-02	Swivel base lock	1	2	2	0	M	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0
BF006-03	Swivel base lock spring	1	2	1	0	L	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0
BF006-04	Base plate lock	1	2	1	0	M	1	0	1	0	0	0	1	0	0	0	0	0	0	0	0
BF006-05	LCD & PCB holder	1	3	0	0	L	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
BF006-06	Power button	1	2	1	1	L	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1
BF006-07	Sensor hatch	1	2	1	0	L	1	1	0	0	0	0	0	0	0	0	0	0	1	0	0
BF006-08	Sensor hatch housing	1	2	0	0	L	1	0	1	0	0	0	0	0	1	0	0	0	0	0	0
BF006-09	Wire holder 1	6	6	0	1	L	0	1	1	0	0	0	0	0	0	0	1	0	0	0	0
BF006-10	Wire holder 2	1	1	0	1	L	0	1	1	0	0	0	0	0	0	0	1	0	0	0	0
BF007	Oscillating motor assembly																				
BF007-01	Motor	1	2	1	0	H	1	0	0	0	0	0	1	0	0	0	0	0	0	0	1
BF007-02	Cam	1	2	1	1	L	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0
BF007-03	180-swivel arm	1	2	1	1	M	1	1	1	0	0	0	0	0	0	0	1	0	0	0	0
BF008	Oscillating base assembly																				
BF008-01	Base plate	1	2	0	0	M	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
BF008-02	Oscillating base	1	5	1	0	M	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
BF008-03	Bearing	1	2	1	0	L	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0
BF008-04	Power cord retainer	1	2	0	1	L	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
BF008-05	Rubber grip	5	1	0	1	L	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0
	Totals	48	118	25	11	0	27	14	17	9	5	0	9	0	5	8	4	8	4	0	3
	Design for Assembly Metrics	**75.2596**		**52.1%**	←Theor. Effy. Pract. Effy.→		**56.3%**	**1.24**		**2.00**			**0.88**				**0.76**				
	Targets																				

Figure 21: Initial DFA Analysis Worksheet

The above chart Figure 21 has a complete DFA analysis. We can observe that the Design for Assembly Metrics score is approximately 75.26. This is a relatively high score. The total number

of parts, including repetitions, is 48, with 118 interfaces. Some reasons for the high score are the many complex parts in various assemblies and the use of many fasteners where they were not needed. We then decided to set a goal to reduce the score to 65 by eliminating parts and making other minor design changes.

Functional Analysis:

The functional analysis assesses the role of each part in the assembly, identifying components that can be standardized or eliminated. In this Dyson Bladeless Fan DFA analysis, a theoretical efficiency of 60% is often the target in many industries. The practical minimum part count was calculated to be lower, with several components potentially being combined or removed to streamline the product without degrading performance. The fan is relatively efficient, but there remains room for improvement. Opportunities to reduce part counts include combining components like wire holders and integrating specific fasteners directly into parts.

Error Proofing:

Error proofing in this assembly involves analyzing the likelihood of mistakes during the assembly process, such as incorrect part alignment or installation. The Dyson Bladeless Fan assembly performed moderately well, with an error-proofing score of 0.86. However, there are opportunities to improve this by reducing the number of fasteners and making components more straightforward to install in the correct order and orientation.

Handling:

Handling refers to the ease with which parts can be manipulated during assembly, considering weight, flexibility, and fragility. The Dyson Bladeless Fan received a handling score of 2.00 in this analysis, indicating moderate difficulty. Some parts, such as the impeller and the base, require careful handling due to their size and alignment requirements. The goal here would be to reduce the number of handling steps or redesign parts for easier manipulation, especially by reducing the number of fasteners.

Insertion:

Insertion examines how easily parts can be placed into their final location within the assembly. This includes considerations such as alignment, visibility, and whether components need to be held in place before fastening. The Dyson Bladeless Fan scored 0.88 for insertion, which is a strong result, indicating that parts generally fit well without excessive force or precision alignment required. However, further improvements could reduce this score by simplifying alignment features or increasing component visibility, allowing for faster, more intuitive assembly.

Secondary Operations:

Secondary operations refer to tasks performed after the main assembly, such as trimming, drilling, or soldering. A lower score indicates fewer operations, meaning a more streamlined process. The

Dyson Bladeless Fan assembly has a secondary operations score of 0.28, reflecting minimal additional steps after completing the main assembly. This low score indicates that the design is efficient in this regard. Still, there are always opportunities to further minimize these operations, mainly by reducing the need for screws or additional fastenings that may require re-orientation or testing.

FORMULAS USED IN DFA ANALYSIS

- DFA Metric= $\sqrt{Number\ of\ Parts * Number\ of\ Interfaces}$
- Theoretical Efficiency= $\frac{Theoretical\ Minimum\ Part}{Number\ of\ Parts}$
- Practical Efficiency= $\frac{Practical\ Minimum\ Part}{Number\ of\ Parts}$
- Error Proofing= $\frac{Assemble\ WronWW\ Part\ or\ OOmit\ Part + Assemble\ Part\ WronWW\ Way\ Around}{Theoretical\ Minimum\ Part}$
- Handling= $\frac{(TanWWle, Nest\ or\ Stick\ ToWWether) + (Flexible, FraWWile, Sharp\ or\ Slippery) + (Pliers, TTTeezers, or\ MaWWnifyinWW\ Glass\ Needed)}{Theoretical\ Minimum\ Part}$
- Insertion= $\frac{Difficult\ to\ AliWWn + HoldinW\ W\ DoTTn\ Required + Resistance\ to\ Insertion + OObstructed\ Access}{Theoretical\ Minimum\ Part}$
- Secondary Operation= $\frac{Reorient\ \mathbf{WW}orkpiece + (Scre\mathbf{SS}, Drill, T\mathbf{SS}ist, Rivet, Bend\ or\ Crimp) + (\mathbf{WW}eld, Solder\ or\ GLue) + (Paint, Lube, Heat, Apply\ Li\mathbf{LL}uid\ or\ Gas) + (Test, Measure\ or\ Ad\mathbf{AA}ust)}{Theoretical\ Minimum\ Part}$

ECONOMIC ANALYSIS

Bladeless fans have higher manufacturing costs due to advanced technology and materials. Due to the unique design and perceived benefits, these fans are marketed at higher prices. So, the analysis is done based on the various costs involved.

After material and process selection is accomplished, cost analysis is done to make sound technological choices and judge the suitability of suppliers' bids. This analysis helped us determine whether our redesign made financial sense. Various cost estimators were considered to determine the possible product cost.

The cost estimators that were considered are as follows:

Order of Magnitude (OME) Estimates:

This was used to roughly guess the price of each component according to the 1:3:9 rule. Initially, material costs for components were calculated using the weight, price per pound of the material, and waste factor as follows:

Material Cost = (Waste Factor * Weight * Price per pound)

Manufacturing Cost = (3 * Material Cost) Selling Price = (9 * Material Cost)

The tables below, Table 2 and Table 3, show a detailed analysis of each part's OME and break-even values. We assumed that we would have fixed costs of $80000.

FORMULAS

ECONOMIC ANALYSIS

- $C_m=\frac{m*c_m}{1-f}$, where **C_m= Unit cost of material**, **m= mass of part, c_m=cost of material, f=fraction of material scrap**.
- $C_L=\frac{c_{ww}}{n'}$, where **C_L= Unit Labor Cost, c_w= hourly cost of wages and benefits,** n' = ppppppppppppppppppp pprrpprr.
- $C_T=\frac{ct \cdot k}{n}$ where, **C_T= Tooling Cost, c_t= cost of making tools, n= production run, k= tooling wear factor** (**k= n**/by life of tooling raised to next highest integer).
- $C_E=(\frac{1}{n'}) \cdot (\frac{c_e}{Lt_{wwo}}) \cdot q$, **where c_e= cost of capital equipment, n'= production rate, L= Load Factor, t_{wo}= capital write-off time, q= fraction of equipment sharing between products.**
- $C_{OH}=\frac{cOOH}{n'}$, **where c_{OH}= overhead hourly rate, n'= production rate**.
- **C_U= Cm+C_L+C_T+C_E+C_{OH}**

BREAK-EVEN ANALYSIS

$$Q_b=\frac{f}{P-v}$$

Where, Q_b= Break-Even Point

f = Fixed Costs

v= Variable Costs

P= Sales Price

Assumptions made in analysis:

For the initial economic analysis, we assumed fixed costs of $80000 for all the components to calculate the break-even points of individual components and the entire product. All the values are shown in Tables 2 and 3 below. For capital write-off time in the sheet, we assumed operations for 45 hours each for 52 weeks, and we amortized the components over a period of 5 and 10 years based on their complexity and size. We also assumed a total product profit of $200 for product break-even analysis.

Table 2: OME and Breakeven Analysis

Parts	Part nos.	Material cost	Labor cost	Manufacturing cost	Selling price	Break-even analysis
Amplifier head	BF001-01	2.02	0.63	5.9	17.69	532
Amplifier housing	BF001-02	1.69	0.56	4.6	13.71	698
Release lock Housing	BF001-03	0.02	0.06	0.4	1.09	7919
Release lock	BF001-04	0.03	0.33	1.5	4.58	1897
Spring	BF001-05	0.00	0.05	0.3	0.77	11144
Motor Bucket retaining ring	BF002	0.11	0.13	0.7	2.07	4362
Impeller	BF003-01	0.33	0.63	4.6	13.77	624
Impeller top housing	BF003-02	0.52	0.63	3.7	11.04	808
Impeller bottom housing	BF003-03	0.33	0.63	3.4	10.18	867
Hub base	BF003-04	0.09	0.25	1.1	3.41	2608
Stator housing	BF003-05	0.33	0.63	3.5	10.47	841
Stationary plastic blade	BF003-06	0.18	0.63	3.6	10.92	791
Spring Holder	BF003-07	0.01	0.42	1.4	4.34	2046
Spring	BF003-09	0.00	0.33	1.6	4.82	1786
Filter Housing	BF004-01	0.35	0.36	2.5	7.37	1200
Filter Panel	BF004-02	1.78	0.63	5.0	15.00	635
Mesh-Filter	BF004-03	1.75	0.56	4.7	14.07	680
Electronic housing	BF005-01	1.38	0.56	4.2	12.65	747
Swivel base	BF005-02	1.58	0.63	5.1	15.40	606
Lower body housing	BF006-01	0.35	0.63	3.3	9.99	887
Swivel base lock	BF006-02	0.00	0.45	2.0	6.14	1407
Swivel base lock spring	BF006-03	0.00	0.42	1.7	5.17	1684
Base plate lock	BF006-04	0.16	0.42	2.1	6.35	1386
LCD & PCB holder	BF006-05	2.02	0.66	5.6	16.82	566
Power button	BF006-06	0.01	0.45	1.9	5.65	1543
Sensor hatch	BF006-07	0.11	0.45	2.3	6.94	1255
Sensor hatch housing	BF006-08	0.11	0.45	2.3	6.94	1255
Wire holder 1	BF006-09	0.01	0.42	1.8	5.29	1643
Wire holder 2	BF006-10	0.01	0.42	1.8	5.29	1643
Cam	BF007-02	0.01	0.42	1.7	5.18	1680
180-swivel arm	BF007-03	0.03	0.50	1.9	5.65	1561
Base plate	BF008-01	0.53	0.42	2.5	7.53	1216
Oscillating base	BF008-02	1.70	0.63	4.9	14.56	654
Bearing	BF008-03	0.00	0.45	1.9	5.63	1546
Power cord retainer	BF008-04	0.01	0.13	0.6	1.68	5149
Rubber grip	BF008-05	0.00	0.04	0.3	0.98	8466
	Total	**17.56**	**15.94**	**96.4**	**289.14**	

Table 3: Break-even Analysis for Product

Overall assembly unit	
Assumption	
Fixed cost ($)	80000
Variable cost ($)	96
Sales price Incl. Profit ($)	489
Break-even (Units)	204 units

MATERIAL ANALYSIS

Once we finished the economic analysis, we conducted a material analysis to determine the materials used in our fan. We found that most of the materials in our product were PC/ABS. A few components are made of POM and ABS. Off-the-shelf parts, like springs and the filter, are made of Borosilicate, Aluminum, and Stainless steel.

A comparison analysis was done to determine if other materials could be used in the product. We used an Ashby chart to determine materials similar in strength that could be used in our product. The Ashby charts for the different materials are shown in Figures 22-27 below.

- $\frac{\sigma_f^{22/33}}{\rho}$
- $\frac{E^{1/22}}{\rho}$

E= Modulus of Elasticity, $\sigma_f = Failure\ Strength, \rho = Density$

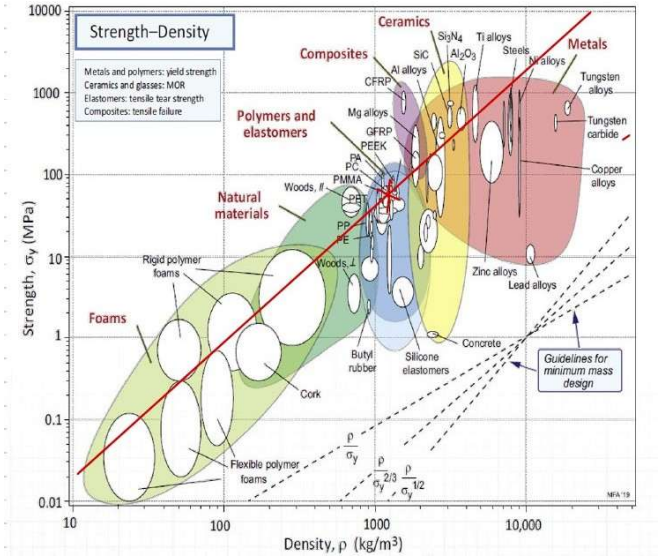


Figure 22: Ashby Chart-PC/ABS

- $\frac{\sigma_f^{22/33}}{\rho}$
- $\frac{E^{1/22}}{\rho}$

E= Modulus of Elasticity, $\sigma_f = Failure\ Strength, \rho = Density$

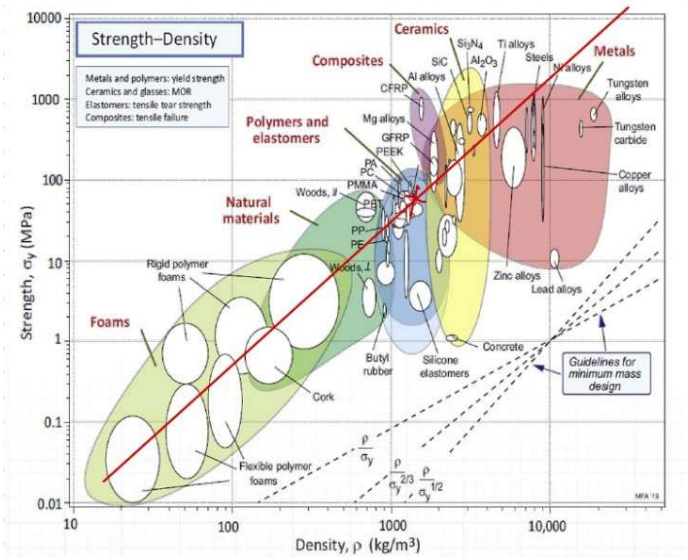


Figure 23: Ashby Chart-POM

- $\frac{\sigma_f^{22/33}}{\rho}$
- $\frac{E^{1/22}}{\rho}$

E= Modulus of Elasticity, $\sigma_f = Failure\ Strength, \rho = Density$

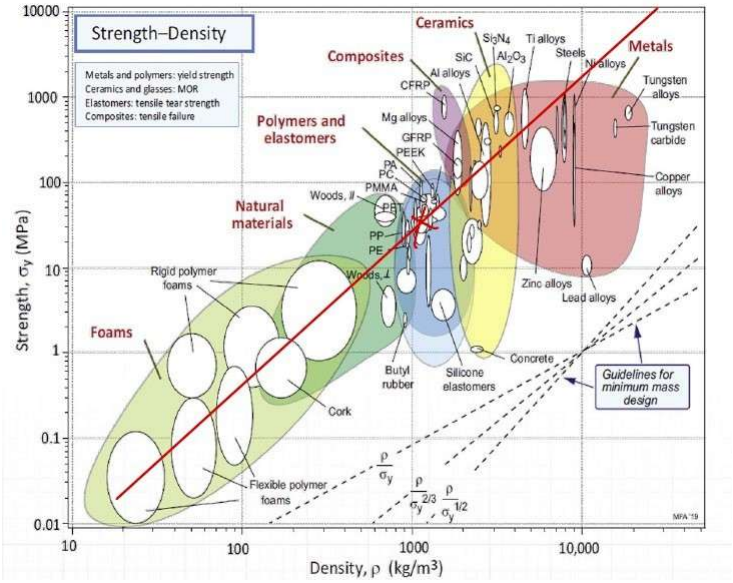


Figure 24: Ashby Chart-ABS

- $\frac{\sigma_f^{22/33}}{\rho}$
- $\frac{E^{1/22}}{\rho}$

E= Modulus of Elasticity, $\sigma_f = Failure\ Strength, \rho = Density$

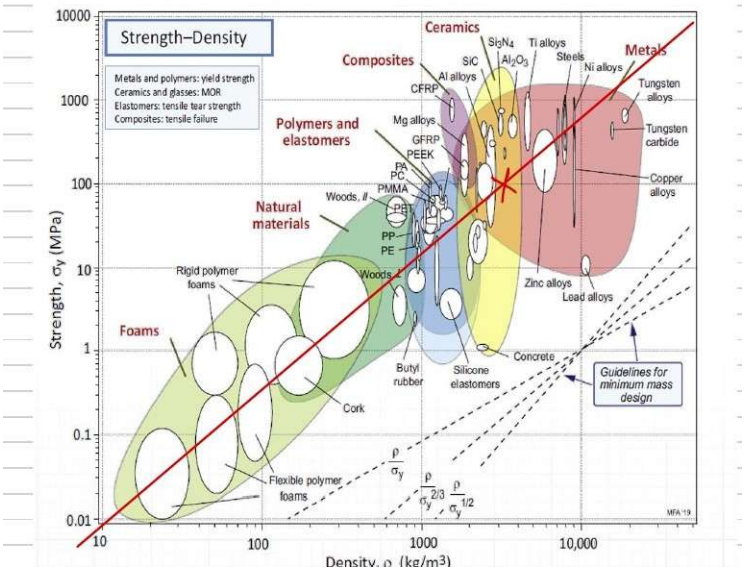


Figure 25: Ashby Chart-Aluminum

- $\frac{\sigma_f^{22/33}}{\rho}$
- $\frac{E^{1/22}}{\rho}$

E= Modulus of Elasticity, $\sigma_f = Failure\ Strength, \rho = Density$

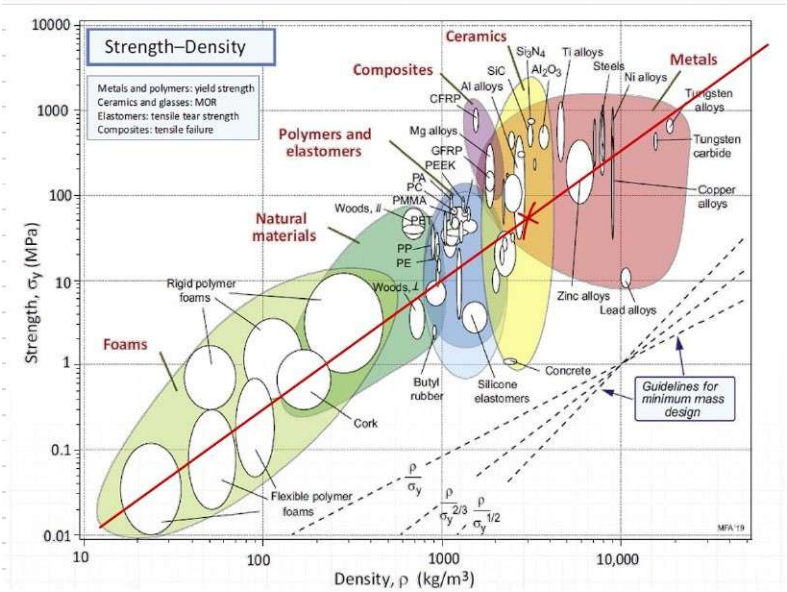


Figure 26: Ashby Chart-Borosilicate

- $\frac{\sigma_f^{22/33}}{\rho}$
- $\frac{E^{1/22}}{\rho}$

E= Modulus of Elasticity, $\sigma_f = Failure\ Strength, \rho = Density$

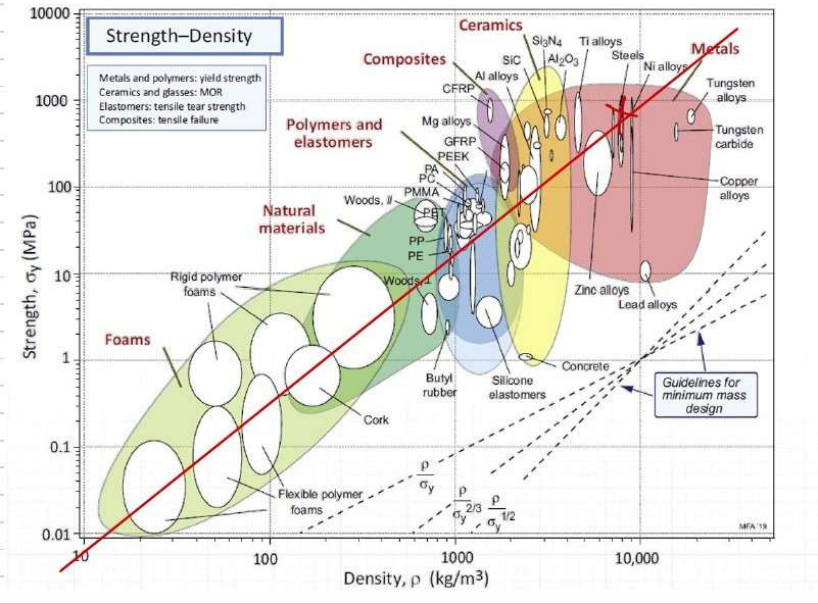


Figure 27: Ashby Chart-Stainless Steel

1. Amplifier Service Assembly (PC/ABS)

Materials Considered:

- PC/ABS (Baseline)
- Polypropylene (PP)
- Nylon
- Polycarbonate (PC)
- Acrylic (PMMA)
- Polyethylene (PE)
- Polyurethane (PU)
- ABS

Ashby Chart Ratio (Strength-to-Weight Formula):

Table 4: Ashby Chart Ratio-Amplifier Assembly

Material	Strength (MPa)	Density (kg/m³)	Ashby Ratio (strength^2/3 / density)
PC/ABS	60	1100	0.099
Polypropylene (PP)	40	900	0.094
Nylon	80	1150	0.116
Polycarbonate (PC)	70	1200	0.110
Acrylic (PMMA)	65	1180	0.107
Polyethylene (PE)	35	950	0.083
Polyurethane (PU)	50	1200	0.089
ABS	45	1040	0.094

Decision Matrix:

Table 5: Decision Matrix- Amplifier Assembly

Material	Strength-to-Weight	Cost	Manufacturability	Durability	Corrosion Resistance	Weighted Score
PC/ABS	8	7	9	8	7	7.9
PP	6	8	8	6	5	6.6
Nylon	9	6	7	9	8	7.8
PC	7	5	6	7	6	6.2
PMMA	6	5	7	6	5	5.8
PE	5	9	9	5	6	6.8
PU	7	4	5	7	7	6.0
ABS	6	7	9	7	6	7.0

Pugh Matrix:

Table 6: Pugh Matrix-Amplifier Assembly

Criteria	Baseline (PC/ABS)	PP	Nylon	PC	PMMA	PE	PU	ABS
Strength-to-Weight	0	-	+	-	-	-	-	-

Cost	0	+	-	-	-	+	-	0
Manufacturability	0	-	-	-	-	0	-	0
Durability	0	-	+	-	-	-	-	-
Corrosion Resistance	0	-	+	-	-	-	0	-
Overall	**Reference**	-	+	-	-	-	-	-

Conclusion regarding Amplifier Service Assembly:

From Figure 22, PC/ABS remains the optimal material for the amplifier housing due to its superior balance of strength, manufacturability, cost, and durability. While materials like nylon offer higher strength and corrosion resistance, PC/ABS provides the best overall performance in production feasibility and cost-effectiveness.

2. **Motor and Bucket Assembly**

Materials Considered for Impeller:

- PC/ABS (Baseline)
- Nylon 6
- Polyoxymethylene (POM)
- Aluminum
- Stainless Steel
- Polypropylene (PP)
- PEEK
- ABS

Ashby Chart Ratio (Strength-to-Weight Formula):

Table 7: Ashby Chart Ratio- Motor and Bucket Assembly

Material	**Strength (MPa)**	**Density (kg/m³)**	**Ashby Ratio (strength^2/3 / density)**
PC/ABS	60	1100	0.099
Nylon 6	80	1150	0.116
POM	70	1400	0.100
Aluminum	270	2700	0.170

Stainless Steel	600	7900	0.111
PP	35	950	0.083
PEEK	100	1320	0.120
ABS	45	1040	0.094

Decision Matrix for MB Assembly:

Table 8: Decision Matrix- Motor and Bucket Assembly

Material	Strength-to Weight	Cost	Manufacturability	Durability	Corrosion Resistance	Weighted Score
PC/ABS	7	8	9	8	7	7.8
Nylon 6	8	6	7	9	7	7.5
POM	7	7	8	8	6	7.3
Aluminum	10	5	6	9	9	7.8
Stainless Steel	9	4	5	10	10	7.6
Polypropylene	5	9	8	6	6	6.8
PEEK	9	4	6	10	10	7.8
ABS	6	8	9	7	6	7.2

Pugh Matrix for MB Assembly:

Table 9: Pugh Matrix- Motor and Bucket Assembly

Criteria	Baseline (PC/ABS)	Nylon 6	POM	Aluminum	Stainless Steel	PP	PEEK	ABS
Strength-to-Weight	**0**	+	**0**	++	++	-	++	-
Cost	**0**	-	**0**	-	--	++	--	**0**
Manufacturability	**0**	**0**	+	-	-	**0**	**0**	+

Durability	**0**	+	**0**	+	++	-	++	-
Corrosion Resistance	**0**	**0**	-	+	++	-	++	-
Overall	**Reference**	+	**0**	++	++	**0**	++	-

Conclusion: For the motor and bucket assembly, from Figure 25, **Aluminum** and **PEEK** present strong alternatives to **PC/ABS** due to their high strength and durability, but their higher cost and more complex manufacturability make **PC/ABS** or **POM** more practical choices in terms of balancing performance with cost-effectiveness. **Nylon 6** also shows good promise for parts requiring additional toughness.

3. Filter Assembly

Components:

- Filter (Borosilicate Glass Fibers)
- Filter Holder (Aluminum)
- Seal (Rubber)

3.1 Filter Material Options and Ashby Chart Ratio Calculation:

Table 10: Ashby Chart Ratios-Filter

Material	Strength (MPa)	Density (kg/m³)	Ashby Ratio (strength^2/3 / density)
Borosilicate Glass	50	2230	0.063
Fiberglass	30	2550	0.045
Quartz	70	2650	0.077
Aluminum Oxide	150	3950	0.091

Decision Matrix for Filter:

Table 11: Decision Matrix- Filter

Material	Strength-to-Weight	Cost	Manufacturability	Durability	Corrosion Resistance	Weighted Score
Borosilicate Glass	7	5	6	10	10	7.2
Fiberglass	6	6	8	7	8	7.0
Quartz	9	4	5	9	10	7.4
Aluminum Oxide	10	2	4	10	9	7.0

Pugh Matrix for Filter:

Table 12: Pugh Matrix-Filter

Criteria	Baseline (Borosilicate Glass)	Fiberglass	Quartz	Aluminum Oxide
Strength-to-Weight	**0**	**-**	+	++
Cost	**0**	+	**-**	**--**
Manufacturability	**0**	+	**-**	**-**
Durability	**0**	**-**	++	++
Corrosion Resistance	**0**	**0**	+	+
Overall	**Reference**	**0**	+	**0**

Conclusion:

From Figure 26, Borosilicate Glass Fiber remains the best option due to its high durability, corrosion resistance, and suitability for the filter's role in high-temperature environments. Alternatives like fiberglass or aluminum oxide might offer higher strength, but borosilicate balances cost and performance.

3.2 Filter Holder (Aluminum) Material Options

Material Options and Ashby Chart Ratio Calculation:

Table 13: Ashby Chart Ratios- Filter Holder

Material	Strength (MPa)	Density (kg/m³)	Ashby Ratio (strength^2/3 / density)
Aluminum 6061	270	2700	0.170

Magnesium	160	1800	0.177
Titanium	1000	4500	0.222
Steel	600	7900	0.111

Decision Matrix for Filter Holder:

Table 14: Decision Matrix-Filter Holder

Material	Strength-to-Weight	Cost	Manufacturability	Durability	Corrosion Resistance	Weighted Score
Aluminum (6061)	8	6	9	8	9	7.8
Magnesium	7	7	8	7	6	7.0
Titanium	10	3	6	10	10	7.8
Steel	9	4	6	9	8	7.2

Pugh Matrix for Filter Holder:

Table 15: Pugh Matrix- Filter Holder

Criteria	Baseline (Aluminum)	Magnesium	Steel	Titanium
Strength-to-Weight	**0**	+	-	++
Cost	**0**	**0**	+	--
Manufacturability	**0**	**0**	-	-
Durability	**0**	-	+	++
Corrosion Resistance	**0**	-	**0**	++
Overall	**Reference**	**0**	**0**	+

Conclusion:

Aluminum 6061 remains the preferred choice due to its excellent strength-to-weight ratio, cost-effectiveness, and ease of manufacturability. Titanium, while offering greater strength, is cost-prohibitive for this application, and magnesium, while lightweight, poses challenges in durability.

4. Lower Body Housing Assembly

Materials Considered:

- PC/ABS (Baseline)

- POM
- Polypropylene (PP)
- ABS
- Nylon 6
- Aluminum
- Polycarbonate (PC)

Ashby Chart Ratio (Strength-to-Weight Formula):

Table 16: Ashby Chart Ratios- Lower Body Housing Assembly

Material	Strength (MPa)	Density (kg/m³)	Ashby Ratio (strength^2/3 / density)
PC/ABS	60	1100	0.099
POM	70	1400	0.100
Polypropylene	35	950	0.083
ABS	45	1040	0.094
Nylon 6	80	1150	0.116
Aluminum	270	2700	0.170
PC	70	1200	0.110

Decision Matrix for Lower Body Housing:

Table 17: Decision Matrix-Lower Body Housing Assembly

Material	Strength -to- Weight	Cost	Manufacturability	Durability	Corrosion Resistance	Weighted Score
PC/ABS	7	8	9	8	7	7.8
POM	7	7	8	8	6	7.3
Polypropylene	5	9	8	6	6	6.8
ABS	6	8	9	7	6	7.2
Nylon 6	8	6	7	9	7	7.5
Aluminum	10	5	6	9	9	7.8

Polycarbonate	7	5	8	7	6	6.6

Pugh Matrix for Lower Body Housing:

Table 18: Pugh Matrix-Lower Body Housing Assembly

Criteria	Baseline (PC/ABS)	POM	PP	ABS	Nylon 6	Aluminum	PC
Strength-to-Weight	**0**	+	-	-	+	++	+
Cost	**0**	**0**	++	+	-	--	**0**
Manufacturability	**0**	**0**	+	+	**0**	-	**0**
Durability	**0**	+	-	**0**	+	++	**0**
Corrosion Resistance	**0**	**0**	**0**	-	**0**	++	-
Overall	**Reference**	+	**0**	**0**	+	++	**0**

Conclusion: PC/ABS is the most balanced choice for the lower body housing assembly due to its ease of manufacturability and cost. Alternatives like POM and Nylon 6 could be considered for parts requiring higher toughness, but Aluminum, while strong, is less practical due to higher cost and manufacturing complexity.

5. Upper Body Housing Assembly

Materials Considered:

- PC/ABS (Baseline)
- Polycarbonate (PC)
- ABS
- Polypropylene (PP)
- Nylon 6
- Aluminum

Ashby Chart Ratio (Strength-to-Weight Formula):

Table 19: Ashby Chart Ratios- Upper Body Housing Assembly

Material	Strength (MPa)	Density (kg/m³)	Ashby Ratio (strength^2/3 / density)
PC/ABS	60	1100	0.099
Polycarbonate	70	1200	0.110
ABS	45	1040	0.094
Polypropylene	35	950	0.083
Nylon 6	80	1150	0.116
Aluminum	270	2700	0.170

Decision Matrix for Upper Body Housing:

Table 20: Decision Matrix-Upper Body Housing Assembly

Material	Strength-to-Weight	Cost	Manufacturability	Durability	Corrosion Resistance	Weighted Score
PC/ABS	7	8	9	8	7	7.8
Polycarbonate	7	6	8	7	6	6.8
ABS	6	8	9	7	6	7.2
Polypropylene	5	9	8	6	6	6.8
Nylon 6	8	6	7	9	7	7.5
Aluminum	10	5	6	9	9	7.8

Pugh Matrix for Upper Body Housing:

Table 21: Pugh Matrix- Upper Body Housing Assembly

Criteria	Baseline (PC/ABS)	Polycarbonate	ABS	Polypropylene	Nylon 6	Aluminum
Strength-to-Weight	**0**	+	-	-	+	++
Cost	**0**	-	+	++	-	--
Manufacturability	**0**	**0**	+	+	**0**	-

Durability	**0**	**-**	**0**	**-**	+	++
Corrosion Resistance	**0**	**0**	**-**	**-**	**0**	++
Overall	**Reference**	**-**	**0**	+	+	++

Conclusion: PC/ABS remains the most practical material for upper body housing due to its manufacturability, cost, and balanced strength properties. Nylon 6 and Aluminum offer better strength and durability but are more expensive or difficult to manufacture, making them less ideal for large-scale production.

6. Oscillating Base Assembly

Materials Considered:

- PC/ABS (Baseline)
- Aluminum
- POM
- Polypropylene (PP)
- ABS

Ashby Chart Ratio (Strength-to-Weight Formula):

Table 22: Ashby Ratios- Oscillating Base Assembly

Material	Strength (MPa)	Density (kg/m³)	Ashby Ratio (strength^2/3 / density)
PC/ABS	60	1100	0.099
Aluminum	270	2700	0.170
POM	70	1400	0.100
Polypropylene	35	950	0.083
ABS	45	1040	0.094

Decision Matrix for Oscillating Base:

Table 23: Decision Matrix-Oscillating Base Assembly

Material	Strength-to-Weight	Cost	Manufacturability	Durability	Corrosion Resistance	Weighted Score
PC/ABS	7	8	9	8	7	7.8
Aluminum	10	5	6	9	9	7.8
POM	7	7	8	8	6	7.3
Polypropylene	5	9	8	6	6	6.8
ABS	6	8	9	7	6	7.2

Pugh Matrix for Oscillating Base:

Table 24: Pugh Matrix-Oscillating Base Assembly

Criteria	Baseline (PC/ABS)	Aluminum	POM	Polypropylene	ABS
Strength-to-Weight	**0**	++	+	-	-
Cost	**0**	--	**0**	++	+
Manufacturability	**0**	-	+	**0**	+
Durability	**0**	++	**0**	-	**0**
Corrosion Resistance	**0**	++	-	-	-
Overall	**Reference**	++	+	**0**	**0**

Conclusion: PC/ABS remains the most efficient material for the oscillating base due to its balance of strength, cost, and manufacturability. Aluminum offers superior strength and durability but is more expensive and complicated to manufacture.

7. Oscillating Motor Assembly

Materials Considered:

- PC/ABS (Baseline)
- Aluminum
- POM
- ABS
- Nylon 6

Ashby Chart Ratio (Strength-to-Weight Formula):

Table 25: Ashby Ratios- Oscillating Motor Assembly

Material	Strength (MPa)	Density (kg/m³)	Ashby Ratio (strength^2/3 / density)
PC/ABS	60	1100	0.099
Aluminum	270	2700	0.170
POM	70	1400	0.100
ABS	45	1040	0.094
Nylon 6	80	1150	0.116

Decision Matrix for Oscillating Motor:

Table 26: Decision Matrix- Oscillating Motor Assembly

Material	Strength -to- Weight	Cost	Manufacturability	Durability	Corrosion Resistance	Weighted Score
PC/ABS	**7**	**8**	**9**	**8**	**7**	**7.8**
Aluminum	**10**	**5**	**6**	**9**	**9**	**7.8**
POM	**7**	**7**	**8**	**8**	**6**	**7.3**
ABS	**6**	**8**	**9**	**7**	**6**	**7.2**
Nylon 6	**8**	**6**	**7**	**9**	**7**	**7.5**

Pugh Matrix for Oscillating Motor:

Table 27: Pugh Matrix-Oscillating Motor Assembly

Criteria	Baseline (PC/ABS)	Aluminum	POM	ABS	Nylon 6
Strength-to-Weight	**0**	++	+	-	+
Cost	**0**	--	**0**	+	-
Manufacturability	**0**	-	+	+	**0**
Durability	**0**	++	**0**	**0**	+
Corrosion Resistance	**0**	++	-	-	**0**
Overall	**Reference**	++	+	**0**	+

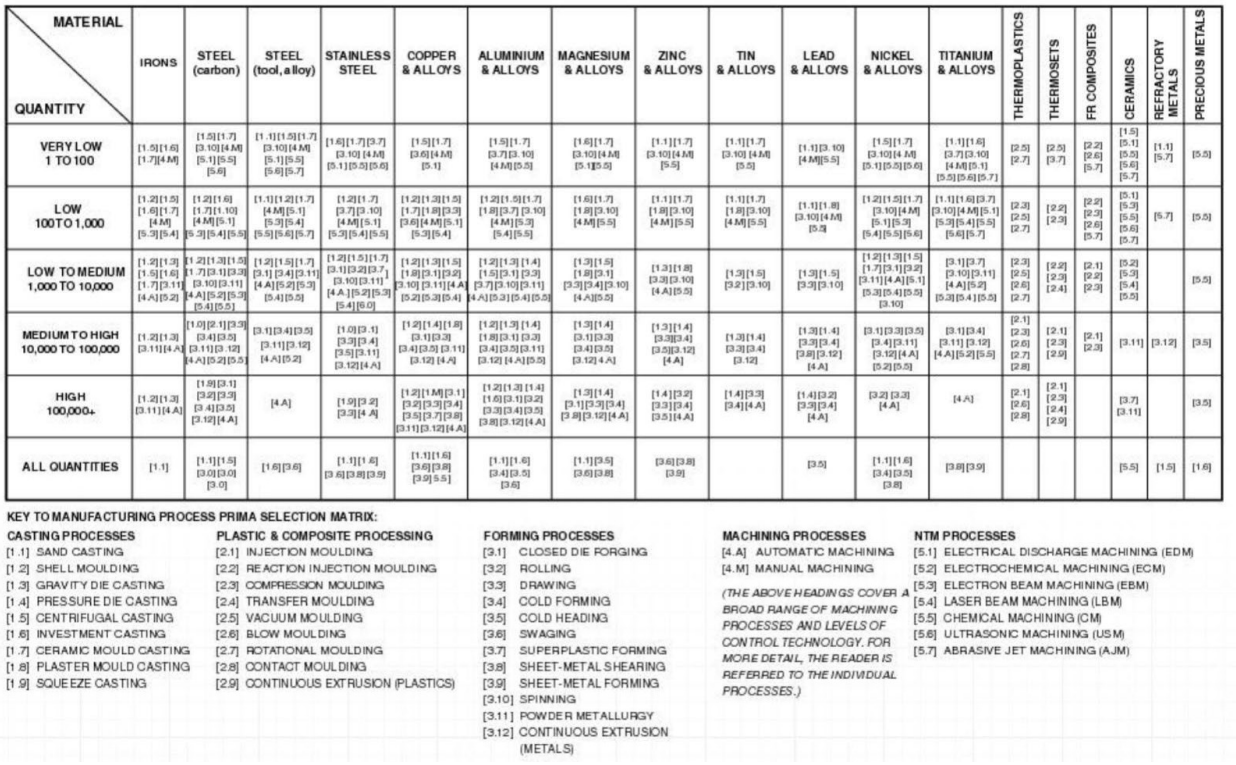
Conclusion: PC/ABS remains the most balanced material for the oscillating motor assembly due to its manufacturability and cost. Aluminum provides superior strength but at a higher price and complexity of manufacturing, making it less practical.

MANUFACTURING PROCESS ANALYSIS

The manufacturing process analysis for the Dyson Bladeless Fan highlights the strategic use of materials and manufacturing techniques to optimize both performance and cost efficiency. Injection molding is the primary method for producing complex plastic components made from PC/ABS and POM, chosen for their durability, ease of molding, and cost-effectiveness in mass production. Stainless steel, used in springs and bearings, undergoes wire drawing and forging to ensure strength and precision for parts that endure high mechanical stress. Borosilicate glass components, like the mesh filter, are formed and annealed to provide excellent thermal and chemical resistance. Overall, the manufacturing processes selected align with the functional demands of the product, emphasizing efficiency, high-volume production, and quality control.

Based on the chart below, injection molding remains the optimal choice for PC/ABS components like housing and impeller due to its suitability for high-volume production (economic batch size 10^3 to 10^7) and ability to form complex shapes. However, **rotational molding** could be an option for more significant components with thicker walls, although it is more economical for lower production volumes (10^2 to 10^4). For **stainless steel** parts such as the springs and bearings, **forging** is a durable option for high-strength components with batch sizes ranging from 10^3 to 10^5; investment **casting** could be considered for smaller, more intricate parts. **Borosilicate glass**

components, such as the mesh filter, are typically manufactured through **glass forming and annealing**. Still, powder methods may be viable for small-scale production, especially if more complex geometries or precise control over porosity are required. Each process aligns with the specific material properties and desired production scales for the different subassemblies. A detailed manufacturing process analysis is given below for each of the other materials we have in our product.



MATERIAL / QUANTITY	IRONS	STEEL (carbon)	STEEL (tool, alloy)	STAINLESS STEEL	COPPER & ALLOYS	ALUMINIUM & ALLOYS	MAGNESIUM & ALLOYS	ZINC & ALLOYS	TIN & ALLOYS	LEAD & ALLOYS	NICKEL & ALLOYS	TITANIUM & ALLOYS	THERMOPLASTICS	THERMOSETS	FR COMPOSITES	CERAMICS	REFRACTORY METALS	PRECIOUS METALS
VERY LOW 1 TO 100	[1.5] [1.6] [1.7] [4.M]	[1.5] [1.7] [3.10] [4.M] [5.1] [5.5] [5.6]	[1.1] [1.5] [1.7] [3.10] [4.M] [5.1] [5.5] [5.6] [5.7]	[1.6] [1.7] [3.7] [3.10] [4.M] [5.1] [5.5] [5.6]	[1.5] [1.7] [3.6] [4.M] [5.1]	[1.5] [1.7] [3.7] [3.10] [4.M] [5.5]	[1.6] [1.7] [3.10] [4.M] [5.1] [5.5]	[1.1] [1.7] [3.10] [4.M] [5.5]	[1.1] [1.7] [3.10] [4.M] [5.5]	[1.1] [3.10] [4.M] [5.5]	[1.5] [1.7] [3.10] [4.M] [5.1] [5.5] [5.6]	[1.1] [1.6] [3.7] [3.10] [4.M] [5.1] [5.5] [5.6] [5.7]	[2.5] [2.7]	[2.5] [3.7]	[2.2] [2.6] [5.7]	[1.5] [5.1] [5.5] [5.6] [5.7]	[1.1] [5.7]	[5.5]
LOW 100 TO 1,000	[1.2] [1.5] [1.6] [1.7] [4.M] [5.3] [5.4]	[1.2] [1.6] [1.7] [1.10] [4.M] [5.1] [5.3] [5.4] [5.5]	[1.1] [1.2] [1.7] [4.M] [5.1] [5.3] [5.4] [5.5] [5.6] [5.7]	[1.2] [1.7] [3.7] [3.10] [4.M] [5.1] [5.3] [5.4] [5.5]	[1.2] [1.3] [1.5] [1.7] [1.8] [3.3] [3.6] [4.M] [5.1] [5.3] [5.4]	[1.2] [1.5] [1.7] [1.8] [3.7] [3.10] [4.M] [5.3] [5.4] [5.5]	[1.6] [1.7] [1.8] [3.10] [4.M] [5.5]	[1.1] [1.7] [1.8] [3.10] [4.M] [5.5]	[1.1] [1.7] [1.8] [3.10] [4.M] [5.5]	[1.1] [1.8] [3.10] [4.M] [5.3]	[1.2] [1.5] [1.7] [3.10] [4.M] [5.1] [5.3] [5.4] [5.5] [5.6]	[1.1] [1.6] [3.7] [3.10] [4.M] [5.1] [5.3] [5.4] [5.5] [5.6] [5.7]	[2.3] [2.5] [2.7]	[2.2] [2.3]	[2.2] [2.3] [2.6] [5.7]	[5.1] [5.3] [5.5] [5.6] [5.7]	[5.7]	[5.5]
LOW TO MEDIUM 1,000 TO 10,000	[1.2] [1.3] [1.5] [1.6] [1.7] [3.11] [4.A] [5.2]	[1.2] [1.3] [1.5] [1.7] [3.1] [3.3] [3.10] [3.11] [4.A] [5.2] [5.3] [5.4] [5.5]	[1.2] [1.5] [1.7] [3.1] [3.4] [3.11] [4.A] [5.2] [5.3] [5.4] [5.5]	[1.2] [1.5] [1.7] [3.1] [3.2] [3.7] [3.10] [3.11] [4.A] [5.2] [5.3] [5.4] [6.0]	[1.2] [1.3] [1.5] [1.8] [3.1] [3.2] [3.10] [3.11] [4.A] [5.2] [5.3] [5.4]	[1.2] [1.3] [1.4] [1.5] [3.1] [3.3] [3.7] [3.10] [3.11] [4.A] [5.3] [5.4] [5.5]	[1.3] [1.5] [1.8] [3.1] [3.3] [3.4] [3.10] [4.A] [5.5]	[1.3] [1.8] [3.3] [3.10] [4.A] [5.5]	[1.3] [1.5] [3.2] [3.10]	[1.3] [1.5] [3.3] [3.10]	[1.2] [1.3] [1.5] [1.7] [3.1] [3.2] [3.11] [4.A] [5.1] [5.3] [5.4] [5.5] [3.10]	[3.1] [3.7] [3.10] [3.11] [4.A] [5.2] [5.3] [5.4] [5.5]	[2.3] [2.5] [2.6] [2.7]	[2.2] [2.3] [2.4]	[2.1] [2.2] [2.3]	[5.2] [5.3] [5.4] [5.5]		[5.5]
MEDIUM TO HIGH 10,000 TO 100,000	[1.2] [1.3] [3.11] [4.A]	[1.0] [2.1] [3.3] [3.4] [3.5] [3.11] [3.12] [4.A] [5.2] [5.5]	[3.1] [3.4] [3.5] [3.11] [3.12] [4.A] [5.2]	[1.0] [3.1] [3.3] [3.4] [3.5] [3.11] [3.12] [4.A]	[1.2] [1.4] [1.8] [3.1] [3.3] [3.4] [3.5] [3.11] [3.12] [4.A]	[1.2] [1.3] [1.4] [1.8] [3.1] [3.3] [3.4] [3.5] [3.11] [3.12] [4.A] [5.5]	[1.3] [1.4] [3.1] [3.3] [3.4] [3.5] [3.12] [4.A]	[1.3] [1.4] [3.3] [3.4] [3.5] [3.12] [4.A]	[1.3] [1.4] [3.3] [3.4] [3.12]	[1.3] [1.4] [3.3] [3.4] [3.9] [3.12] [4.A]	[3.1] [3.3] [3.5] [3.4] [3.11] [3.12] [4.A] [5.2] [5.5]	[3.1] [3.4] [3.11] [3.12] [4.A] [5.2] [5.5]	[2.1] [2.3] [2.6] [2.7] [2.8]	[2.1] [2.3] [2.9]	[2.1] [2.3]	[3.11]	[3.12]	[3.5]
HIGH 100,000+	[1.2] [1.3] [3.11] [4.A]	[1.9] [3.1] [3.2] [3.3] [3.4] [3.5] [3.12] [4.A]	[4.A]	[1.9] [3.2] [3.3] [4.A]	[1.2] [1.M] [3.1] [3.2] [3.3] [3.4] [3.5] [3.7] [3.8] [3.11] [3.12] [4.A]	[1.2] [1.3] [1.4] [1.6] [3.1] [3.2] [3.3] [3.4] [3.5] [3.8] [3.12] [4.A]	[1.3] [1.4] [3.1] [3.3] [3.4] [3.8] [3.12] [4.A]	[1.4] [3.2] [3.3] [3.4] [3.5] [4.A]	[1.4] [3.3] [3.4] [4.A]	[1.4] [3.2] [3.3] [3.4] [4.A]	[3.2] [3.3] [4.A]	[4.A]	[2.1] [2.6] [2.8]	[2.1] [2.3] [2.4] [2.9]		[3.7] [3.11]		[3.5]
ALL QUANTITIES	[1.1]	[1.1] [1.5] [3.0] [3.0] [3.0]	[1.6] [3.6]	[1.1] [1.6] [3.6] [3.8] [3.9]	[1.1] [1.6] [3.6] [3.8] [3.9] [5.5]	[1.1] [1.6] [3.4] [3.5] [3.6]	[1.1] [3.5] [3.6] [3.8]	[3.6] [3.8] [3.9]		[3.5]	[1.1] [1.6] [3.4] [3.5] [3.8]	[3.8] [3.9]				[5.5]	[1.5]	[1.6]

KEY TO MANUFACTURING PROCESS PRIMA SELECTION MATRIX:

CASTING PROCESSES
[1.1] SAND CASTING
[1.2] SHELL MOULDING
[1.3] GRAVITY DIE CASTING
[1.4] PRESSURE DIE CASTING
[1.5] CENTRIFUGAL CASTING
[1.6] INVESTMENT CASTING
[1.7] CERAMIC MOULD CASTING
[1.8] PLASTER MOULD CASTING
[1.9] SQUEEZE CASTING

PLASTIC & COMPOSITE PROCESSING
[2.1] INJECTION MOULDING
[2.2] REACTION INJECTION MOULDING
[2.3] COMPRESSION MOULDING
[2.4] TRANSFER MOULDING
[2.5] VACUUM MOULDING
[2.6] BLOW MOULDING
[2.7] ROTATIONAL MOULDING
[2.8] CONTACT MOULDING
[2.9] CONTINUOUS EXTRUSION (PLASTICS)

FORMING PROCESSES
[3.1] CLOSED DIE FORGING
[3.2] ROLLING
[3.3] DRAWING
[3.4] COLD FORMING
[3.5] COLD HEADING
[3.6] SWAGING
[3.7] SUPERPLASTIC FORMING
[3.8] SHEET-METAL SHEARING
[3.9] SHEET-METAL FORMING
[3.10] SPINNING
[3.11] POWDER METALLURGY
[3.12] CONTINUOUS EXTRUSION (METALS)

MACHINING PROCESSES
[4.A] AUTOMATIC MACHINING
[4.M] MANUAL MACHINING
(THE ABOVE HEADINGS COVER A BROAD RANGE OF MACHINING PROCESSES AND LEVELS OF CONTROL TECHNOLOGY. FOR MORE DETAIL, THE READER IS REFERRED TO THE INDIVIDUAL PROCESSES.)

NTM PROCESSES
[5.1] ELECTRICAL DISCHARGE MACHINING (EDM)
[5.2] ELECTROCHEMICAL MACHINING (ECM)
[5.3] ELECTRON BEAM MACHINING (EBM)
[5.4] LASER BEAM MACHINING (LBM)
[5.5] CHEMICAL MACHINING (CM)
[5.6] ULTRASONIC MACHINING (USM)
[5.7] ABRASIVE JET MACHINING (AJM)

Figure 28: Influence of Materials on Process Selection

The above Figure 28 is a matrix used for selecting manufacturing processes based on the type of material and the quantity required. It categorizes materials such as iron, steel, aluminum, and various alloys, as well as plastics and composites, ceramics, and precious metals. For each material type, it suggests the most suitable manufacturing processes based on production volume (ranging from very low to high quantities). The chart also provides a selection of processes under various categories like casting, plastic processing, forming, machining, and non-traditional manufacturing methods (NTM). This matrix is designed to assist in determining the appropriate production methods tailored to specific materials and production scales.

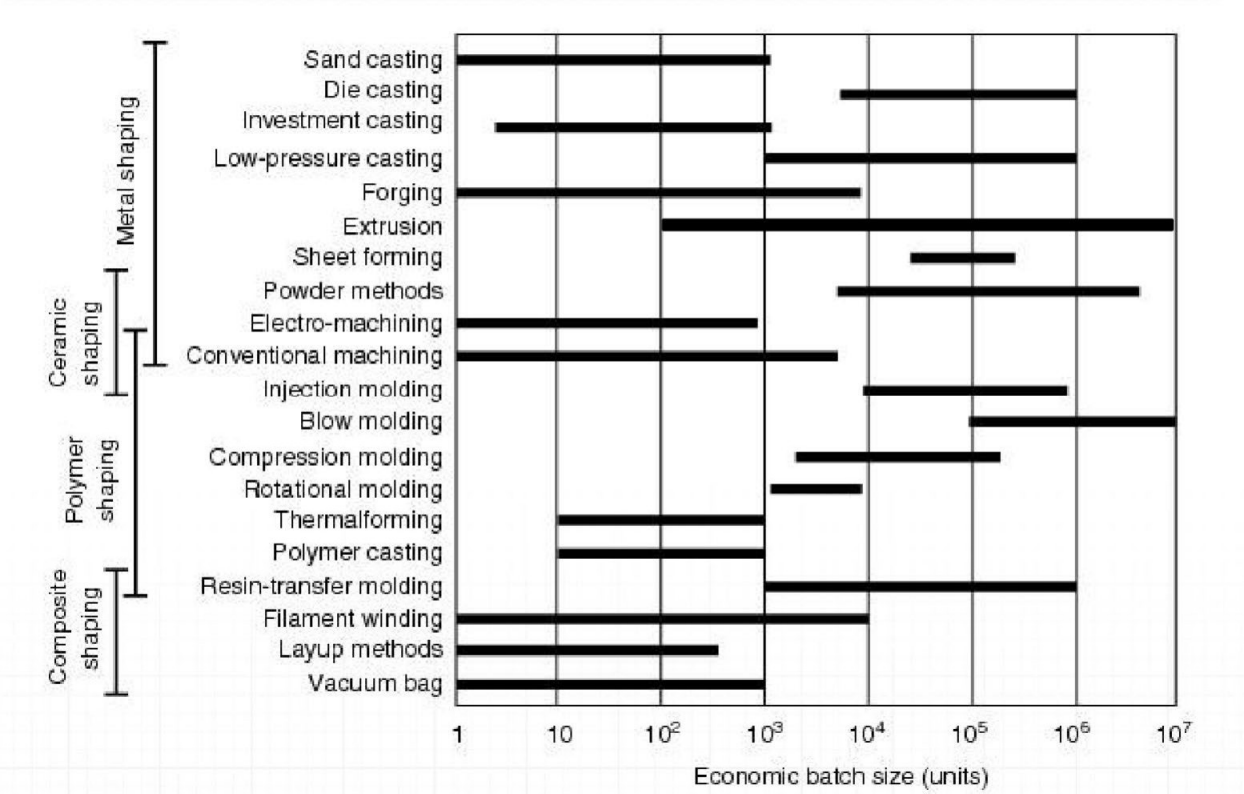


Figure 29: Manufacturing Process V/s Economic Batch Size

1. PC/ABS Components (Amplifier Housing, Lower Body Housing, and Impeller Components)

Material Overview:

- **PC/ABS** is a thermoplastic blend that combines the high impact resistance of polycarbonate (PC) and the ease of processing of acrylonitrile butadiene styrene (ABS). It is used for most housing and casing components due to its excellent durability and ease of manufacturing through injection molding.

Manufacturing Process:

- **Injection Molding** is the preferred manufacturing technique for these components. Injection molding is ideal for high-volume production runs (as seen in the report for components like the front body). This process enables the production of complex geometries with tight tolerances, making it suitable for intricate parts like the amplifier housing and impeller components. Additionally, the ability to mass-produce these parts efficiently at a lower cost justifies the selection of PC/ABS.

2. Stainless Steel Components (Springs and Bearings)

Material Overview:

- **Stainless Steel** is utilized for components such as springs and bearings due to its excellent mechanical strength and corrosion resistance. This ensures that these components can withstand the wear and mechanical stress encountered during the fan's operation.

Manufacturing Process:

- **Wire Drawing** is commonly used to produce springs, where a steel wire is pulled through a die to achieve the desired diameter. Wire drawing provides excellent surface finish and strength to the steel. Forging and machining processes are typically used for bearings. Forging offers the high strength required for the bearings to endure the rotational stresses in the oscillating base assembly, while machining allows for precise tolerance control.

3. Borosilicate Glass (Mesh Filter)

Material Overview:

- **Borosilicate Glass** is known for its excellent thermal and chemical resistance, making it ideal for filtration components like the mesh filter. It is commonly used where exposure to high temperatures or corrosive environments is expected.

Manufacturing Process:

- **Glass Forming and Annealing** are the standard processes for creating borosilicate glass components. The glass is heated until malleable, formed into the desired shape, and then annealed to relieve internal stresses. This process ensures the mesh filter remains durable and susceptible to cracking under thermal loads.

4. POM (Base Plate Lock and Moving Parts)

Material Overview:

- **Polyoxymethylene (POM)** is used for components requiring low friction and high wear resistance, such as the base plate lock. It is highly crystalline, providing excellent dimensional stability and toughness, critical for components that must endure repeated mechanical stress.

Manufacturing Process:

- **Injection Molding** is also preferred for POM components, like PC/ABS. Due to its high wear resistance, POM is particularly suited for moving parts like the lock and mechanical interfaces. Its low friction coefficient makes it ideal for such applications without additional lubrication.

DESIGN CHANGES

The bladeless fans have many complicated injection-molded parts. The Swivel base is one of them, and it has many provisions for other moving and stationary parts to accommodate. After analyzing the swivel and its assembled parts, we reduced certain parts that don't impact performance or function.

1. Reduce the swivel support from 3 to 2

We designed the experiment quickly to check whether the oscillating base could rotate with two swivel supports and move as before. The provision to the supports and one of the swivel supports were removed.

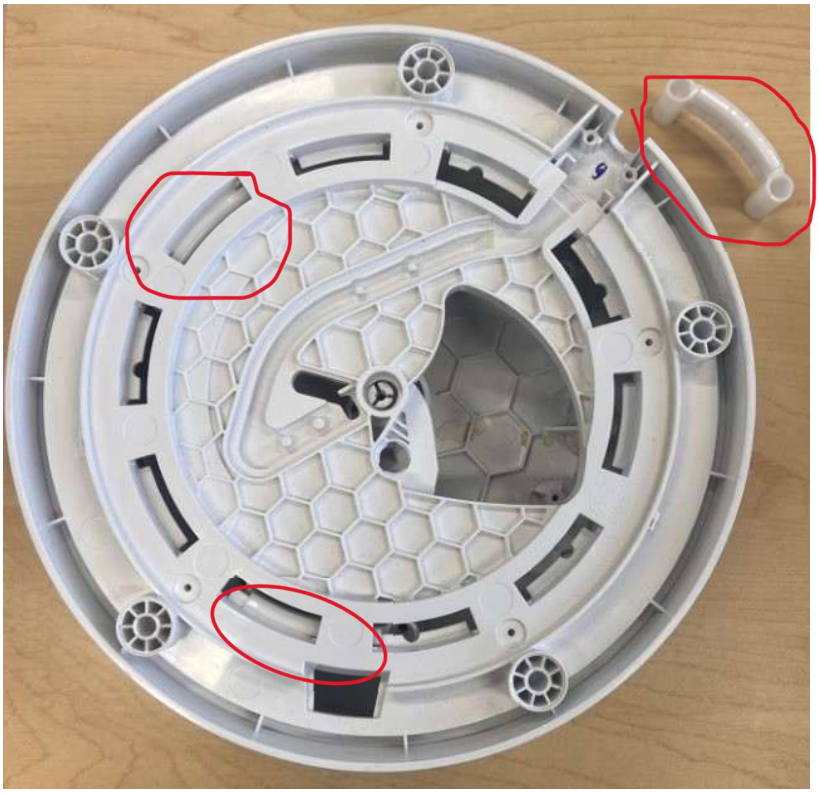


Figure 30: Swivel Base

Before

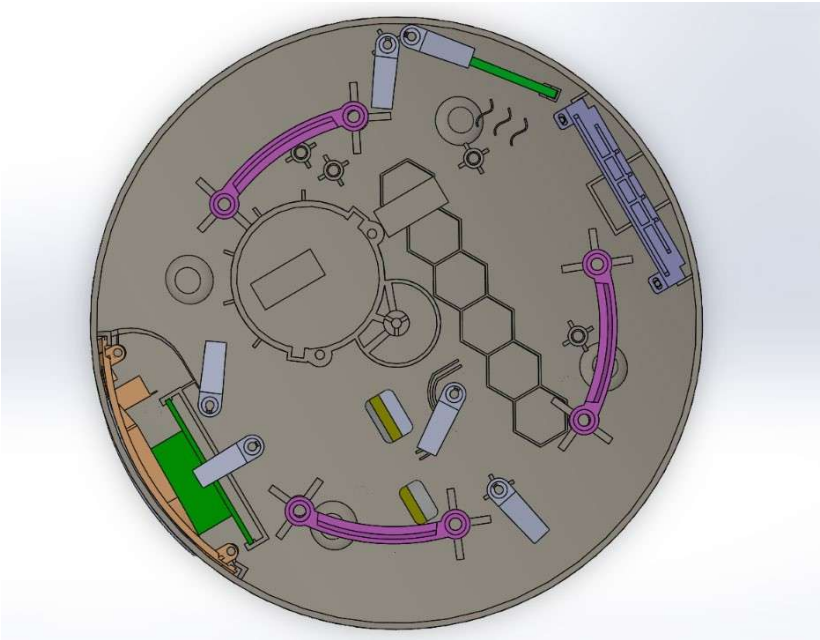


Figure 31: Swivel Base Design Before

After

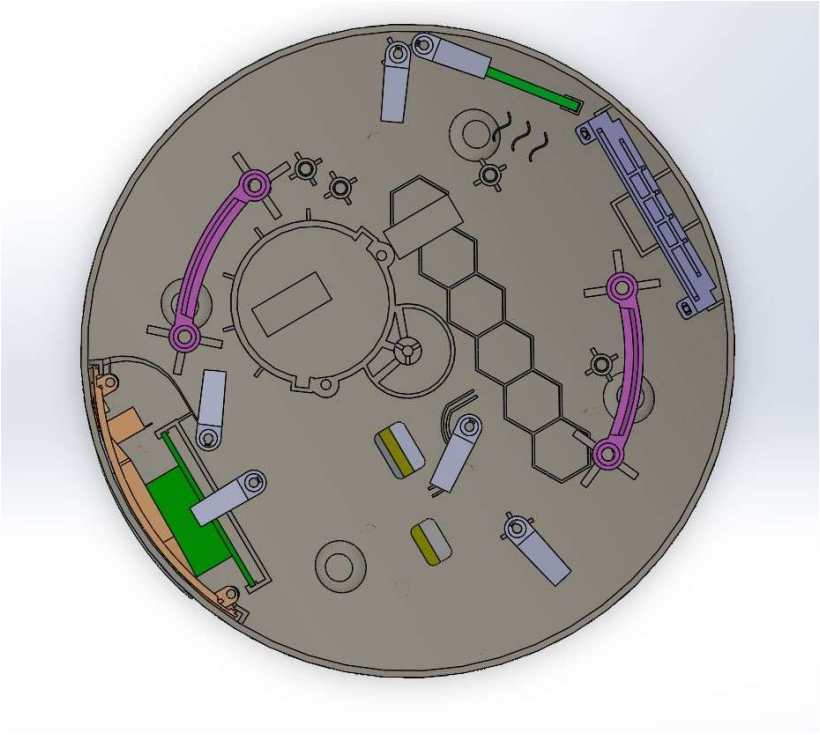


Figure 32: Swivel Base Design After

2. Eliminating three wire holders

There are six wire holders. Two hold the PCBs (green), which cannot be removed. One holder keeps the wire intact, but the other two holders' position is where guided provisions are already given. We propose to remove that. Guides are provided to hold and direct the cables at the zone where changes are made.

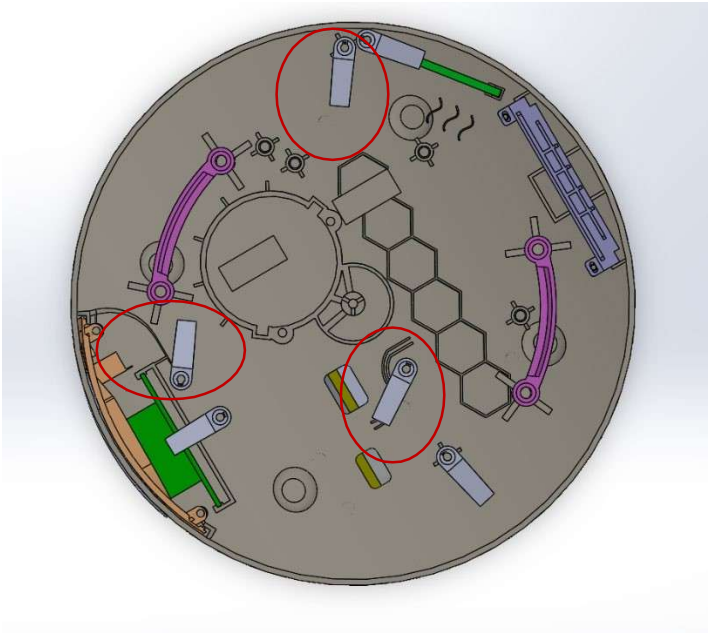


Figure 33: Swivel Base with Wire Holders Before

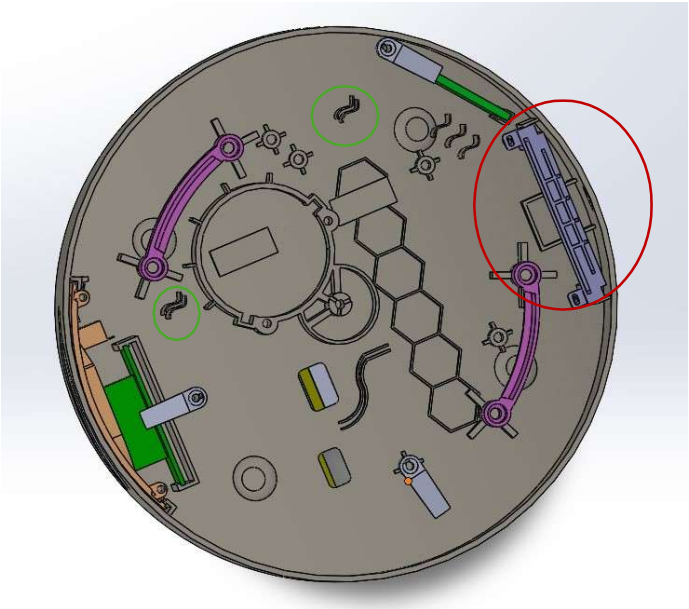


Figure 34: Swivel Base with Wire Holders After

The two small indents marked within the green circles in Figure 34 were added to the base to better guide the wires. They can also hold down the cables without needing any additional screw-based installations.

3. Removal of LED frame

Dissimilar to other PCB zones, where wire holders hold the PCB plate, the LED-PCB is held using a separate frame part. The part can be eliminated by giving the positive lock at the top of the provision already in the swivel base. This could be a wild design change as the lock features offered at the top might be difficult while jumping over the tool, but we believe there is a possibility.

Before

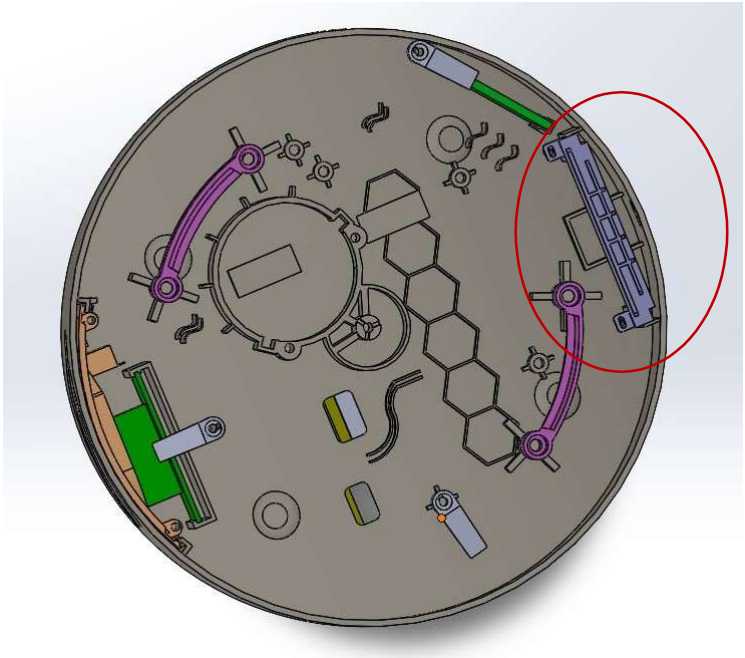


Figure 35: LED Frame Before

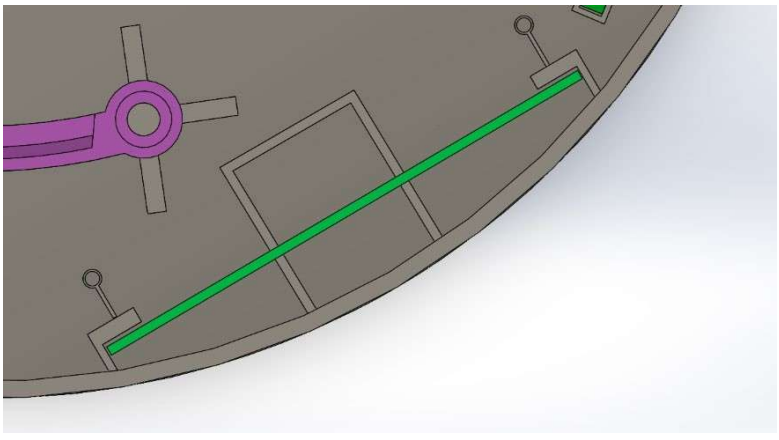


Figure 36: LED Frame Zoomed in Before

After

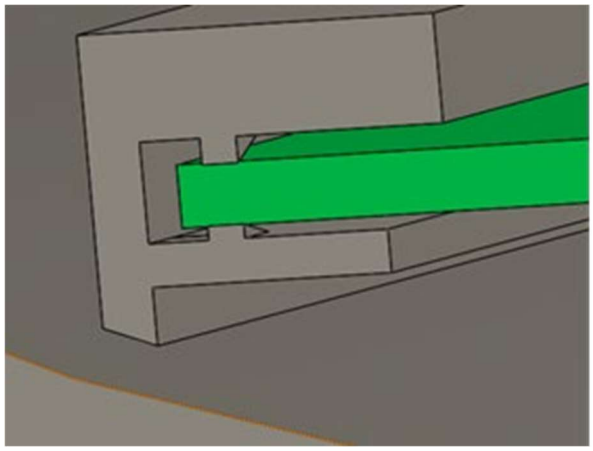


Figure 37: LED Frame Zoomed in After

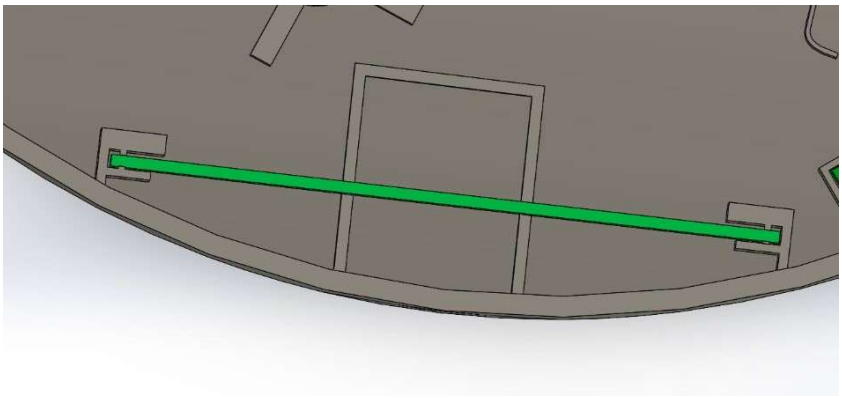


Figure 38: LED Frame After

4. Removal of Sensor hatch and housing

After analysis, it was found that the swivel base-guided provision itself holds the PCB attached to the sensor (humidity). Both parts are provided to house the sensor inside. We propose to remove both parts as they are understood to be unaffected.

Before

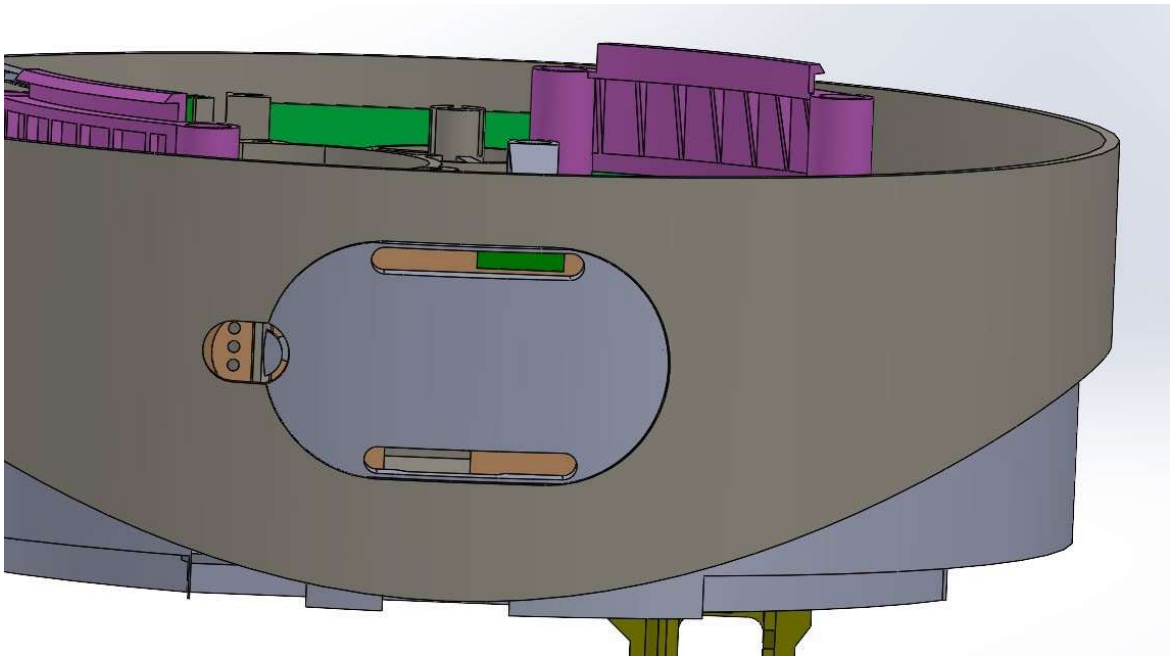


Figure 39: Sensor Hatch Before

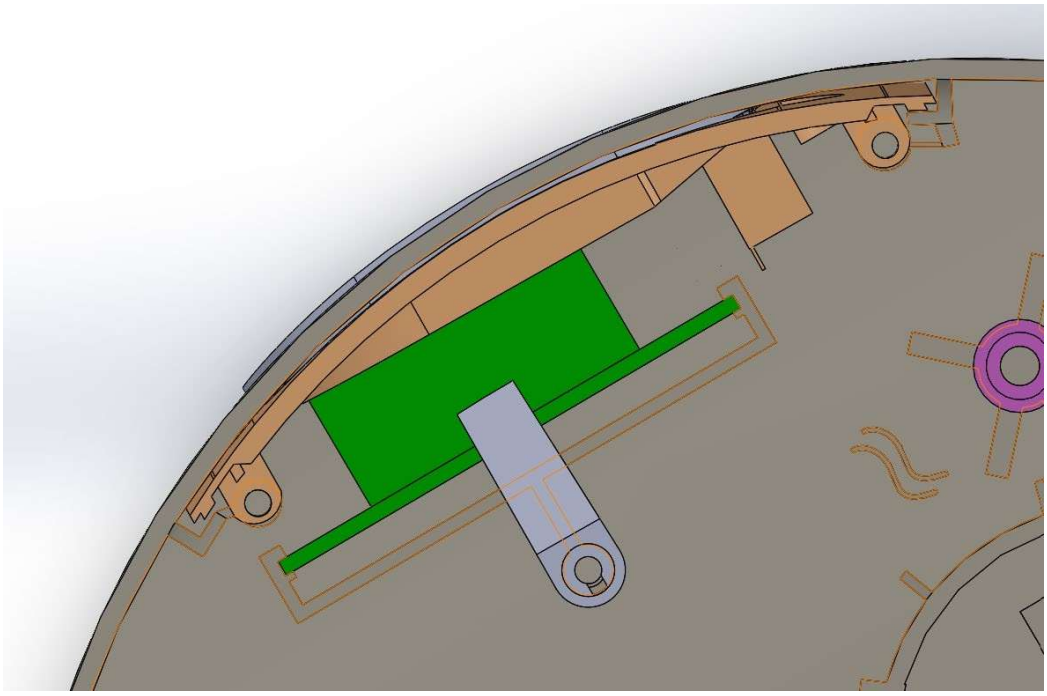


Figure 40: Sensor Hatch Housing Before

After

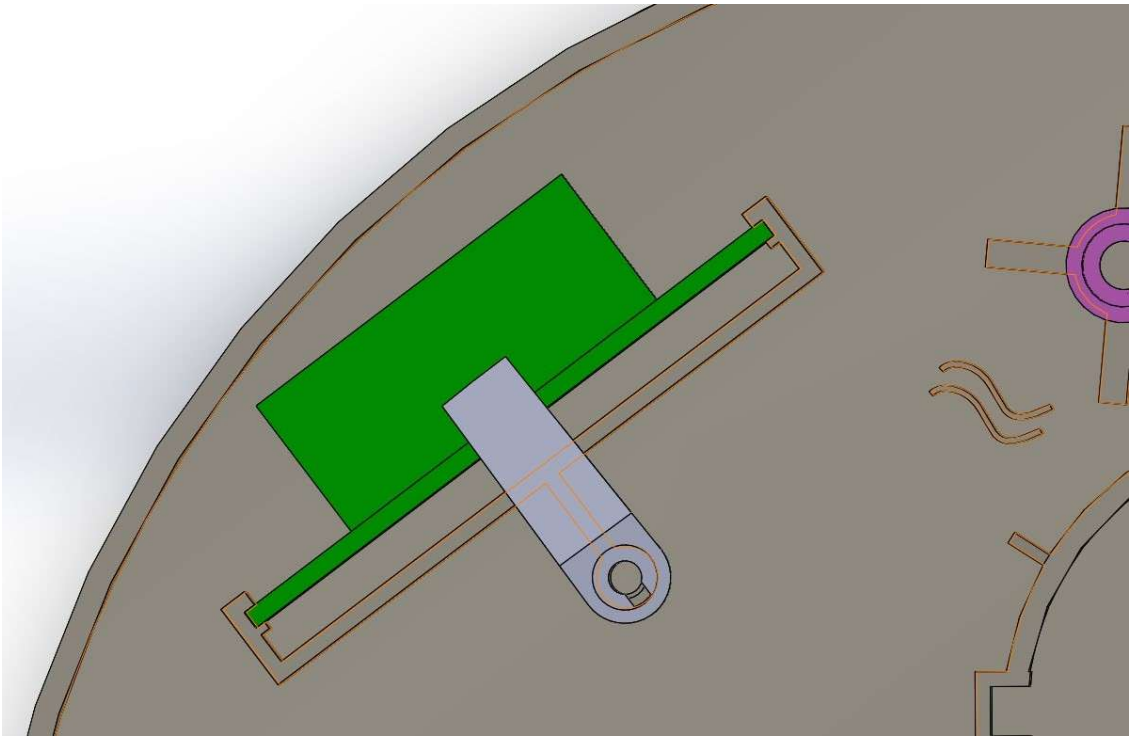


Figure 41: Sensor Housing After

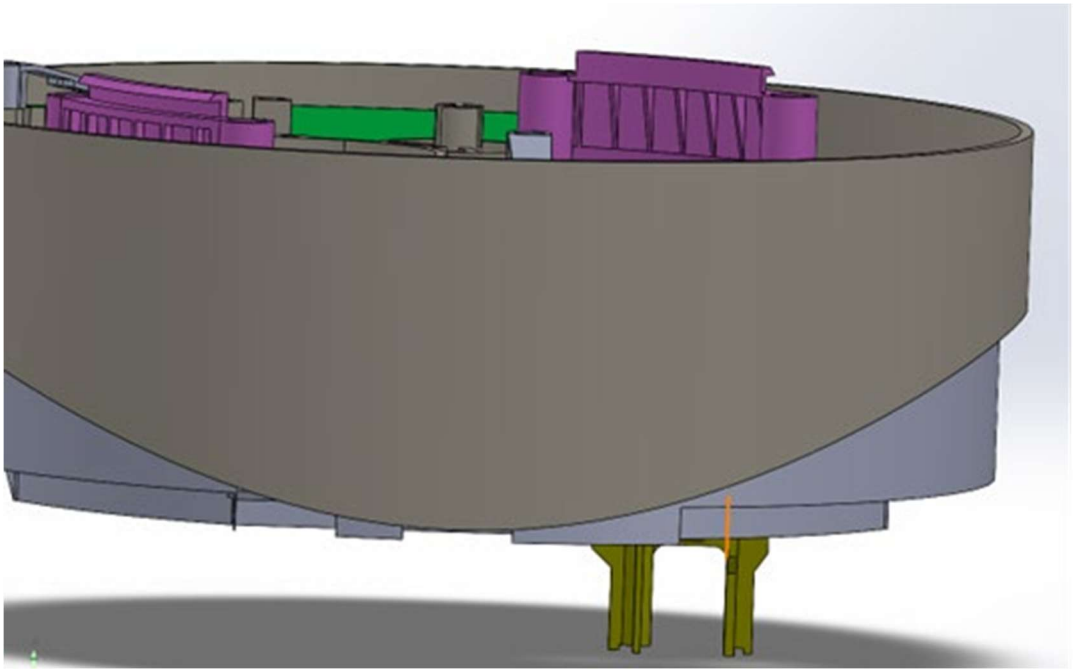


Figure 42: Swivel Base Sensor Hatch Location After

NEW DFA ANALYSIS

Based on the design changes, a DFA analysis, shown in Figure 38, was done for the product with reduced components and their impact on the DFA metrics score. We set a target of 65 to achieve. The image below shows the new DFA metrics score.

DFA Analysis Worksheet

Assembly Name: Dyson Bladeless Fan- DP-01 **Team:** Group 6 **Date:** 30-09-2024

If the answer is Yes to any of the metrics or questions enter a 1. If the answer is No then enter 0. Each cell must have a number.

Part		Complexity		Functional Analysis / Redesign Opportunity				Error Proofing		Handling			Insertion				Secondary Operations				
Part Number	Part Name	Number of Parts (N_p)	Number of Interfaces (N_i)	Theoretical Minimum Part	Part Can Be Standardized (if not already standard)	Cost (Low/Medium/High)	Practical Minimum Part	Assemble Wrong Part/ Omit Part	Assemble Part Wrong Way Around	Tangle, Nest, or Stick Together	Flexible, Fragile, Sharp or Slippery	Pliers, Tweezers, or Magnifying Glass Needed	Difficult to Align/ Locate	Holding Down Required	Resistance to Insertion	Obstructed Access/ Visibility	Re-orient Workpiece	Screw, Drill, Twist, Rivet, Bend, or Crimp	Weld, Solder, or Glue	Paint, Lube, Heat, Apply Liquid or Gas	Test, Measure or Adjust
BF001	Amplifier Service assembly																				
BF001-01	Amplifier head	1	1	1	0	M	1	0	0	1	0	0	0	0	0	0	0	0	1	0	0
BF001-02	Amplifier housing	1	6	1	0	M	1	0	0	1	0	0	0	0	0	0	0	0	1	0	0
BF001-03	Release lock Housing	2	3	0	0	L	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
BF001-04	Release lock	2	4	0	0	L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
BF001-05	Spring	2	2	0	0	L	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0
BF002	Motor Bucket retaining ring	1	3	1	0	L	1	0	0	1	0	0	1	0	1	0	1	1	0	0	0
BF003	Motor & Bucket assembly																				
BF003-01	Impeller	1	3	1	1	M	1	0	1	1	1	0	1	0	0	1	0	0	0	0	1
BF003-02	Impeller top housing	1	3	1	0	M	1	0	0	0	0	0	1	0	0	1	0	1	0	0	0
BF003-03	Impeller bottom housing	1	8	1	0	M	1	0	1	0	0	0	1	0	0	1	0	1	0	0	0
BF003-04	Hub base	1	1	1	0	L	1	0	1	1	0	0	0	0	0	1	0	0	1	0	0
BF003-05	Stator housing	1	3	1	0	L	1	0	0	1	0	0	1	0	0	1	0	1	0	0	0
BF003-06	Stationary plastic blade	1	3	1	1	M	1	0	0	1	1	0	0	0	0	1	0	1	0	0	0
BF003-07	Spring Holder	3	6	0	0	L	0	1	1	0	0	0	0	0	0	1	0	0	0	0	0
BF003-08	Motor	1	2	1	0	H	1	0	0	1	0	0	1	0	0	0	0	0	0	0	0
BF003-09	Spring	3	6	0	0	L	0	1	1	0	1	0	1	0	1	0	0	0	0	0	0
BF004	Filter assembly																				
BF004-01	Filter Housing	1	5	1	0	M	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
BF004-02	Filter Panel	2	7	1	0	L	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0
BF004-03	Mesh-Filter	1	2	1	1	M	1	1	0	1	1	0	0	0	0	0	0	0	1	0	0
BF005	Upper body housing assembly																				
BF005-01	Electronic housing	1	7	1	0	M	1	0	0	1	0	0	0	0	0	0	0	1	1	0	0
BF005-02	Swivel base	1	5	1	0	M	1	0	0	0	0	0	0	0	1	0	1	0	0	0	0
BF005-03	Electronics Component Holder	1	1	0	0	L	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0
BF005-04	Rubber Seal	1	2	1	1	L	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0
BF006	Lower body housing assembly																				
BF006-01	Lower body housing	1	14	1	0	M	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0
BF006-02	Swivel base lock	1	2	2	0	M	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0
BF006-03	Swivel base lock spring	1	2	1	0	L	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0
BF006-04	Base plate lock	1	2	1	0	M	1	0	1	0	0	0	1	0	0	0	0	0	0	0	0
BF006-06	Power button	1	2	1	1	L	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1
BF006-09	Wire holder 1	3	3	0	1	L	0	1	1	0	0	0	0	0	0	0	1	0	0	0	0
BF007	Oscillating motor assembly																				
BF007-01	Motor	1	2	1	0	H	1	0	0	0	0	0	1	0	0	0	0	0	0	0	1
BF007-02	Cam	1	2	1	1	L	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0
BF007-03	180-swivel arm	1	2	1	1	M	1	1	1	0	0	0	0	0	0	0	1	0	0	0	0
BF008	Oscillating base assembly																				
BF008-02	Oscillating base	1	5	1	0	M	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
BF008-03	Bearing	1	2	1	0	L	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0
BF008-04	Power cord retainer	1	2	0	1	L	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
BF008-05	Rubber grip	5	1	0	1	L	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0
	Totals	40	105	24	10	0	24	12	14	9	5	0	9	0	4	8	3	7	3	0	3
	Design for Assembly Metrics	64.8074		60.0%	←Theor. Effy. Pract. Effy.→		60.0%	1.08		2.00			0.88				0.67				
	Targets	65																			

Figure 43: New DFA Analysis Worksheet

We achieved our target and even exceeded the target and reduced the DFA metrics score by 10.8 points. The new DFA metric score was 64.8 meaning we achieved our target. We reduced the parts to 40 and the interfaces to 105. We achieved this by removing unnecessary screws, revamping the lower body housing assembly and the oscillating base assembly, removing four parts, and integrating their functions directly into the main housing, respectively. Both theoretical and practical efficiencies improved to 60% from 52.1% and 56.3%, respectively.

NEW ECONOMIC ANALYSIS

After conducting a DFA analysis on the impact of changes made, we proceeded to do another economic analysis to determine if the part reductions influenced the cost of manufacturing and sales price. The table below shows the change in these costs. A follow-up calculation was also done to find the break-even point for the product. The tables below, Table 12 and Table 13, show all the values obtained. We assumed that our Fixed Costs were $80000.

FORMULAS

ECONOMIC ANALYSIS

- $C_m=\frac{m*c_m}{1-f}$, where **C_m= Unit cost of material**, **m= mass of part, c_m=cost of material, f=fraction of material scrap**.
- $C_L=\frac{c_{ww}}{n'}$, where **C_L= Unit Labor Cost, c_w= hourly cost of wages and benefits,** n' = ppppppppppppppppppp pprrpprr.
- $C_T=\frac{ct \cdot k}{n}$ where, **C_T= Tooling Cost, c_t= cost of making tools, n= production run, k= tooling wear factor** (**k= n**/by life of tooling raised to next highest integer).
- $C_E=(\frac{1}{n'}) \cdot (\frac{c_e}{Lt_{wwo}}) \cdot q$, **where c_e= cost of capital equipment, n'= production rate, L= Load Factor, t_{wo}= capital write-off time, q= fraction of equipment sharing between products.**
- $C_{OH}=\frac{cOOH}{n'}$, **where c_{OH}= overhead hourly rate, n'= production rate**.
- **C_U= Cm+C_L+C_T+C_E+C_{OH}**

BREAK-EVEN ANALYSIS

$$Q_b=\frac{f}{P-v}$$

Where, Q_b= Break-Even Point

f = Fixed Costs

v= Variable Costs

P= Sales Price

Assumptions made in analysis:

For the final economic analysis, we assumed fixed costs of $80000 for all the components to calculate the break-even points of individual components and the entire product. All the values are shown in Tables 2 and 3 below. For capital write-off time in the sheet, we assumed operations for 45 hours each for 52 weeks, and we amortized the components over 5 and 10 years based on their complexity and size. We also thought a total product profit of $200 for product break-even analysis.

Table 28: New OME and Breakeven Analysis for each part

Parts	Part nos.	Material cost	Labor cost	Manufacturing cost	Selling price	Break-even analysis (Units)
Amplifier head	BF001-01	2.02	0.63	5.9	17.69	532
Amplifier housing	BF001-02	1.69	0.56	4.6	13.71	698
Release lock Housing	BF001-03	0.02	0.06	0.4	1.09	7919
Release lock	BF001-04	0.03	0.33	1.5	4.58	1897
Spring	BF001-05	0.00	0.05	0.3	0.77	11144
Motor Bucket retaining ring	BF002	0.11	0.13	0.7	2.07	4362
Impeller	BF003-01	0.33	0.63	4.6	13.77	624
Impeller top housing	BF003-02	0.52	0.63	3.7	11.04	808
Impeller bottom housing	BF003-03	0.33	0.63	3.4	10.18	867
Hub base	BF003-04	0.09	0.25	1.1	3.41	2608
Stator housing	BF003-05	0.33	0.63	3.5	10.47	841
Stationary plastic blade	BF003-06	0.18	0.63	3.6	10.92	791
Spring Holder	BF003-07	0.01	0.42	1.4	4.34	2046
Spring	BF003-09	0.00	0.33	1.6	4.82	1786
Filter Housing	BF004-01	0.35	0.36	2.5	7.37	1200
Filter Panel	BF004-02	1.78	0.63	5.0	15.00	635
Mesh-Filter	BF004-03	1.75	0.56	4.7	14.07	680
Electronic housing	BF005-01	1.38	0.56	4.2	12.65	747
Swivel base	BF005-02	1.58	0.63	5.1	15.40	606
Lower body housing	BF006-01	0.35	0.63	3.3	9.99	887
Swivel base lock	BF006-02	0.00	0.45	2.0	6.14	1407
Swivel base lock spring	BF006-03	0.00	0.42	1.7	5.17	1684

Base plate lock	BF006-04	0.16	0.42	2.1	6.35	1386
Power button	BF006-06	0.01	0.45	1.9	5.65	1543
Cam	BF007-02	0.01	0.42	1.7	5.18	1680
180-swivel arm	BF007-03	0.03	0.50	1.9	5.65	1561
Base plate	BF008-01	0.53	0.42	2.5	7.53	1216
Oscillating base	BF008-02	1.70	0.63	4.9	14.56	654
Bearing	BF008-03	0.00	0.45	1.9	5.63	1546
Power cord retainer	BF008-04	0.01	0.13	0.6	1.68	5149
Rubber grip	BF008-05	0.00	0.04	0.3	0.98	8466
	Total	**15.30**	**13.54**	**82.6**	**247.86**	

Table 29: New Breakeven Analysis for Product

Overall assembly unit	
Assumption	
Fixed cost ($)	**80000**
Variable cost ($)	**83**
Sales price Incl. Profit ($)	**448**
Break-even (Units)	**228**

NEW MATERIAL AND MANUFACTURING PROCESS ANALYSIS

The Dyson Bladeless Fan's redesign has focused on optimizing the assembly process, improving error-proofing, and reducing the overall part count while maintaining its core functionality and high-performance standards. After a detailed analysis of the materials and manufacturing processes, it became evident that the original selection of materials and processes remains optimal even after the redesign. These choices are maintained based on their performance, ease of manufacturing, and economic viability, making them suitable for the revised design without requiring significant changes in production methodology.

Material Selection Overview:

The materials used in the Dyson Bladeless Fan cover a range of properties, from lightweight polymer components to robust metals and glass. The primary materials include **PC/ABS (polycarbonate/acrylonitrile butadiene sterile), stainless steel, borosilicate glass, and POM (polyoxymethylene)**. These materials were initially chosen for their unique properties, such as high impact resistance, ease of manufacturability, and superior durability, which are still ideal for the redesigned components.

- **PC/ABS** continues to be the most suitable material for the fan's housing and structural components, providing the necessary toughness and dimensional stability at a competitive cost. Its ability to be efficiently molded through **injection molding** and its compatibility with the fan’s complex geometries make it irreplaceable in the original and redesigned models.
- **Stainless Steel** is retained for high-stress components like springs and bearings due to its strength, corrosion resistance, and ability to endure mechanical stress. It is produced using **wire drawing** for springs and **forging** or **machining** for the bearings. These processes ensure that the material’s mechanical properties are optimized for long-term performance.
- **Borosilicate Glass**, used for the mesh filter, offers excellent thermal and chemical resistance, making it perfect for air filtration and ensuring the fan operates efficiently even in fluctuating temperatures. The glass forming and annealing manufacturing process remains suitable, ensuring dimensional accuracy and durability in the original and redesigned components.
- **POM** is used for parts such as the base plate lock and other moving parts that require low friction and high wear resistance. **Injection molding** continues to be the best manufacturing method for this material, allowing for precise tolerance and smooth mechanical operation.

Manufacturing Process Overview:

The original manufacturing processes used for the fan's components, primarily **injection molding**, **wire drawing**, **forging**, **machining**, and **glass forming**, have proven efficient and cost-effective. These processes have been tested for suitability in high-volume production, making them ideal for the redesigned model. A thorough analysis revealed that no significant modifications to these processes are necessary after the redesign.

1. **Injection Molding**:
    - Injection molding remains the primary process for producing **PC/ABS** and **POM** components, such as the fan's housing, impeller, and base assemblies. This process is ideal for high-volume production due to its speed, repeatability, and the ability to produce intricate shapes with tight tolerances. The fan's redesign did not introduce geometries or features that would necessitate a shift from injection molding. This method still allows for efficient mass production with minimal changes to tooling or setup.
2. **Wire Drawing and Forging**:
    - For **stainless steel** components like the springs and bearings, **wire drawing** and **forging** remain the best manufacturing methods. These processes offer excellent control over the mechanical properties of the parts, ensuring they maintain the necessary strength and resistance to fatigue. The redesign did not significantly alter the geometry or function of these components, allowing these traditional metalworking processes to continue being employed without modification.
3. **Glass Forming and Annealing**:
    - **Borosilicate glass** components, such as the mesh filter, are formed and annealed using traditional glass working techniques. These processes ensure that the glass is solid and resistant to thermal stress and chemical corrosion. The redesign did not introduce any new requirements for the glass components, so **glass forming** remains the most suitable process for producing these parts.
4. **Machining**:
    - **Machining** continues to be used for precision components where high tolerances are necessary, such as metal fittings and moving parts. The redesign did not require any alterations to the materials or shapes that would necessitate switching to a different manufacturing process.

Consistency in Material and Process Use Post-Redesign:

One of the significant findings from the redesigning process is that the initial selection of materials and manufacturing processes was already highly optimized for the fan's performance and assembly

needs. The slight modifications made to some components, particularly the wire holders and base assemblies, did not introduce any new material or manufacturing requirements. This is a testament to the robustness of the original design choices.

The consistency of using **PC/ABS** for significant structural components, **stainless steel** for high-stress mechanical parts, and **borosilicate glass** for filtration ensures that the manufacturing process remains streamlined. **Injection molding** and **wire drawing** processes can continue without modification, maintaining both the cost-effectiveness and the production scalability needed for high-volume manufacturing. Additionally, these materials and methods allow for flexibility in the event of future design changes, as they can be easily adjusted to accommodate further improvements in the fan's geometry or functionality.

Final Remarks:

In conclusion, after a comprehensive review of the materials and the manufacturing processes used in the Dyson Bladeless Fan, the original decisions were highly efficient and remain applicable post-redesign. The same materials—**PC/ABS**, **POM**, **stainless steel**, and **borosilicate glass**—continue to provide the desired mechanical and thermal properties, while manufacturing processes such as **injection molding**, **wire drawing**, **forging**, and **glass forming** still offer the best combination of cost-effectiveness and production capability. This consistency across materials and processes ensures that the Dyson Bladeless Fan can be produced efficiently, even with the minor changes introduced during the redesigning phase.

ETHICAL CONSIDERATION AND SAFETY

Several ethical and safety factors must be considered when reverse engineering a product like the Dyson Bladeless Fan. Ethically, it is essential to respect intellectual property rights, including patents, trademarks, and designs, ensuring that the reverse engineering process does not lead to infringement or replication of proprietary technologies without appropriate permissions. From a safety perspective, disassembling the fan requires careful handling of components, particularly electrical parts, to avoid accidents such as electric shock, injuries from sharp edges, or damage to sensitive parts like the motor or electronics. It's crucial to follow proper protocols, such as powering down and discharging capacitors before teardown, to ensure the safety of those involved. Additionally, any insights gained from reverse engineering should be used responsibly, ensuring that any redesign or reimplementation maintains or improves consumer safety standards, especially regarding the product's structural integrity, electrical insulation, and operational safety. This ensures that future designs inspired by the original fan prioritize user safety and comply with industry standards and regulations.

CONCLUSIONS

The reverse engineering project of the Dyson Bladeless Fan was an extensive process that allowed for a thorough exploration of the product's design, materials, and manufacturing processes. The primary objective was to understand the fan's inner workings, evaluate its components' efficiency, and identify areas where improvements could be made in manufacturability, cost, and overall performance. Throughout the project, we disassembled the fan to gain detailed insights into its structure, modeled the components using CAD software, and performed various analyses, including Design for Assembly (DFA) and material evaluation.

One of the key objectives of this project was to optimize the assembly process and explore potential cost reductions while maintaining or enhancing the fan's performance. We conducted a deep dive into the materials used and assessed whether they were still the best options after redesigning specific components. The analysis confirmed that the materials chosen offered the best balance of strength, durability, and manufacturability, particularly given the product's high-volume production needs. Initially, the manufacturing processes chosen, such as injection molding, forging, and glass forming, were highly efficient and did not require significant changes, even after the component was redesigned.

The project also focused on improving the overall assembly efficiency by reducing part complexity and simplifying specific assembly steps. The DFA analysis highlighted areas for improvement, such as combining components like wire holders or reducing the number of fasteners. We performed three design changes to reduce parts and assembly time, minimize errors during the assembly process, and ultimately lower production costs without compromising the product's integrity or functionality.

Ultimately, this reverse engineering project provided valuable insights into how the Dyson Bladeless Fan is designed and manufactured. This type of project was also the first for all the team members. All members enjoyed the project and the opportunity presented to model a unique product and, in doing so, fully understand the methodology and principles behind innovation in a product that has a straightforward function.

ACKNOWLEDGEMENTS

The team would like to thank Dr. Dan Riffell for his guidance throughout the project. He taught us the concepts behind DFMA and gave us all insight into the difference between good and bad engineering.

REFERENCES

[1] Custom Part Net. (2018). Custompartnet.com

[2] Parts Warehouse. Partswarehouse.com

APPENDICES

- Initial Economic Analysis.
- Final Economic Analysis.
- Dyson Operations Manual.
- All the drawings and Assembly Drawing with BOM.
- Design Change Drawings.

INITIAL ECONOMIC ANALYSIS

Formula	Cost Element	Variables	Units	BF001-01	BF001-02	BF001-03	BF001-04	BF001-05
				Amplifier head	Amplifier housing	Release lock Housing	Release lock	Spring
	Material			PC/ABS	PC/ABS	PC/ABS	PC/ABS	Spring steel
Cm=m*cm/1-f	Material cost	Cm	$/lb	3	3	3	3	1
	Fraction of process that is scrap	f		0.05	0.05	0.05	0.05	0.05
	Mass of part	m	lb	0.639	0.535	0.007	0.009	0.0034
	C_m unit cost of material			**2.0178G4737**	**1.68G473684**	**0.022105263**	**0.028421053**	**0.003578G47**
Cl=cw/n	Labour cost	Cw	$/h	25	25	25	25	20
	Production rate	n	units/h	40	45	400	75	400
	C_l unit cost of labour			**0.625**	**0.555555556**	**0.0625**	**0.333333333**	**0.05**
Ct=ct*k/n	Tooling cost	Ct	$/set	70000	50000	30000	30000	3000
	Total production run	n	units	100000	100000	100000	100000	100000
	Tooling life	nt	units	100000	100000	200000	200000	200000
	Sets of tooling required	k		1	1	0.5	0.5	0.5
	C_t unit cost of tooling			**0.7**	**0.5**	**0.15**	**0.15**	**0.015**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	50000	50000	50000	25000	10000
	Capital write-off time	Two	years	23400	23400	23400	23400	23400
	Load fraction	L		1	1	1	1	1
	Load sharing fraction	q		1	1	1	1	1
	C_e unit cost of capital equipment			**0.053418803**	**0.047483381**	**0.00534188**	**0.014245014**	**0.001068376**
Coh=coh/n	Factory overhead	coh	$/h	100	80	50	75	75
	Production rate	n	units/h	40	45	400	75	400
	C_{oh} unit cost of factory overhead			**2.5**	**1.777777778**	**0.125**	**1**	**0.1875**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**5.8G631354**	**4.5702G03G8**	**0.364G47143**	**1.525GGG4**	**0.257147323**
				6.053684211	5.068421053	0.066315789	0.085263158	0.010736842
				OME achieved!				

Formula	Cost Element	Variables	Units	BF002	BF003-01	BF003-02	BF003-03	BF003-04
				Motor Bucket retaining ring	Impeller	Impeller top housing	Impeller bottom housing	Hub base
Material				PC/ABS	PC/ABS	PC/ABS	PC/ABS	PC/ABS
Cm=m*cm/1-f	Material cost	Cm	$/lb	3	3	3	3	3
	Fraction of process that is scrap	f		0.05	0.05	0.05	0.05	0.05
	Mass of part	m	lb	0.035	0.06	0.165	0.106	0.029
	C_m unit cost of material			**0.110526316**	**0.328421053**	**0.521052632**	**0.334736842**	**0.0G1578G47**
Cl=cw/n	Labour cost	Cw	$/h	25	25	25	25	25
	Production rate	n	units/h	200	40	40	40	100
	C_l unit cost of labour			**0.125**	**0.625**	**0.625**	**0.625**	**0.25**
Ct=ct*k/n	Tooling cost	Ct	$/set	15000	80000	50000	40000	7500
	Total production run	n	units	100000	100000	100000	100000	100000
	Tooling life	nt	units	200000	50000	100000	100000	200000
	Sets of tooling required	k		0.5	2	1	1	0.5
	C_t unit cost of tooling			**0.075**	**1.6**	**0.5**	**0.4**	**0.0375**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	20000	75000	75000	75000	17000
	Capital write-off time	Two	years	23400	11700	11700	11700	23400
	Load fraction	L		1	1	1	1	1
	Load sharing fraction	q		1	1	1	1	1
	C_e unit cost of capital equipment			**0.004273504**	**0.16025641**	**0.16025641**	**0.16025641**	**0.007264G57**
Coh=coh/n	Factory overhead	coh	$/h	75	75	75	75	75
	Production rate	n	units/h	200	40	40	40	100
	C_{oh} unit cost of factory overhead			**0.375**	**1.875**	**1.875**	**1.875**	**0.75**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**0.68G7GG82**	**4.588677463**	**3.68130G042**	**3.3G4GG3252**	**1.136343G05**
				0.331578947	0.985263158	1.563157895	1.004210526	0.274736842
				OME achieved!				

Formula	Cost Element	Variables	Units	BF003-05	BF003-06	BF003-07	BF003-09	BF004-01
				Stator housing	Stationary plastic blade	Spring Holder	Spring	Filter Housing
Material				PC/ABS	PC/ABS	PC/ABS	Spring steel	ABS
Cm=m*cm/1-f	Material cost	Cm	$/lb	3	3	3	1	3
	Fraction of process that is scrap	f		0.05	0.05	0.05	0.05	0.05
	Mass of part	m	lb	0.104	0.057	0.0034	0.0034	0.11
	C_m unit cost of material			**0.328421053**	**0.18**	**0.010736842**	**0.003578G47**	**0.347368421**
Cl=cw/n	Labour cost	Cw	$/h	25	25	25	20	25
	Production rate	n	units/h	40	40	60	60	70
	C_l unit cost of labour			**0.625**	**0.625**	**0.416666667**	**0.333333333**	**0.357142857**
Ct=ct*k/n	Tooling cost	Ct	$/set	25000	40000	3000	3000	20000
	Total production run	n	units	100000	100000	100000	100000	100000
	Tooling life	nt	units	50000	50000	200000	200000	30000
	Sets of tooling required	k		2	2	0.5	0.5	3.333333333
	C_t unit cost of tooling			**0.5**	**0.8**	**0.015**	**0.015**	**0.666666667**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	75000	75000	5000	5000	25000
	Capital write-off time	Two	years	11700	11700	23400	23400	23400
	Load fraction	L		1	1	1	1	1
	Load sharing fraction	q		1	1	1	1	1
	C_e unit cost of capital equipment			**0.16025641**	**0.16025641**	**0.003561254**	**0.003561254**	**0.015262515**
Coh=coh/n	Factory overhead	coh	$/h	75	75	60	75	75
	Production rate	n	units/h	40	40	60	60	70
	C_{oh} unit cost of factory overhead			**1.875**	**1.875**	**1**	**1.25**	**1.071428571**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**3.488677463**	**3.64025641**	**1.445G64762**	**1.605473534**	**2.45786G032**
				0.985263158	0.54	0.032210526	0.010736842	1.042105263
				OME achieved!				

Formula	Cost Element	Variables	Units	BF004-02	BF004-03	BF005-01	BF005-02	BF006-01
				Filter Panel	Mesh-Filter	Electronic housing	Swivel base	Lower body housing
Material				Al	Borosilicate	PC/ABS	PC/ABS	PC/ABS
Cm=m*cm/1-f	Material cost	Cm	$/lb	3	2.5	3	3	3
	Fraction of process that is scrap	f		0.05	0.05	0.05	0.05	0.05
	Mass of part	m	lb	0.564	0.665	0.438	0.5	0.11
	C_m unit cost of material			**1.781052632**	**1.75**	**1.3831578G5**	**1.578G47368**	**0.347368421**
Cl=cw/n	Labour cost	Cw	$/h	25	25	25	25	25
	Production rate	n	units/h	40	45	45	40	40
	C_l unit cost of labour			**0.625**	**0.555555556**	**0.555555556**	**0.625**	**0.625**
Ct=ct*k/n	Tooling cost	Ct	$/set	30000	35000	35000	30000	15000
	Total production run	n	units	100000	100000	100000	100000	100000
	Tooling life	nt	units	100000	100000	100000	30000	40000
	Sets of tooling required	k		1	1	1	3.333333333	2.5
	C_t unit cost of tooling			**0.3**	**0.35**	**0.35**	**1**	**0.375**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	40000	35000	40000	25000	50000
	Capital write-off time	Two	years	23400	23400	23400	11700	11700
	Load fraction	L		1	1	1	1	1
	Load sharing fraction	q		1	1	1	1	1
	C_e unit cost of capital equipment			**0.042735043**	**0.033238367**	**0.037G86705**	**0.053418803**	**0.106837607**
Coh=coh/n	Factory overhead	coh	$/h	90	90	85	75	75
	Production rate	n	units/h	40	45	45	40	40
	C_{oh} unit cost of factory overhead			**2.25**	**2**	**1.88888888G**	**1.875**	**1.875**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**4.GG8787674**	**4.6887G3G22**	**4.21558G044**	**5.132366172**	**3.32G206028**
				5.343157895	5.25	4.149473684		
				OME achieved!				

Formula	Cost Element	Variables	Units	BF006-02	BF006-03	BF006-04	BF006-05	BF006-06
				Swivel base lock	Swivel base lock spring	Base plate lock	LCD & PCB holder	Power button
Material				PC	Stainless steel	POM	PC/ABS	PC/ABS
Cm=m*cm/1-f	Material cost	Cm	$/lb	3.5	1	3	3	3
	Fraction of process that is scra	f		0.05	0.05	0.05	0.05	0.05
	Mass of part	m	lb	0.0002	0.0034	0.052	0.639	0.002
	C_m unit cost of material			**0.000736842**	**0.003578G47**	**0.164210526**	**2.0178G4737**	**0.00631578G**
Cl=cw/n	Labour cost	Cw	$/h	25	25	25	25	25
	Production rate	n	units/h	55	60	60	38	55
	C_l unit cost of labour			**0.454545455**	**0.416666667**	**0.416666667**	**0.6578G4737**	**0.454545455**
Ct=ct*k/n	Tooling cost	Ct	$/set	30000	5000	28000	20000	5000
	Total production run	n	units	100000	100000	100000	100000	100000
	Tooling life	nt	units	200000	100000	100000	40000	100000
	Sets of tooling required	k		0.5	1	1	2.5	1
	C_t unit cost of tooling			**0.15**	**0.05**	**0.28**	**0.5**	**0.05**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	100000	5000	10000	55000	10000
	Capital write-off time	Two	years	23400	23400	23400	23400	23400
	Load fraction	L		1	1	1	1	1
	Load sharing fraction	q		1	1	1	1	1
	C_e unit cost of capital equipment			**0.077700078**	**0.003561254**	**0.007122507**	**0.061853351**	**0.007770008**
Coh=coh/n	Factory overhead	coh	$/h	75	75	75	90	75
	Production rate	n	units/h	55	60	60	38	55
	C_{oh} unit cost of factory overhead			**1.363636364**	**1.25**	**1.25**	**2.368421053**	**1.363636364**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**2.046618738**	**1.723806868**	**2.117GGG7**	**5.606063878**	**1.882267615**
				4.736842105	1.042105263	0.002210526	0.010736842	0.492631579
				OME achieved!				

Formula	Cost Element	Variables	Units	BF006-07	BF006-08	BF006-09	BF006-10	BF007-02
				Sensor hatch	Sensor hatch housing	Wire holder 1	Wire holder 2	Cam
Material				PC/ABS	PC/ABS	PC/ABS	POM	PC/ABS
Cm=m*cm/1-f	Material cost	Cm	$/lb	3	3	3	3	3
	Fraction of process that is scra	f		0.05	0.05	0.05	0.05	0.05
	Mass of part	m	lb	0.035625	0.035625	0.00220462	0.00220462	0.002
	C_m unit cost of material			**0.1125**	**0.1125**	**0.006G61G58**	**0.006G61G58**	**0.00631578G**
Cl=cw/n	Labour cost	Cw	$/h	25	25	25	25	25
	Production rate	n	units/h	55	55	60	60	60
	C_l unit cost of labour			**0.454545455**	**0.454545455**	**0.416666667**	**0.416666667**	**0.416666667**
Ct=ct*k/n	Tooling cost	Ct	$/set	15000	15000	6500	6500	5000
	Total production run	n	units	100000	100000	100000	100000	100000
	Tooling life	nt	units	40000	40000	80000	80000	100000
	Sets of tooling required	k		2.5	2.5	1.25	1.25	1
	C_t unit cost of tooling			**0.375**	**0.375**	**0.08125**	**0.08125**	**0.05**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	10000	10000	13000	13000	7500
	Capital write-off time	Two	years	23400	23400	23400	23400	23400
	Load fraction	L		1	1	1	1	1
	Load sharing fraction	q		1	1	1	1	1
	C_e unit cost of capital equipment			**0.007770008**	**0.007770008**	**0.00G25G25G**	**0.00G25G25G**	**0.00534188**
Coh=coh/n	Factory overhead	coh	$/h	75	75	75	75	75
	Production rate	n	units/h	55	55	60	60	60
	C_{oh} unit cost of factory overhead			**1.363636364**	**1.363636364**	**1.25**	**1.25**	**1.25**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**2.313451826**	**2.313451826**	**1.764137884**	**1.764137884**	**1.728324336**
				6.053684211	0.018947368	0.3375	0.3375	0.020885874
				OME achieved!				

Formula	Cost Element	Variables	Units	BF007-03	BF008-01	BF008-02	BF008-03	BF008-04
				180-swivel arm	Base plate	Oscillating base	Bearing	Power cord retainer
Material				Al	PC/ABS	PC/ABS	Stainless steel	PC/ABS
Cm=m*cm/1-f	Material cost	Cm	$/lb	3	3	3	0.145	3
	Fraction of process that is scra	f		0.1	0.05	0.381	0.05	0.05
	Mass of part	m	lb	0.008	0.168	0.35	0.00110231	0.002
	C_m unit cost of material			**0.026666667**	**0.530526316**	**1.6G628433**	**0.000168247**	**0.00631578G**
Cl=cw/n	Labour cost	Cw	$/h	30	25	25	25	25
	Production rate	n	units/h	60	60	40	55	200
	C_l unit cost of labour			**0.5**	**0.416666667**	**0.625**	**0.454545455**	**0.125**
Ct=ct*k/n	Tooling cost	Ct	$/set	10000	28000	28000	5000	5000
	Total production run	n	units	100000	100000	100000	100000	100000
	Tooling life	nt	units	100000	100000	50000	100000	100000
	Sets of tooling required	k		1	1	2	1	1
	C_t unit cost of tooling			**0.1**	**0.28**	**0.56**	**0.05**	**0.05**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	10000	45000	45000	10000	25000
	Capital write-off time	Two	years	23400	23400	11700	23400	23400
	Load fraction	L		1	1	1	1	1
	Load sharing fraction	q		1	1	1	1	1
	C_e unit cost of capital equipment			**0.007122507**	**0.032051282**	**0.0G6153846**	**0.007770008**	**0.00534188**
Coh=coh/n	Factory overhead	coh	$/h	75	75	75	75	75
	Production rate	n	units/h	60	60	40	55	200
	C_{oh} unit cost of factory overhead			**1.25**	**1.25**	**1.875**	**1.363636364**	**0.375**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**1.88378G174**	**2.50G244265**	**4.852438176**	**1.876120073**	**0.56165767**
				0.020885874	0.018947368	0.08	1.591578947	5.088852989
				OME achieved!				

Formula	Cost Element	Varia	Units	BF008-05		
				Rubber grip		
	Material			Silicon		
Cm=m*cm/1-f	Material cost	Cm	$/lb	2		
	Fraction of process that is scra	f		0.05		
	Mass of part	m	lb	0.001		
	C_m unit cost of material			**0.002105263**		
Cl=cw/n	Labour cost	Cw	$/h	15		
	Production rate	n	units/h	400		
	C_l unit cost of labour			**0.0375**		
Ct=ct*k/n	Tooling cost	Ct	$/set	20000		
	Total production run	n	units	200000		
	Tooling life	nt	units	200000		
	Sets of tooling required	k		1		
	C_t unit cost of tooling			**0.1**		
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	10000		
	Capital write-off time	Two	years	23400		
	Load fraction	L		1		
	Load sharing fraction	q		1		
	C_e unit cost of capital equipment			**0.001068376**		
Coh=coh/n	Factory overhead	coh	$/h	75		
	Production rate	n	units/h	400		
	C_{oh} unit cost of factory overhead			**0.1875**		
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**0.32817363G**		
				0.000504742	0.018947368	0.006315789
				OME achieved!		

FINAL ECONOMIC ANALYSIS

Formula	Cost Element	Variables	Units	BF001-01	BF001-02	BF001-03	BF001-04	BF001-05
				Amplifier head	Amplifier housing	Release lock Housing	Release lock	Spring
	Material			PC/ABS	PC/ABS	PC/ABS	PC/ABS	Spring steel
Cm=m*cm/1-f	Material cost	Cm	$/lb	3	3	3	3	1
	Fraction of process that is scrap	f		0.05	0.05	0.05	0.05	0.05
	Mass of part	m	lb	0.639	0.535	0.007	0.009	0.0034
	C_m unit cost of material			**2.0178G4737**	**1.68G473684**	**0.022105263**	**0.028421053**	**0.003578G47**
Cl=cw/n	Labour cost	Cw	$/h	25	25	25	25	20
	Production rate	n	units/h	40	45	400	75	400
	C_l unit cost of labour			**0.625**	**0.555555556**	**0.0625**	**0.333333333**	**0.05**
Ct=ct*k/n	Tooling cost	Ct	$/set	70000	50000	30000	30000	3000
	Total production run	n	units	100000	100000	100000	100000	100000
	Tooling life	nt	units	100000	100000	200000	200000	200000
	Sets of tooling required	k		1	1	0.5	0.5	0.5
	C_t unit cost of tooling			**0.7**	**0.5**	**0.15**	**0.15**	**0.015**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	50000	50000	50000	25000	10000
	Capital write-off time	Two	years	23400	23400	23400	23400	23400
	Load fraction	L		1	1	1	1	1
	Load sharing fraction	q		1	1	1	1	1
	C_e unit cost of capital equipment			**0.053418803**	**0.047483381**	**0.00534188**	**0.014245014**	**0.001068376**
Coh=coh/n	Factory overhead	coh	$/h	100	80	50	75	75
	Production rate	n	units/h	40	45	400	75	400
	C_{oh} unit cost of factory overgead			**2.5**	**1.777777778**	**0.125**	**1**	**0.1875**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**5.8G631354**	**4.5702G03G8**	**0.364G47143**	**1.525GGG4**	**0.257147323**
				6.053684211	5.068421053	0.066315789	0.085263158	0.010736842
				OME achieved!				

Formula	Cost Element	Variables	Units	BF002	BF003-01	BF003-02	BF003-03	BF003-04
				Motor Bucket retaining ring	Impeller	Impeller top housing	Impeller bottom housing	Hub base
	Material			PC/ABS	PC/ABS	PC/ABS	PC/ABS	PC/ABS
Cm=m*cm/1-f	Material cost	Cm	$/lb	3	3	3	3	3
	Fraction of process that is scrap	f		0.05	0.05	0.05	0.05	0.05
	Mass of part	m	lb	0.035	0.06	0.165	0.106	0.029
	C_m unit cost of material			**0.110526316**	**0.328421053**	**0.521052632**	**0.334736842**	**0.0G1578G47**
Cl=cw/n	Labour cost	Cw	$/h	25	25	25	25	25
	Production rate	n	units/h	200	40	40	40	100
	C_l unit cost of labour			**0.125**	**0.625**	**0.625**	**0.625**	**0.25**
Ct=ct*k/n	Tooling cost	Ct	$/set	15000	80000	50000	40000	7500
	Total production run	n	units	100000	100000	100000	100000	100000
	Tooling life	nt	units	200000	50000	100000	100000	200000
	Sets of tooling required	k		0.5	2	1	1	0.5
	C_t unit cost of tooling			**0.075**	**1.6**	**0.5**	**0.4**	**0.0375**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	20000	75000	75000	75000	17000
	Capital write-off time	Two	years	23400	11700	11700	11700	23400
	Load fraction	L		1	1	1	1	1
	Load sharing fraction	q		1	1	1	1	1
	C_e unit cost of capital equipment			**0.004273504**	**0.16025641**	**0.16025641**	**0.16025641**	**0.007264G57**
Coh=coh/n	Factory overhead	coh	$/h	75	75	75	75	75
	Production rate	n	units/h	200	40	40	40	100
	C_{oh} unit cost of factory overgead			**0.375**	**1.875**	**1.875**	**1.875**	**0.75**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**0.68G7GG82**	**4.588677463**	**3.68130G042**	**3.3G4GG3252**	**1.136343G05**
				0.331578947	0.985263158	1.563157895	1.004210526	0.274736842
				OME achieved!				

Formula	Cost Element	Variables	Units	BF003-05	BF003-06	BF003-07	BF003-09	BF004-01
				Stator housing	Stationary plastic blade	Spring Holder	Spring	Filter Housing
	Material			PC/ABS	PC/ABS	PC/ABS	Spring steel	ABS
Cm=m*cm/1-f	Material cost	Cm	$/lb	3	3	3	1	3
	Fraction of process that is scrap	f		0.05	0.05	0.05	0.05	0.05
	Mass of part	m	lb	0.104	0.057	0.0034	0.0034	0.11
	C_m unit cost of material			**0.328421053**	**0.18**	**0.010736842**	**0.003578G47**	**0.347368421**
Cl=cw/n	Labour cost	Cw	$/h	25	25	25	20	25
	Production rate	n	units/h	40	40	60	60	70
	C_l unit cost of labour			**0.625**	**0.625**	**0.416666667**	**0.333333333**	**0.357142857**
Ct=ct*k/n	Tooling cost	Ct	$/set	25000	40000	3000	3000	20000
	Total production run	n	units	100000	100000	100000	100000	100000
	Tooling life	nt	units	50000	50000	200000	200000	30000
	Sets of tooling required	k		2	2	0.5	0.5	3.333333333
	C_t unit cost of tooling			**0.5**	**0.8**	**0.015**	**0.015**	**0.666666667**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	75000	75000	5000	5000	25000
	Capital write-off time	Two	years	11700	11700	23400	23400	23400
	Load fraction	L		1	1	1	1	1
	Load sharing fraction	q		1	1	1	1	1
	C_e unit cost of capital equipment			**0.16025641**	**0.16025641**	**0.003561254**	**0.003561254**	**0.015262515**
Coh=coh/n	Factory overhead	coh	$/h	75	75	60	75	75
	Production rate	n	units/h	40	40	60	60	70
	C_{oh} unit cost of factory overgead			**1.875**	**1.875**	**1**	**1.25**	**1.071428571**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**3.488677463**	**3.64025641**	**1.445G64762**	**1.605473534**	**2.45786G032**
				0.985263158	0.54	0.032210526	0.010736842	1.042105263
				OME achieved!				

Formula	Cost Element	Variables	Units	BF004-02	BF004-03	BF005-01	BF005-02	BF006-01
				Filter Panel	Mesh-Filter	Electronic housing	Swivel base	Lower body housing
	Material			Al	Borosilicate	PC/ABS	PC/ABS	PC/ABS
Cm=m*cm/1-f	Material cost	Cm	$/lb	3	2.5	3	3	3
	Fraction of process that is scrap	f		0.05	0.05	0.05	0.05	0.05
	Mass of part	m	lb	0.564	0.665	0.438	0.5	0.11
	C_m unit cost of material			**1.781052632**	**1.75**	**1.3831578G5**	**1.578G47368**	**0.347368421**
Cl=cw/n	Labour cost	Cw	$/h	25	25	25	25	25
	Production rate	n	units/h	40	45	45	40	40
	C_l unit cost of labour			**0.625**	**0.555555556**	**0.555555556**	**0.625**	**0.625**
Ct=ct*k/n	Tooling cost	Ct	$/set	30000	35000	35000	30000	15000
	Total production run	n	units	100000	100000	100000	100000	100000
	Tooling life	nt	units	100000	100000	100000	30000	40000
	Sets of tooling required	k		1	1	1	3.333333333	2.5
	C_t unit cost of tooling			**0.3**	**0.35**	**0.35**	**1**	**0.375**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	40000	35000	40000	25000	50000
	Capital write-off time	Two	years	23400	23400	23400	11700	11700
	Load fraction	L		1	1	1	1	1
	Load sharing fraction	q		1	1	1	1	1
	C_e unit cost of capital equipment			**0.042735043**	**0.033238367**	**0.037G86705**	**0.053418803**	**0.106837607**
Coh=coh/n	Factory overhead	coh	$/h	90	90	85	75	75
	Production rate	n	units/h	40	45	45	40	40
	C_{oh} unit cost of factory overgead			**2.25**	**2**	**1.88888888G**	**1.875**	**1.875**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**4.GG8787674**	**4.6887G3G22**	**4.21558G044**	**5.132366172**	**3.32G206028**
				5.343157895	5.25	4.149473684	4.736842105	1.042105263
				OME achieved!				

Formula	Cost Element	Variables	Units	BF006-02	BF006-03	BF006-04	BF006-06	BF006-09
				Swivel base lock	Swivel base lock spring	Base plate lock	Power button	Wire holder 1
	Material			PC	Stainless steel	POM	PC/ABS	PC/ABS
Cm=m*cm/1-f	Material cost	Cm	$/lb	3.5	1	3	3	3
	Fraction of process that is scrap	f		0.05	0.05	0.05	0.05	0.05
	Mass of part	m	lb	0.0002	0.0034	0.052	0.002	0.00220462
	C_m unit cost of material			**0.000736842**	**0.003578G47**	**0.164210526**	**0.00631578G**	**0.006G61G58**
Cl=cw/n	Labour cost	Cw	$/h	25	25	25	25	25
	Production rate	n	units/h	55	60	60	55	60
	C_l unit cost of labour			**0.454545455**	**0.416666667**	**0.416666667**	**0.454545455**	**0.416666667**
Ct=ct*k/n	Tooling cost	Ct	$/set	30000	5000	28000	5000	6500
	Total production run	n	units	100000	100000	100000	100000	100000
	Tooling life	nt	units	200000	100000	100000	100000	80000
	Sets of tooling required	k		0.5	1	1	1	1.25
	C_t unit cost of tooling			**0.15**	**0.05**	**0.28**	**0.05**	**0.08125**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	100000	5000	10000	10000	13000
	Capital write-off time	Two	years	23400	23400	23400	23400	23400
	Load fraction	L		1	1	1	1	1
	Load sharing fraction	q		1	1	1	1	1
	C_e unit cost of capital equipment			**0.077700078**	**0.003561254**	**0.007122507**	**0.007770008**	**0.00G25G25G**
Coh=coh/n	Factory overhead	coh	$/h	75	75	75	75	75
	Production rate	n	units/h	55	60	60	55	60
	C_{oh} unit cost of factory overgead			**1.363636364**	**1.25**	**1.25**	**1.363636364**	**1.25**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**2.046618738**	**1.723806868**	**2.117GGG7**	**1.882267615**	**1.764137884**
				0.002210526	0.010736842	0.492631579	0.018947368	0.020885874
				OME achieved!				

Formula	Cost Element	Variables	Units	BF007-02	BF007-03	BF008-01	BF008-02	BF008-03
				Cam	180-swivel arm	Base plate	Oscillating base	Bearing
	Material			PC/ABS	Al	PC/ABS	PC/ABS	Stainless steel
Cm=m*cm/1-f	Material cost	Cm	$/lb	3	3	3	3	0.145
	Fraction of process that is scrap	f		0.05	0.1	0.05	0.381	0.05
	Mass of part	m	lb	0.002	0.008	0.168	0.35	0.00110231
	C_m unit cost of material			**0.00631578G**	**0.026666667**	**0.530526316**	**1.6G628433**	**0.000168247**
Cl=cw/n	Labour cost	Cw	$/h	25	30	25	25	25
	Production rate	n	units/h	60	60	60	40	55
	C_l unit cost of labour			**0.416666667**	**0.5**	**0.416666667**	**0.625**	**0.454545455**
Ct=ct*k/n	Tooling cost	Ct	$/set	5000	10000	28000	28000	5000
	Total production run	n	units	100000	100000	100000	100000	100000
	Tooling life	nt	units	100000	100000	100000	50000	100000
	Sets of tooling required	k		1	1	1	2	1
	C_t unit cost of tooling			**0.05**	**0.1**	**0.28**	**0.56**	**0.05**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	7500	10000	45000	45000	10000
	Capital write-off time	Two	years	23400	23400	23400	11700	23400
	Load fraction	L		1	1	1	1	1
	Load sharing fraction	q		1	1	1	1	1
	C_e unit cost of capital equipment			**0.00534188**	**0.007122507**	**0.032051282**	**0.0G6153846**	**0.007770008**
Coh=coh/n	Factory overhead	coh	$/h	75	75	75	75	75
	Production rate	n	units/h	60	60	60	40	55
	C_{oh} unit cost of factory overgead			**1.25**	**1.25**	**1.25**	**1.875**	**1.363636364**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**1.728324336**	**1.88378G174**	**2.50G244265**	**4.852438176**	**1.876120073**
				0.018947368	0.08	1.591578947	5.088852989	0.000504742
				OME achieved!				

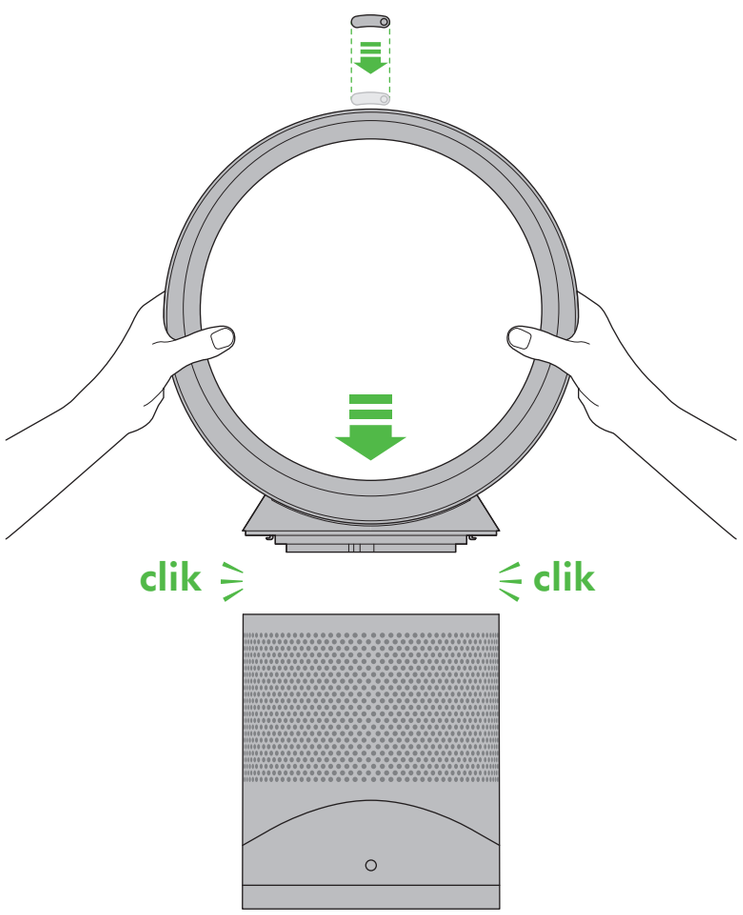
Formula	Cost Element	Variables	Units	BF008-04	BF008-05
				Power cord retainer	Rubber grip
	Material			PC/ABS	Silicon
Cm=m*cm/1-f	Material cost	Cm	$/lb	3	2
	Fraction of process that is scrap	f		0.05	0.05
	Mass of part	m	lb	0.002	0.001
	C_m unit cost of material			**0.00631578G**	**0.002105263**
Cl=cw/n	Labour cost	Cw	$/h	25	15
	Production rate	n	units/h	200	400
	C_l unit cost of labour			**0.125**	**0.0375**
Ct=ct*k/n	Tooling cost	Ct	$/set	5000	20000
	Total production run	n	units	100000	200000
	Tooling life	nt	units	100000	200000
	Sets of tooling required	k		1	1
	C_t unit cost of tooling			**0.05**	**0.1**
Ce=(1/n)*(ce/L*Two)*q	Capital cost	ce	$	25000	10000
	Capital write-off time	Two	years	23400	23400
	Load fraction	L		1	1
	Load sharing fraction	q		1	1
	C_e unit cost of capital equipment			**0.00534188**	**0.001068376**
Coh=coh/n	Factory overhead	coh	$/h	75	75
	Production rate	n	units/h	200	400
	C_{oh} unit cost of factory overgead			**0.375**	**0.1875**
Cm+Cl+Ct+Ce+Coh	**Total unit cost**			**0.56165767**	**0.32817363G**
				0.018947368	0.006315789
				OME achieved!	

dyson

See “Setting up Dyson Link app”
Consulte la sección “Cómo instalar la aplicación Dyson Link”

Operating manual
Instrucciones de seguridad

Assembly
Armado



Register today
Register your machine to be the first to receive Dyson updates and news.

Regístrate hoy
Registra tu máquina para ser el primero en recibir actualizaciones y noticias de Dyson.

IMPORTANT SAFETY INSTRUCTIONS

BEFORE USING THIS MACHINE READ ALL INSTRUCTIONS AND CAUTIONARY MARKINGS IN THIS MANUAL AND ON THE MACHINE

When using an electrical machine, basic precautions should always be followed, including the following:

⚠ WARNING

THE FAN AND THE REMOTE CONTROL BOTH CONTAIN MAGNETS.

1. Pacemakers and defibrillators may be affected by strong magnetic fields. If you or someone in your household has a pacemaker or defibrillator, avoid placing the remote control in a pocket or near to the device.
2. Credit cards and electronic storage media may also be affected by magnets and should be kept away from the remote control and the top of the machine.

⚠ WARNING

These warnings apply to the machine, and also where applicable, to all tools, accessories, chargers or mains adaptors.

TO REDUCE THE RISK OF FIRE, ELECTRIC SHOCK, OR INJURY:

1. This Dyson machine is not intended for use by persons (including children) with reduced physical, sensory or reasoning capabilities, or lack of experience and knowledge, unless they have been given supervision or instruction concerning use of the machine by a person responsible for their safety.
2. Ensure machine is fully assembled in line with the instructions before use. Do not dismantle the machine or use without the loop amplifier fitted.
3. Do not allow to be used as a toy. Close attention is necessary when used by or near children. Children should be supervised to ensure that they do not play with the machine.
4. Use only as described in this Dyson Operating Manual. Do not carry out any maintenance other than that shown in this manual, or advised by the Dyson Helpline.
5. Suitable for dry location ONLY. Do not use outdoors or on wet surfaces and do not expose to water or rain.
6. Do not handle any part of the plug or machine with wet hands.
7. Do not operate any machine with a damaged cord or plug. Discard the machine or return to an authorized service facility for examination and/or repair. If the supply cord is damaged it must be replaced by Dyson, its service agent or similarly qualified person in order to avoid a hazard.
8. If the machine is not working as it should, has received a sharp blow, has been dropped, damaged, left outdoors, or dropped into water, do not use and contact the Dyson Helpline.
9. Do not stretch the cord or place the cord under strain. Keep cord away from heated surfaces.
10. Do not run cord under carpeting. Do not cover cord with throw rugs, runners or similar coverings. Do not route cord under furniture or appliances. Arrange cord away from traffic area and where it will not be tripped over.
11. Do not unplug by pulling on the cord. To unplug, grasp the plug, not the cord. The use of an extension cord is not recommended.
12. Do not put any object into openings or the inlet grill. Do not use with any opening blocked; keep free of dust, lint, hair, and anything that may reduce airflow.
13. Do not use any cleaning agents or lubricants on this machine. Unplug before cleaning or carrying out any maintenance.
14. Always carry the machine by the base, do not carry it by the loop amplifier.
15. Turn off all controls before unplugging. Unplug from outlet when not in use for extended periods. To avoid a tripping hazard, safely coil the cord.
16. Do not use near furnaces, fireplaces, stoves or other high temperature heat sources.
17. Do not use in conjunction with or directly next to an air freshener or similar products. Do not spray or apply perfumes directly on or near the filter. Keep essential oils and chemicals away from the machine.

18. This product employs overload protection (fuse). A blown fuse indicates an overload or short-circuit situation. If the fuse blows, unplug the product from the outlet. Replace the fuse as per the user servicing instructions (follow product marking for proper fuse rating) and check the product. If the replacement fuse blows, a short-circuit may be present and the product should be discarded or returned to an authorized service facility for examination and/or repair.
19. WARNING: Chemical Burn and Choking Hazard. Keep batteries away from children. This product contains a lithium button/coin cell battery. If a new or used lithium button/coin cell battery is swallowed or enters the body, it can cause severe internal burns and can lead to death in as little as 2 hours. Always completely secure the battery compartment. If the battery compartment does not close securely, stop using the product, remove the batteries, and keep it away from children. If you think batteries might have been swallowed or placed inside any part of the body, seek immediate medical attention.
20. When removing or replacing the filter, take care to keep your fingers clear between the filter and the purifier base.

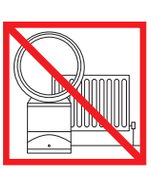
READ AND SAVE THESE INSTRUCTIONS

THIS DYSON MACHINE IS INTENDED FOR HOUSEHOLD USE ONLY

POLARIZATION INSTRUCTIONS

To reduce the risk of electric shock, this machine has a polarized plug (one blade is wider than the other) as a safety feature. This plug will fit in a polarized outlet only one way. If the plug does not fit fully in the outlet, reverse the plug. If it still does not fit, contact a qualified electrician to install the proper outlet. Do not attempt to defeat this safety feature and do not change the plug in any way.


Do not pull on the cord.


Do not store near heat sources.


Do not use near open flames.


Do not carry by the loop amplifier.


Do not spray scented products such as air freshener or perfume near the filter or the machine.

INSTRUCCIONES DE SEGURIDAD IMPORTANTES

ANTES DE UTILIZAR ESTA MÁQUINA, LEA TODAS LAS INSTRUCCIONES Y LA INDICACIONES DE PRECAUCIÓN QUE VIENEN TANTO EN ESTE MANUAL COMO EN LA MÁQUINA

Cuando se usa un máquina eléctrico, deben seguirse las precauciones básicas para reducir riesgos de incendio, descarga eléctrica y lesiones personales, que incluyen lo siguiente:

⚠ PELIGRO

El ventilador y el control remoto contienen imanes.

1. Los marcapasos y desfibriladores puedan verse afectados por fuertes campos magnéticos. Si usted o alguien en su casa tiene un marcapasos o un desfibrilador, evite colocar el control remoto en un bolsillo o cerca del máquina.
2. Es posible que los imanes también dañen las tarjetas de crédito y los medios de almacenamiento electrónico, por lo que deben mantenerse alejados del control remoto y de la parte superior del máquina.

⚠ PELIGRO

Estas advertencias se aplican al máquina, y cuando corresponde, a todas las herramientas, accesorios, cargadores y adaptadores de corriente.

PARA DISMINUIR EL RIESGO DE PROVOCAR UN INCENDIO, RECIBIR UNA DESCARGA ELÉCTRICA O LESIONES:

1. Este máquina Dyson no está destinada para el uso por parte de personas (incluso niños) con capacidades físicas, cognitivas o mentales reducidas, o con falta de experiencia y conocimiento, salvo que hubieran recibido supervisión o instrucciones acerca del uso del máquina por parte de una persona responsable por su seguridad.
2. Antes de utilizar el máquina, asegúrese de que esté completamente armado según las instrucciones. No desarme el máquina ni lo utilice sin colocar el bucle amplificador.
3. No permita que se utilice como un juguete. Es necesaria la atención de un adulto cuando es utilizada por o cerca a un niño. Los niños deben ser supervisados para asegurarse de que no jueguen con la máquina.
4. Utilícela solamente como se describe en este manual de funcionamiento Dyson. No realice ningún otro mantenimiento que no sea el mostrado en este manual, o el recomendado por la línea de ayuda de servicio al cliente Dyson.
5. Adecuado ÚNICAMENTE para lugares secos. No lo use al aire libre o sobre superficies húmedas ni lo exponga al agua o la lluvia.
6. No manipule el enchufe o de la máquina con las manos mojadas.
7. No ponga el máquina en funcionamiento si algún cable o enchufe está dañado. Deje de utilizar el máquina o llévelo a un centro de servicio técnico autorizado para que lo revisen y/o reparen. Si el cable de alimentación está dañado, deberá reemplazarlo Dyson, un agente autorizado u otra persona capacitada a fin de evitar cualquier peligro.
8. Si la máquina no funciona como es debido, ha recibido un golpe fuerte o se ha caído, dañado, dejado a la intemperie, o sumergido en agua, no lo utilice y comuníquese con la línea de ayuda de servicio al cliente Dyson.
9. No estire el cable ni lo someta a tensión. Mantenga el cable apartado de superficies calientes.
10. No pase el cable por debajo de alfombras. No cubra el cable con tapetes, alfombrillas u otras cubiertas similares. No pase el cable por debajo de muebles o aparatos. Mantenga el cable alejado de las áreas transitadas y de lugares donde pueda causar tropiezos.
11. No desconecte la máquina tirando del cable. Para desconectar, tome el enchufe y no el cable. No es recomendable utilizar una extensión de cable.
12. No coloque ningún objeto en aberturas o en la rejilla de entrada. No lo use si una abertura está obstruida; manténgalo libre de polvo, pelusa, cabellos y cualquier objeto que pueda debilitar el flujo de aire.
13. No utilice productos de limpieza o lubricantes en el máquina. Desconéctelo antes de limpiarlo o realizar tareas de mantenimiento.
14. Para el traslado, sujete el máquina por la base; nunca por el bucle amplificador.

15. Apague todos los controles antes de desconectarlo. Desenchúfelo de la toma si no lo usa durante períodos largos de tiempo. Para evitar el riesgo de tropezarse, enrolle de manera segura el cable.
16. No lo use cerca de hornos, chimeneas, estufas u otras fuentes térmicas de alta temperatura.
17. No lo utilice en combinación o directamente cerca de un aromatizador de aire o de productos similares. No rocíe ni aplique perfumes directamente sobre el filtro o cerca de él. Mantenga los aceites esenciales y los productos químicos lejos del máquina.
18. Este producto emplea una protección contra sobrecargas (fusible). Un fusible fundido indica que se produjo una sobrecarga o un cortocircuito. Si el fusible se funde, desconecte el producto de la toma de corriente. Reponga el fusible conforme a las instrucciones de servicio para el usuario (siga las indicaciones del producto en cuanto a la nominación correcta del fusible) y revise el producto. Si el fusible de repuesto se funde, puede existir un cortocircuito y se debe desechar el producto o llevarlo a examinar y/o reparar a un establecimiento de servicio autorizado.
19. ADVERTENCIA: Peligro de quemadura química y atragantamiento. Mantenga las baterías lejos de los niños. Este producto contiene una batería de litio tipo botón/moneda . Si una batería de litio tipo botón/moneda se traga o ingresa en el cuerpo, puede causar quemaduras internas graves y conducir a la muerte en menos de 2 horas. Siempre cierre de manera segura y por completo el compartimento de baterías. Si el compartimento de baterías no cierra de manera segura, deje de utilizar el producto, quite las baterías y manténgalas lejos de los niños. Si cree que una batería ha sido tragada o introducida en cualquier parte del cuerpo, busque atención médica de inmediato.
20. Cuando quite o reemplace el filtro, tenga cuidado de mantener los dedos alejados del espacio entre el filtro y la base del purificador.

LEA Y GUARDE ESTAS INSTRUCCIONES

ESTA MÁQUINA DYSON ES PARA USO DOMÉSTICO SOLAMENTE

INSTRUCIONES DE POLARIZACIÓN

Para reducir el riesgo de choque eléctrico, este aparato tiene un enchufe polarizado (una clavija es más ancha que la otra) como una característica de seguridad. Este enchufe encajará en un tomacorriente polarizado de una sola manera. Si el enchufe no entra completamente en el tomacorriente, invierta su posición. Si aun así no encaja correctamente, comuníquese con un electricista calificado para instalar el tomacorriente adecuado. No intente anular la característica de seguridad y no modifique el enchufe de ninguna manera.



No jale el cable.



No guarde la máquina cerca de fuentes de calor.



No utilice la máquina cerca de una llama.

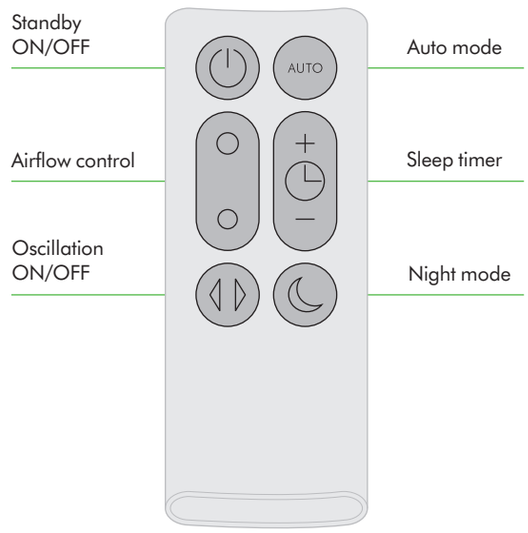


No lo transporte sujetando el bucle amplificador.

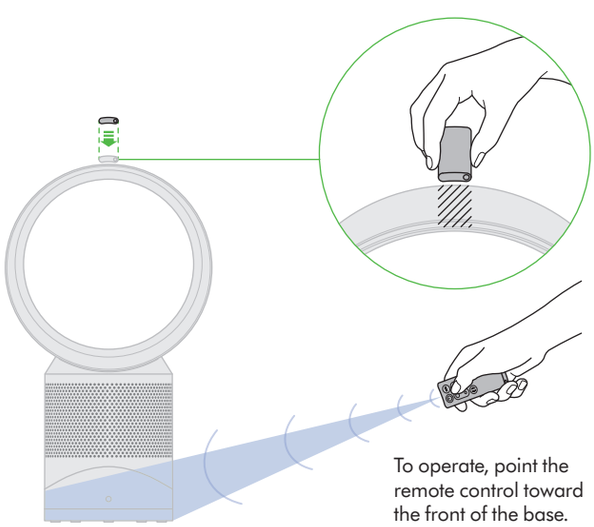


No rocíe productos con aromas, como aromatizadores de aire o perfumes, cerca del filtro o del máquina.

Remote control



Attach the remote control to the top of the machine with the buttons facing down.



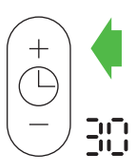
To operate, point the remote control toward the front of the base.

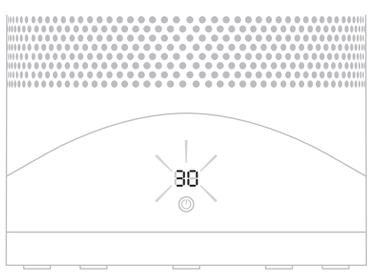
Display icons



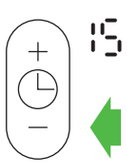
Flashing green = Wi-Fi is ready to connect to your Dyson Link app	Solid white = cleaning/ air quality target has not been met
Flashing white = Wi-Fi connecting	Solid green = clean/air quality target has been met
Solid white = Wi-Fi connected	Off = machine is not in auto cleaning mode
Off = Wi-Fi disabled	Solid white = machine is in night mode
	Off = machine is not in night mode

Sleep timer

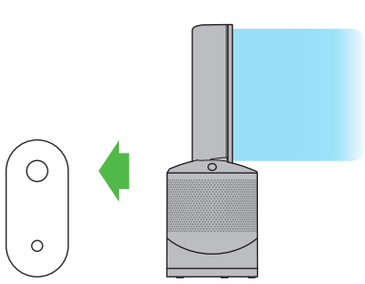




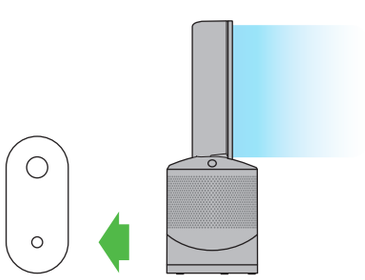
To set the sleep timer, select the desired time. When time reaches zero, the machine will go into standby mode. To cancel, decrease the time until two dashes appear.



Airflow

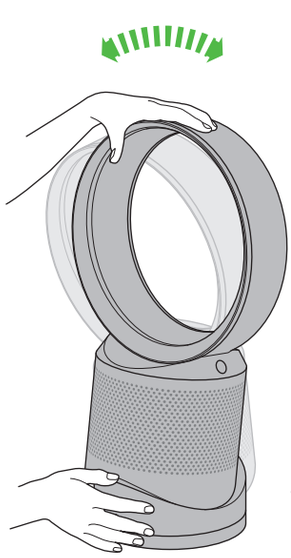


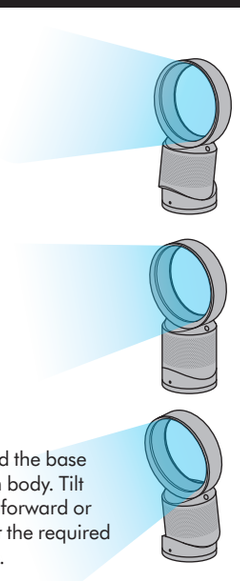
Select a high flow of air.



Select a low flow of air.

Tilt function





Carefully hold the base and the main body. Tilt the amplifier forward or backward for the required airflow angle.

Night mode

The night mode will dim the display and run quietly, making it ideal for use while sleeping.

When night mode is enabled (indicated by an illuminated half moon symbol in the top right corner of the display), the display will dim and after 5 seconds will fade to off. The half moon symbol remains illuminated.



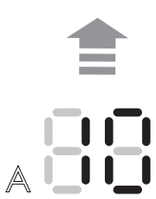
If the machine is not in auto mode, the airflow speed will reduce to level 4. When night mode is activated, if it is already below level 4, it will remain the same. This can be manually changed if desired. The display will illuminate briefly, then fade as before. If in auto mode the airflow speed will be reduced automatically.

Auto mode

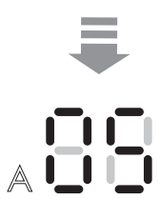
AUTO

Selecting auto mode adjusts the airflow according to the air quality in the room.

Poor air quality



Air quality improving



Air quality target has been met.



When auto mode is enabled (indicated by an A symbol in the bottom left corner of the display), the on-board sensors will intelligently adjust the speed of the machine according to the air quality.

If the air is clean the A symbol will turn green and the numerical display will switch off. If the air quality worsens the A symbol will turn white and the machine will automatically restart cleaning the room. While cleaning, the numerical display will show the airflow speed.

For additional control the Dyson Link app can be used to further adjust the sensitivity of the machine.

A period of 6 days is required after the machine is first used for the sensor to calibrate. During this period the machine may be more sensitive to VOCs (such as odors) than normal.

Setting up Dyson Link app

The machine can be enhanced when connected to the Dyson Link app on your smart device.

dyson

Download the Dyson Link app, available on iOS and Android.



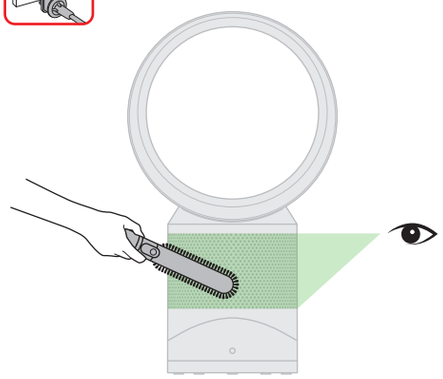
Select 'Add a Product' from within the app and follow the on screen instructions.



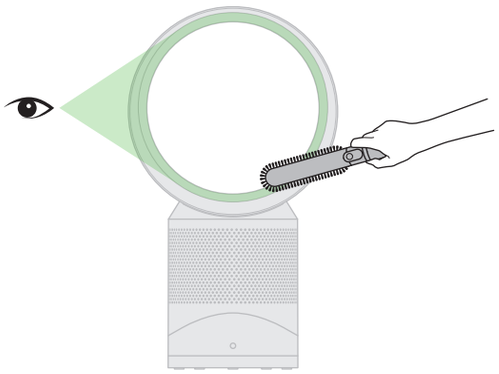
In order to connect the machine, you will need the information on the front of this manual.

Clearing blockages

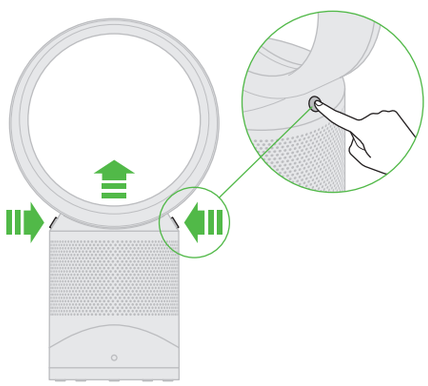




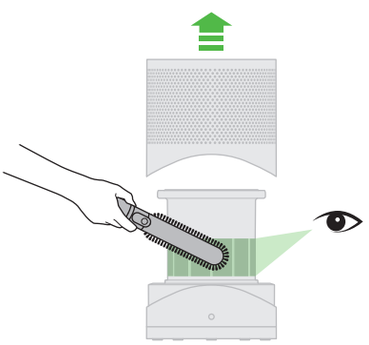
Ensure the machine is unplugged. Look for blockages in the air inlet holes on the filter. Use a soft brush to remove debris.



Look for blockages in the small aperture inside the loop amplifier. Use a soft brush to remove debris.

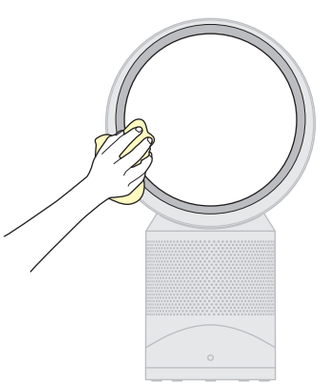


Press buttons to lift off loop amplifier and lift off filter.

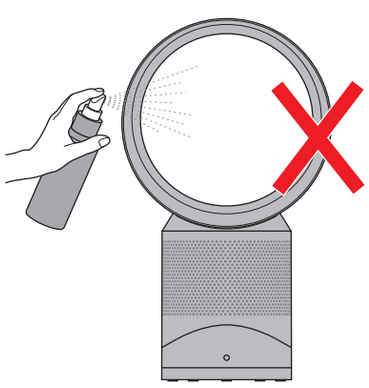


Look for blockages in the air inlet holes under the filter and use a soft brush to remove debris. Replace the filter and then the loop amplifier.

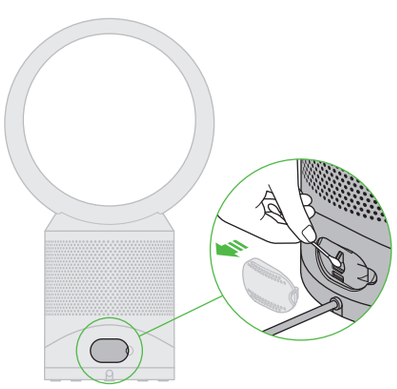
Cleaning



Ensure the machine is unplugged. Dust may accumulate on the surface of the machine. Wipe dust from the loop amplifier, filter unit and other parts with a dry or damp cloth when dust accumulates.



Do not use detergents or polishes to clean the machine.

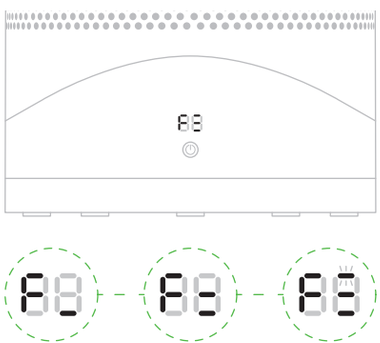


Clean the hatch every time the filter is replaced. Open the small hatch on the side of the machine. Use a cotton swab to clean inside and remove any debris. This maintains sensor accuracy and ensures the continuing high performance of the machine.

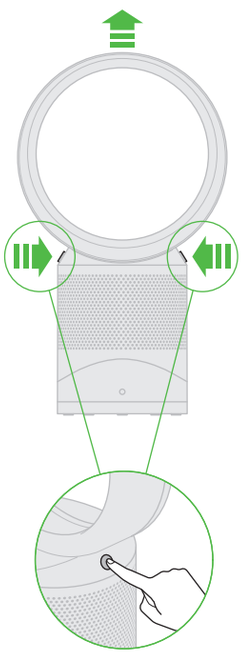
Replacing the filter unit

1

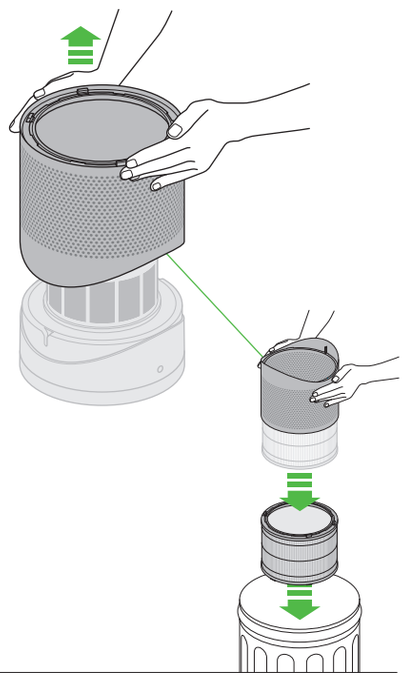
When your non-washable filter unit needs replacing the machine will display the pattern shown. To replace the non-washable filter unit, follow these steps. Ensure your machine is unplugged. Your Dyson machine is programmed to remind you to replace the filter unit after 12 months based on 12 hours usage a day. Please note your indicator will activate sooner with more use. The indicator is only a guide. Use in heavily polluted environments may require the filter unit to be replaced more regularly. Outside air quality can be checked through the Dyson Link app.



2

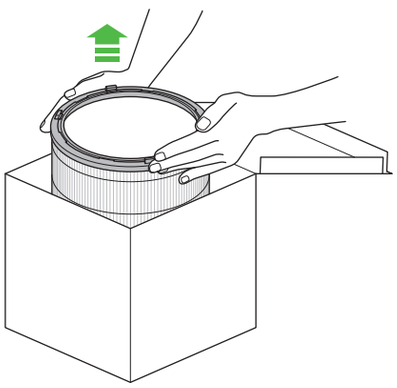


3

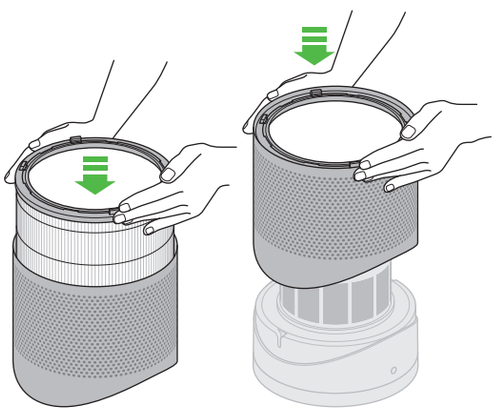


4

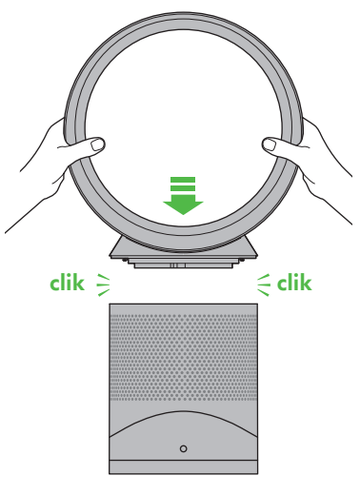
When disposing of the exhausted filter unit please do so in accordance with local regulations.



5

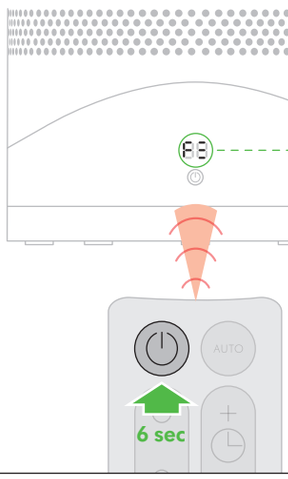


6



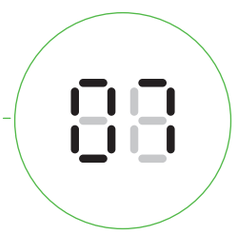
7

When you have re-assembled your machine, press and hold the 'Standby ON/OFF' button for 6 seconds on the remote control to reset the filter indicator.





A clockwise-rotating loading pattern will be shown on the LED display.

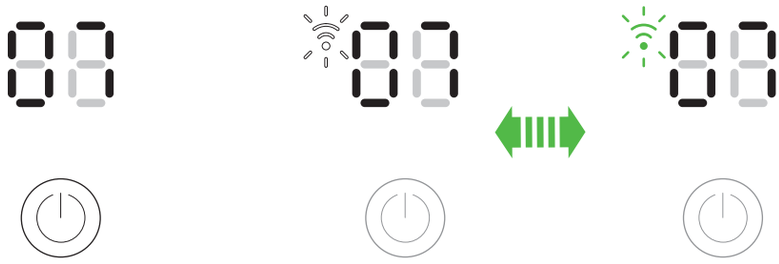


Once the loading pattern is complete, the machine will return to its normal usage.

Clearing your Wi-Fi settings

It is possible to remove your network details from the machine.

21 sec



Press and hold the Standby ON/OFF button on the machine for longer than 21 seconds.

The Wi-Fi symbol will flash alternately between white and green to indicate that your network details have been removed.

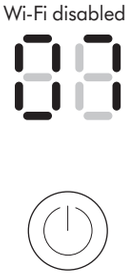
If you clear your network settings the machine will not connect to the Dyson Link app.

Enabling or Disabling Wi-Fi

Wi-Fi can be enabled or disabled by pressing and holding the ON/OFF button on the machine for between 10 and 20 seconds.

10-20 sec

Wi-Fi disabled



Wi-Fi enabled



Solid white = Wi-Fi connected

Flashing white = Wi-Fi connecting

Flashing green = Wi-Fi is ready to connect to your Dyson Link app

Off = Wi-Fi disabled

If Wi-Fi is disabled, the machine will not connect to the Dyson Link app.

Troubleshooting



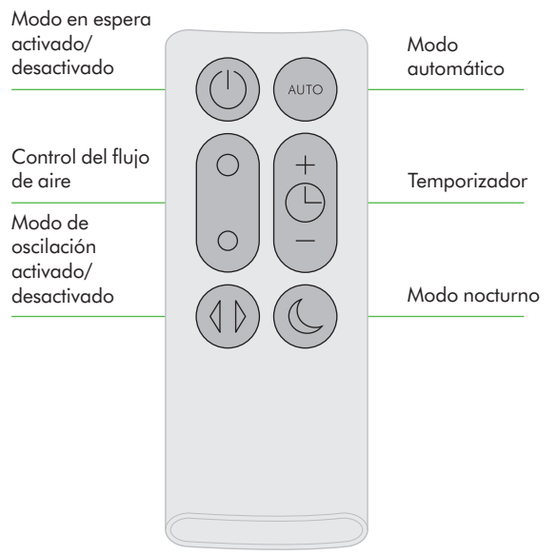
If you see a flashing 'E' when trying to use the machine, it is unable to do what has been requested.



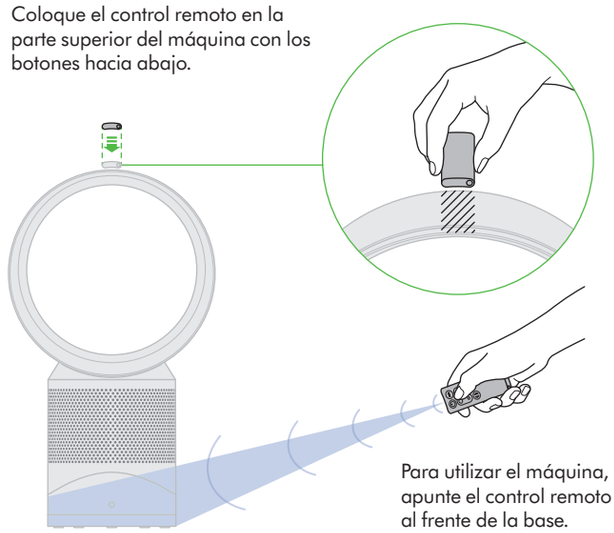
If you see a solid 'E' when trying to use the machine, it shows the machine has stopped functioning correctly.

Make sure that you have checked the following:
The room temperature is above 5°C/41°F
There are no blockages in the inlet grille or loop amplifier outlet.
The filter has been correctly assembled and doesn't need to be replaced.
The machine is correctly plugged in directly to the wall outlet.
If the message still displays, unplug and leave for 10 seconds before turning on again.
If the problem persists, please call the Dyson Helpline at 1-866-693-9766.

Control remoto



Coloque el control remoto en la parte superior del máquina con los botones hacia abajo.



Para utilizar el máquina, apunte el control remoto al frente de la base.

Mostrar iconos



Blanco sólido = el objetivo de limpieza/calidad del aire no se ha cumplido

Luz verde intermitente = el Wi-Fi ya está conectado con la aplicación Dyson Link

Luz verde sólida = limpio/ se cumplió el objetivo de calidad del aire

Luz blanca intermitente = el Wi-Fi está conectándose

Apagado = el máquina no está en modo de limpieza automática

Luz blanca sólida = el Wi-Fi está conectado

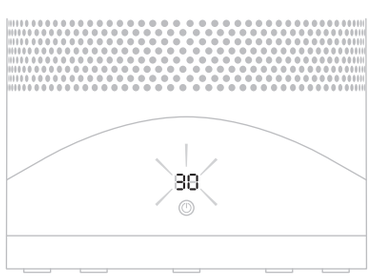
Luz blanca sólida = el máquina está en modo nocturno

Apagado = el Wi-Fi está desactivado

Apagado = el máquina no está en modo nocturno

Temporizador

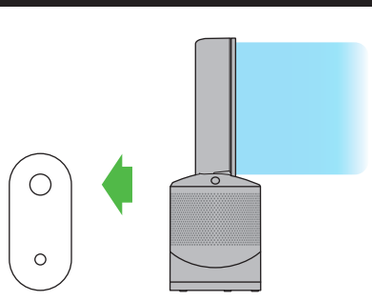




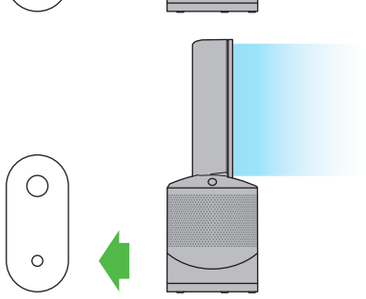
Para configurar el temporizador, seleccione el tiempo deseado. Cuando el tiempo llegue a cero, el máquina pasará al modo en espera. Para cancelar, reduzca el tiempo hasta que aparezcan dos rayas.



Flujo de aire



Seleccione un flujo de aire alto.



Seleccione un flujo de aire bajo.

Función de inclinación



Sujete con cuidado la base y el cuerpo principal. Incline hacia adelante o hacia atrás para obtener el ángulo de flujo del aire que desea.

Modo nocturno

El modo nocturno atenuará la pantalla y hará que funcione silenciosamente, lo cual es ideal para usar el máquina mientras se duerme.

Cuando se activa el modo nocturno (indicado por un símbolo de media luna iluminado en la esquina superior derecha de la pantalla), la pantalla se atenuará y a los 5 segundos se apagará. El símbolo de la media luna permanece iluminado.



Si el máquina no está en modo automático, la velocidad del flujo de aire se reducirá al nivel 4. Cuando se activa el modo nocturno, si ya está por debajo del nivel 4, permanecerá así. Esto se puede cambiar manualmente si lo desea. La pantalla se iluminará por un instante y luego se apagará como antes. Si está en modo automático, la velocidad del flujo de aire se reducirá automáticamente.
Si está en modo automático, la energía se reducirá automáticamente.

Modo automático

AUTO

La selección del modo automático ajusta el flujo de aire de acuerdo con la calidad de aire de la habitación.

Baja calidad del aire



La calidad del aire necesita mejorar



Se cumplió con el objetivo de calidad del aire



Cuando se activa el modo automático (indicado por un símbolo A en la esquina inferior izquierda de la pantalla), los sensores de a bordo ajustarán de manera inteligente la velocidad del máquina según la calidad del aire.

Si el aire está limpio, el símbolo A se volverá verde y la pantalla numérica se apagará. Si la calidad del aire empeora, el símbolo A se volverá blanco y el máquina volverá a empezar a limpiar la habitación automáticamente. Durante la limpieza, los números en la pantalla indicarán la velocidad del flujo de aire.

Para un control adicional, la aplicación del vínculo Dyson se puede utilizar para realizar más ajustes a la sensibilidad del máquina.
Se requiere un período de 6 días después del primer uso de la máquina para que el sensor se calibre. Durante ese período, la máquina puede ser más sensible que lo normal a los compuestos orgánicos volátiles (VOC) (como los olores).

Cómo configurar la aplicación Dyson Link

El máquina puede ser mejorado al conectarlo con la aplicación Dyson Link en su equipo inteligente.

Descargue la aplicación Dyson Link, disponible en iOS y Android.



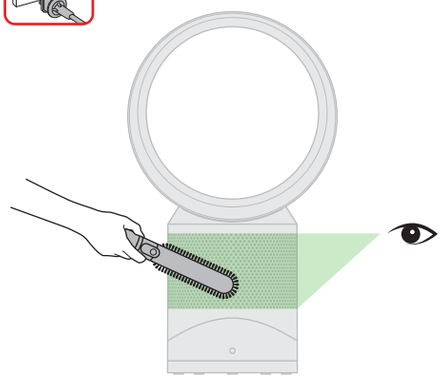
Seleccione "Add a Product" (Agregar un producto) dentro de la aplicación y siga las instrucciones en pantalla.



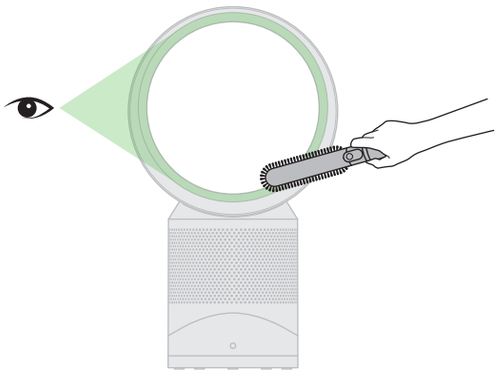
Para conectar el máquina, necesitará la información que aparece al frente de este manual.

Para eliminar bloqueos

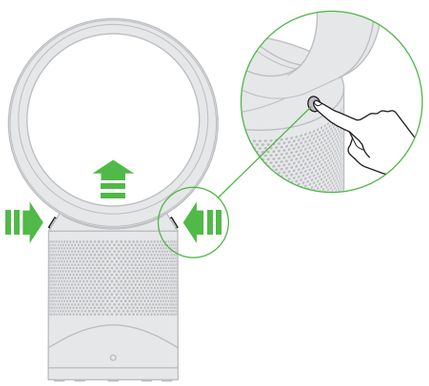




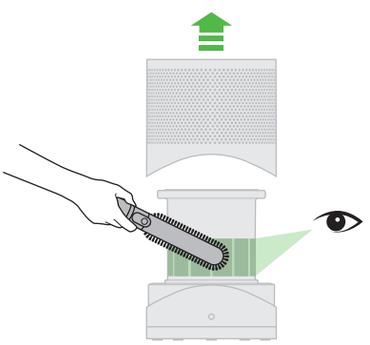
Asegúrese de que el máquina esté desconectado. Verifique que no haya obstrucciones en las entradas de aire del filtro. Utilice un cepillo suave para eliminar cualquier desecho.



Verifique que no haya obstrucciones en la pequeña apertura dentro del bucle amplificador. Utilice un cepillo suave para eliminar cualquier desecho.

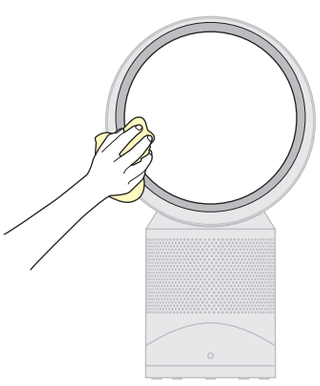


Oprima los botones para levantar el bucle amplificador y levantar el filtro.

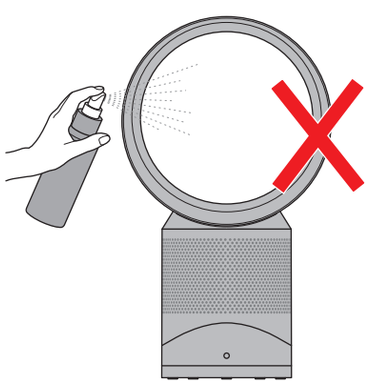


Verifique que no haya obstrucciones en las entradas de aire debajo del filtro y use un cepillo suave para eliminar desechos. Cambie el filtro y luego el bucle amplificador.

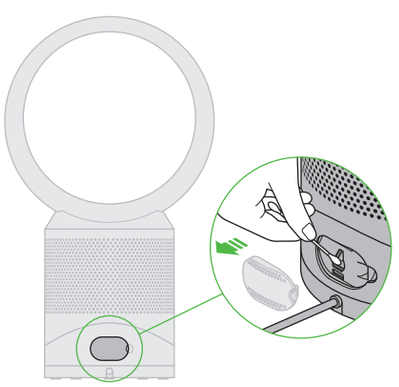
Limpieza



Asegúrese de que el máquina esté desconectado. Se puede acumular polvo en la superficie del máquina. Cuando se acumule polvo, quítelo del bucle amplificador, del filtro de la unidad y de otras partes con un paño seco o húmedo.



No utilice detergentes ni pulidores para limpiar el máquina.



Limpie la trampilla cada vez que cambie el filtro. Abra la trampilla pequeña al costado del máquina. Use una bola de algodón para limpiar el interior y quitar los desechos. Esto mantiene la exactitud del sensor y asegura un alto desempeño continuo del máquina.

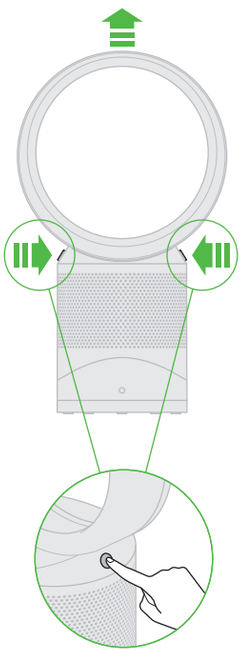
Reemplazo de la unidad de filtro

1

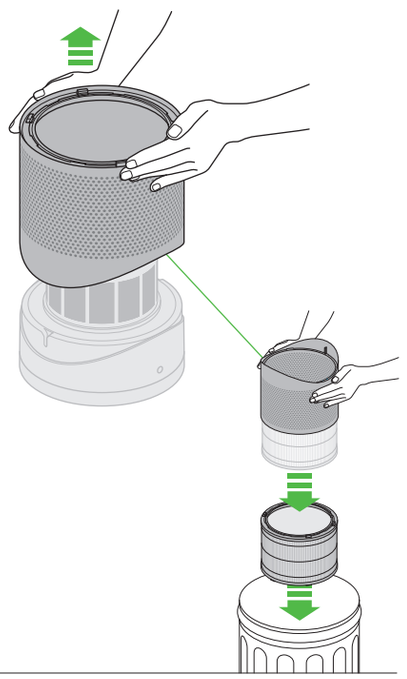
Cuando la unidad de filtro no lavable debe reemplazarse, el máquina exhibirá el patrón que se muestra. Para reemplazar la unidad de filtro no lavable, siga estos pasos. Asegúrese de que el aparato esté desenchufado. Su máquina Dyson está programada para recordarle que reemplace la unidad de filtro después de 12 meses sobre la base de un uso de 12 horas por día. Observe que el indicador se activará antes si hay un mayor uso. El indicador es sólo una guía. El uso en ambientes altamente contaminados puede requerir que la unidad de filtro se reemplace más regularmente. La calidad del aire exterior se puede controlar mediante la aplicación Dyson Link.



2

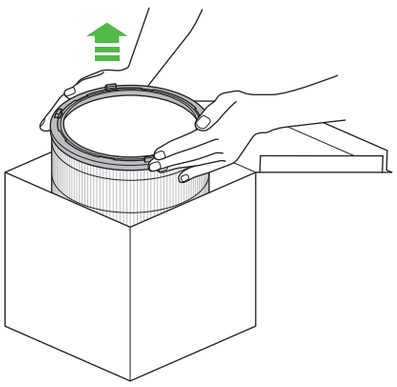


3

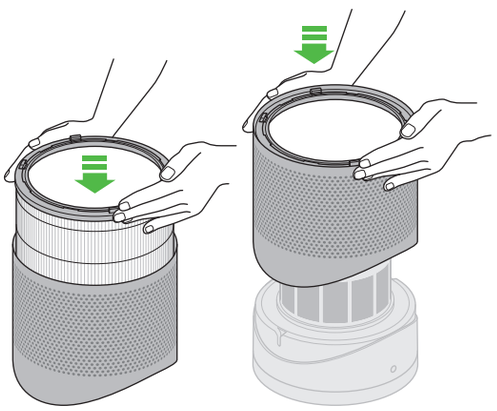


4

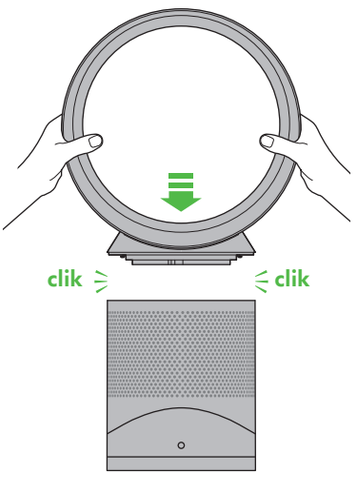
Cuando deseche la unidad de filtro gastada, hágalo según las regulaciones locales.



5

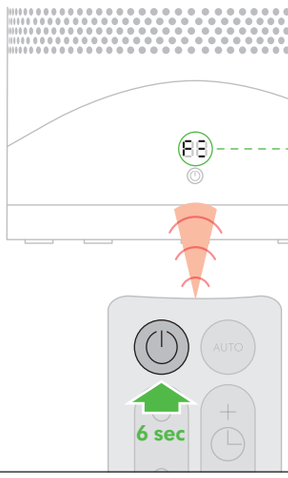


6



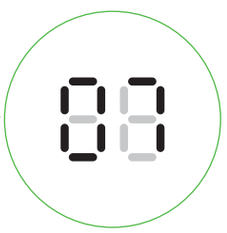
7

Cuando haya vuelto a montar el máquina, mantenga presionado el botón de encendido durante 6 segundos en el control remoto para restablecer el indicador del filtro.





Se mostrará en la pantalla LED un patrón de carga que gira en el sentido de las agujas del reloj.

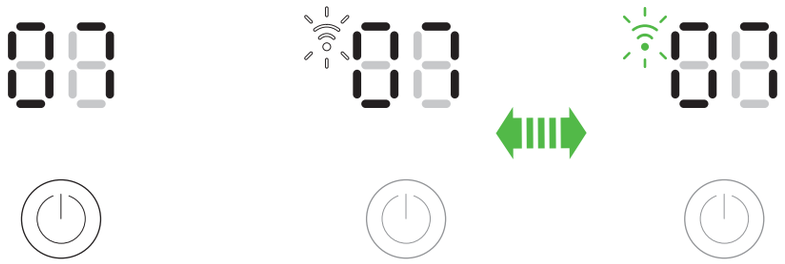


Una vez que se haya completado el patrón de carga, el máquina volverá a su uso normal.

Cómo restablecer la configuración del Wi-Fi

Es posible eliminar sus detalles de red del máquina.

21 sec



Mantenga presionado el botón ON/OFF (Encendido/Apagado) de la máquina durante más de 21 segundos.

El símbolo de Wi-Fi parpadeará en forma intermitente entre blanco y verde para indicar que los detalles de la red han sido eliminados.

Si borra sus configuraciones de red, el máquina no se conectará a la aplicación del vínculo Dyson.

Cómo activar o desactivar el Wi-Fi

El Wi-Fi se puede activar o desactivar manteniendo presionado el botón ON / OFF (Encendido / Apagado) del máquina entre 10 y 20 segundos.

10-20 sec

El Wi-Fi está desactivado



Compatible con Wi-Fi



Luz blanca sólida = el Wi-Fi está conectado

Luz blanca intermitente = el Wi-Fi está conectándose

Luz verde intermitente = el Wi-Fi ya está conectado con la aplicación Dyson Link

Apagado = el Wi-Fi está desactivado

Si el Wi-Fi está desactivado, el máquina no podrá conectarse a la aplicación Dyson Link.

Resolución de problemas



Si ve una "E" intermitente mientras trata de usar el máquina, es porque no puede hacer lo que se le ha solicitado.



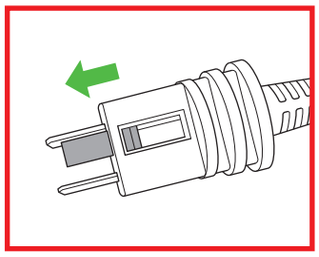
Si ve una "E" de color sólido mientras trata de usar el máquina, es para indicar que el máquina dejó de funcionar correctamente.

Asegúrese de haber revisado lo siguiente:
La temperatura de la habitación supera los 5 °C/ 41°F
No hay obstrucciones en la rejilla de entrada o en la salida del bucle amplificador.
El filtro se instaló correctamente y no debe ser reemplazado.
La máquina está enchufada en forma apropiada y directamente en el tomacorrientes de pared.
Si aún aparece el mensaje, desenchufe la máquina y espere 10 segundos antes de volver a encenderla. Si el problema persiste, llame a la línea directa de Asistencia al cliente de Dyson, 1-866-693-9766.

USER SERVICING INSTRUCTIONS

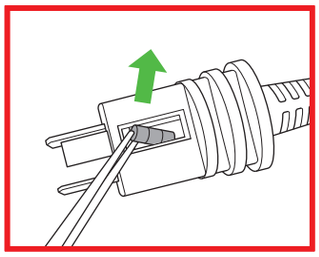
DO NOT REPLACE ATTACHMENT PLUG

Risk of fire. Do not replace attachment plug. Contains a safety device (fuse) that should not be removed. Discard product if the attachment plug is damaged.

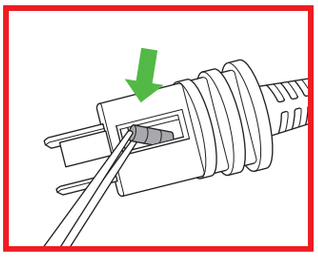


Grasp plug and remove from the receptacle or other outlet device. Do not unplug by pulling on cord.

Slide open the fuse access cover on top of the plug towards blades.

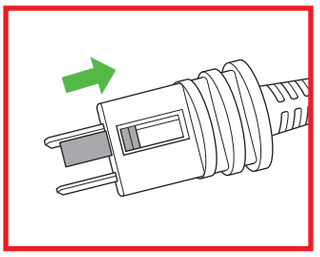


Remove fuse carefully.



Risk of fire.

Replace fuse only with 5 Amp, 125 Volt fuse.



Close the fuse cover.

Slide closed the fuse access cover on top of the attachment plug.

USING YOUR DYSON MACHINE

PLEASE READ THE 'IMPORTANT SAFETY INSTRUCTIONS' IN YOUR DYSON USER MANUAL BEFORE PROCEEDING.

LOOKING AFTER YOUR DYSON MACHINE

- Do not carry out any maintenance or repair work other than that shown and described in your Dyson User manual, or advised by the Dyson Helpline.
- Always disconnect the plug from the mains outlet before inspecting for problems. If the machine will not operate, first check that the mains outlet has an electricity supply and that the plug is properly inserted into the outlet.
- If the fuse needs replacing, please check 'User servicing instructions'.

CONTROL WITHOUT REMOTE

- Your machine can be controlled through the Dyson Link app.

REPLACEABLE PARTS

BATTERY REPLACEMENT

⚠ CAUTION

- Unscrew the battery compartment on the remote control. Loosen the base and pull to remove the battery.
- Do not install backwards or short circuit the batteries.
- Do not attempt to dismantle or charge the batteries. Keep away from fire.
- Follow battery manufacturers' instructions when installing new batteries (battery type CR 2032).
- Always replace the screw in the remote control and refer to the battery hazard in the warning section.

NON-WASHABLE FILTER UNIT

- Your filter unit is non-washable and non-recyclable.
- To replace your filter unit follow the steps as shown.
- Failure to replace the filter when prompted may result in changes to product performance and appearance.
- A new filter unit can be purchased at www.dyson.com/filter.

AUTO MODE

- A period of six days is required after your machine is first used for the sensor to calibrate. During this period, your machine may be more sensitive to VOCs (such as odors) than normal.

DISPOSAL INFORMATION

- Dispose of or recycle the battery in accordance with local ordinances or regulations.
- Keep the used batteries away from children as these can still harm children if swallowed.
- Dyson products are made from high grade recyclable materials. Recycle where possible.
- Your filter unit is non-washable and non-recyclable.
- The battery should be removed from the machine before disposal.
- Dispose of the exhausted filter in accordance with local ordinances or regulations.
- California: This product includes a separate remote control device with a coin cell battery. The coin cell battery contains Perchlorates. Special handling may apply. See www.dtsc.ca.gov/hazardouswaste/perchlorate

DYSON CUSTOMER CARE

THANK YOU FOR CHOOSING TO BUY A DYSON MACHINE.

We don't stop caring about our machines once they're yours. With your free 2-year warranty, your Dyson purifying fan will be covered for parts and labor (excluding filters) for two years from the date of purchase, subject to the terms of the warranty. Even after your warranty has ended, we'll still be on hand to help.If you have any questions regarding your Dyson machine, visit www.dyson.com/support for online help, general tips and useful information about Dyson.
Alternatively, you can call the Dyson Helpline with your serial number and details of where and when you bought the machine.
If your Dyson machine needs a repair, call the Dyson Helpline so we can discuss the available options. If your Dyson machine is under warranty, and it is a covered repair, it will be repaired at no cost.
Your serial number can be found on your rating plate which is on the base of your machine.

REGISTER YOUR MACHINE TO BE THE FIRST TO RECEIVE DYSON UPDATES AND NEWS

To help us ensure you receive a prompt and efficient service, please register as a Dyson machine owner. There are four ways to do this:

- Online at www.dyson.com/register
- Telephone the Dyson Helpline on 1-866-693-9766.
- Complete the enclosed Warranty Form and mail it to us.
- Smartphone. Download the Dyson Link app and you will be taken through registration as part of the set up.

This will confirm ownership of your Dyson machine in the event of an insurance loss, and enable us to contact you if necessary.

LIMITED 2-YEAR WARRANTY

TERMS AND CONDITIONS OF THE DYSON 2-YEAR LIMITED WARRANTY

WHAT IS COVERED

This limited warranty is subject to the following terms:

- Your Dyson machine is warranted against original defects in materials and workmanship for a period of 2 years from the date of purchase, when used for normal domestic purposes in accordance with the Dyson Operating Manual. Please retain your proof of purchase. If you do not have your proof of purchase, your warranty will start 90 days after the date of manufacture, according to Dyson's records. This warranty provides, at no extra cost to you, all labor and parts necessary to ensure that your machine is in proper operating condition during the warranty period. The replacement of any part of your machine under warranty will not extend the period of warranty; however, your warranty will be extended for any time your machine is being repaired or retained for work being performed under the warranty. Any parts which are returned and replaced by Dyson will become the property of Dyson. Please call the Dyson Helpline on 1-866-693-9766 for more information.
- For refurbished machines, length of warranty may differ, please see warranty letter included with machine for warranty terms.
- This warranty is only valid if the machine is used in the country in which it was sold.

WHAT IS NOT COVERED

- Replacement filters. The machine's filter is not covered by the warranty.

Dyson shall not be liable for costs, damages or repairs incurred as a result of, due to, or from:

- Damage caused by not carrying out the recommended machine maintenance.
- Purchases of machines from unauthorized dealers.
- Careless operation or handling, misuse, abuse and/or lack of maintenance or use not in accordance with the Dyson User manual.
- Use of the machine other than for normal domestic purposes within the United States, e.g. for commercial or rental purposes.
- Use of parts not in accordance with the Dyson User manual.
- Use of parts and accessories other than those produced or recommended by Dyson.

- External sources such as weather, electrical outages or power surges.
- Repairs or alterations carried out by unauthorized parties or agents.
- Blockages – please refer to the illustrations and the section 'Clearing blockages' in this Dyson User manual for details on how to look for and clear blockages.
- Normal wear and tear (e.g. fuse etc.).
- Damage as a result of use not in accordance with the rating plate.
- Additionally, this limited warranty does not apply to any machine that has had the serial number altered or removed.
- If there is any doubt as to what is covered by the warranty, please contact Dyson.

WARRANTY LIMITATIONS AND EXCLUSIONS

- To the fullest extent permitted by applicable law, any implied warranties relating to your machine including but not limited to warranty of merchantability or warranty of fitness for a particular purpose, are limited to the duration of this warranty.
- To the fullest extent permitted by applicable law, your remedy for breach of this warranty is limited to the warranty service described below. Dyson will not be liable for any consequential damages or incidental damages you may incur in connection with your purchase and use of your machine.
- This limited warranty gives you specific legal rights. You may also have other rights which vary by State.
- This limited warranty is not transferable.
- The limitations or exclusions of liability stated herein apply in the state of New Jersey, to the fullest extent permitted by applicable law.

WARRANTY SERVICE

- Before a service can take place, the terms of the warranty require that you contact the Dyson Helpline with your model name/number, serial number, date and place of purchase.
- Please keep your purchase receipt in a safe place to ensure that you have this information.
- All work will be coordinated by Dyson or its authorized agents.
- Any replaced defective parts will become the property of Dyson.
- Service under this warranty will not extend the period of this warranty; however, your warranty will be extended for any time your machine is being repaired or retained for work being performed under the warranty.
- If your machine is not functioning properly, please call the Dyson Helpline on 1-866-693-9766. Most issues can be rectified over the telephone by our trained Helpline staff.

IMPORTANT DATA PROTECTION INFORMATION

You will need to provide us with basic contact information when you register your Dyson machine or via the Dyson Link app.
When registering your Dyson machine:

- You will need to provide us with basic contact information to register your machine and enable us to support your warranty. Failure to register will not diminish your warranty rights.

When registering via the Dyson Link app:

- You will need to provide us with basic contact information to register the Dyson Link app. This enables us to securely link your machine to your instance of the app.
- When you register, you will have the opportunity to choose whether you would like to receive communications from us. If you opt-in to communications from Dyson, we will send you details of special offers and news of our latest innovations.
- We never sell your information to third parties and only use information that you share with us as defined by our privacy policies which are available on our website at privacy.dyson.com.

CONFORMITY INFORMATION

This device complies with Part 15 of the FCC rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.
This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation.
If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

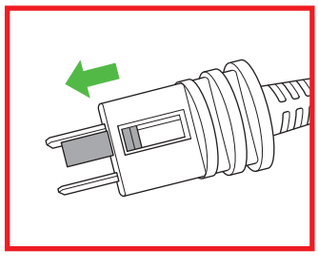
- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

This equipment complies with FCC radiation exposure limits set forth for an uncontrolled environment. In order to avoid the possibility of exceeding the FCC radio frequency exposure limits, human proximity to the antenna shall not be less than 7.9 in (20 cm) during normal operation.
Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

INSTRUCCIONES DE SERVICIO PARA EL USUARIO

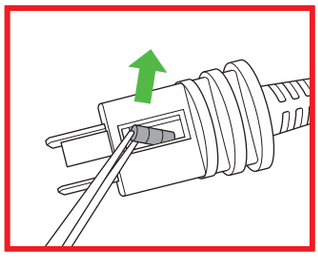
NO REEMPLACE EL ENCHUFE ACCESORIO

Riesgo de incendio. No reemplace el enchufe accesorio. Contiene un dispositivo de seguridad (fusible) que no se debe quitar. Deje de utilizar el producto si el enchufe accesorio está dañado.

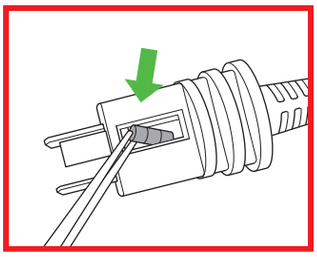


Agarre el enchufe y retírelo del receptáculo u otro dispositivo de toma de corriente. No lo desenchufe jalando el cable.

Deslice la tapa de acceso al fusible situada sobre el enchufe, hacia las aletas.

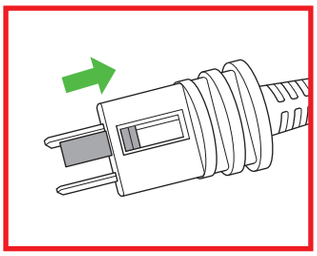


Extraiga con cuidado el fusible.



Riesgo de incendio.

Reemplace utilizando solamente un fusible de 5 amperios, 125 voltios.



Cierre la tapa de fusible.

Deslice de regreso la tapa de acceso al fusible sobre el enchufe, para cerrar.

USO DE SU MÁQUINA DYSON

ANTES DE CONTINUAR, SÍRVASE LEER LAS "INSTRUCCIONES DE SEGURIDAD IMPORTANTES" DE ESTE MANUAL DE FUNCIONAMIENTO DYSON.

CÓMO CUIDAR SU MÁQUINA DYSON

- No realice ningún tipo de mantenimiento o reparaciones que no sean las indicadas en la Guía del usuario de Dyson o en la Ayuda en línea de Dyson.
- Antes de revisar la máquina para detectar cualquier problema, desconecte el enchufe de la red eléctrica. Si la máquina no funciona, verifique primero que la toma de corriente de la red eléctrica tenga suministro y que el enchufe esté colocado correctamente en la toma de corriente.
- Si es necesario cambiar el fusible, consulte las 'Instrucciones de servicio para el usuario'.

SIN CONTROL REMOTO

- La máquina puede ser controlada mediante la aplicación Dyson Link.

PIEZAS REEMPLAZABLES

REEMPLAZO DE BATERÍA

⚠ PRECAUCIÓN:

- Desatornille el compartimento de la batería del control remoto. Desajuste la base y tire para quitar la batería.
- No instale al revés ni ponga en cortocircuito las baterías.
- No intente desmontar o cargar las baterías. Mantenga alejado del fuego.
- Siga las instrucciones del fabricante de las baterías al instalar nuevas baterías (batería tipo CR 2032).
- Vuelva a colocar siempre el tornillo en el control remoto y consulte los peligros relacionados con la batería en la sección de advertencias.

UNIDAD DE FILTRO NO LAVABLE

- La unidad de filtro no es lavable ni reciclable.
- Para reemplazar la unidad de filtro, siga los pasos según se muestra.
- No sustituir el filtro cuando se indica puede dar lugar a cambios en el rendimiento y el aspecto del producto.
- Puede comprar una unidad de filtro nueva en www.dyson.com/filter.

MODO AUTOMÁTICO

- Se requiere un período de 6 días después del primer uso de la máquina para que el sensor se calibre. Durante ese período, la máquina puede ser más sensible que lo normal a los compuestos orgánicos volátiles (VOC) (como los olores).

INFORMACIÓN DE ELIMINACIÓN DE RESIDUOS

- Los productos Dyson están hechos a partir de materiales reciclables de alta calidad. Recíclelos siempre que sea posible.
- Deseche o recicle la batería de acuerdo a ordenanzas o regulaciones locales.
- Mantenga las baterías usadas lejos de los niños porque aún pueden causar daño si se tragan.
- Las unidades de filtro no son lavables ni reciclables.
- Deseche las unidades de filtro agotadas conforme lo indiquen las ordenanzas y reglamentaciones locales.
- Debe retirarse la batería antes de desechar el producto.
- California: Este producto contiene un control remoto separado con una batería tipo moneda. La batería tipo moneda contiene percloratos. Es posible que se aplique manejo especial. Vea www.dtsc.ca.gov/hazardouswaste/perchlorate.

SERVICIO DE ATENCIÓN AL CLIENTE DE DYSON

GRACIAS POR ELEGIR COMPRAR UNA MÁQUINA DYSON.

No dejamos de preocuparnos por nuestros dispositivos, ni siquiera cuando ya son tuyos. Con tu garantía gratuita, las piezas y la mano de obra (salvo los filtros) del purificador ventilador Dyson estarán cubiertas durante dos años desde la fecha de compra, sujeto a los términos de la garantía. Incluso después de que tu garantía haya terminado, estaremos a tu disposición para ayudarte. Si tiene preguntas sobre su máquina Dyson, visite www.dyson.com para obtener ayuda, consejos generales e información útil en línea sobre Dyson.
Alternativamente, puede llamar a la Línea directa de asistencia al cliente de Dyson con el número de serie e información de cuándo y dónde compró la máquina.
Si su máquina Dyson necesita reparación llame a la Línea de ayuda de Dyson para poder analizar las opciones disponibles. Si su máquina tiene garantía y la reparación necesaria está incluida dentro de esa garantía, se reparará sin costo alguno.

REGISTRA TU MÁQUINA PARA SER EL PRIMERO EN RECIBIR ACTUALIZACIONES Y NOTICIAS DE DYSON

Para ayudarnos a asegurarle que reciba un servicio rápido y eficaz, por favor regístrese como propietario de una máquina Dyson. Hay tres maneras de hacerlo:

- En línea en www.dyson.com
- Llamando por teléfono a la Línea de ayuda de Dyson, 1-866-693-9766.
- Enviando por correo el formulario que se anexa, dirigido a nuestra dirección.
  Esto confirmará que la máquina Dyson es de su propiedad en caso de que ocurra una pérdida cubierta por el seguro, y nos permitirá contactarnos con usted si fuese necesario.
- La falta de registro no disminuirá sus derechos de garantía.

2 AÑOS DE GARANTÍA LIMITADA

TÉRMINOS Y CONDICIONES DE LA GARANTÍA LIMITADA DE 2 AÑOS DE DYSON

QUÉ ESTÁ CUBIERTO

- Su máquina Dyson está garantizada contra defectos originales de materiales o mano de obra por un período de 2 años a partir de la fecha de compra, cuando se utilice para fines domésticos normales y de acuerdo con el Manual de Instrucciones de Dyson. Conserve su factura de compra. Si no tiene la factura de compra, la validez de la garantía empezará a contar desde los 90 días posteriores a la fecha de fabricación, según los registros de Dyson. Esta garantía proporciona, sin costo para usted, toda la mano de obra y piezas para poner su máquina en perfectas condiciones de funcionamiento durante el plazo de garantía. El reemplazo de cualquier pieza de la máquina con garantía no extiende el período de la garantía. Cualquier parte que sea devuelta y reemplazada por Dyson pasará a ser propiedad de Dyson; sin embargo, su garantía se extenderá durante todo el tiempo en que su máquina esté siendo reparada o retenida por el trabajo realizado en virtud de la garantía.Todas las piezas que se reemplacen o se devuelvan a Dyson se convertirán en propiedad de Dyson. Comuníquese con la Línea Directa de Asistencia al Cliente de Dyson al 1-866-693-9766 para obtener más información.
- Esta garantía solo tendrá validez si la máquina se utiliza en el país en el que se realizó la venta.
- Esta garantía está sujeta a los siguientes términos:

QUÉ NO ESTÁ CUBIERTO

- Filtros de reemplazo. Los filtros de la máquina no están cubiertos por la garantía.
- Daño causado por no llevar a cabo el mantenimiento recomendado de la máquina.
  Dyson no será responsable de los costos, daños o reparaciones originados como resultado de o debido a lo siguiente:
- Compras de máquinas de un proveedor no autorizado.

- El funcionamiento o manejo descuidado, uso indebido y/o falta de mantenimiento o el uso de forma contraria al Manual de Instrucciones de Dyson.
- El uso de piezas que no estén contempladas en el Manual de Instrucciones de Dyson.
- El uso de piezas y accesorios que no sean los producidos o recomendados por Dyson.
- Fuentes externas como el clima, las interrupciones eléctricas o las sobrecargas de energía.
- Reparaciones o alteraciones llevadas a cabo por partes o agentes no autorizados.
- Eliminación de obstrucciones en el aparato.
- Desgaste y deterioro normal (p. ej. el fusible, etc.).
- Daño como resultado de un uso que contravenga las indicaciones de la placa de especificaciones.
- Además, esta garantía limitada no se aplica a ninguna máquina cuyo número de serie se haya alterado o quitado.
- Si hay alguna duda sobre la cobertura de la garantía, comuníquese con Dyson.

LIMITACIONES Y EXCLUSIONES DE LA GARANTÍA

- En la máxima medida permitida por la ley vigente, cualquier garantía implícita relacionada con su máquina, que incluye, entre otras, la garantía de comercialización o la garantía de adecuación para un propósito particular, se limita a la duración de esta garantía.
- En la máxima medida permitida por la ley vigente, su recurso por el incumplimiento de esta garantía se limita al servicio de garantía que se describe a continuación. Dyson no es responsable por ningún daño indirecto o daño inherente en el que pueda incurrir en conexión con la adquisición y uso de su máquina.
- Esta garantía le otorga derechos legales específicos. Usted podría también tener otros derechos los cuales varían de acuerdo al estado.
- Esta garantía limitada no es transferible.
- Las limitaciones o exclusiones de responsabilidad establecidas en este documento rigen en el estado de Nueva Jersey, en la máxima medida permitida por la ley vigente.

SERVICIO DE GARANTÍA

- Antes de llevar a cabo el servicio, los términos de la garantía limitada requieren que usted se ponga en contacto con el servicio de Atención al Cliente de Dyson con el nombre o número de su modelo, el número de serie, así como el lugar y la fecha de compra.
- Por favor mantenga la factura de compra en un sitio seguro para asegurarse de tener esta información.
- Todo trabajo será coordinado por Dyson o sus agentes autorizados.
- Cualquier pieza defectuosa que se reponga es propiedad de Dyson.
- El servicio en virtud de esta garantía no extenderá el período de esta garantía; sin embargo, su garantía se extenderá durante todo el tiempo en que su máquina esté siendo reparada o retenida por el trabajo realizado en virtud de la garantía.
- Si su máquina no funciona correctamente, comuníquese a la línea de ayuda de Dyson al 1-866-693-9766. Nuestro personal de Atención al cliente podrá rectificar telefónicamente la mayoría de los problemas; sin embargo, su garantía se extenderá durante todo el tiempo en que su máquina esté siendo reparada o retenida por el trabajo realizado en virtud de la garantía.

INFORMACIÓN IMPORTANTE SOBRE LA PROTECCIÓN DE DATOS PERSONALES

Cuando registre su aparato Dyson:

- Deberá proporcionarnos información de contacto básica para registrar su máquina y permitirnos respaldar su garantía.
- Cuando se registre, tendrá la posibilidad de elegir si quiere recibir comunicaciones de nosotros. Si acepta recibir comunicaciones de Dyson, le enviaremos detalles de ofertas especiales y noticias de nuestras últimas innovaciones.
- Nunca venderemos su información a terceros. Usaremos la información que comparta con nosotros únicamente de la manera que se define en nuestras políticas de privacidad, que están disponibles en nuestro sitio web privacy.dyson.com

THANK YOU FOR CHOOSING TO BUY A DYSON MACHINE

We don't stop caring about our machines once they're yours. With your free 2-year warranty, your Dyson purifying fan will be covered for parts and labor (excluding filters) for two years from the date of purchase, subject to the terms of the warranty. Even after your warranty has ended, we'll still be on hand to help.
If you have any questions about your Dyson machine, call the Dyson Helpline with your serial number and details of where and when you bought the machine. Most questions can be solved over the phone by one of our trained Dyson Helpline staff.
Call 1-866-693-9766 for general machine questions or 866-861-2521 for questions on using the Dyson Link app.
Your serial number can be found on your rating plate which is on the base of your machine.
Visit www.dyson.com/support for online help, support videos, general tips and useful information about Dyson.

GRACIAS POR ELEGIR COMPRAR UNA MÁQUINA DYSON.

No dejamos de preocuparnos por nuestros dispositivos, ni siquiera cuando ya son tuyos. Con tu garantía gratuita, las piezas y la mano de obra (salvo los filtros) del purificador ventilador Dyson estarán cubiertas durante dos años desde la fecha de compra, sujeto a los términos de la garantía. Incluso después de que tu garantía haya terminado, estaremos a tu disposición para ayudarte.Si tiene alguna duda acerca de su producto Dyson, llame a la Línea directa de asistencia al cliente de Dyson con el número de serie e información de cuándo y dónde compró el equipo. La mayoría de las dudas pueden resolverse por teléfono con la ayuda de un miembro de nuestro personal capacitado de la Línea directa de asistencia al cliente.
Llame al 1-866-693-9766 para preguntas generales sobre la máquina o al 866-861-2521 para preguntas sobre cómo utilizar la aplicación Dyson Link.
Su número de serie se encuentra en la placa de especificaciones situada en la base de la máquina.
Como alternativa, visite www.dyson.com/support para obtener ayuda, consejos generales e información útil en línea sobre Dyson.

Note your serial number for future reference.
Anote su número de serie para futuras consultas.



This illustration is for example purposes only.
Esta figura es únicamente para mostrar un ejemplo.

JN.114466 PN.306236-03-04 23.08.21

REGISTRATION FORM CONTINUED
CONTINUACIÓN DEL FORMULARIO DE REGISTRO

Please take a few moments to answer our questions.
They will help us develop new products for the future.

1. Do you own any other air movement products?
   Check all that apply.
   - ☐ Air conditioner
   - ☐ Ceiling fan
   - ☐ Pedestal fan
   - ☐ Table fan
   - ☐ Tower fan
   - ☐ Floor fan
   - ☐ Humidifier
   - ☐ Air purifier
   - ☐ Fan heater
   - ☐ Dehumidifier
   - ☐ None of the above
   - ☐ Other, please specify _____

2. Why did you choose to buy a Dyson fan?
   Please select up to 5 of the following choices below.
   - ☐ Frustration with existing fan
   - ☐ Latest fan technology
   - ☐ Dyson brand reputation
   - ☐ Smooth, powerful airflow
   - ☐ Previous experience of Dyson
   - ☐ Safety
   - ☐ Features – oscillation, touch-tilt
   - ☐ Easy to clean
   - ☐ Price/special offer
   - ☐ Store staff recommendation
   - ☐ Aesthetics
   - ☐ Friend/family recommendation
   - ☐ Other

3. Where did you first hear about Dyson fans?
   - ☐ From friends and family
   - ☐ Saw it on the internet
   - ☐ From a TV/magazine advertisement
   - ☐ Email from Dyson
   - ☐ Saw it in a store/catalog
   - ☐ News item or article
   - ☐ Recommended by salesperson

4. Do you own any other Dyson products?
   - ☐ Upright vacuum
   - ☐ Canister vacuum
   - ☐ Handheld vacuum
   - ☐ Airblade hand dryer

Tómese unos instantes para responder a nuestras preguntas.
Esto nos ayudará a desarrollar nuevos productos en el futuro.

1. ¿Posee algún otro producto de movimiento de aire?
   Seleccione todos los que apliquen.
   - ☐ Aire acondicionado
   - ☐ Ventilador de techo
   - ☐ Ventilador de pedestal
   - ☐ Ventilador de mesa
   - ☐ Ventilador tipo torre
   - ☐ Ventilador de piso
   - ☐ Humidificador
   - ☐ Purificador de aire
   - ☐ Ventilador de aire caliente
   - ☐ Deshumidificador
   - ☐ Ninguno de los anteriores
   - ☐ Otro, favor de especificar _____

2. ¿Por qué eligió comprar un ventilador Dyson?
   Seleccione por favor hasta 5 de lo siguiente opciones de abajo.
   - ☐ Frustración por el ventilador que ya tenía
   - ☐ Tecnología de ventilación más nueva
   - ☐ Prestigio de la marca Dyson
   - ☐ Flujo uniforme y potente
   - ☐ Experiencia anterior con Dyson
   - ☐ Seguridad
   - ☐ Funciones: oscilación, inclinación con un toque
   - ☐ Fácil de limpiar
   - ☐ Precio, oferta especial
   - ☐ Recomendación del personal de la tienda
   - ☐ Apariencia estética
   - ☐ Recomendación de amigos o familiares
   - ☐ Otro

3. ¿Dónde se enteró por primera vez de los ventiladores Dyson?
   - ☐ Por amigos y familiares
   - ☐ Lo vi en Internet
   - ☐ Por un anuncio en televisión/revista
   - ☐ Correo electrónico de Dyson
   - ☐ Lo vi en el catálogo de una tienda
   - ☐ Noticias o articulo de noticias
   - ☐ Recomendación de un vendedor

4. ¿Posee otro aparato Dyson?
   - ☐ Aspiradora vertical
   - ☐ Aspiradora tipo canasta
   - ☐ Aspiradora portátil
   - ☐ Secador de manos Airblade

THANK YOU FOR YOUR TIME • GRACIAS POR DEDICARNOS SU TIEMPO

dyson

4 EASY WAYS TO REGISTER YOUR MACHINE.
4 MANERAS SENCILLAS DE REGISTRAR SU MÁQUINA.

Date of purchase | Fecha de compra ☐☐ / ☐☐ / ☐☐

Serial number | Número de série

Title | Tratamiento

Last name | Apellido

First name | Nombre

Address | Dirección

Zip Code | Código postal

email | email

Telephone | Teléfono

Store of purchase | Tienda de compra

☐

From time to time, we'd like to send you emails about new Dyson technology and services or get your opinion about your purchase. If you'd like us to keep in touch, please check this box.

Ocasionalmente, nos gustaría enviarle correos electrónicos con información sobre los nuevos servicios y tecnologías de Dyson o para saber su opinión sobre su compra. Si desea que nos mantengamos en contacto, seleccione esta casilla.



SMARTPHONE
Download the Dyson Link app and you will be taken through registration as part of the set up.

TELÉFONO INTELIGENTE
Descargue la aplicación del vínculo Dyson y se lo guiará por la registración como parte de la configuración.



ONLINE
Convenient and instant.
www.dyson.com/register

EN LÍNEA
Conveniente y al instante.
www.dyson.com/register



PHONE
Dyson Helpline:
1-866-693-9766 (Mon-Fri 8:00am - 8:00pm CT; Sat 9:00am - 6:00pm CT).

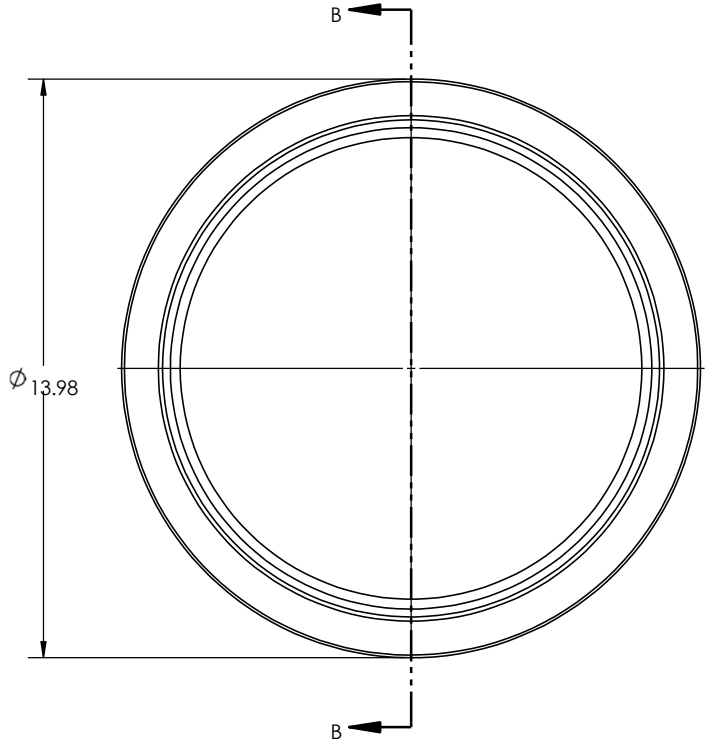
POR TELÉFONO
Servicio de atención al cliente Dyson:
1-866-693-9766 (Lunes a Viernes de 8 a.m. a 8 p.m., hora Centro; Sábados de 9 a.m. a 6 p.m., hora Centro).

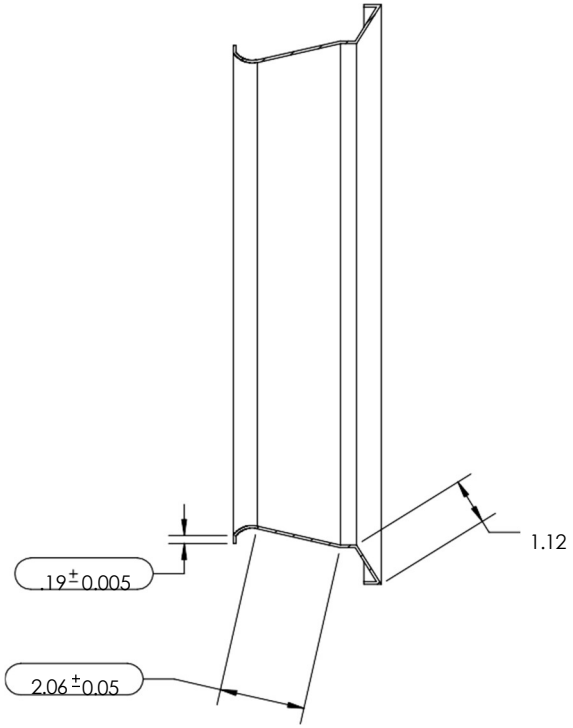


MAIL
Complete and return the form in the envelope supplied.

POR CORREO
Complete este formulario y envíelo.

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B	CHANGED MATERIAL & DELETED DUPLICATE DIMS	10/15/2024

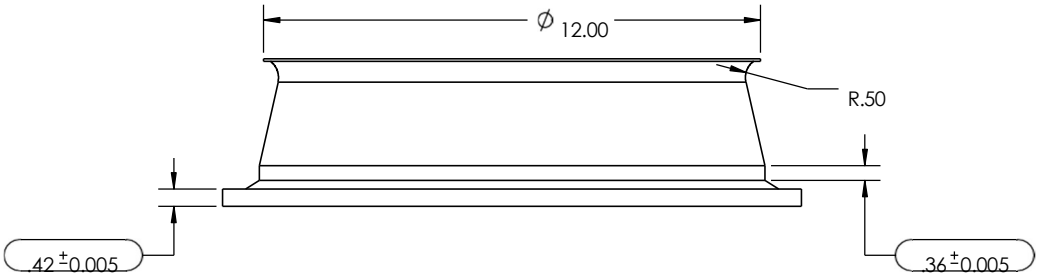




SECTION B-B
SCALE 1 : 3



SCALE 1 : 4



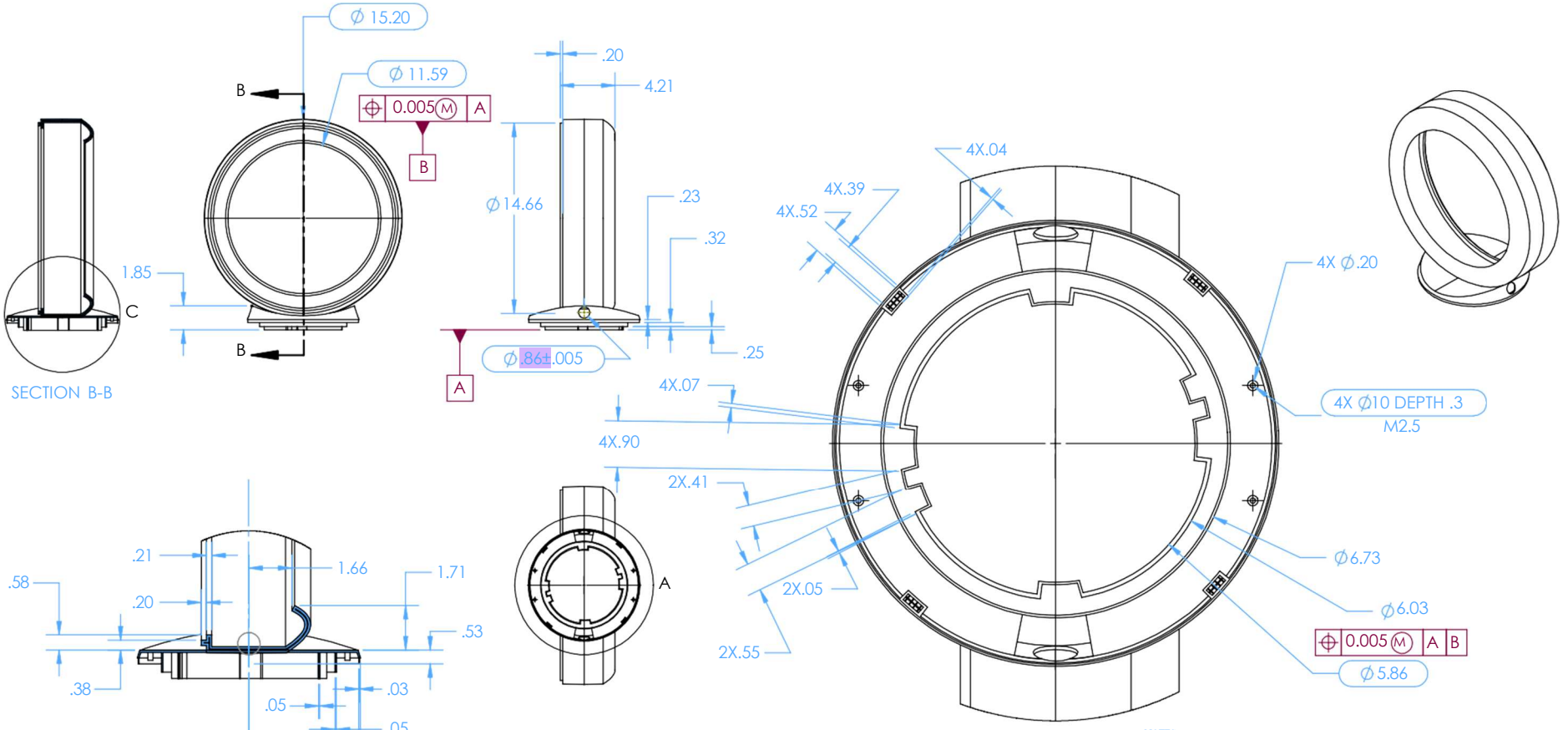
NOTE:
1. UNLESS MENTIONED SCALE 1:3
2. WALL THICKNESS THROUGHOUT IS 0.0650"

TOLERANCES UNLESS NOTED:	X.X ±0.05 X.XX ±0.010 X.XXX ±0.005	UNIVERSITY OF COLORADO BOULDER
UNITS:INCHES	X.X 1°	DESCRIPTION: AMPLIFIER_HEAD
MATERIAL: PC/ABS		
FINISH: NONE	PN: BF001-01	REV: B
		SHEET: 1 of 1

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REV.	DESCRIPTION	DATE	DRAWN BY	APPROVED BY
A	INITIAL RELEASE	10/01/2024	SARAVANA	



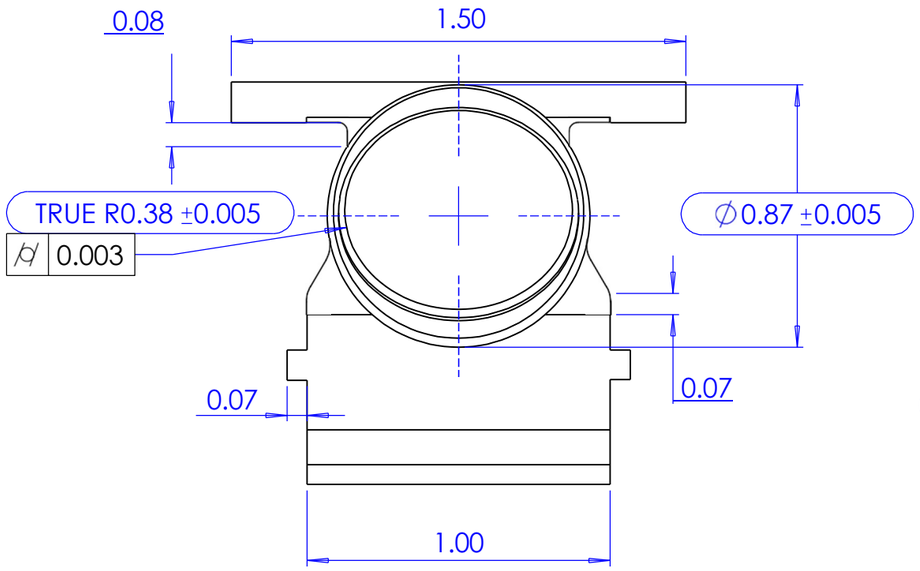
NOTES:
1. PART SHALL BE VISIBLY CLEAN OF ALL DIRT, OIL, AND SOLVENTS.
2. THIS IS A LIMITED DIMENSION DRAWING. UNSPECIFIED DIMENSIONS SHALL BE TAKEN FROM THE ATTACHED COMPUTER-GENERATED SOLID MODEL
3. CRITICAL OR INSPECTION DIMENSIONS ARE ENCLOSED WITH THE SYMBOL

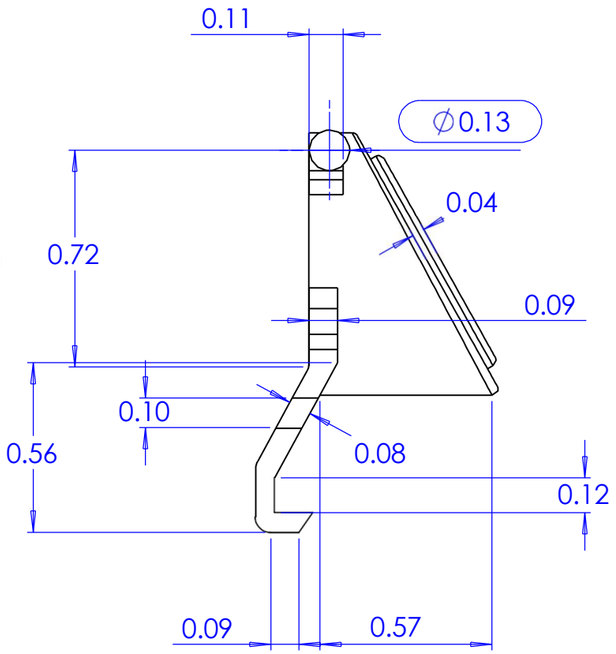
TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005 X.X ° ± 0.5°	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427
MATERIAL: PC/ABS		DESCRIPTION: AMPLIFIER HOUSING
FINISH: N/A	SCALE: 1:8	PN: BF001-01 REV: A SHEET: 1 of 1

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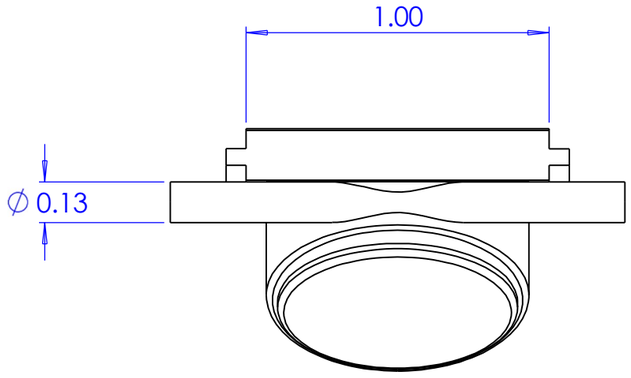
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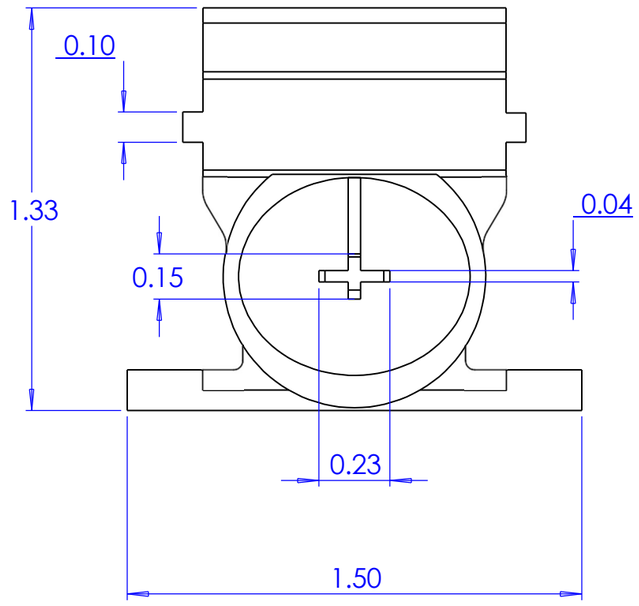
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B	INITIAL RELEASE	09/29/2024

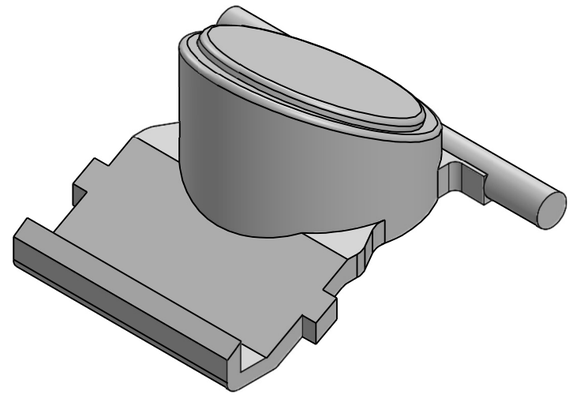




NOTES:
1. CRITICAL OR INSPECTION DIMENSIONS ARE ENCLOSED.
2. THE DIAMETER OF THE BF001-04 RELEASE LOCK MATTERS AS IT LOCKS INTO POSITION OF BF001-01 AMPLIFIER HEAD AND BF001-03





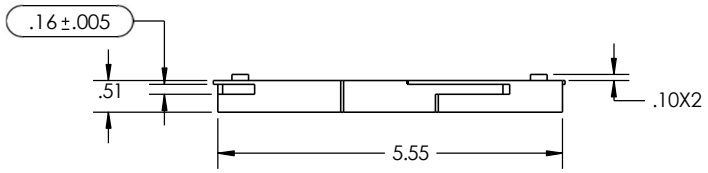


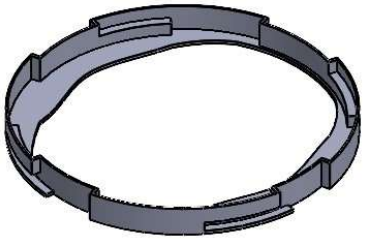
TOLERANCES UNLESS NOTED:	X.X ± 0.050 X.XX ± 0.010 X.XXX ± 0.005	UNIVERSITY OF COLORADO 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
UNITS: INCHES	X.X° ° 0.5		
MATERIAL: ABS		DESCRIPTION: RELEASE LOCK	
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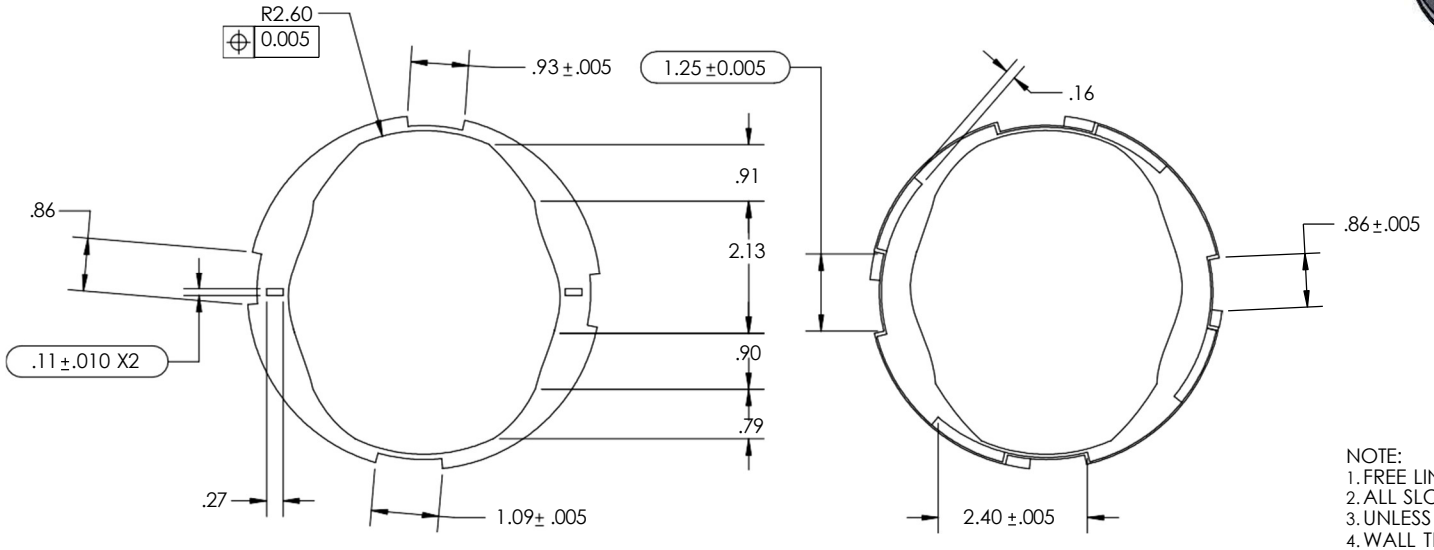
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REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	10/10/24







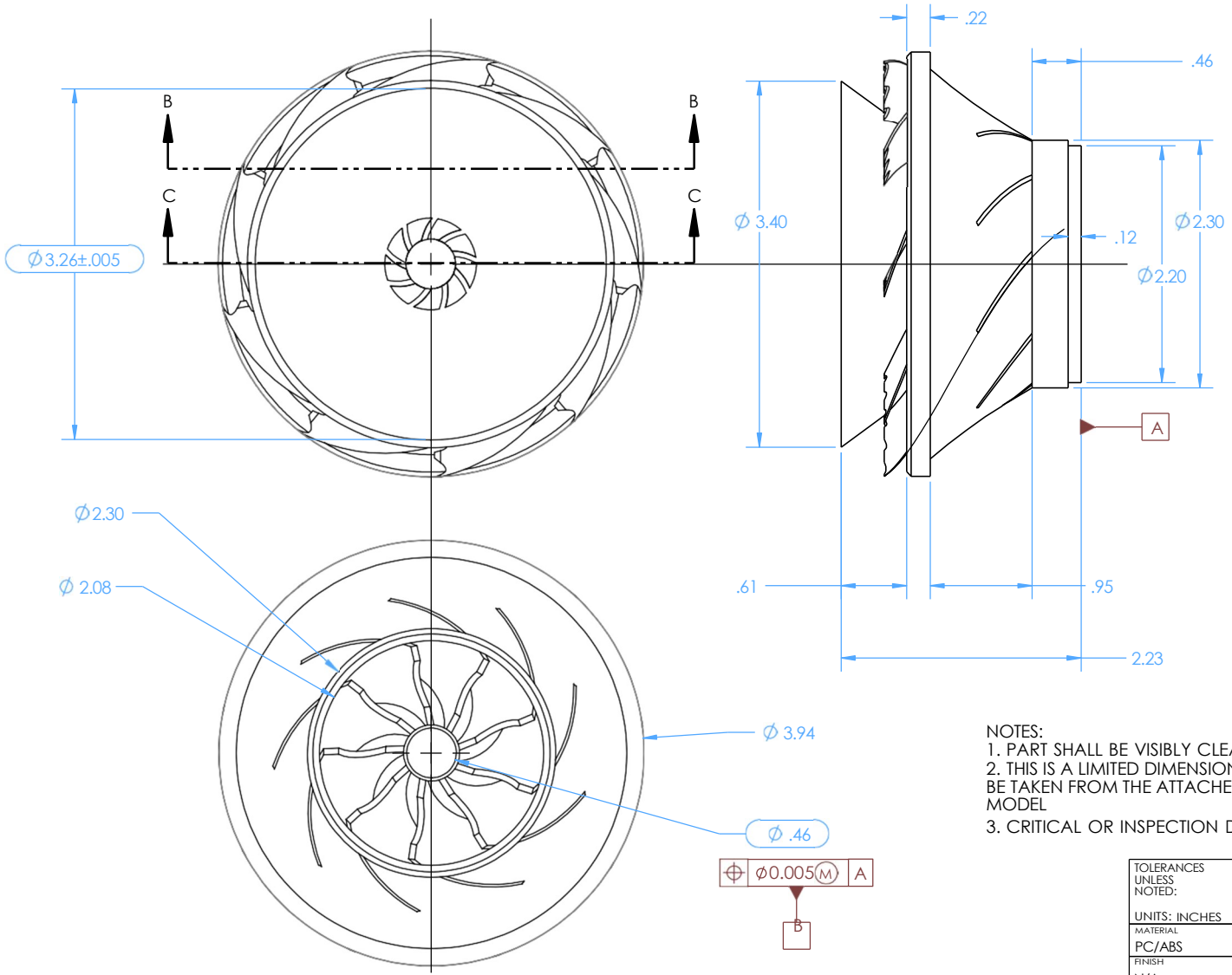
NOTE:
1. FREE LINES ARE SYMMETRIC
2. ALL SLOTS HAVE SAME DIMS
3. UNLESS MENTIONED SCALE IS 1:2
4. WALL THICKNESS THROUGH IS 0.0585"

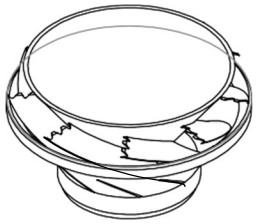
TOLERANCES UNLESS NOTED: UNITS:INCHES	X.X ±0.05 X.XX ±0.010 X.XXX ±0.005 X.X ° °1"	UNIVERSITY OF COLORADO BOULDER	
MATERIAL: PC/ABS		DESCRIPTION: MOTOR BUCKET RETAINING RING	
FINISH: NONE		PN: BF002	REV: A
			SHEET: 1 of 1

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REV.	DESCRIPTION	DATE	DRAWN BY	APPROVED BY
A	INITIAL RELEASE	10/10/2024	SARAVANA	





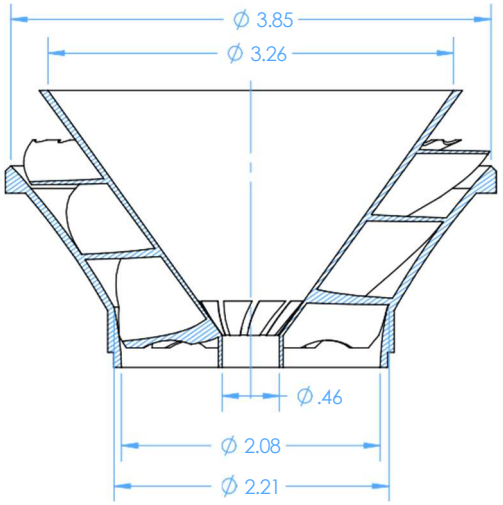
SCALE
1:2

NOTES:
1. PART SHALL BE VISIBLY CLEAN OF ALL DIRT, OIL, AND SOLVENTS.
2. THIS IS A LIMITED DIMENSION DRAWING. UNSPECIFIED DIMENSIONS SHALL BE TAKEN FROM THE ATTACHED COMPUTER-GENERATED SOLID MODEL
3. CRITICAL OR INSPECTION DIMENSIONS ARE ENCLOSED WITH THE SYMBOL

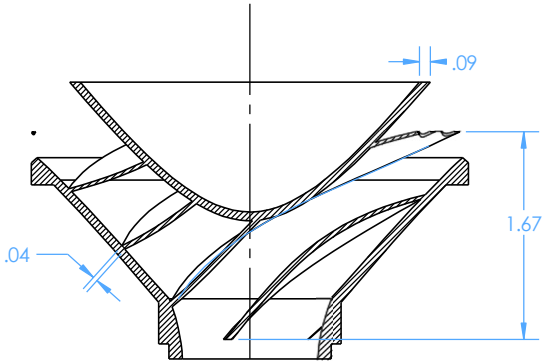
TOLERANCES UNLESS NOTED:	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427
UNITS: INCHES	X.X ° ° 1"	
MATERIAL: PC/ABS		DESCRIPTION: IMPELLER
FINISH: N/A	SCALE: 1:1	PN: BF003-01 REV: A SHEET: 1 of 2

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SECTION C-C
SCALE 1 : 1



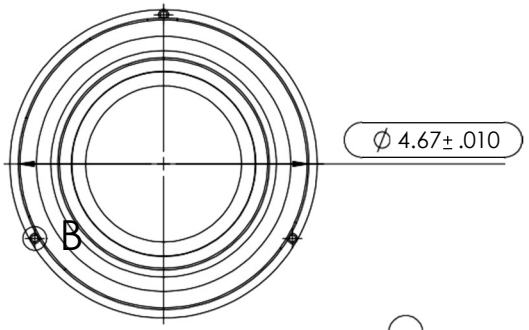
SECTION B-B
SCALE 1 : 1

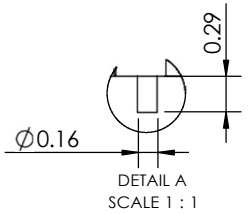
TOLERANCES UNLESS NOTED: UNITS:INCHES	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005 X.X ° ° 1"	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
MATERIAL PC/ABS		DESCRIPTION IMPELLER	
FINISH N/A	SCALE 1:1	PN	REV A
			SHEET 2 of 2

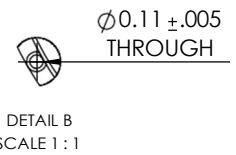
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.

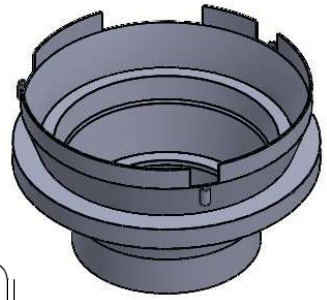
SOLIDWORKS Educational Product. For Instructional Use Only.

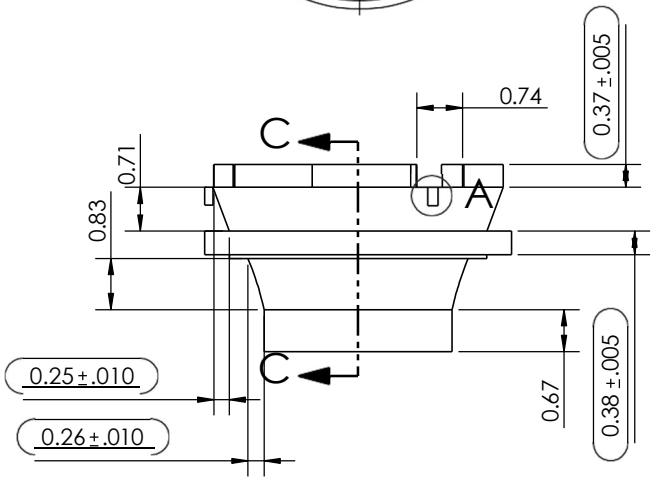
REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	10/7/2024

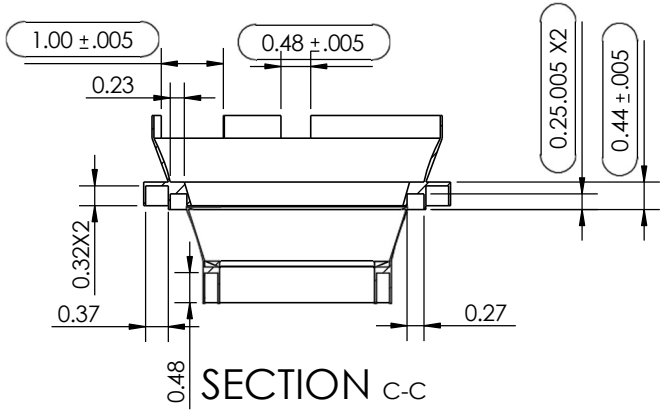




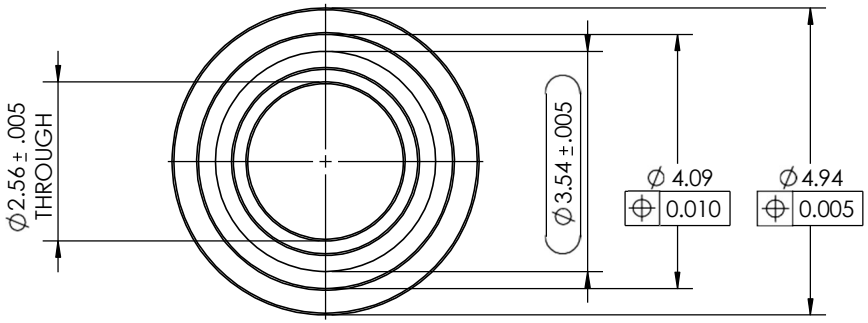








SECTION C-C



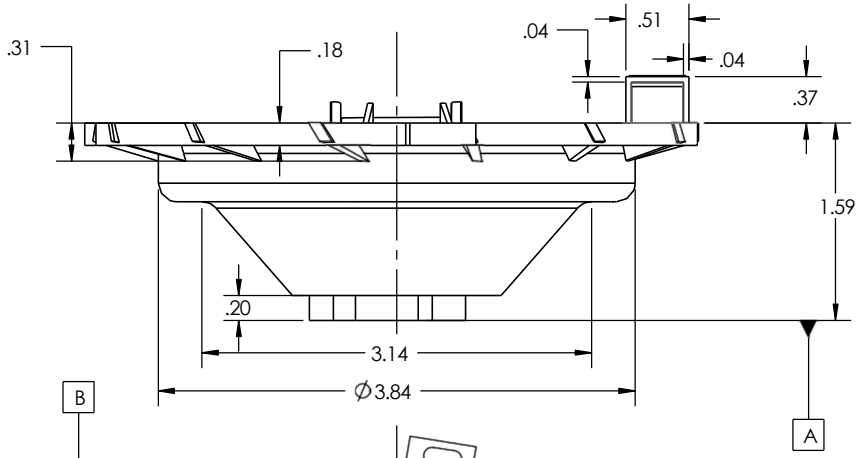
NOTE:
1. UNLESS MENTIONED SCALE IS 1:2
2. WALL THICKNESS THROUGHOUT IS .059"

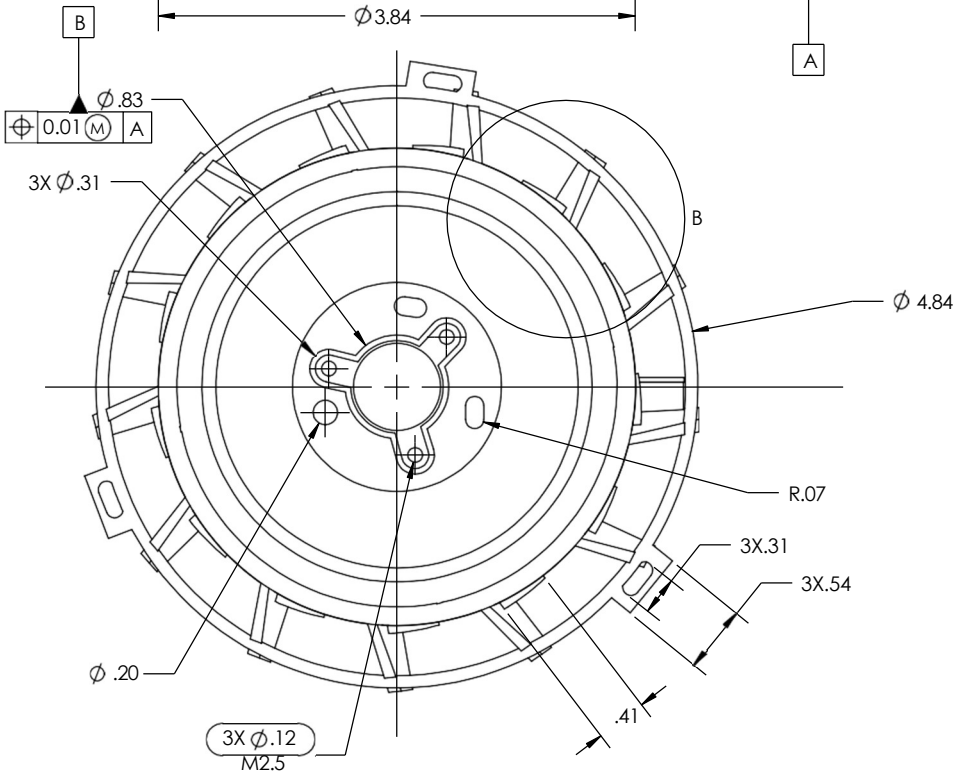
TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ±0.05 X.XX ±0.010 X.XXX ±0.005 X.X ° °1"	UNIVERSITY OF COLORADO BOULDER	
MATERIAL: PC/ABS		DESCRIPTION: IMPELLER BOTTOM HOUSING	
FINISH: NONE		PN: BF003-03	REV: A
			SHEET: 1 of 1

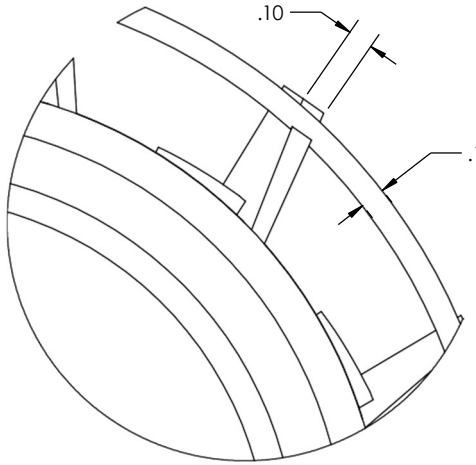
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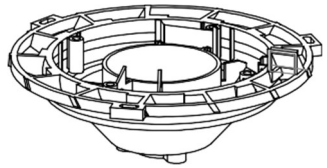
REV.	DESCRIPTION	DATE	DRAWN BY	APPROVED BY
A	INITIAL RELEASE	10/05/2024	SARAVANA	







DETAIL B
SCALE 2 : 1



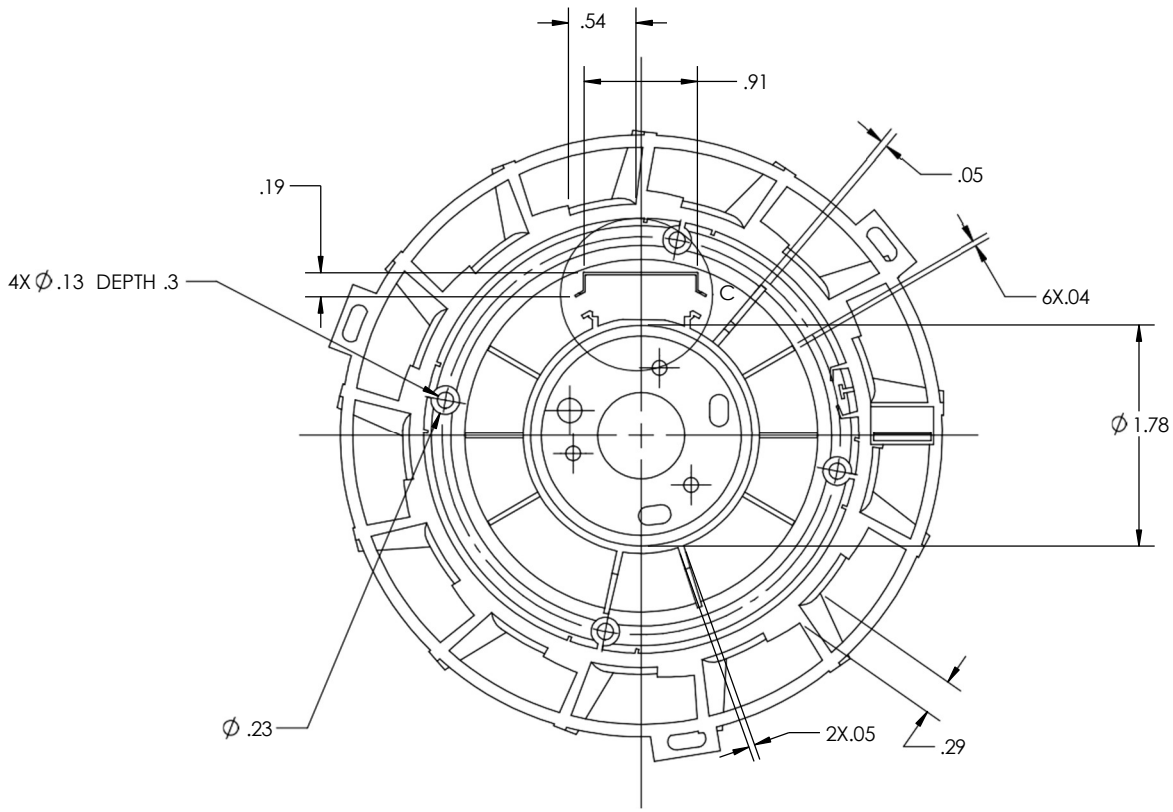
SCALE
1:2

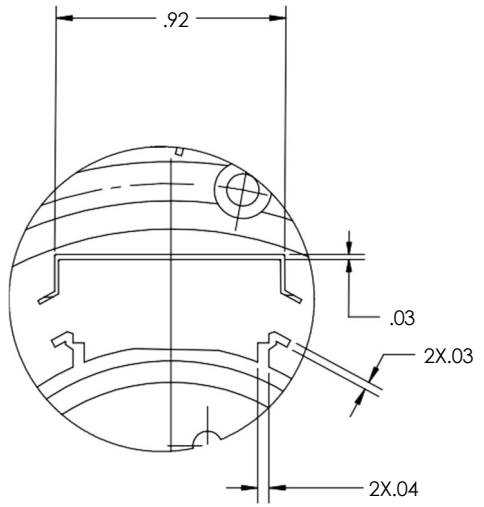
NOTES:
1. PART SHALL BE VISIBLY CLEAN OF ALL DIRT, OIL, AND SOLVENTS.
2. THIS IS A LIMITED DIMENSION DRAWING. UNSPECIFIED DIMENSIONS SHALL BE TAKEN FROM THE ATTACHED COMPUTER-GENERATED SOLID MODEL
3. CRITICAL OR INSPECTION DIMENSIONS ARE ENCLOSED WITH THE SYMBOL

TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ± 0.05 X.XX ±0.010 X.XXX ±0.005 X.X ° ° 1"	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
MATERIAL PC/ABS		DESCRIPTION IMPELLOR BOTTOM HOUSING	
FINISH N/A	SCALE 1:1	PN BF003-03	REV A	SHEET 1 of 2

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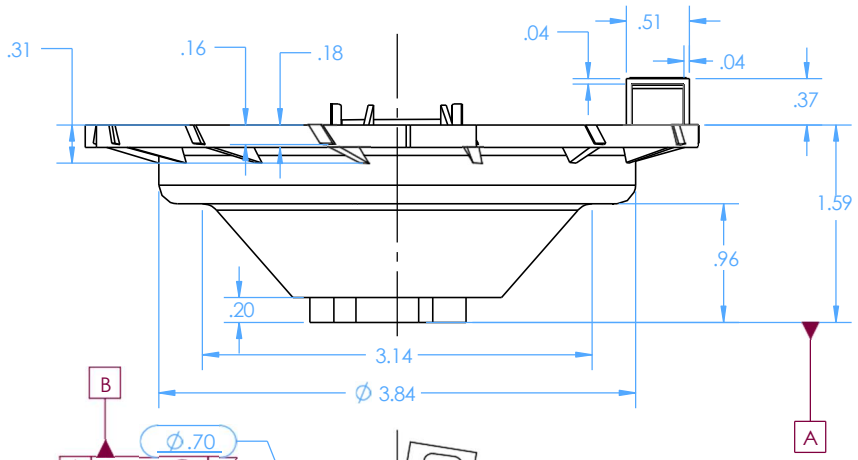
DETAIL C
SCALE 2 : 1

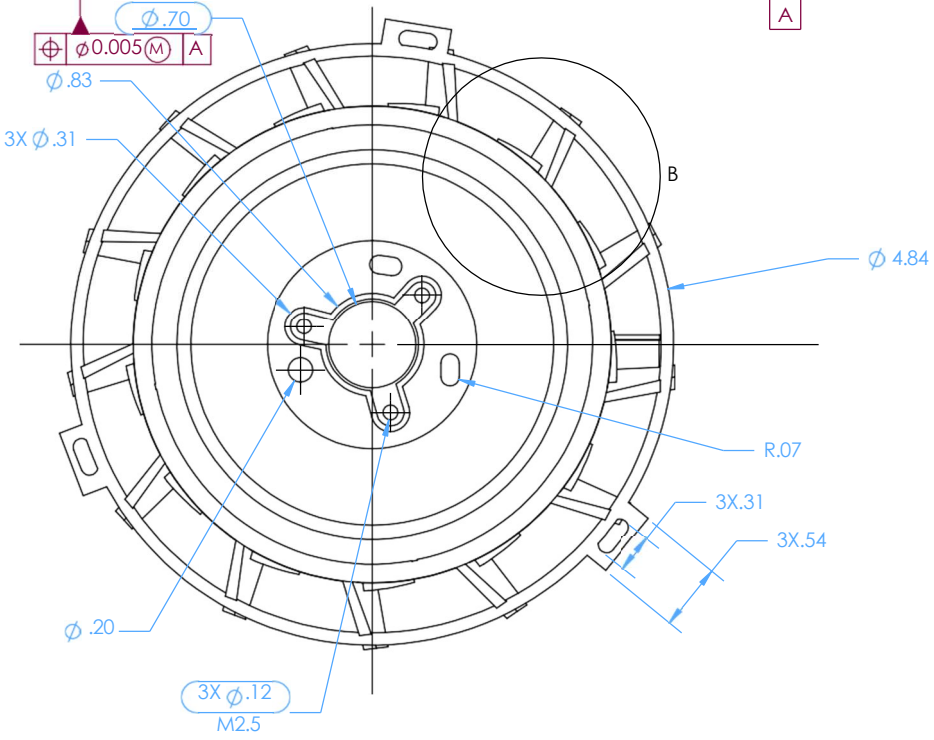
TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005 X.X ° ° 1"	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
MATERIAL PC/ABS		DESCRIPTION IMPELLOR BOTTOM HOUSING	
FINISH N/A	SCALE 1:2	PN BF003-03	REV SHEET 2 of 2

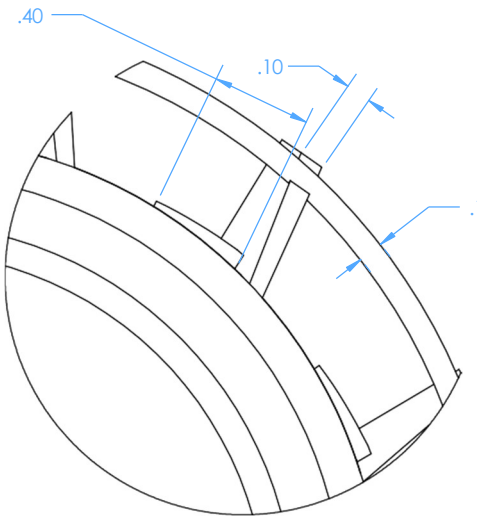
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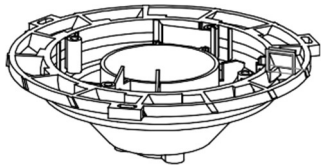
REV.	DESCRIPTION	DATE	DRAWN BY	APPROVED BY
A	INITIAL RELEASE	10/05/2024	SARAVANA	







DETAIL B
SCALE 2 : 1



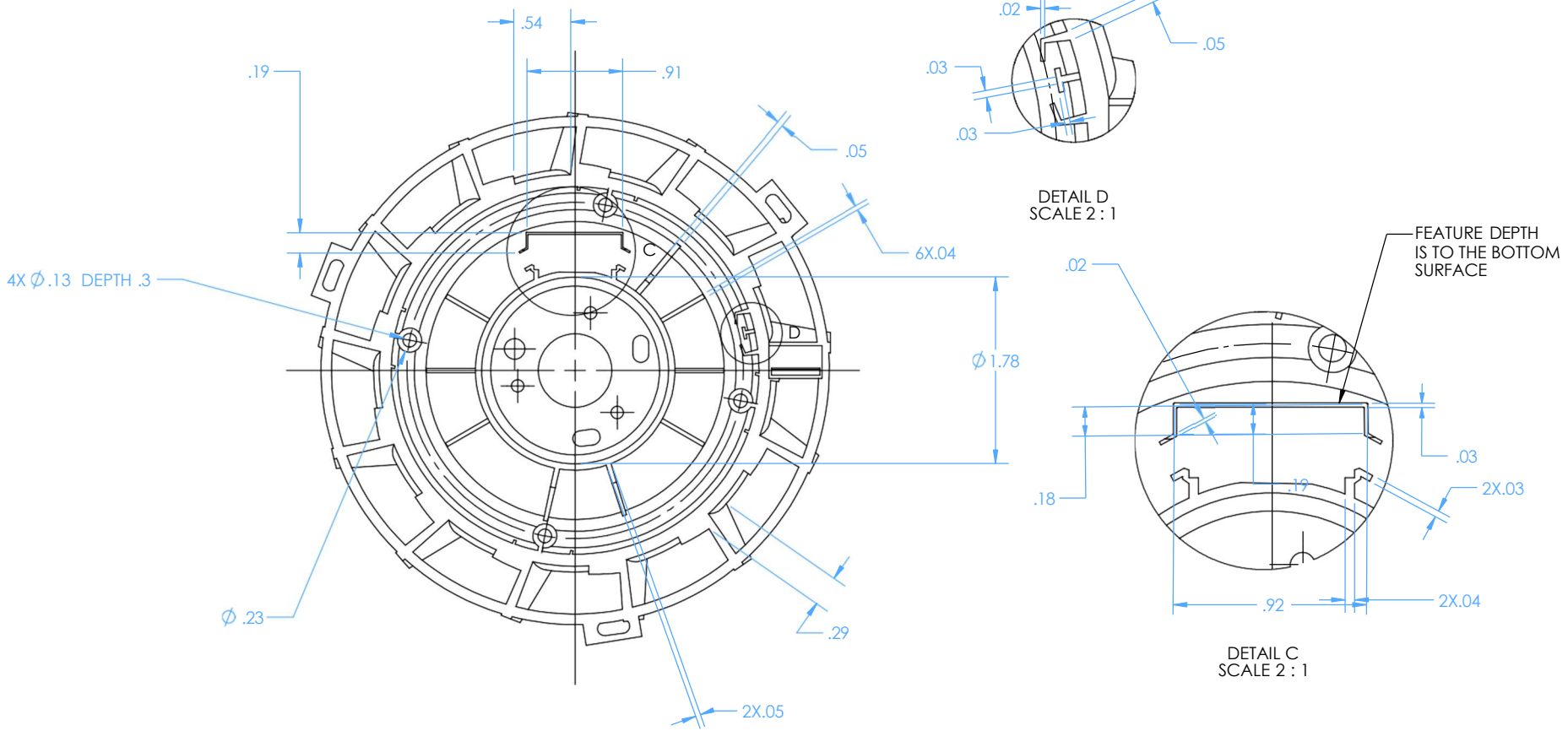
SCALE
1:2

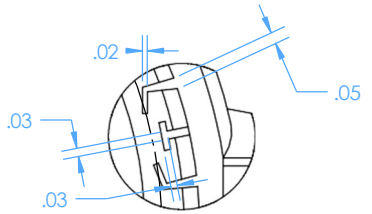
NOTES:
1. PART SHALL BE VISIBLY CLEAN OF ALL DIRT, OIL, AND SOLVENTS.
2. THIS IS A LIMITED DIMENSION DRAWING. UNSPECIFIED DIMENSIONS SHALL BE TAKEN FROM THE ATTACHED COMPUTER-GENERATED SOLID MODEL
3. CRITICAL OR INSPECTION DIMENSIONS ARE ENCLOSED WITH THE SYMBOL ⊖

TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ± 0.05 X.XX ±0.010 X.XXX ±0.005 X.X ° ° 1"	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
MATERIAL: PC/ABS		DESCRIPTION: IMPELLOR BOTTOM HOUSING	
FINISH: N/A	SCALE: 1:1	PN: BF003-03	REV: A
			SHEET: 1 of 2

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DETAIL D
SCALE 2 : 1

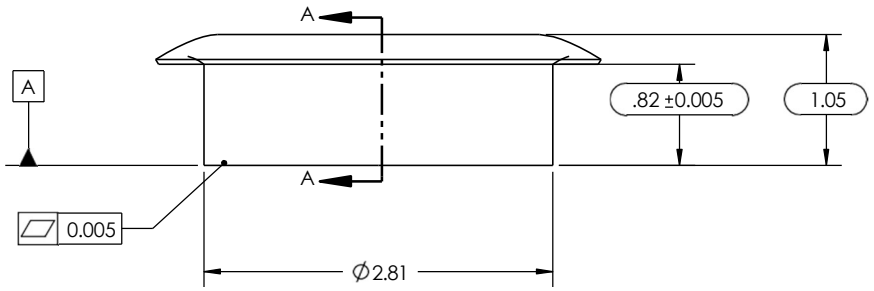
DETAIL C
SCALE 2 : 1

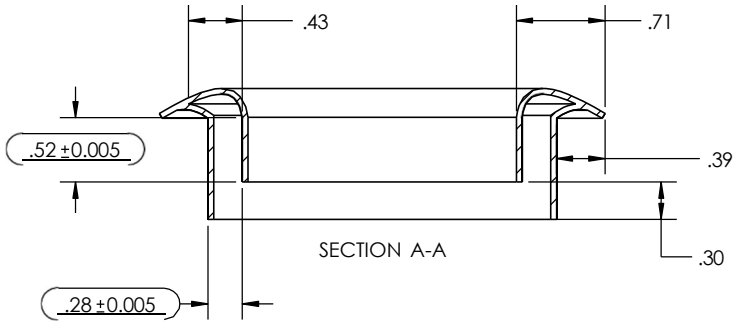
TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005 X.X ° ° 1"	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
MATERIAL PC/ABS		DESCRIPTION IMPELLOR BOTTOM HOUSING	
FINISH N/A	SCALE 1:1	PN BF003-03	REV A
			SHEET 2 of 2

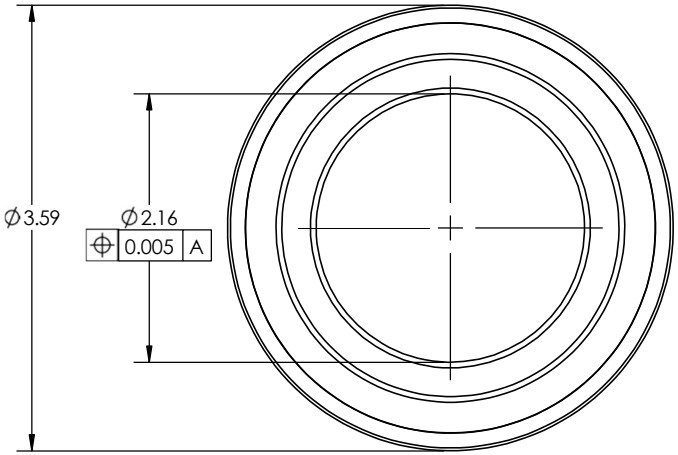
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REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	9/23/2024
B	ADDITION OF GD&T	9/25/2024
C	CHANGED NOTES AND ADDED DIMS	10/15/2024









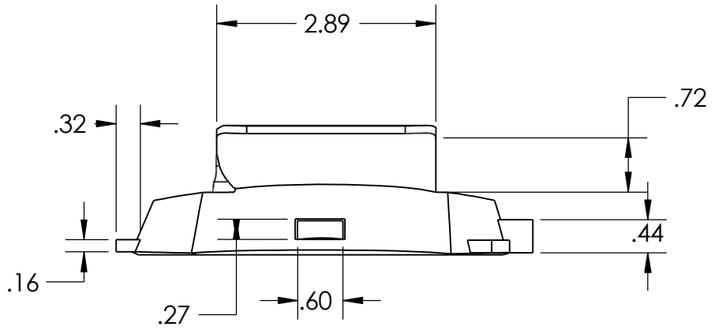
NOTE:[]
1. UNLESS MENTIONED SCALE 1:1
2. WALL THICKESS THROUGHOUT IS 0.0460"

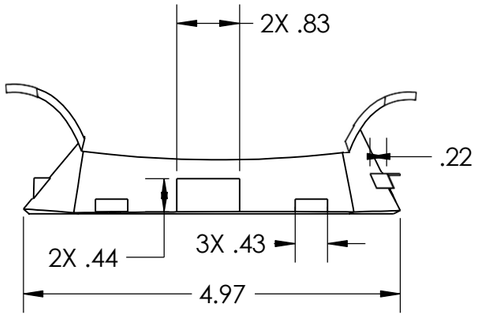
TOLERANCES UNLESS NOTED: UNITS:INCHES	X.X ±0.05 X.XX ±0.010 X.XXX ±0.005 X.X ° °1"	UNIVERSITY OF COLORADO BOULDER
MATERIAL: PC-ABS		DESCRIPTION: HUB BASE
FINISH: NONE		PN: BF003-04 REV: C SHEET: 1 of 1

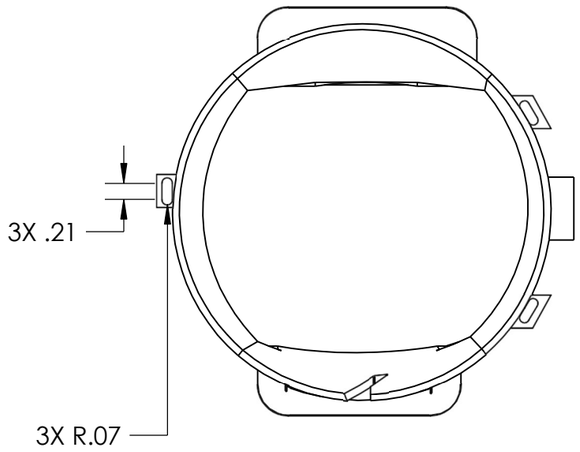
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.

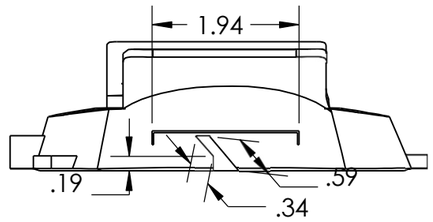
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REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	INITIAL RELEASE	20-10-2024	



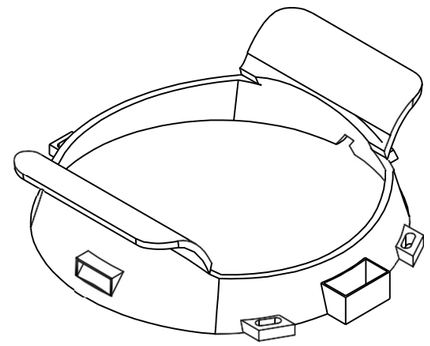






NOTES:

1. ANY UNSPECIFIED DIMENSIONS CAN BE TAKEN FROM THE 3D CAD MODEL, BF003-05.
2. THE PART SHOULD BE VISIBLY CLEAN OF ALL DIRT, SOLVENTS AND OIL.

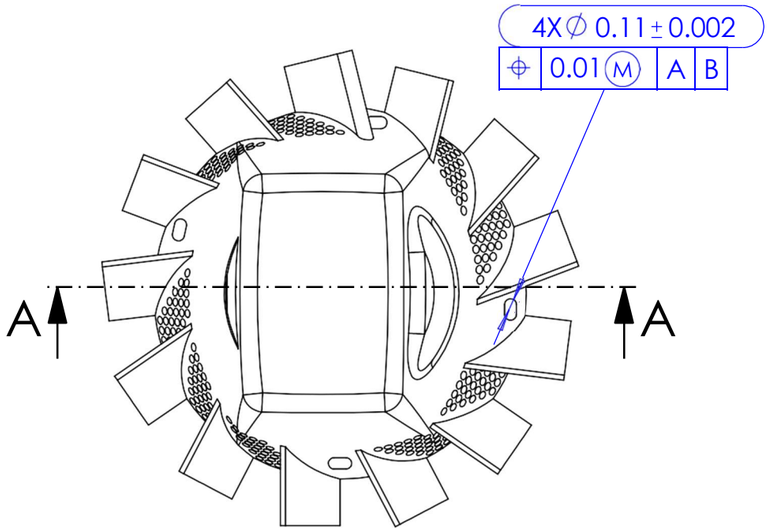


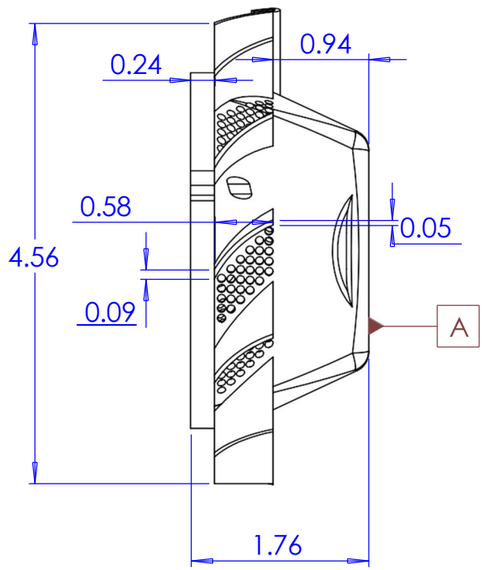
TOLERANCES UNLESS NOTED:	X.X ±0.050 X.XX ±0.010 X.XXX ±0.005	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
UNITS: INCHES	X.X° 0.5°		
MATERIAL: PC/ABS		DESCRIPTION: STATOR FAN HOUSING	
FINISH: BLACK		PN: BF003-05	REV: A — SHEET: 1 of 1

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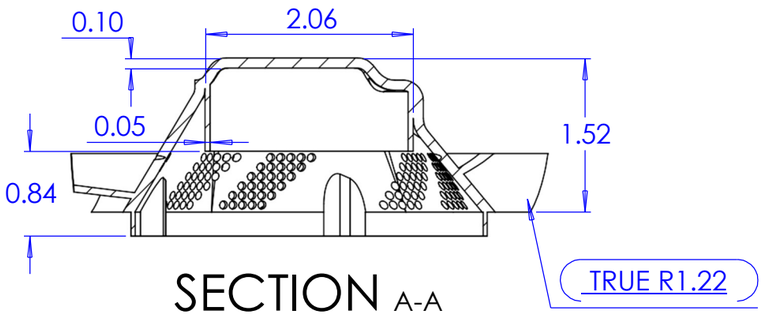
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REV	DESCRIPTION	DATE
A	INITIAL RELEASE	09/29/2024

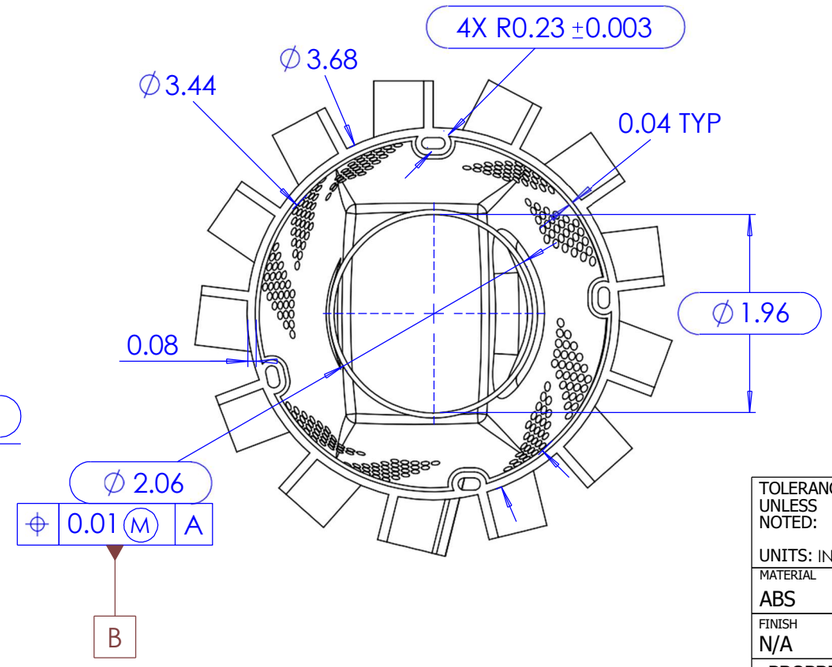


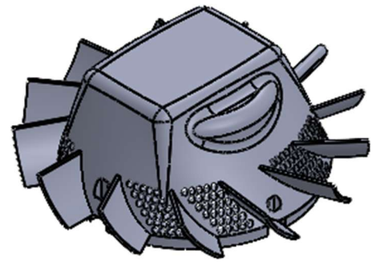


NOTES:
1. PART SHALL BE VISIBLY CLEAN OF ALL DIRT, OIL, AND SOLVENTS.
2. THIS IS A LIMITED DIMENSION DRAWING. UNSPECIFIED DIMENSIONS SHALL BE TAKEN FROM THE ATTACHED COMPUTER-GENERATED SOLID MODEL,BF003-05 STATOR HOUSING.
4. PART FITS BETWEEN BF003-05 AND BF003-04
3. CRITICAL OR INSPECTION DIMENSIONS ARE ENCLOSED WITH THE SYMBOL



SECTION A-A





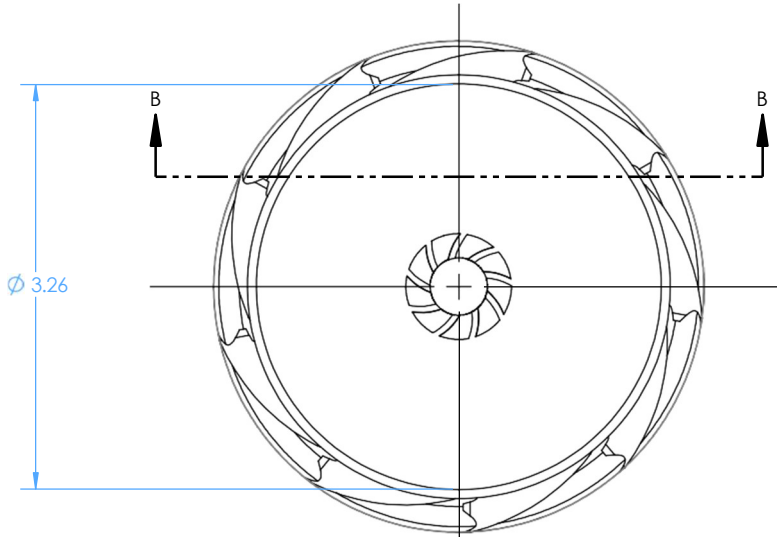
SCALE 1:2

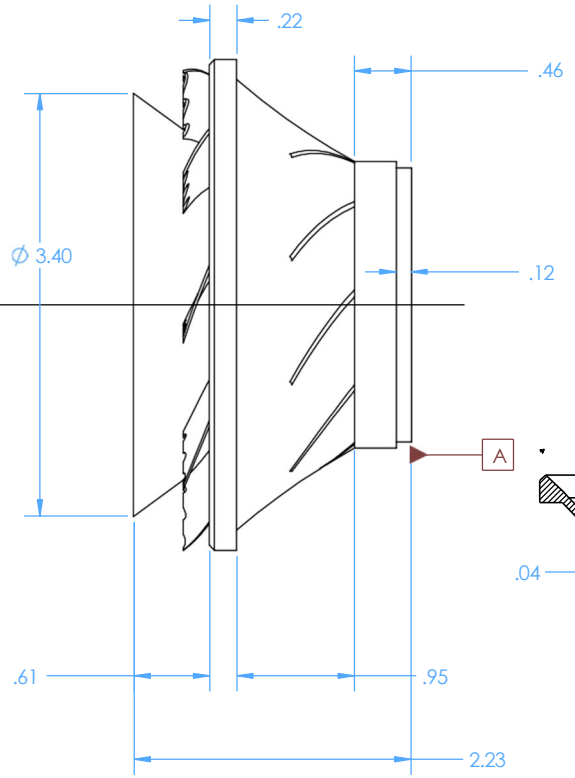
TOLERANCES UNLESS NOTED:	X.X ± 0.050 X.XX ± 0.010 X.XXX ± 0.005	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
UNITS: INCHES	X.X° ° 0.5		
MATERIAL: ABS		DESCRIPTION: STATIONARY PLASTIC BLADE	
FINISH: N/A		PN: BF003-06	REV: A / SHEET: 1 of 1

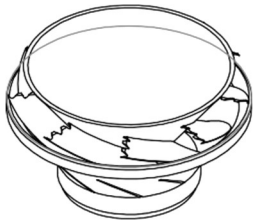
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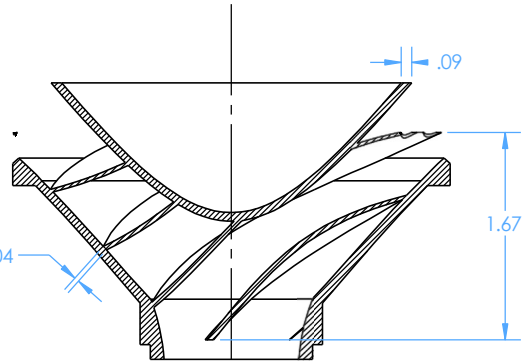
REV.	DESCRIPTION	DATE	DRAWN BY	APPROVED BY
A	INITIAL RELEASE	10/10/2024	SARAVANA	



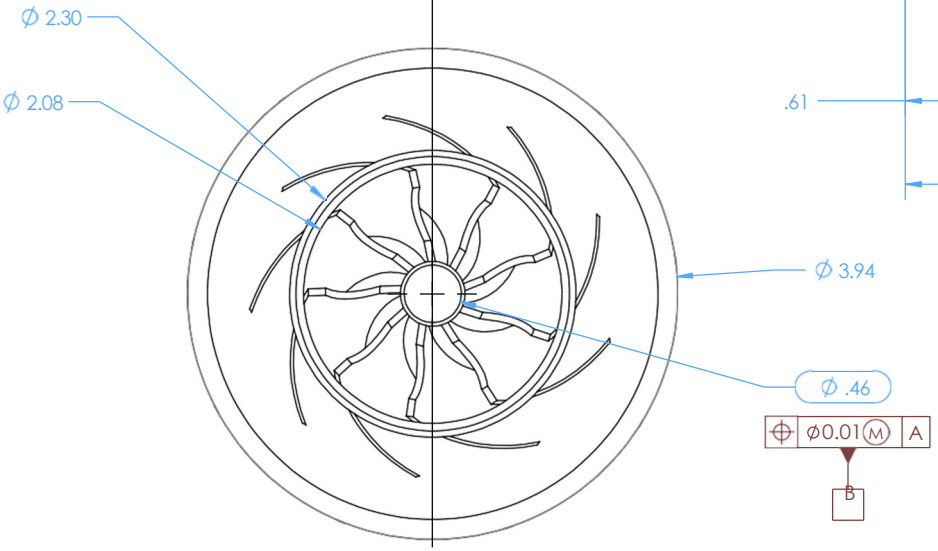




SCALE
1:2



SECTION B-B
SCALE 1 : 1



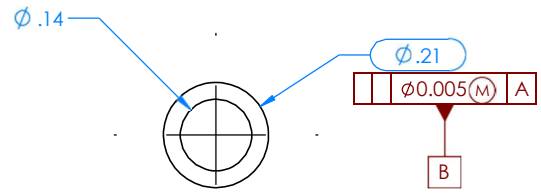
NOTES:
1. PART SHALL BE VISIBLY CLEAN OF ALL DIRT, OIL, AND SOLVENTS.
2. THIS IS A LIMITED DIMENSION DRAWING. UNSPECIFIED DIMENSIONS SHALL BE TAKEN FROM THE ATTACHED COMPUTER-GENERATED SOLID MODEL, F00-01 REV A, BOTTOM WIRE CAP
3. CRITICAL OR INSPECTION DIMENSIONS ARE ENCLOSED WITH THE SYMBOL

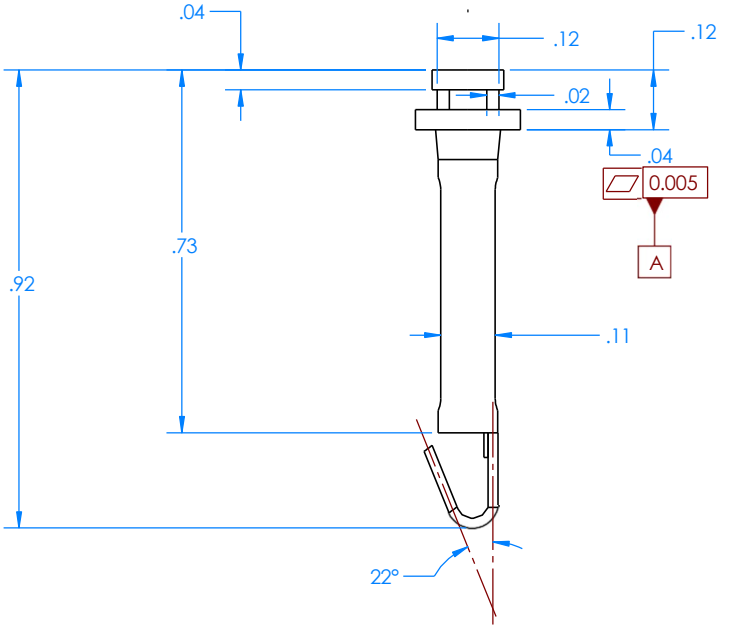
TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005 X.X ° ° 1"	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427
MATERIAL: PC/ABS		DESCRIPTION: STATIONARY PLASTIC BLADE
FINISH: N/A	SCALE: 1:1	PN: BF003-06 REV: A SHEET: 1 of 1

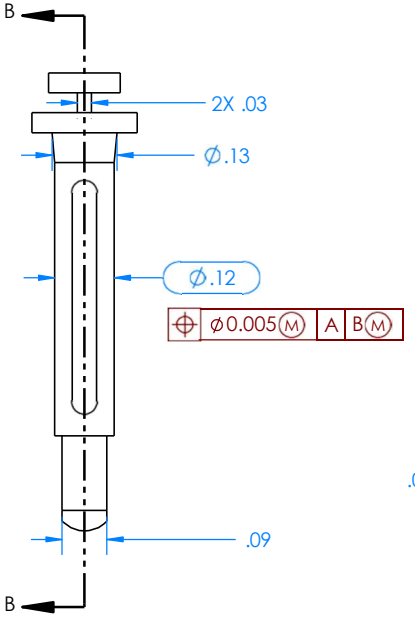
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.

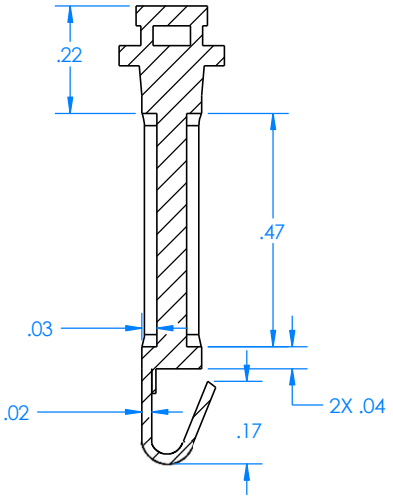
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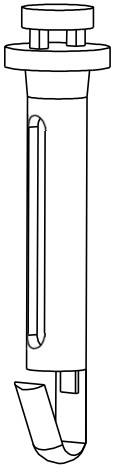
REV.	DESCRIPTION	DATE	DRAWN BY	APPROVED BY
A	INITIAL RELEASE	09/24/2024	SARAVANA	











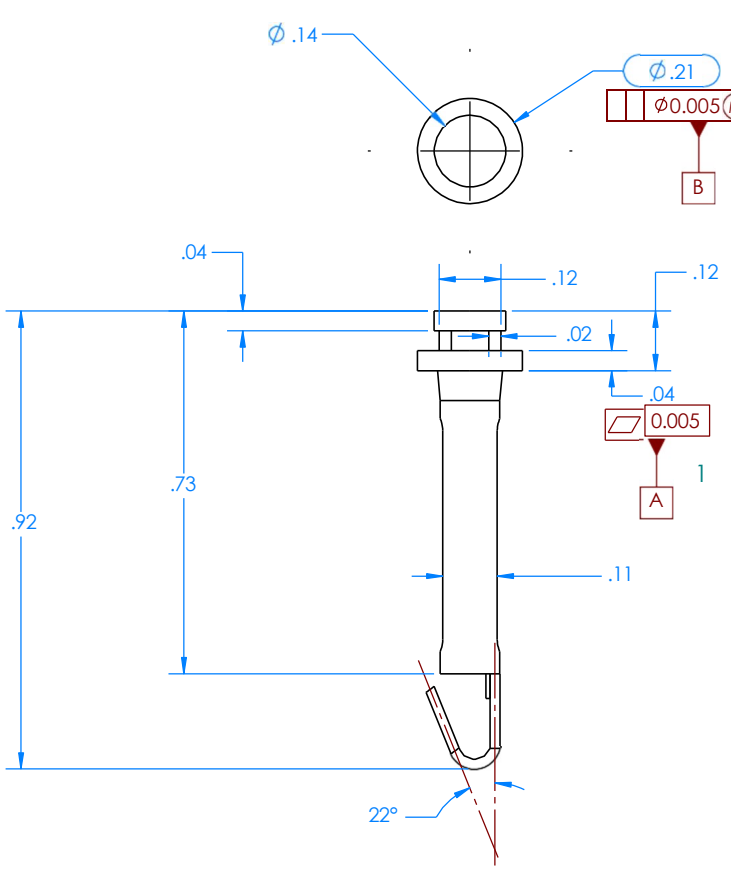
SECTION B-B

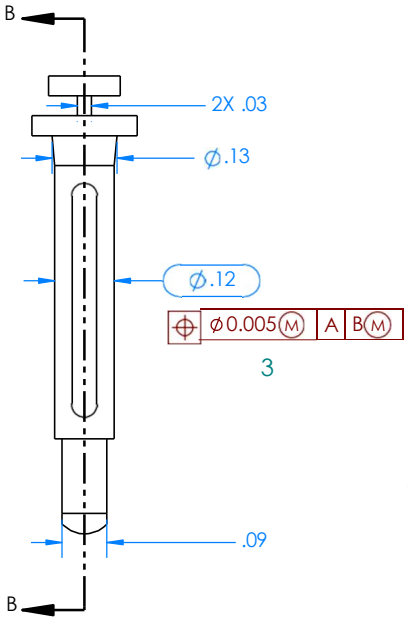
TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005 X.X ° ± 0.5°	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
MATERIAL: PC/ABS		DESCRIPTION: SPRING HOLDER	
FINISH: N/A	SCALE: 4:1	PN: BF003-07	REV: A
			SHEET: 1 of 1

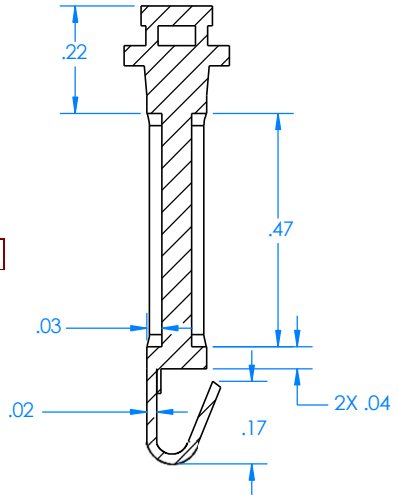
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.

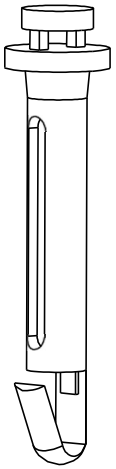
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REV.	DESCRIPTION	DATE	DRAWN BY	APPROVED BY
A	INITIAL RELEASE	09/24/2024	SARAVANA	









SECTION B-B

TOLERANCES UNLESS NOTED:			UNIVERSITY OF COLORADO
UNITS: INCHES	X.X	± 0.05	MCEN 5045
	X.XX	± 0.010	1111 ENGINEERING DRIVE
	X.XXX	± 0.005	BOULDER, CO 80309-0427
	X.X °	± 0.5°	

MATERIAL	DESCRIPTION			
PC/ABS	SPRING HOLDER			
FINISH	SCALE	PN	REV	SHEET
N/A	4:1	BF003-07	A	1 of 2

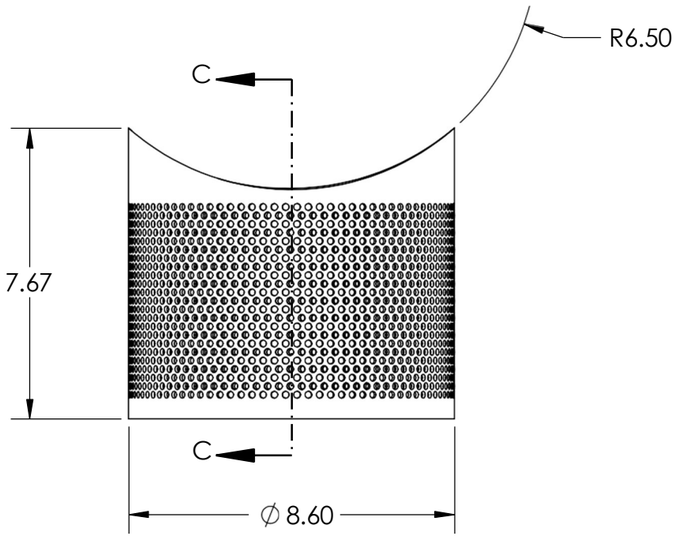
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.

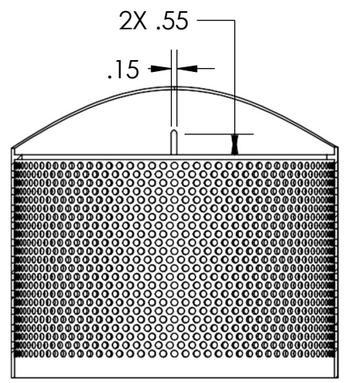
PC/ABS	SPRING HOLDER			
FINISH	SCALE	PN	REV	SHEET
N/A	4:1	F00-09	A	2 of 2

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REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	INITIAL RELEASE	20-10-2024	



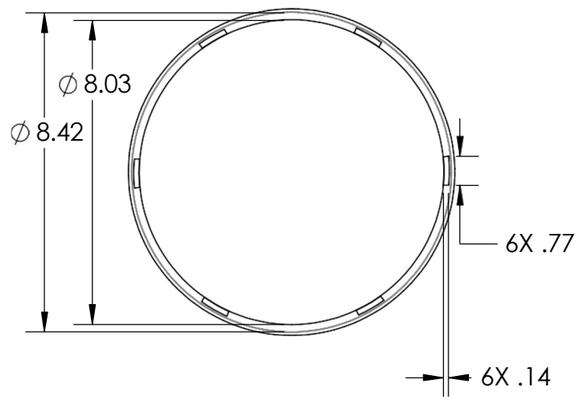


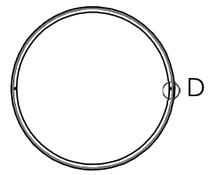
SECTION C-C

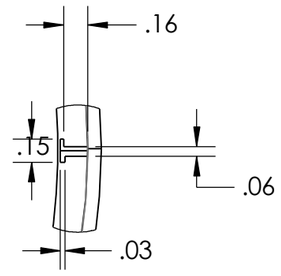


NOTES:

1. ANY UNSPECIFIED DIMENSIONS CAN BE TAKEN FROM THE 3D MODEL BF004-01.
2. THE MODEL SHOULD BE CLEAN OF ALL DIRT, SOLVENTS AND OIL.







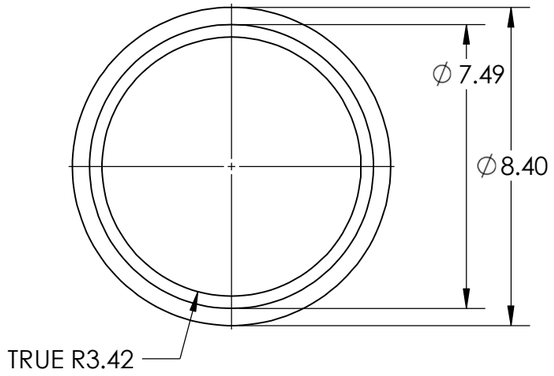
DETAIL D
SCALE 1 : 1

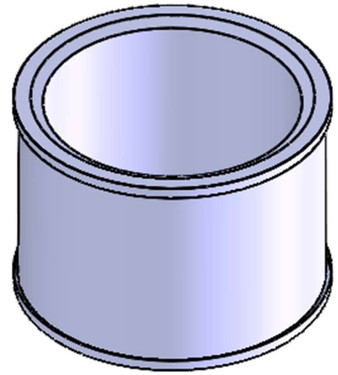
TOLERANCES UNLESS NOTED:	X.X ±0.050 X.XX ±0.010 X.XXX ±0.005	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427
UNITS: INCHES	X.X° 0.5°	
MATERIAL: ABS		DESCRIPTION: FILTER HOUSING
FINISH: GLOSSY SILVER	PN: BF004-01 / REV: A	SHEET: 1 of 1

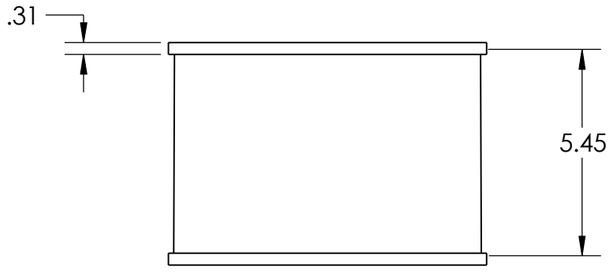
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REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	INITIAL RELEASE	20-10-2024	







NOTES:

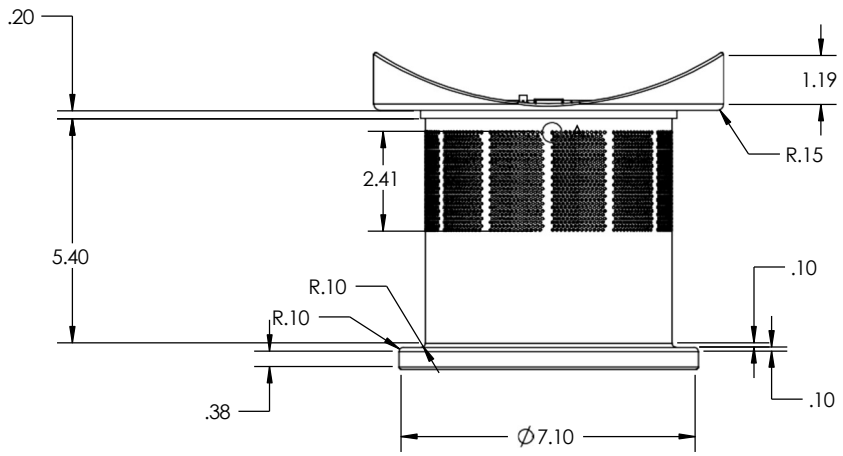
1. ANY UNSPECIFIED DIMENSIONS CAN BE TAKEN FROM THE 3D CAD MODEL BF004-02.
2. THE PART SHOULD BE CLEAN OF DIRT AND SOLVENTS.

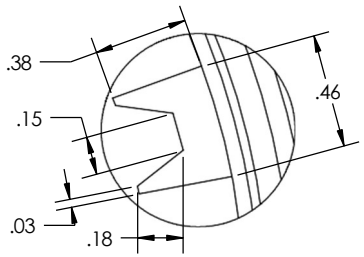
TOLERANCES UNLESS NOTED: X.X ±0.050, X.XX ±0.010, X.XXX ±0.005; UNITS: INCHES X.X° 0.5°	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
MATERIAL: ALUMINUM	DESCRIPTION: FILTER	
FINISH: GLOSSY GREEN	PN: BF004-02	REV: A; SHEET: 1 of 1

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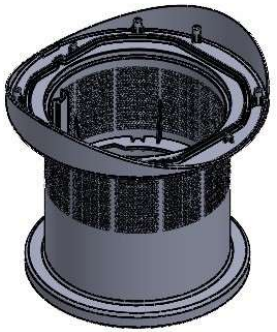
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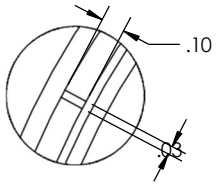
REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	10/5/2024





DETAIL B
SCALE 2 : 1

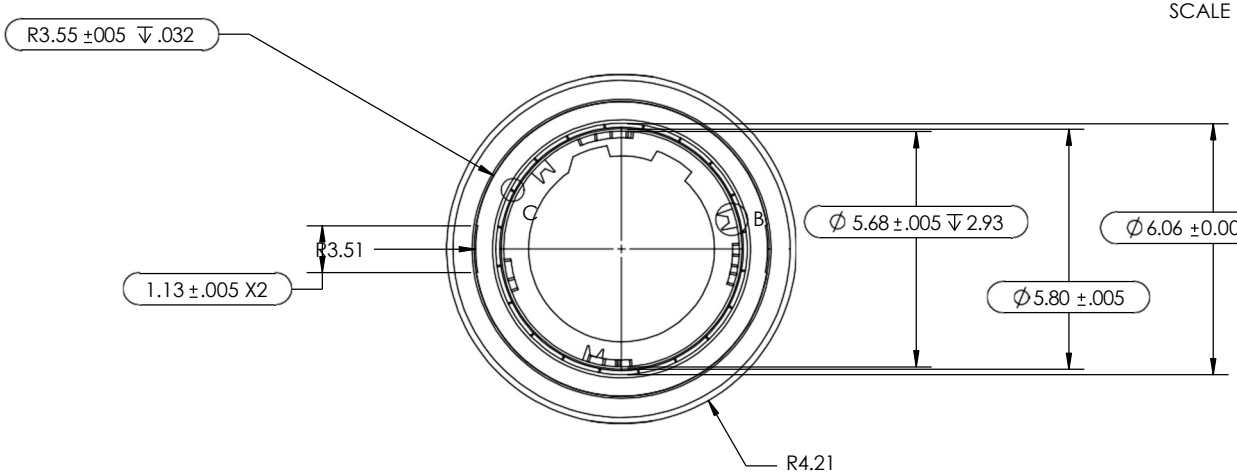




DETAIL C
SCALE 2 : 1



DETAIL A
SCALE 1 : 1



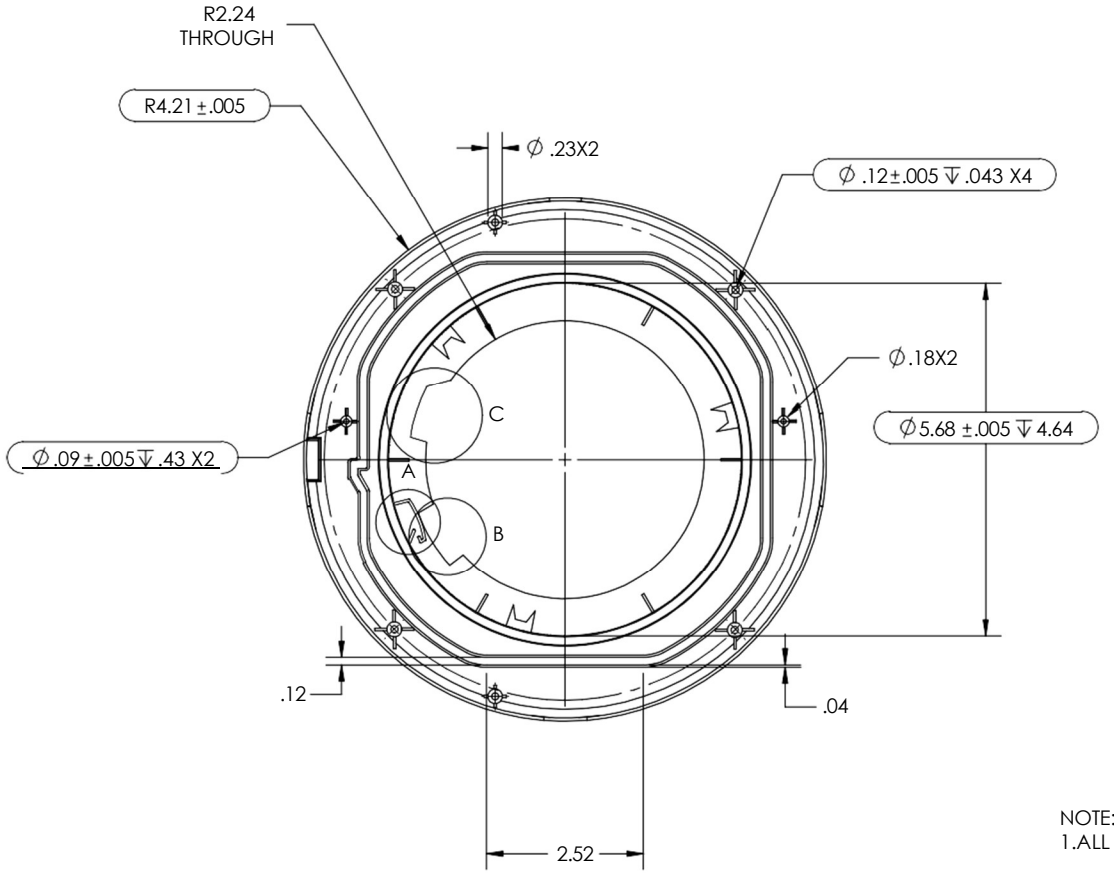
NOTE:
1. UNLESS MENTIONED SCALE IS 1:3
2. HOLES IN VIEW 'A' HAS A DIA OF 0.04" PATTERN OF 13X14 HOLES AND HAS 7 SETS AROUND SURFACE

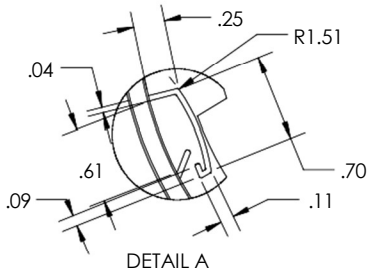
TOLERANCES UNLESS NOTED: UNITS:INCHES	X.X ±0.05 X.XX ±0.010 X.XXX ±0.005 X.X ° ° 1"	UNIVERSITY OF COLORADO BOULDER	
MATERIAL: PC/ABS		DESCRIPTION: ELECTRICAL HOUSING	
FINISH: NONE		PN: BF005-01	REV: A
			SHEET: 1 of 2

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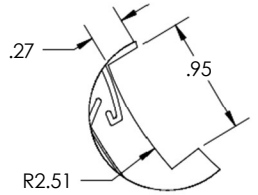
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REVISIONS		
REVISIONS	DESCRIPTION	DATE
A	INITIAL RELEASE	10/5/2024

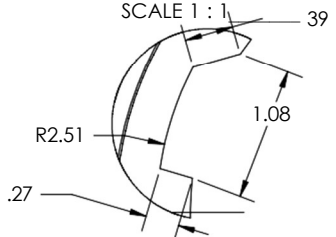




DETAIL A
SCALE 1 : 1



DETAIL B
SCALE 1 : 1



DETAIL C
SCALE 1 : 1

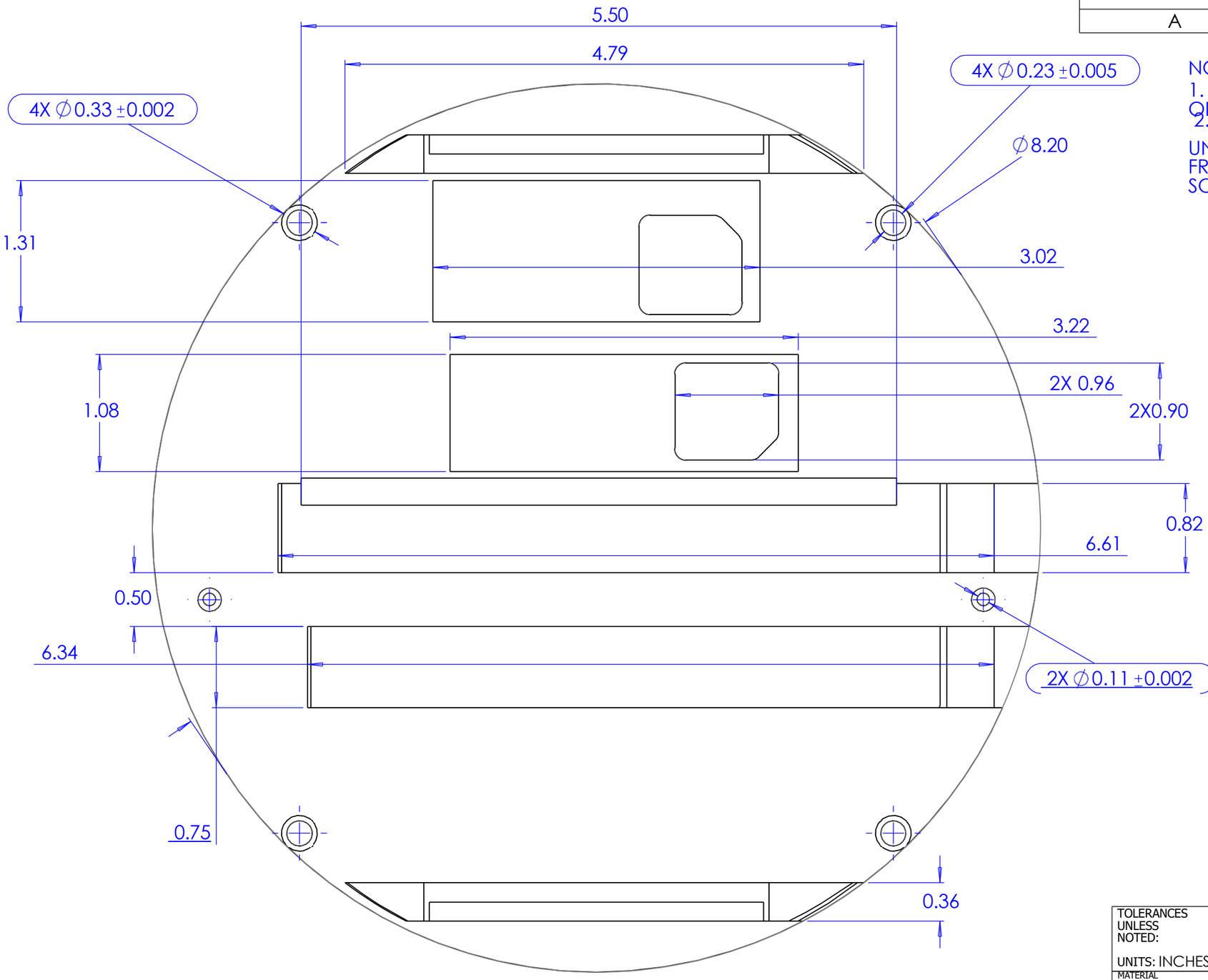
NOTE:
1.ALL RIBS HAVE WALL THICKNESS OF .04"

TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ±0.05 X.XX ±0.010 X.XXX ±0.005 X.X ° °1"	UNIVERSITY OF COLORADO BOULDER	
MATERIAL: PC/ABS		DESCRIPTION: ELECTRICAL HOUSING	
FINISH: NONE		PN: BF005-01	REV: A
			SHEET: 2 of 2

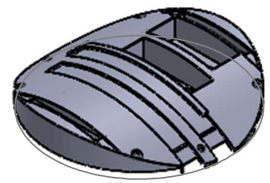
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REV	DESCRIPTION	DATE
A	INITIAL RELEASE	10/10/2024



NOTES:
1. PART SHALL BE VISIBLY CLEAN OF ALL DIRT, OIL, AND SOLVENTS.
2. THIS IS A LIMITED DIMENSION DRAWING. UNSPECIFIED DIMENSIONS SHALL BE TAKEN FROM THE ATTACHED COMPUTER-GENERATED SOLID MODEL.

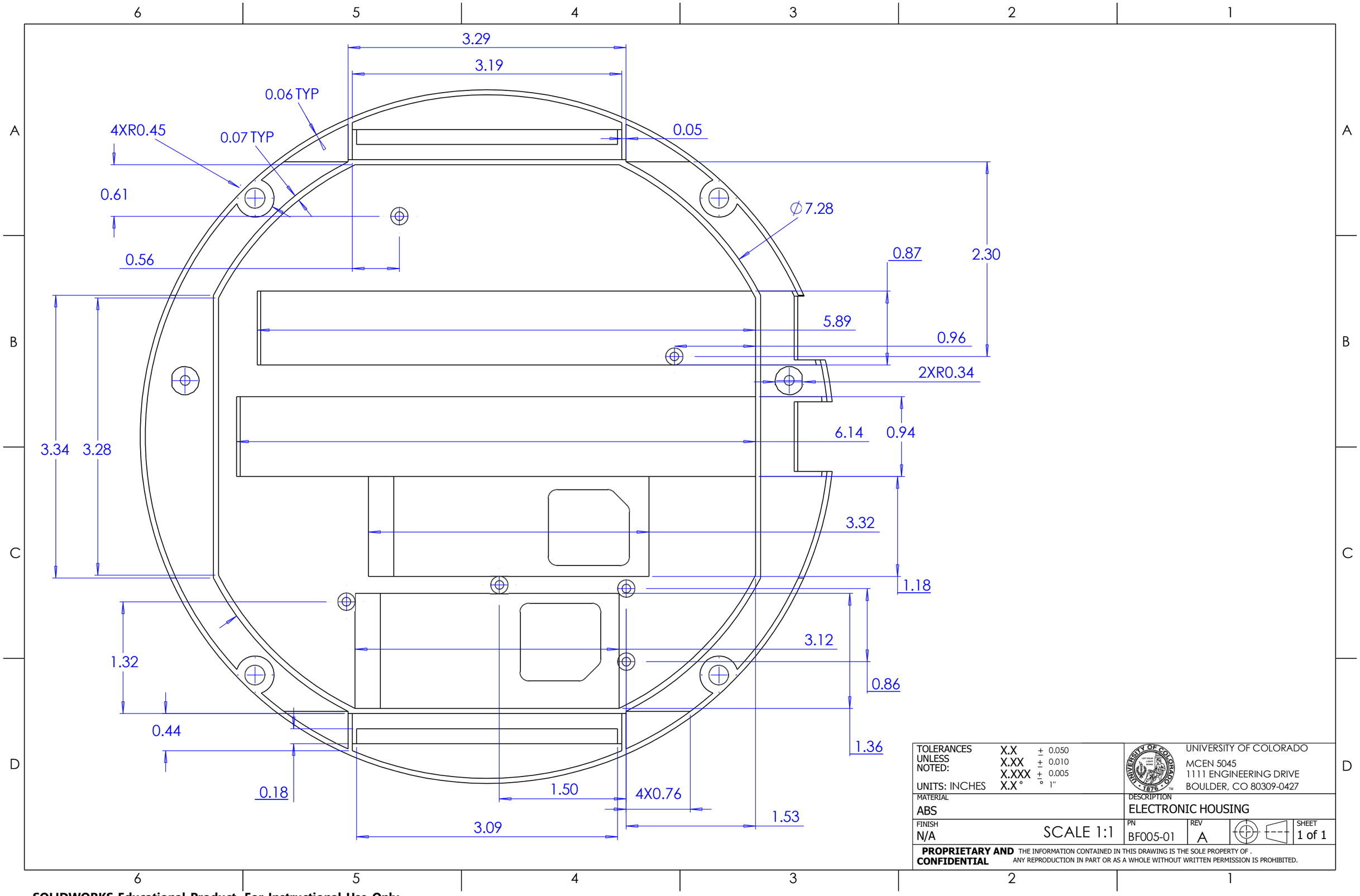


SCALE 1:5

TOLERANCES UNLESS NOTED:	X.X ± 0.050 X.XX ± 0.010 X.XXX ± 0.005	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427
UNITS: INCHES	X.X° ± 1°	
MATERIAL: ABS		DESCRIPTION: ELECTRONIC HOUSING
FINISH: N/A	SCALE 1:1	PN: BF005-01 · REV: A · SHEET: 1 of 1

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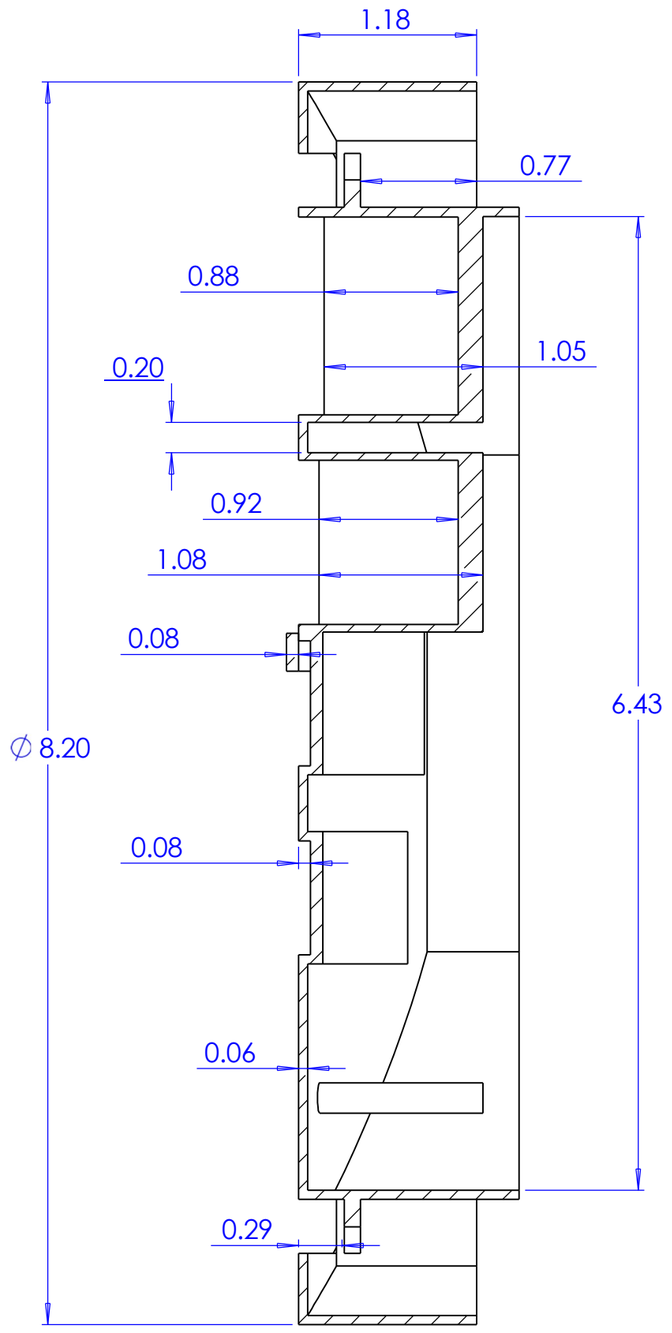
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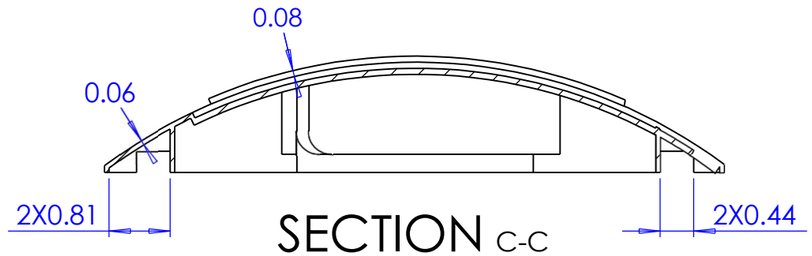
TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ± 0.050 X.XX ± 0.010 X.XXX ± 0.005 X.X° ° 1"	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
MATERIAL ABS		DESCRIPTION ELECTRONIC HOUSING	
FINISH N/A	SCALE 1:1	PN BF005-01	REV A
			SHEET 1 of 1

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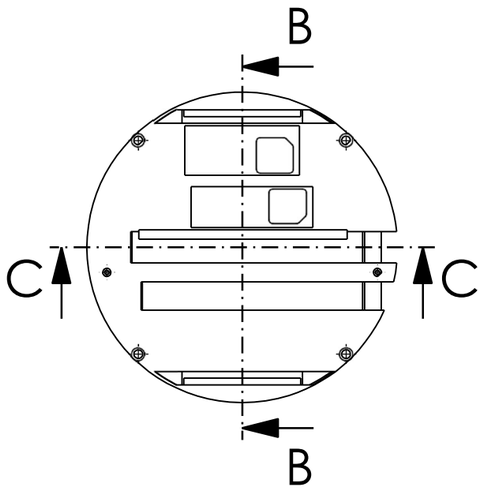


SECTION B-B



SECTION C-C

SCALE 1 : 2



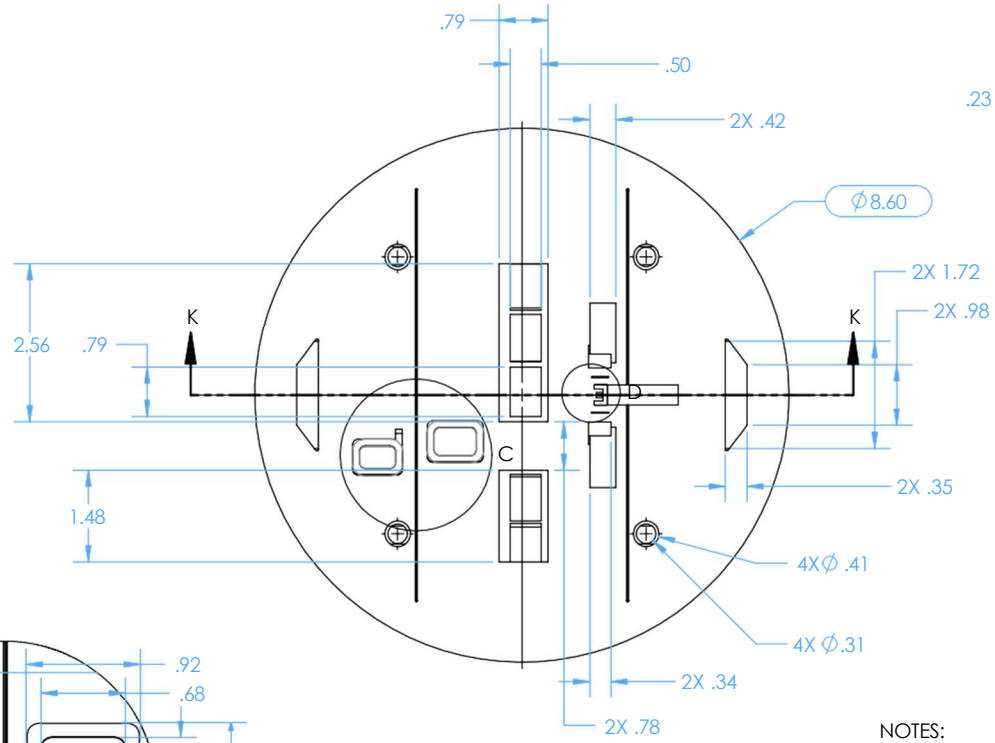
SCALE 1:4

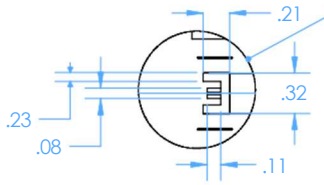
TOLERANCES UNLESS NOTED:	X.X ± 0.050 X.XX ± 0.010 X.XXX ± 0.005	UNIVERSITY OF COLORADO	
UNITS: INCHES	X.X° ° 1"	MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
MATERIAL: ABS		DESCRIPTION: ELECTRONIC HOUSING	
FINISH: N/A	SCALE 1:1	PN: BF005-01	REV: A / SHEET: 1 of 1

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REV.	DESCRIPTION	DATE	DRAWN BY	APPROVED BY
A	INITIAL RELEASE	10/05/2024	SARAVANA	

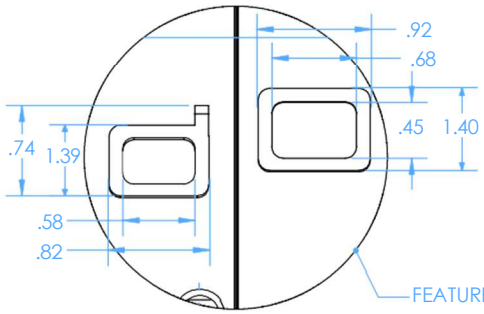




DETAIL D
SCALE 1 : 1



SCALE 1:8



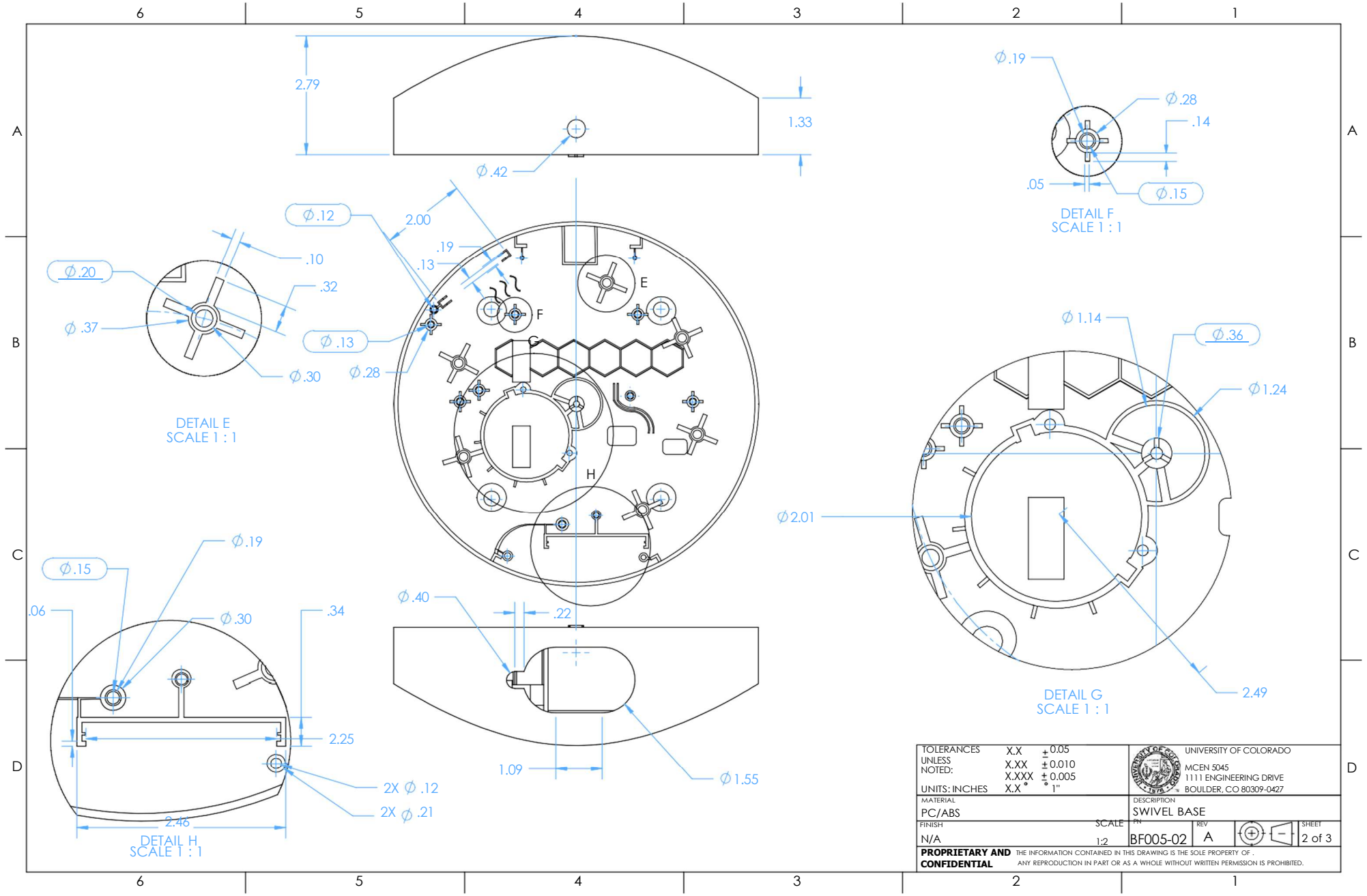
DETAIL C
SCALE 1 : 1

NOTES:
1. PART SHALL BE VISIBLY CLEAN OF ALL DIRT, OIL, AND SOLVENTS.
2. THIS IS A LIMITED DIMENSION DRAWING. UNSPECIFIED DIMENSIONS SHALL BE TAKEN FROM THE ATTACHED COMPUTER-GENERATED SOLID MODEL
3. CRITICAL OR INSPECTION DIMENSIONS ARE ENCLOSED WITH A SYMBOL

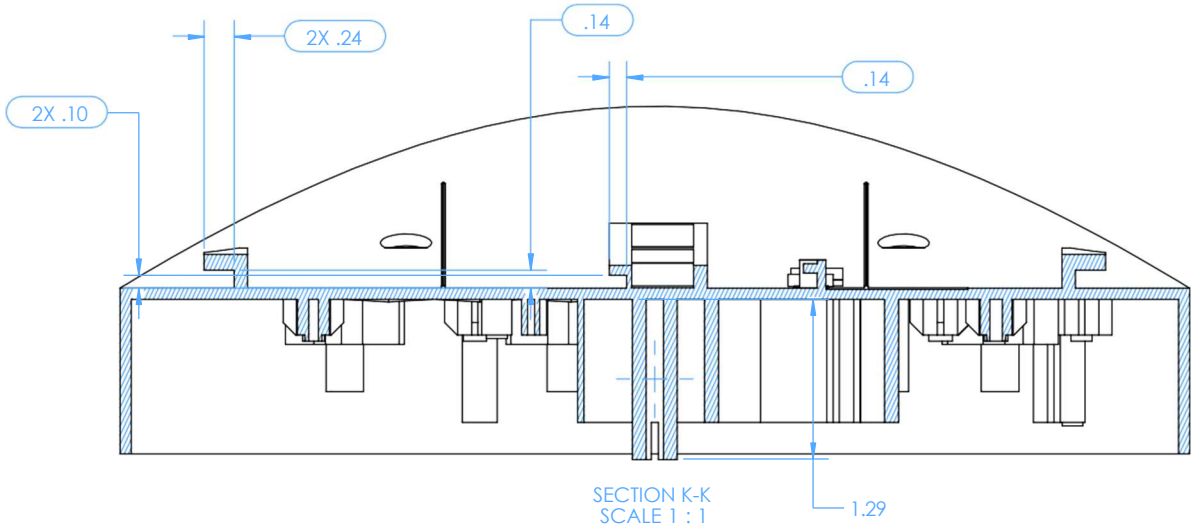
TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005 X.X ° ° 1"	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
MATERIAL: PC/ABS		DESCRIPTION: SWIVEL BASE	
FINISH: N/A	SCALE: 1:2	PN: BF005-02	REV: A / SHEET: 1 of 3

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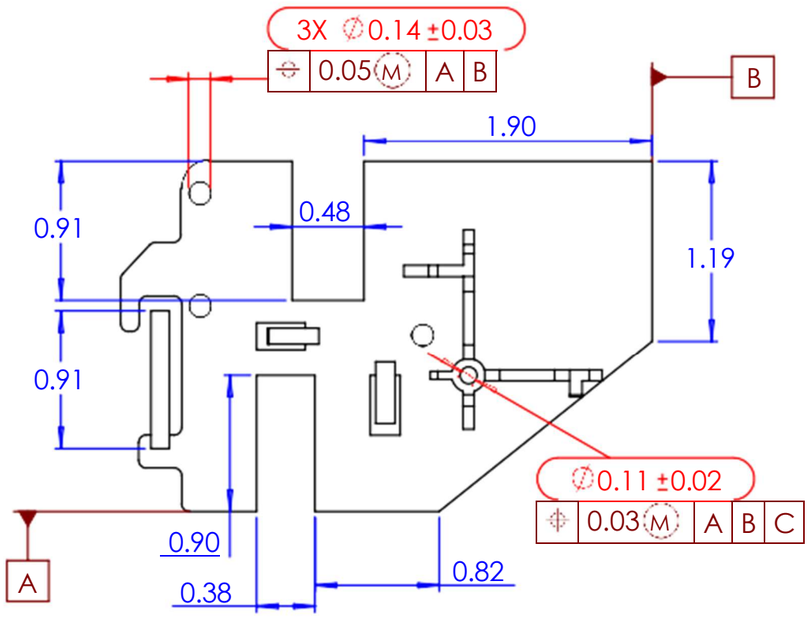
TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005 X.X ° ° 1"	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
MATERIAL PC/ABS		DESCRIPTION SWIVEL BASE	
FINISH N/A	SCALE 1:2	PN BF005-02	REV A
			SHEET 3 of 3

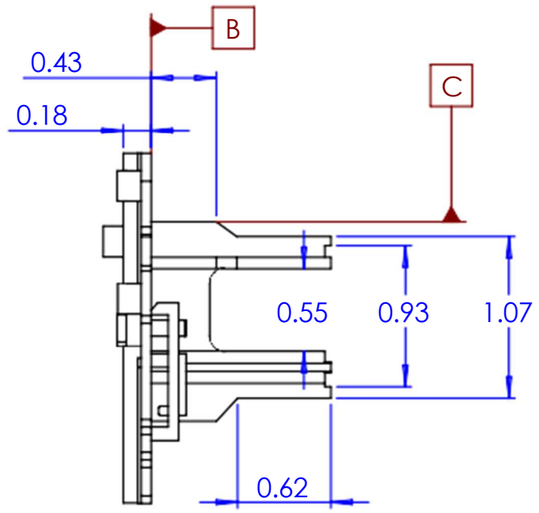
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.

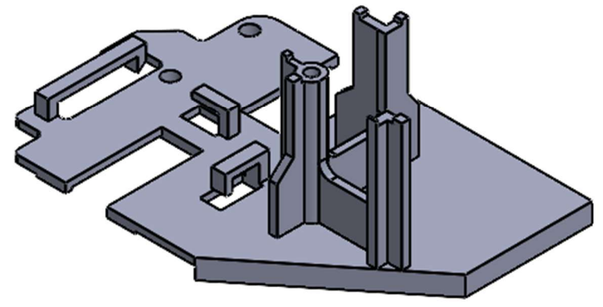
SOLIDWORKS Educational Product. For Instructional Use Only.

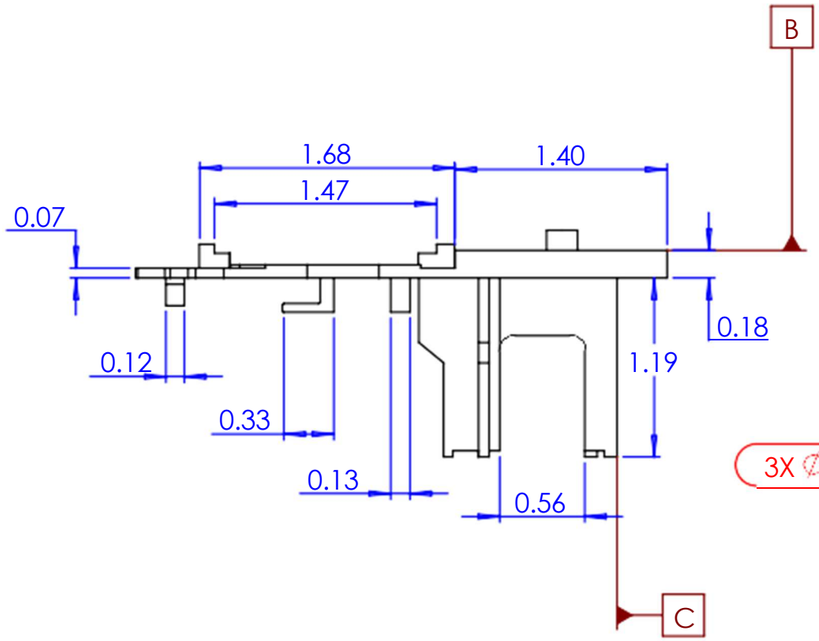
REV.	DESCRIPTION	DATE
C	INITIAL RELEASE	09/29/2024

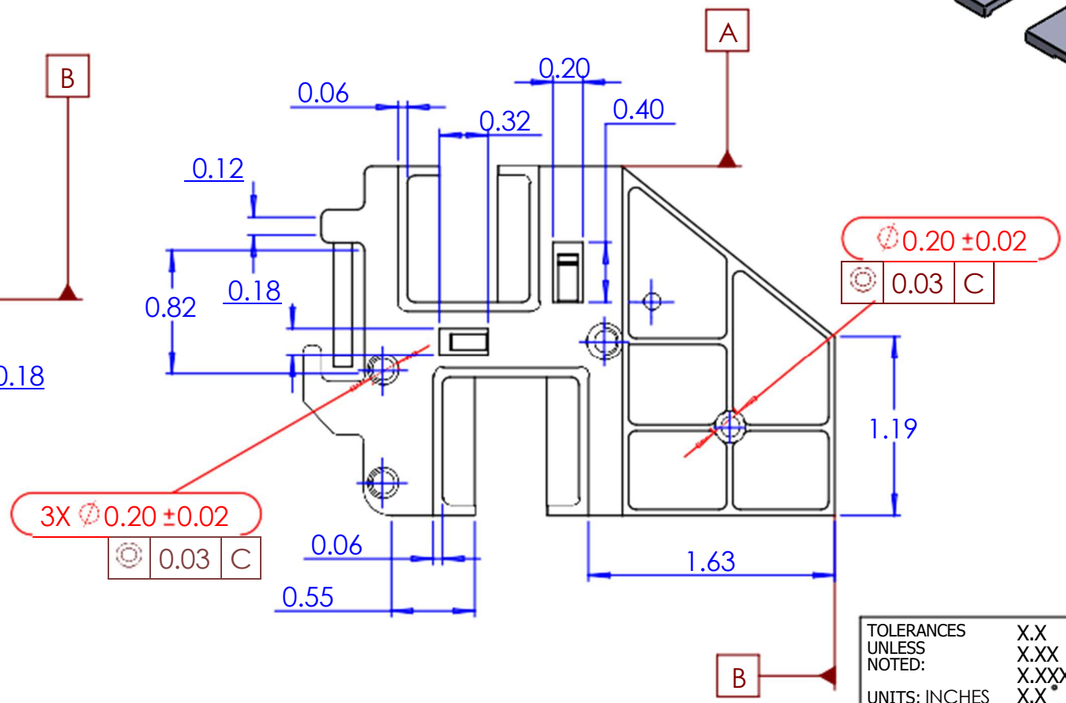
NOTES:
1. CRITICAL OR INSPECTION DIMENSIONS ARE ENCLOSED.
3. PART SHALL BE VISIBLY CLEAN OF ALL DIRT, OIL, AND SOLVENTS.
4. THE LENGTH AND WIDTH OF THE SLOT MATTERS AS IT LOCKS INTO POSITION OF THE BF005-01 ELECTRONIC HOUSING











TOLERANCES UNLESS NOTED:	X.X	±	0.050
	X.XX	±	0.010
	X.XXX	±	0.005
UNITS: INCHES	X.X°	±	0.5

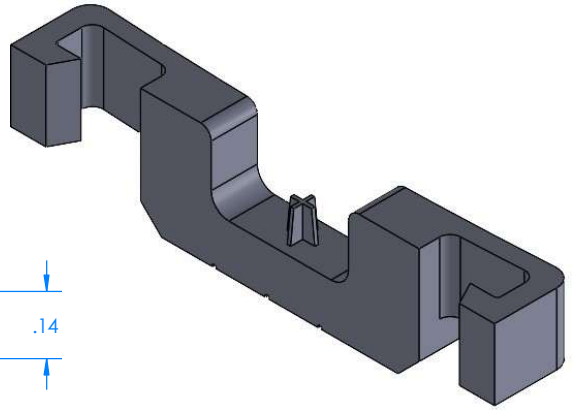
UNIVERSITY OF COLORADO
1111 ENGINEERING DRIVE
BOULDER, CO 80309-0427

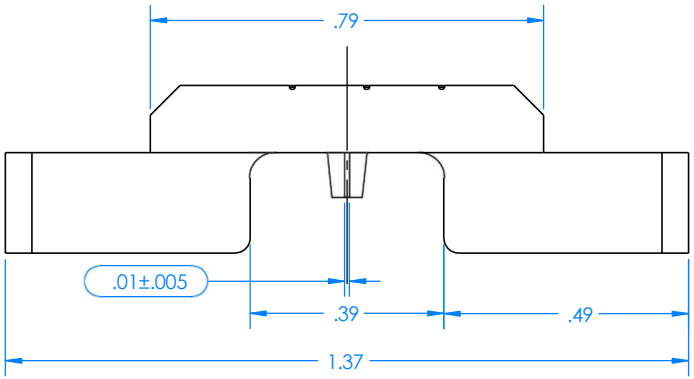
MATERIAL	DESCRIPTION		
ABS	ELECTRONICS COMPONENT HOLDER		
FINISH	PN	REV	SHEET
N/A	BF005-03	C	1 of 1

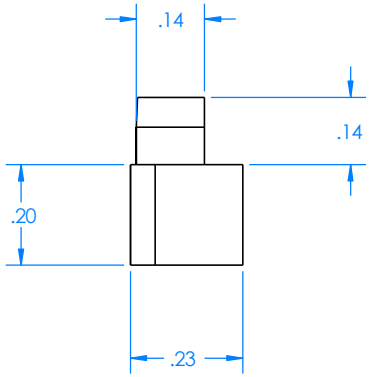
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.

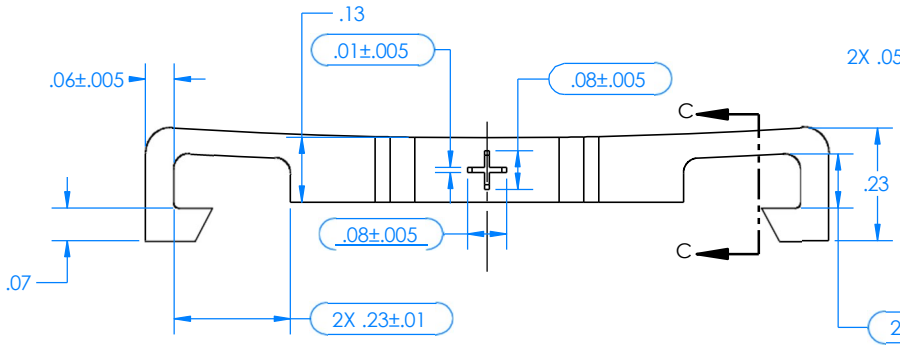
SOLIDWORKS Educational Product. For Instructional Use Only.

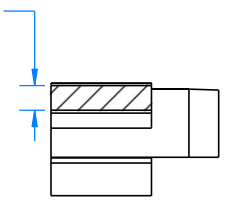
REV.	DESCRIPTION	DATE	DRAWN BY	APPROVED BY
A	INITIAL RELEASE	09/17/2024	SARAVANA	











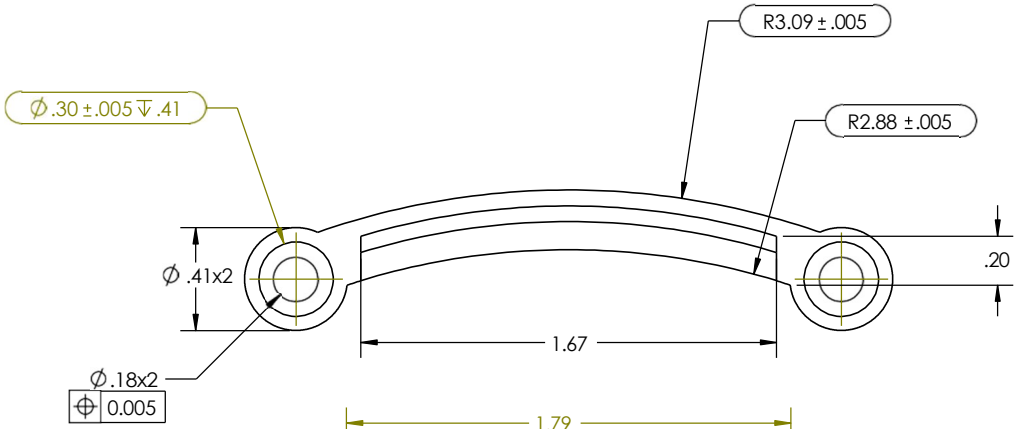
SECTION C-C

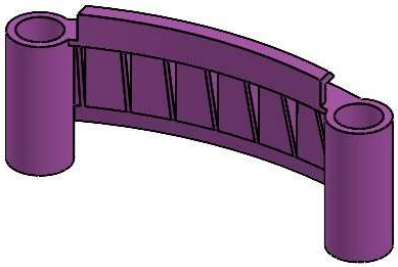
TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005 X.X ° ± 0.5°	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
MATERIAL: POLYCARBONATE		DESCRIPTION: SWIVEL BASE LOCK	
FINISH: N/A	SCALE: 4:1	PN: BF006-02	REV: A	SHEET: 1 of 1

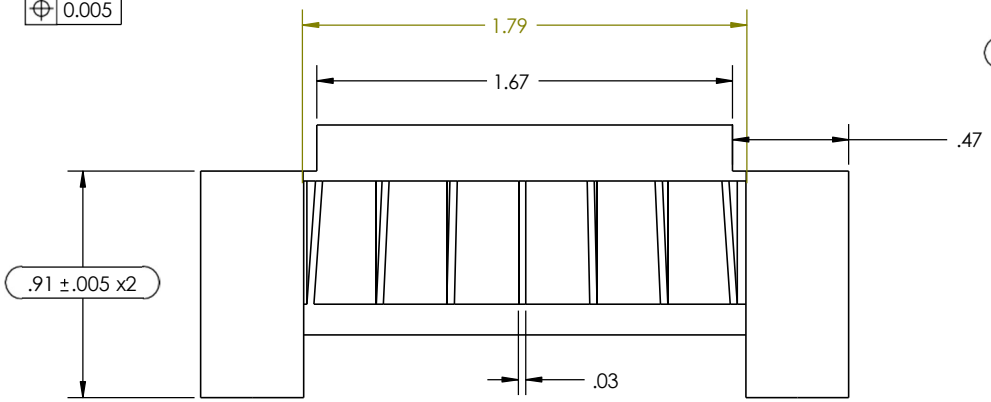
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.

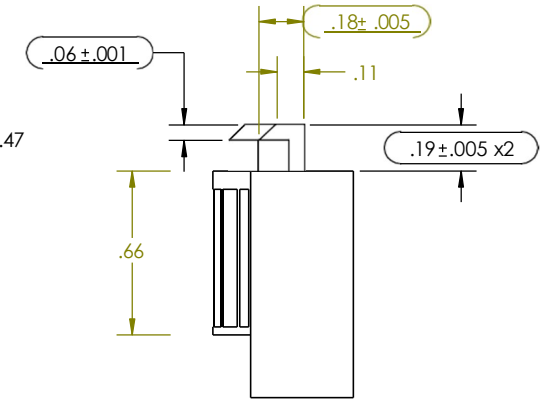
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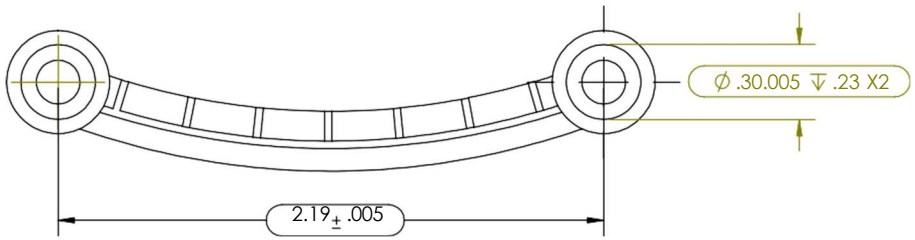
REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	9/27/2024











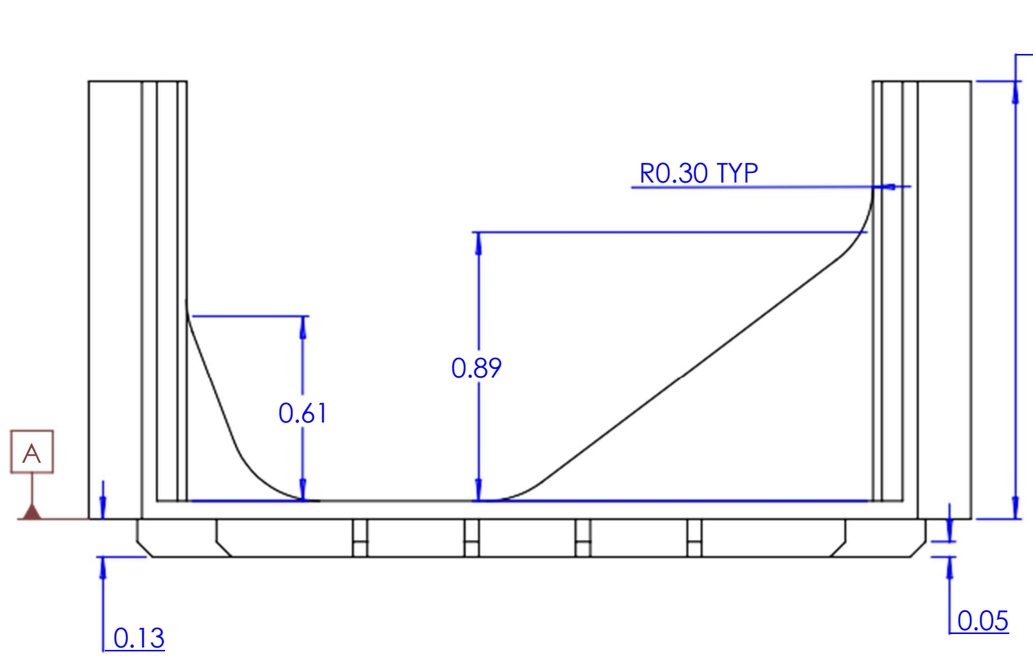
NOTE:
1.UNLESS MENSTIONED SCALE IS 2:1

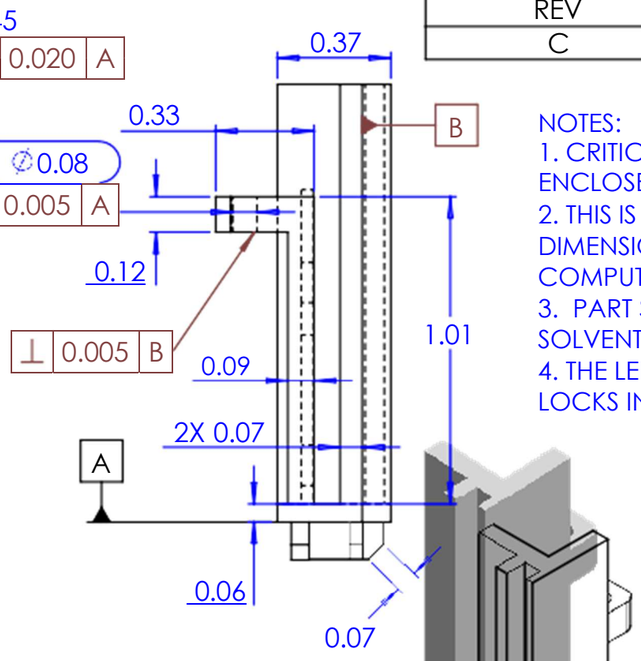
TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ±0.05 X.XX ±0.010 X.XXX ±0.005 X.X ° °1''	UNIVERSITY OF COLORADO BOULDER
MATERIAL: POM		DESCRIPTION: Base plate lock
FINISH: NONE	PN: BF006-04 REV: A	SHEET: 1 of 1

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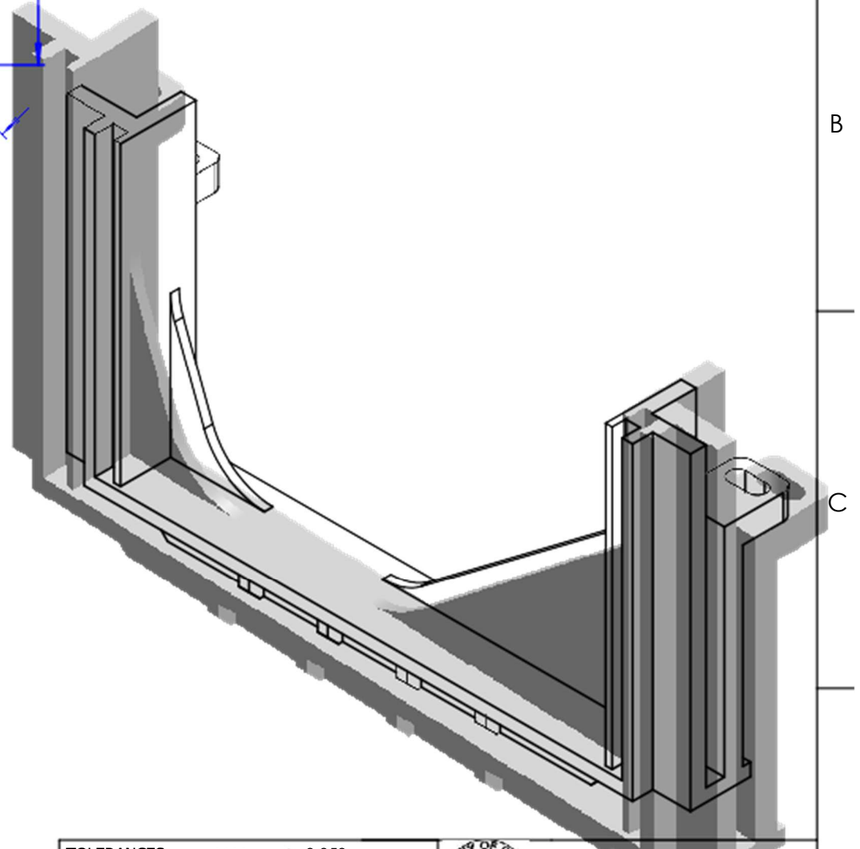
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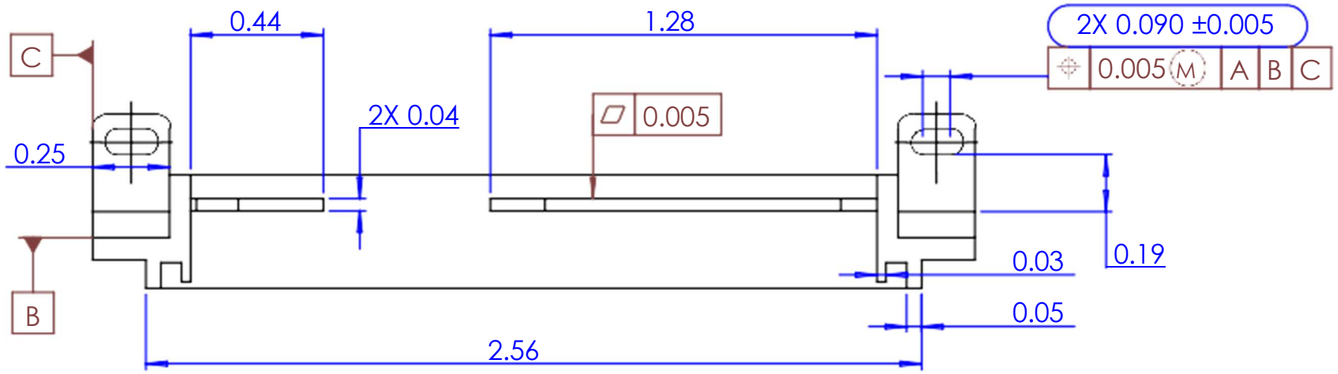
REV	DESCRIPTION	DATE
C	INITIAL RELEASE	09/29/2024

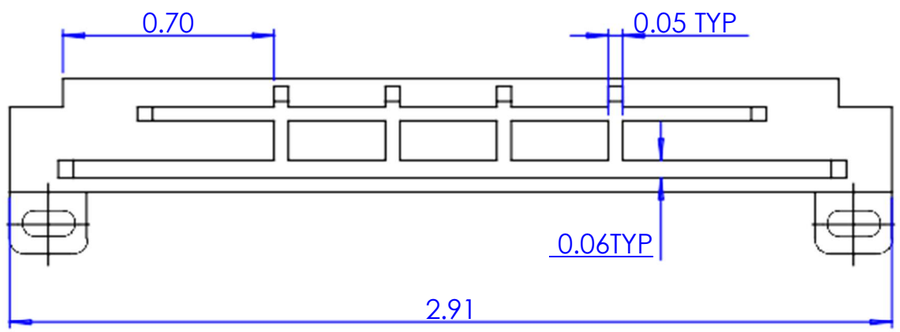




NOTES:
1. CRITICAL OR INSPECTION DIMENSIONS ARE ENCLOSED.
2. THIS IS A LIMITED DIMENSION DRAWING. UNSPECIFIED DIMENSIONS SHALL BE TAKEN FROM THE ATTACHED COMPUTER-GENERATED SOLID MODEL, BF006-05 REV B.
3. PART SHALL BE VISIBLY CLEAN OF ALL DIRT, OIL, AND SOLVENTS.
4. THE LENGTH AND WIDTH OF THE SLOT MATTERS AS IT LOCKS INTO POSITION OF THE BF006-01.





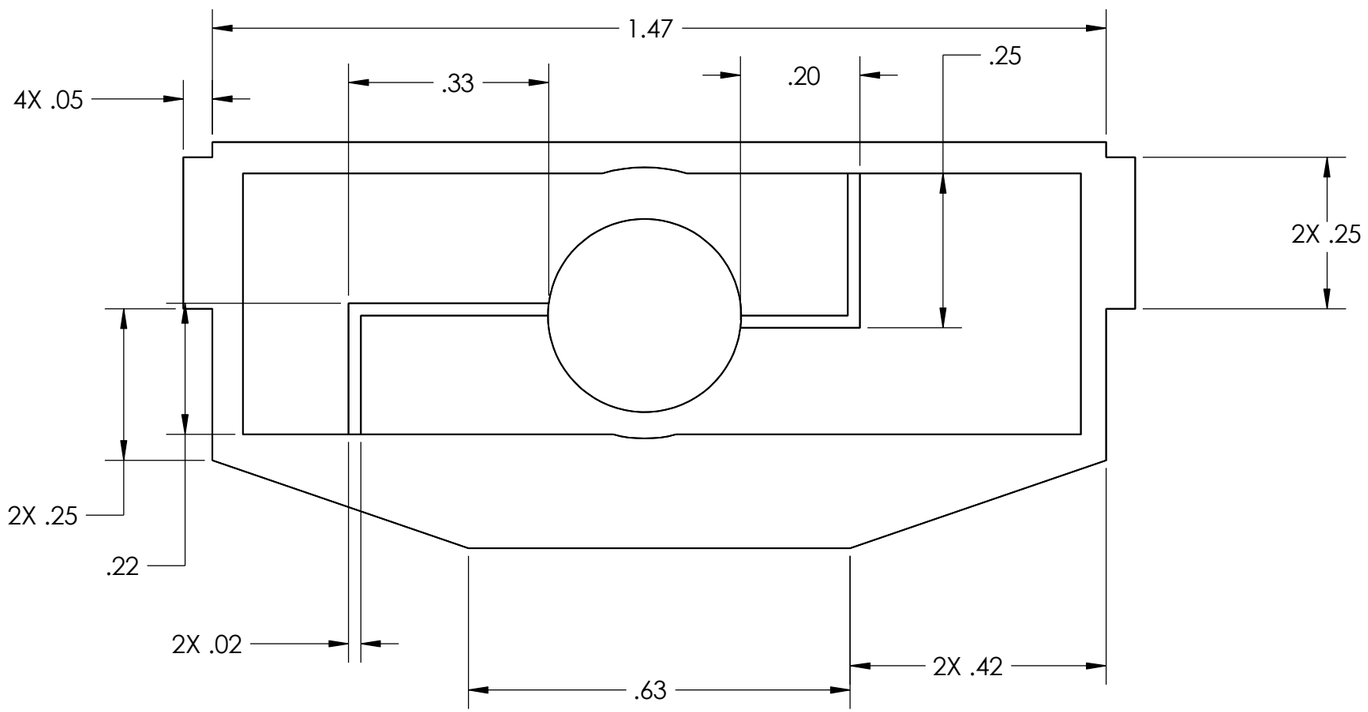


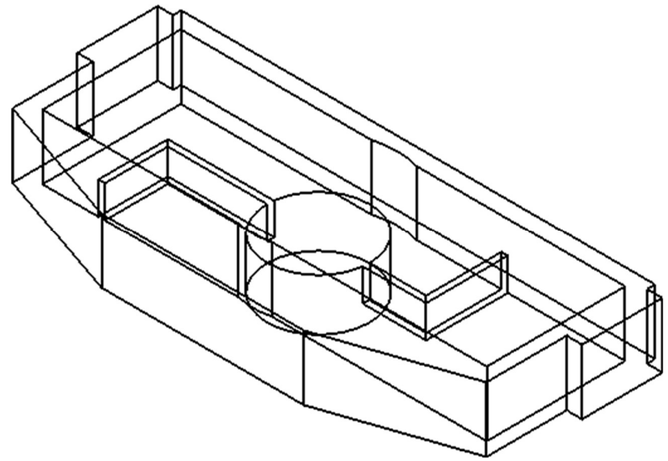
TOLERANCES UNLESS NOTED:	X.X ± 0.050 X.XX ± 0.010 X.XXX ± 0.005	UNIVERSITY OF COLORADO 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
UNITS: INCHES	X.X° ± 0.5°		
MATERIAL: ABS		DESCRIPTION: LCD AND PCB HOLDER	
FINISH: N/A		PN: BF006-05	REV: C
			SHEET: 1 of 1

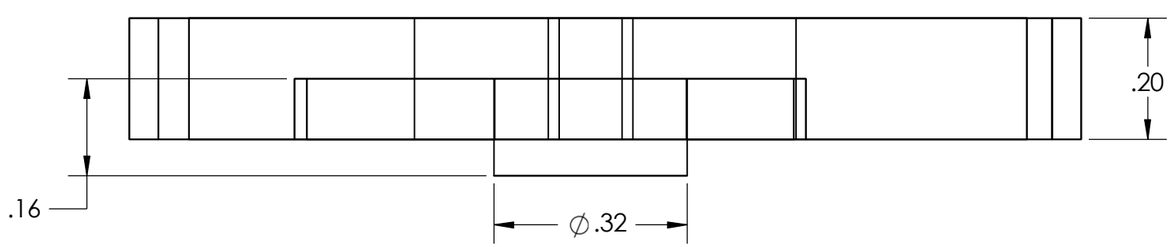
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REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	INITIAL RELEASE	20-10-2024	







NOTES:

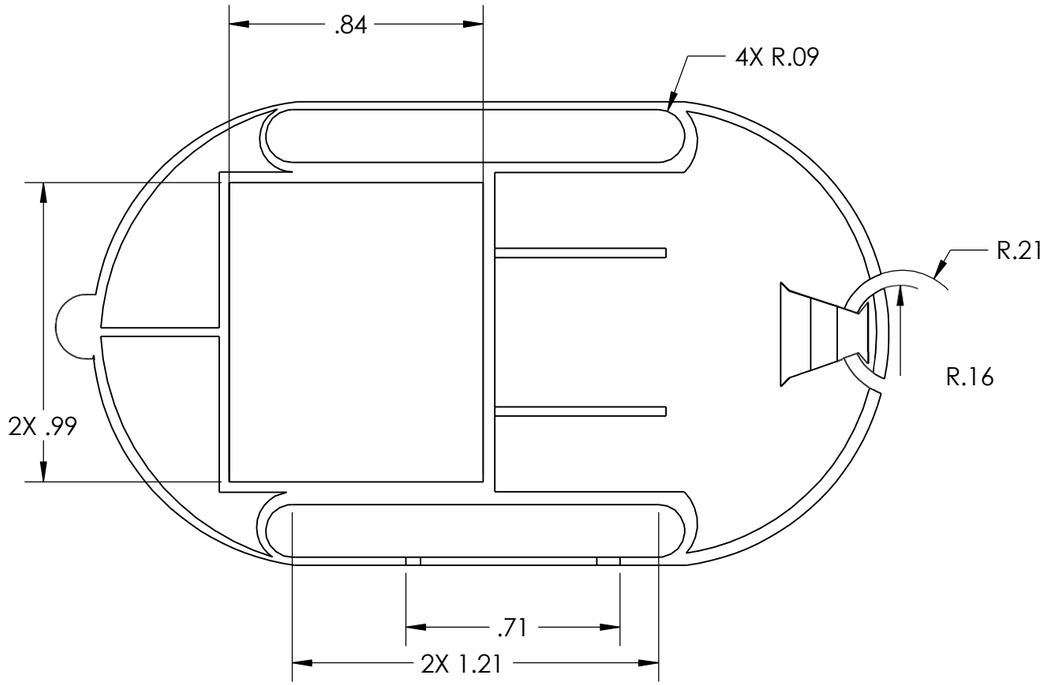
1. ANY UNSPECIFIED DIMENSIONS CAN BE TAKEN FROM THE 3D CAD MODEL, BF006-06.
2. THE PART SHOULD BE TRANSPARENT AND FLEXIBLE.

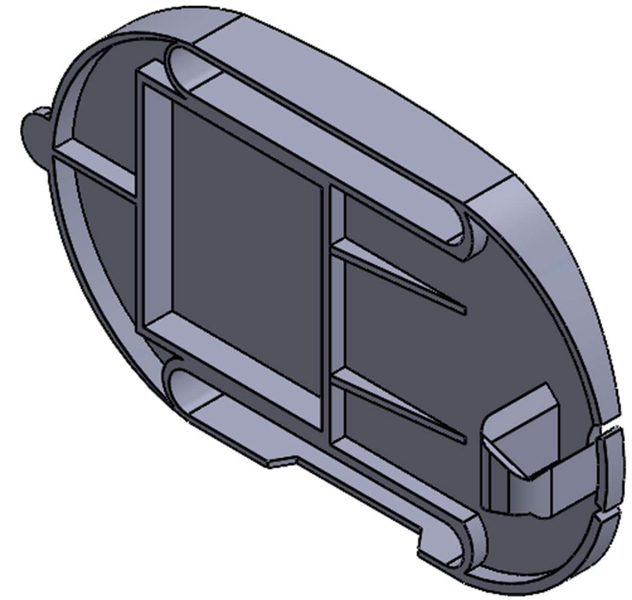
TOLERANCES UNLESS NOTED:	X.X ±0.050 X.XX ±0.010 X.XXX ±0.005	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427
UNITS: INCHES	X.X° 0.5°	
MATERIAL: PC		DESCRIPTION: POWER BUTTON
FINISH: TRANSPARENT		PN: BF006-06 · REV: A · SHEET: 1 of 1

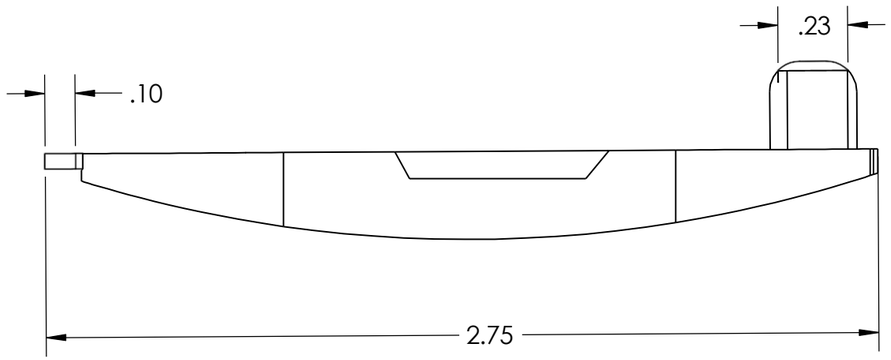
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REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	INITIAL RELEASE	20-10-2024	







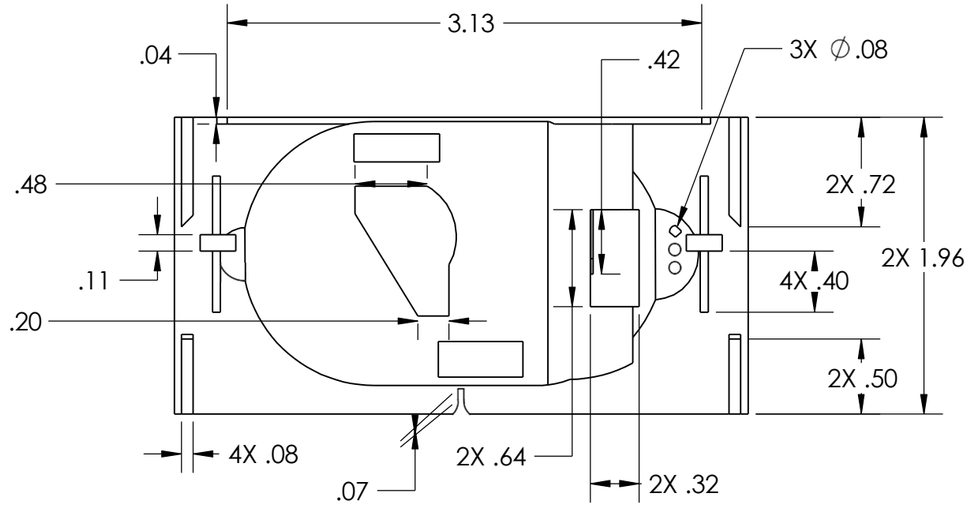
Notes:
ANY DIMENSION UNSPECIFIED CAN BE TAKEN FROM 3D CAD MODEL, BF006-07.
PART SHOULD NOT HAVE ANY DIRT, SOLVENTS OR OIL.

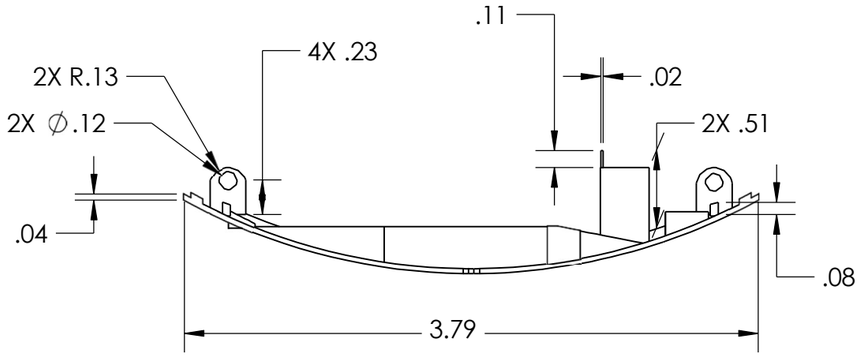
TOLERANCES UNLESS NOTED:	X.X ±0.050 X.XX ±0.010 X.XXX ±0.005	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427
UNITS: INCHES	X.X° 0.5°	
MATERIAL: PC/ABS		DESCRIPTION: SENSOR HATCH
FINISH: GLOSSY WHITE	PN: BF006-07 / REV: A	SHEET: 1 of 1

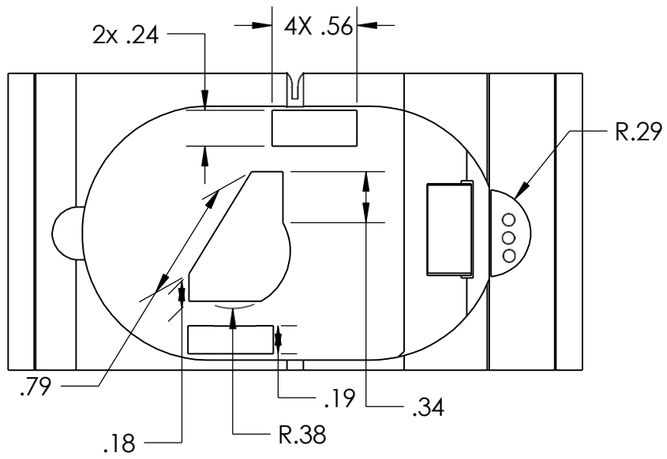
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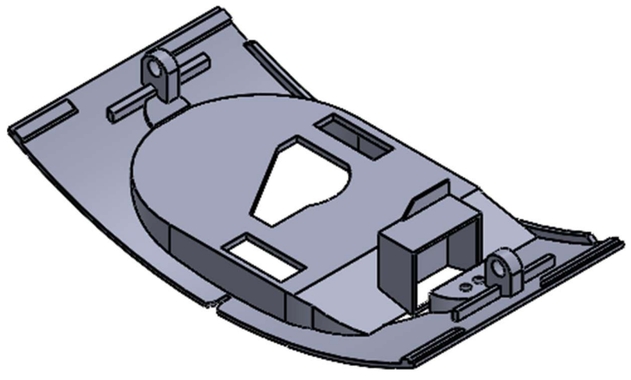
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REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	INITIAL RELEASE	20-10-2024	









NOTES:

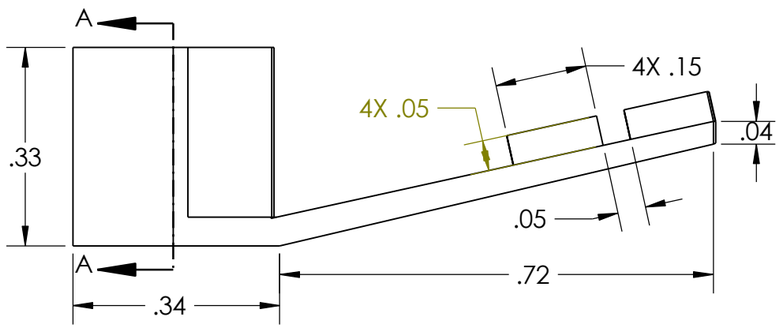
1. ANY OTHER DIMENSIONS CAN BE TAKEN DIRECTLY FROM THE ATTACHED 3D CAD MODEL, BF006-08.
2. PART SHOULD NOT HAVE ANY DIRT OR SOLVENTS.

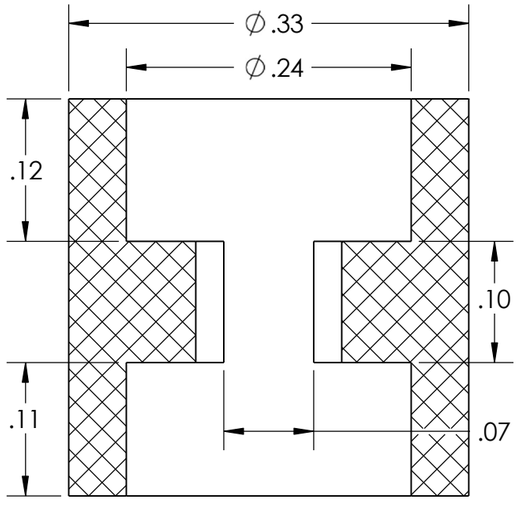
TOLERANCES UNLESS NOTED:	X.X ±0.050 X.XX ±0.010 X.XXX ±0.005	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
UNITS: INCHES	X.X° 0.5°		
MATERIAL: PC/ABS		DESCRIPTION: SENSOR HATCH HOUSING	
FINISH: GLOSSY WHITE		PN: BF006-08	REV: A
			SHEET: 1 of 1

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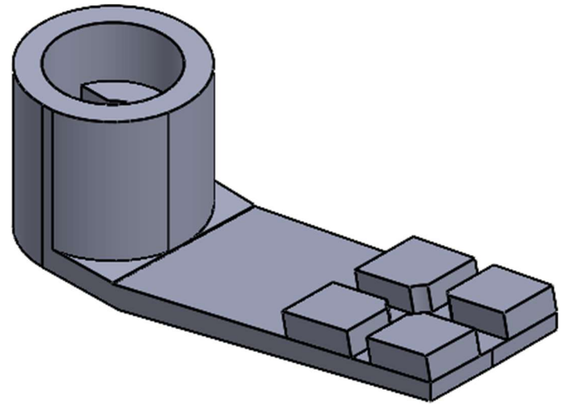
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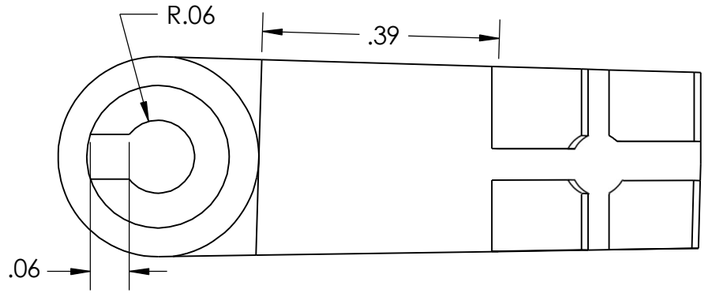
REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	INITIAL RELEASE	21-10-2024	





SECTION A-A
SCALE 8 : 1





NOTES:

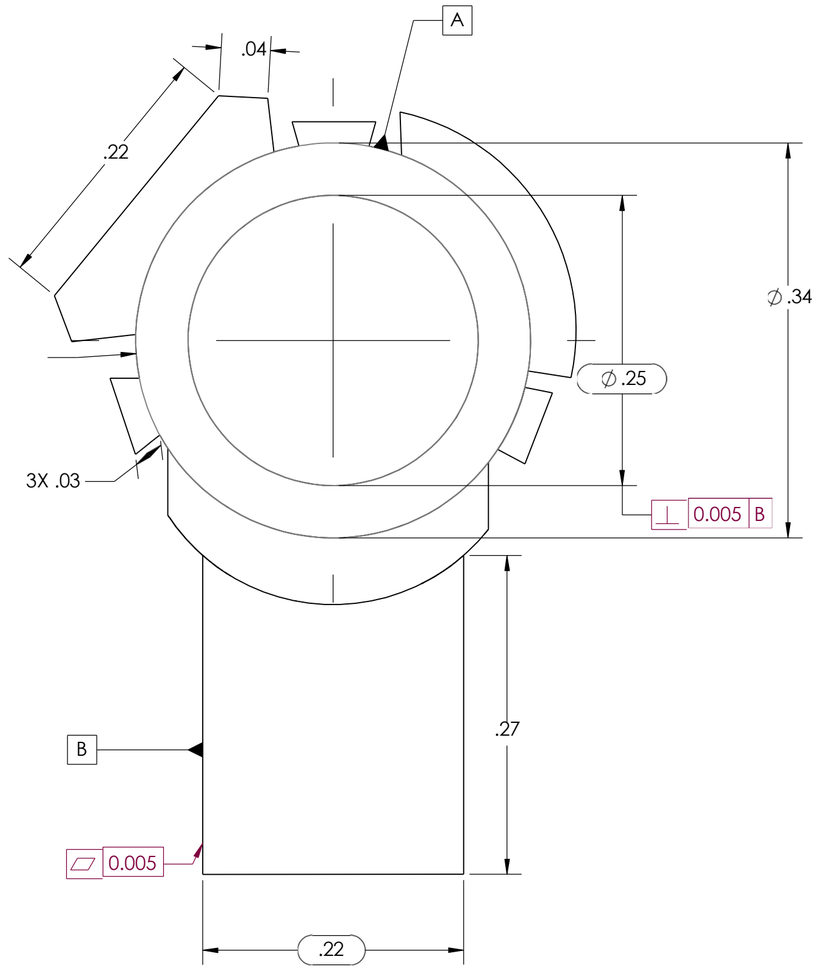
ANY UNSPECIFIED DIMENSIONS CAN BE TAKEN FROM THE 3D CAD MODEL BF006-09.
THE PART SHOULD NOT HAVE ANY DIRT, SOLVENTS OR OIL WHEN FINISHED.

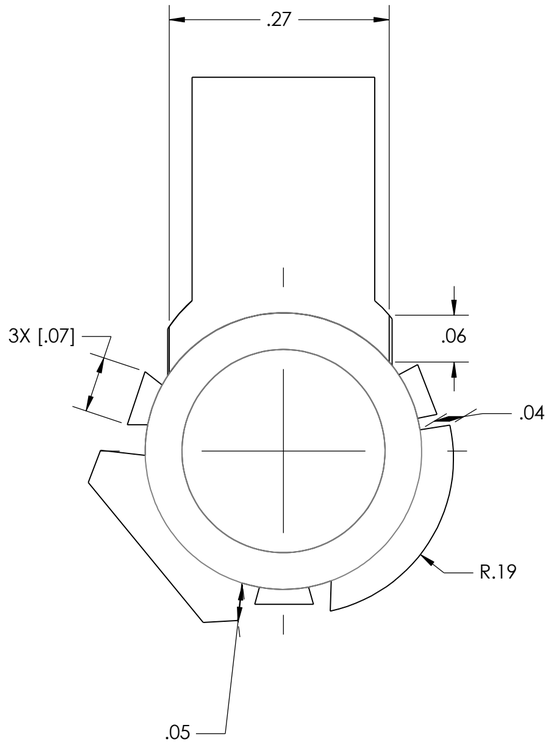
TOLERANCES UNLESS NOTED:	X.X ±0.050 X.XX ±0.010 X.XXX ±0.005	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427
UNITS: INCHES	X.X° 0.5°	
MATERIAL: PC/ABS		DESCRIPTION: WIRE HOLDER
FINISH: MEDIUM GLOSS-BLACK		PN: BF006-09 · REV: A · SHEET: 1 of 1

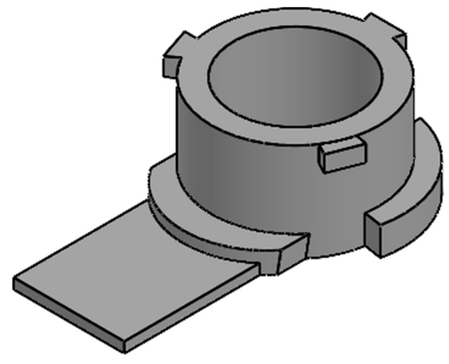
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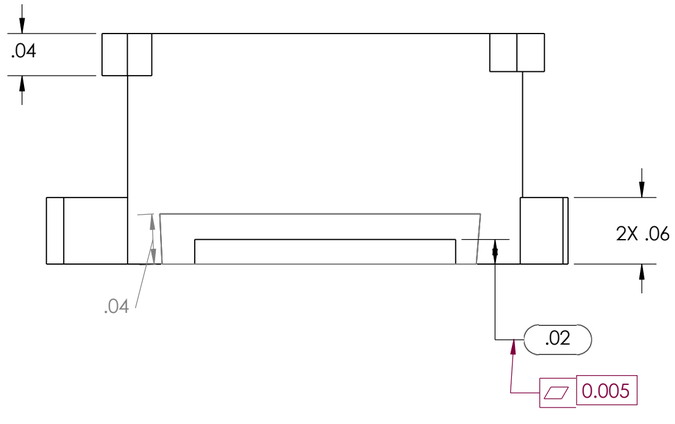
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REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	Initial Design	23-09-2024	
D2	B	(B) GD&T added	25-09-2024	









NOTES:

A: ALL CRITICAL AND INSPECTABLE DIMENSIONS ARE PLACED INSIDE A BOX.
B. THE CRITICAL FEATURES IN THE PART MATE WITH PART NUMBER F00-08 WHERE A SCREW SITS INSIDE THE INNER DIAMETER.
C. THE LENGTH OF THE CLIP MATTERS AS IT LOCKS INTO POSITION AND IT SHOULD ALSO BE INSPECTABLE.
D. ANY UNSPECIFIED DIMENSIONS CAN BE TAKEN FROM THE 3D MODEL, F00-12 REV A, SCREW CLIP

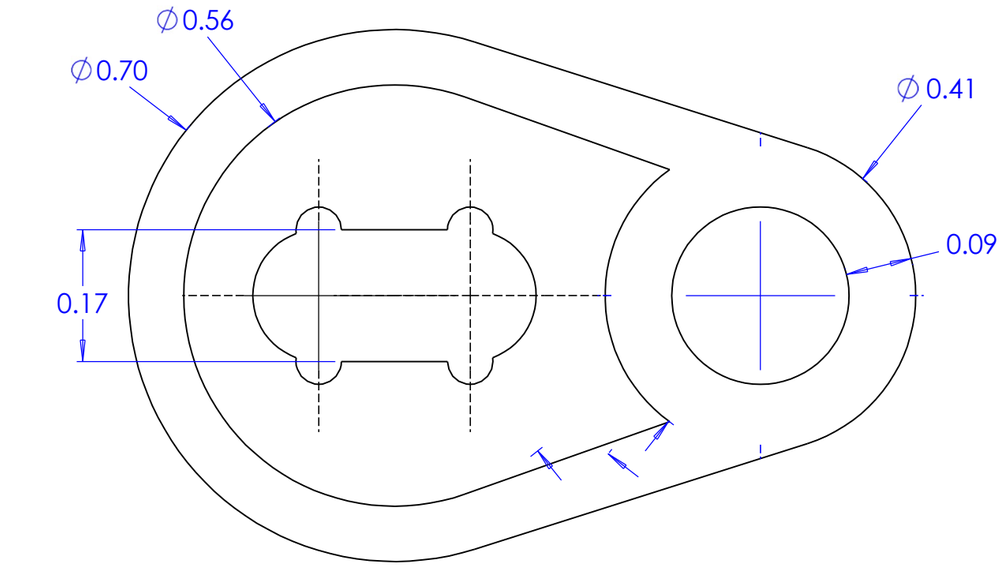
TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ±0.050 X.XX ±0.010 X.XXX ±0.005 X.X° 0.5°	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427
MATERIAL: POLYOXYMETHYLENE		DESCRIPTION: SCREW CLIP
FINISH: MATTE WHITE	PN: BF006-10	REV: B / SHEET: 1 of 1

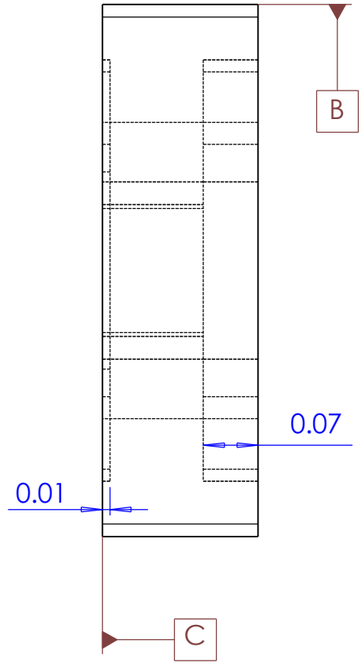
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF UNIVERSITY OF COLORADO. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.

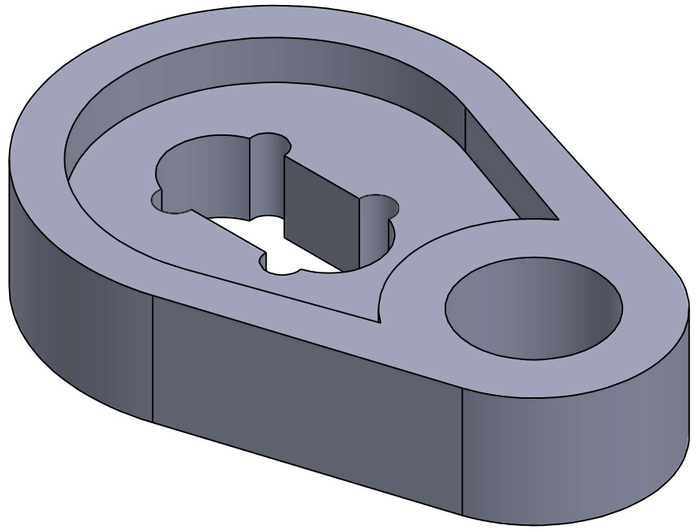
SOLIDWORKS Educational Product. For Instructional Use Only.

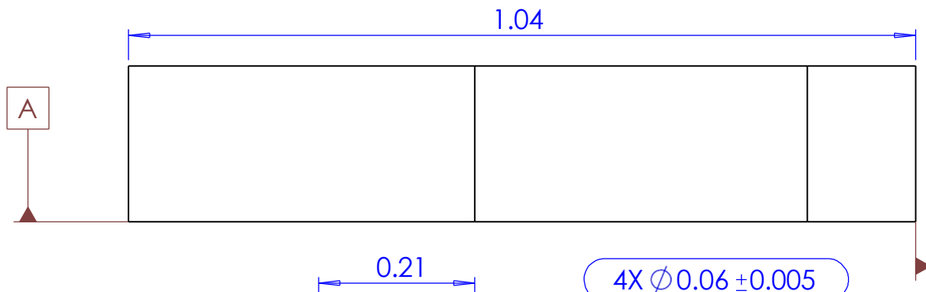
REV	DESCRIPTION	DATE
A	INITIAL RELEASE	09/29/2024

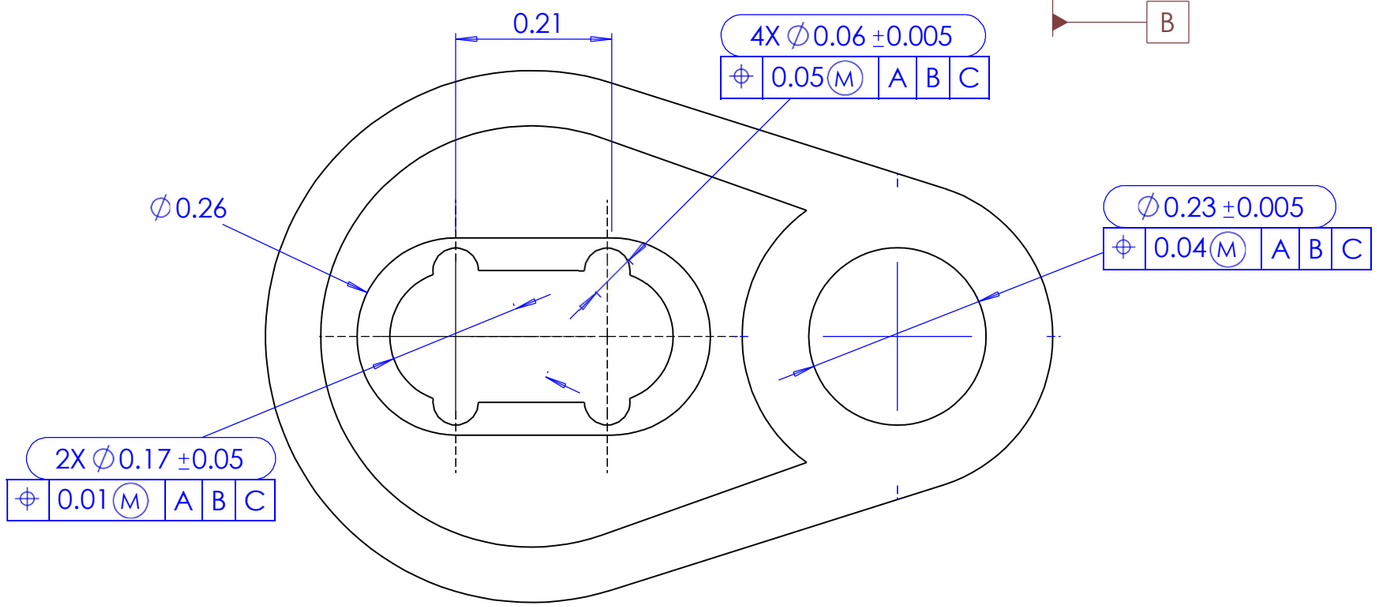
NOTES:
1. CRITICAL OR INSPECTION DIMENSIONS ARE ENCLOSED.
2. PART SHALL BE VISIBLY CLEAN OF ALL DIRT, OIL, AND SOLVENTS.
4. THE LENGTH AND WIDTH OF THE SLOT MATTERS AS IT LOCKS INTO POSITION OF THE BF007-01 MOTOR AND THE BIG HOLE SIZE SHOULD BE PROPER FOR BF007-03 180 SWIVEL ARM











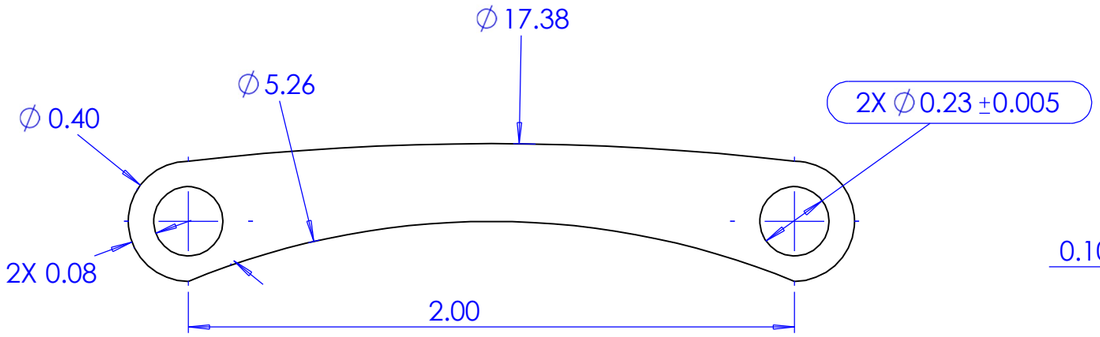
TOLERANCES UNLESS NOTED:	X.X	± 0.050	UNIVERSITY OF COLORADO 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427
	X.XX	± 0.010	
	X.XXX	± 0.005	
UNITS: INCHES	X.X °	° 0.5	

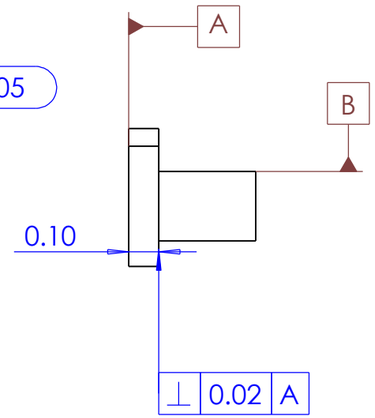
MATERIAL	DESCRIPTION			
ABS	CAM			
FINISH	PN	REV		SHEET
N/A	BF007-02	A		1 of 1

PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.

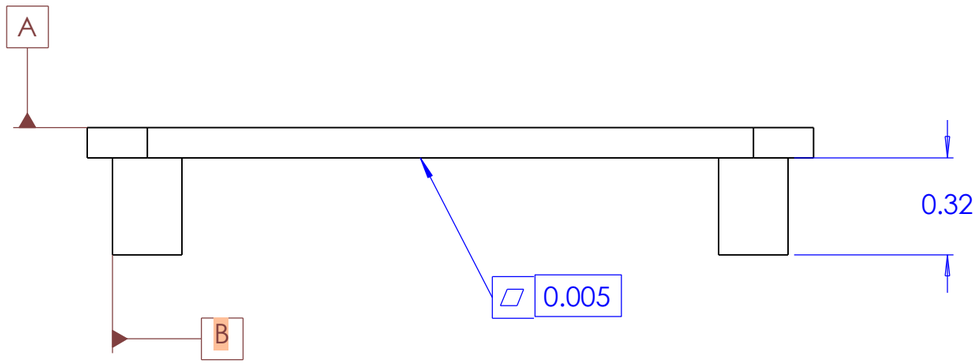
SOLIDWORKS Educational Product. For Instructional Use Only.

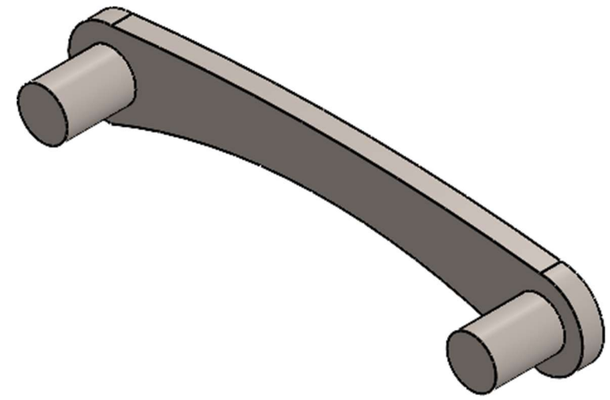
REV	DESCRIPTION	DATE
A	INITIAL RELEASE	10/29/2024





NOTES:
1. CRITICAL OR INSPECTION DIMENSIONS ARE ENCLOSED.
2. THE LENGTH OF THE EXTRUDED CYLINDER MATTERS AS IT GET FITTED ON TO THE BF007-02 CAM, STEPPER MOTOR AND BF006-01 LOWER BODY HOUSING.



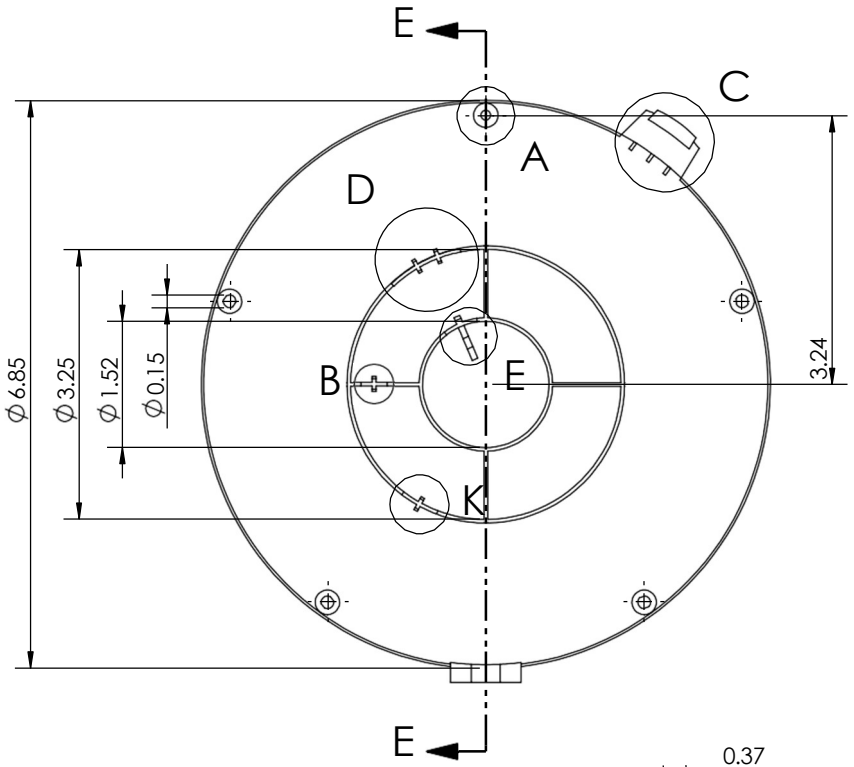


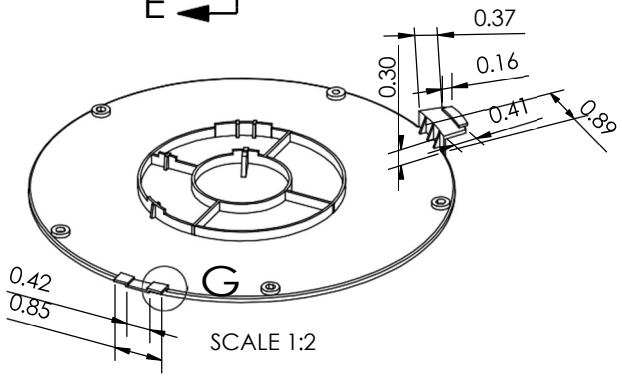
TOLERANCES UNLESS NOTED:	X.X ± 0.050 X.XX ± 0.010 X.XXX ± 0.005 X.X° ± 0.5	UNIVERSITY OF COLORADO 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427
UNITS: INCHES		
MATERIAL: ALLOY STEEL		DESCRIPTION: SWIVEL ARM
FINISH: N/A	PN: BF007-03	REV: A
		SHEET: 1 of 1

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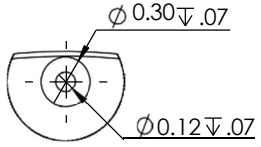
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REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	9/11/2024
B	ADDED DETAILED VIEWS	10/15/2024

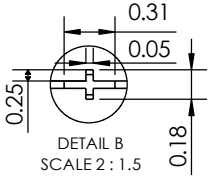




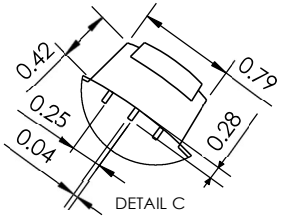
SCALE 1:2



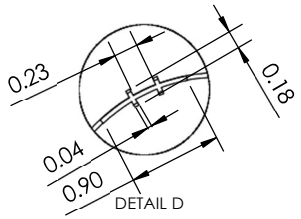
DETAIL A
SCALE 2 : 1.5



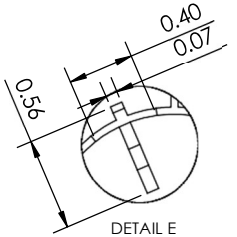
DETAIL B
SCALE 2 : 1.5



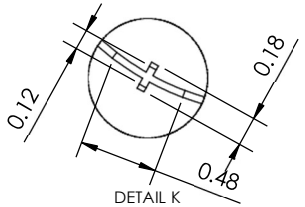
DETAIL C
SCALE 1 : 1



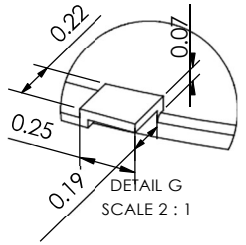
DETAIL D
SCALE 1 : 1.2



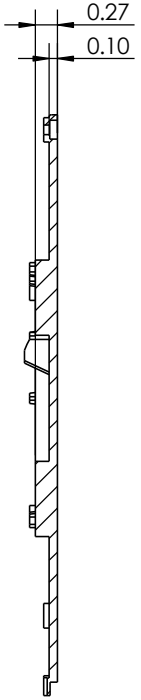
DETAIL E
SCALE 2 : 1.5



DETAIL K
SCALE 2 : 1.5



DETAIL G
SCALE 2 : 1



SECTION E-E

SCALE 1 : 1.5

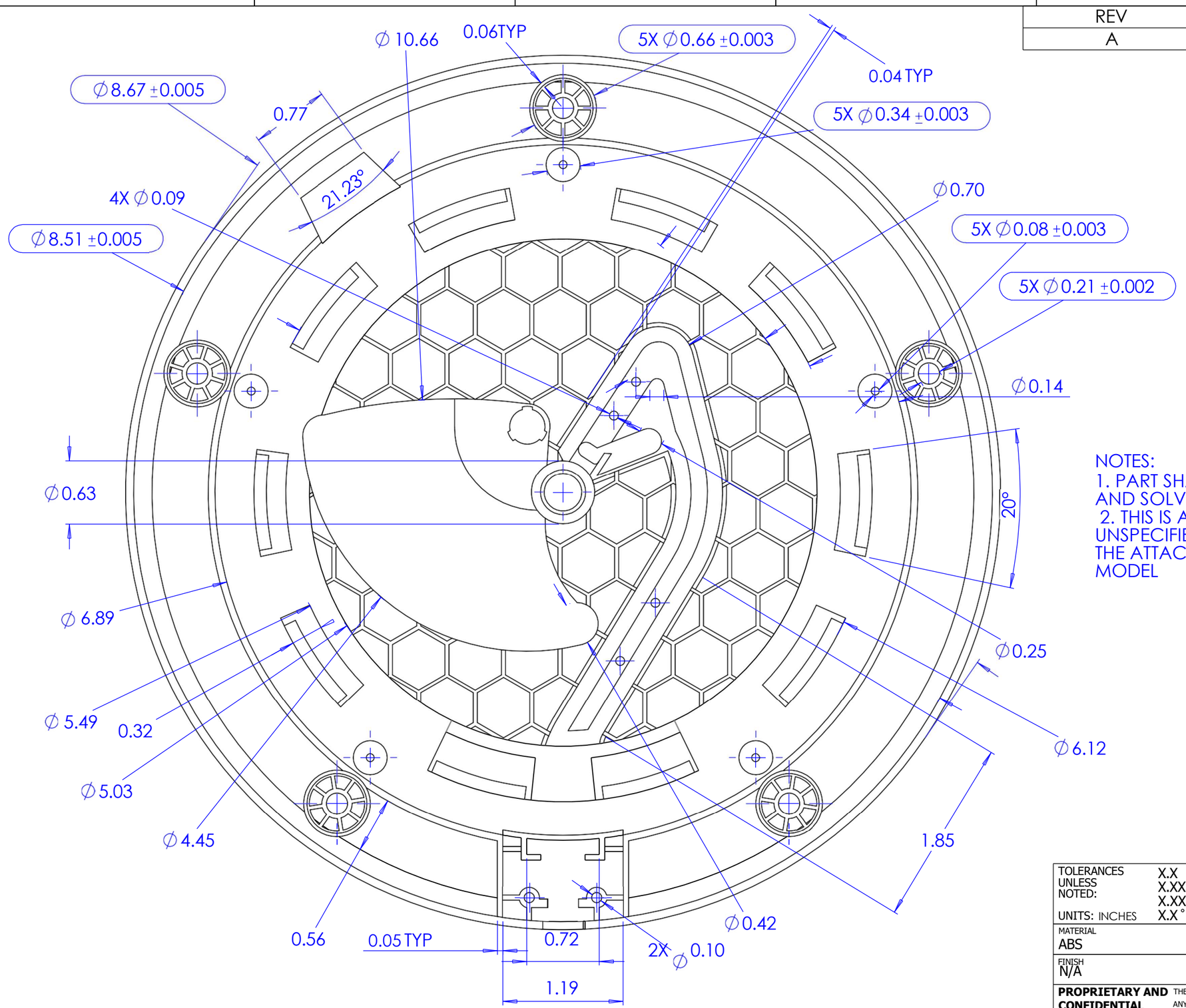
NOTE:
UNLESS OTHERWISE MENTIONED SCALE 2:3

TOLERANCES UNLESS NOTED:	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005	UNIVERSITY OF COLORADO MCEN 5045	
UNITS:	X.X ° ° 1"		
MATERIAL: PC/ABS		DESCRIPTION: BASE PLATE	
FINISH: NONE		PN: BF008-01	REV: B
			SHEET: 1 of 1

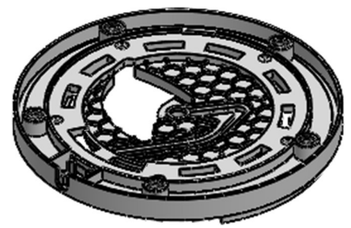
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REV	DESCRIPTION	DATE
A	INITIAL RELEASE	09/30/2024



NOTES:
1. PART SHALL BE VISIBLY CLEAN OF ALL DIRT, OIL, AND SOLVENTS.
2. THIS IS A LIMITED DIMENSION DRAWING. UNSPECIFIED DIMENSIONS SHALL BE TAKEN FROM THE ATTACHED COMPUTER-GENERATED SOLID MODEL

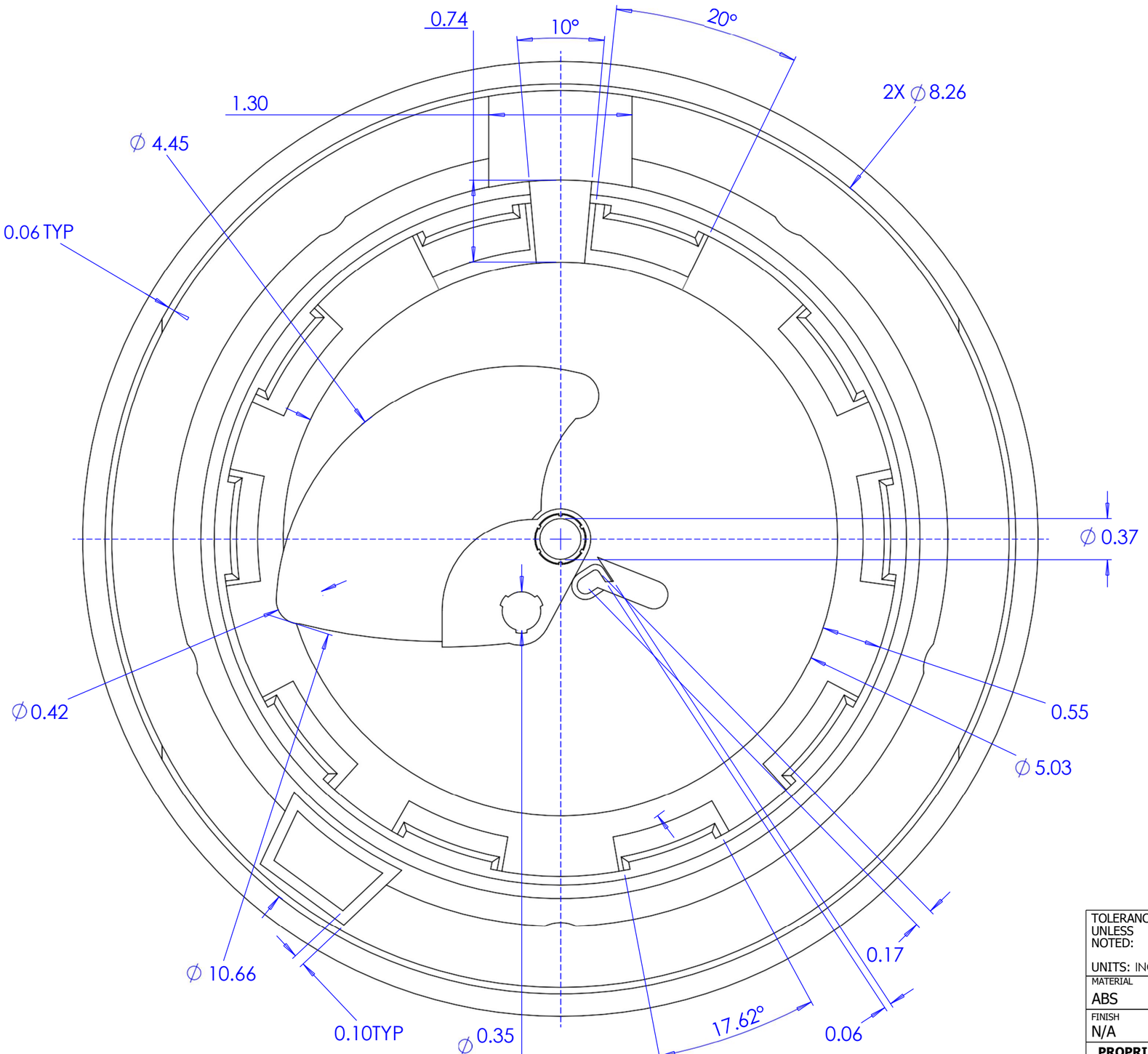


SCALE 1:5

TOLERANCES UNLESS NOTED: X.X ± 0.050, X.XX ± 0.010, X.XXX ± 0.005, X.X° ° 1"	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
UNITS: INCHES		
MATERIAL: ABS	DESCRIPTION: BASE PLATE	
FINISH: N/A — SCALE 1:1	PN: BF008-01 — REV: A	SHEET: 1 of 1

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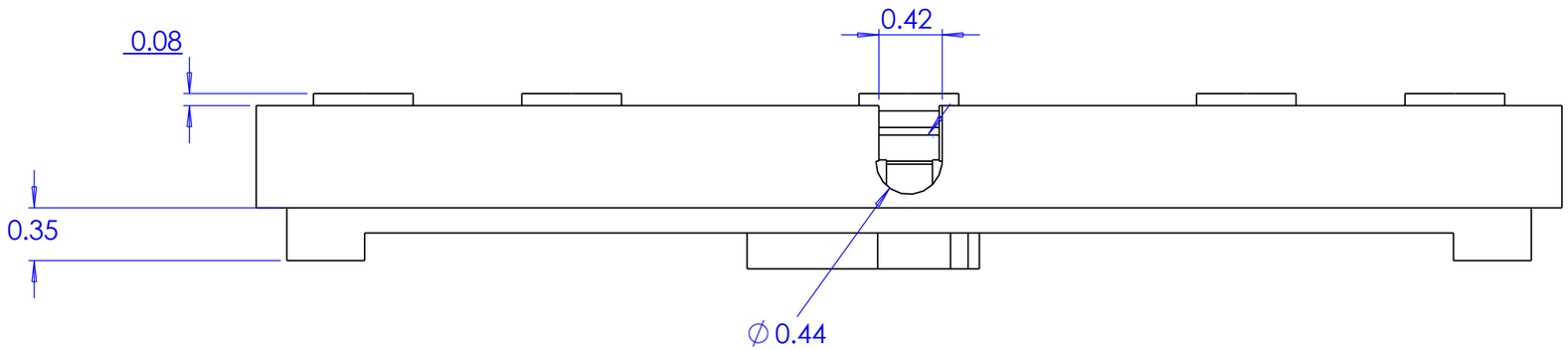
SOLIDWORKS Educational Product. For Instructional Use Only.

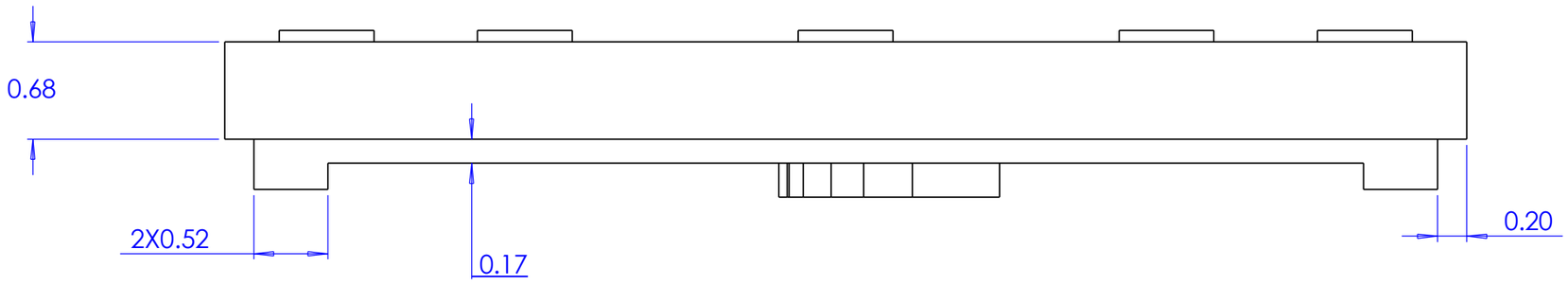


TOLERANCES UNLESS NOTED:	X.X ± 0.050 X.XX ± 0.010 X.XXX ± 0.005	UNIVERSITY OF COLORADO	
UNITS: INCHES	X.X° ° 1"	MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
MATERIAL ABS		DESCRIPTION BASE PLATE	
FINISH N/A		PN BF008-01	REV A
			SHEET 1 of 1

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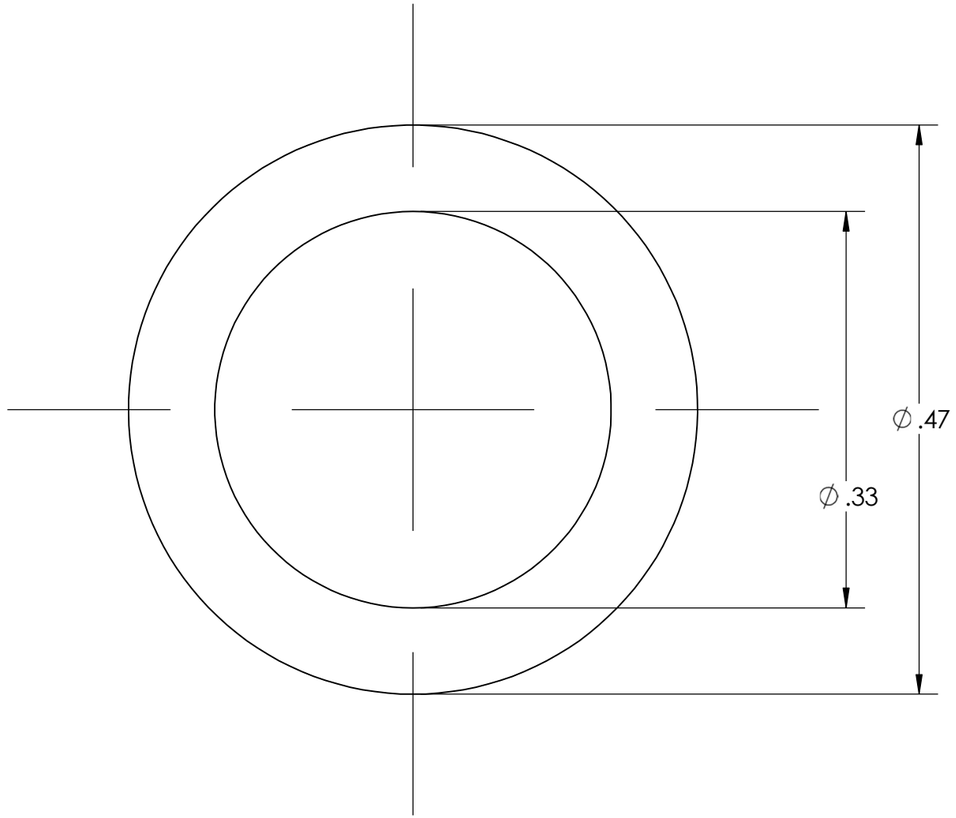


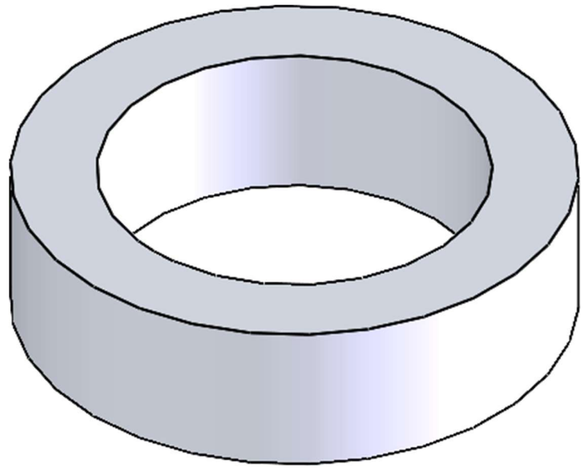
TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ± 0.050 X.XX ± 0.010 X.XXX ± 0.005 X.X° ° 1"	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
MATERIAL ABS		DESCRIPTION BASE PLATE	
FINISH N/A		PN BF008-01	REV A
			SHEET 1 of 1

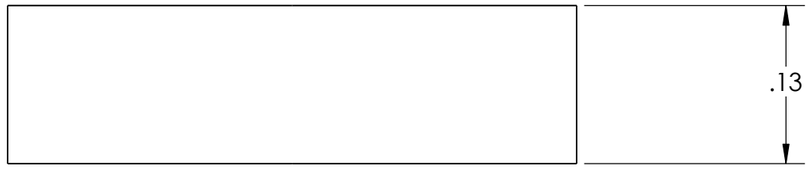
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REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	INITIAL RELEASE	16-10-2024	





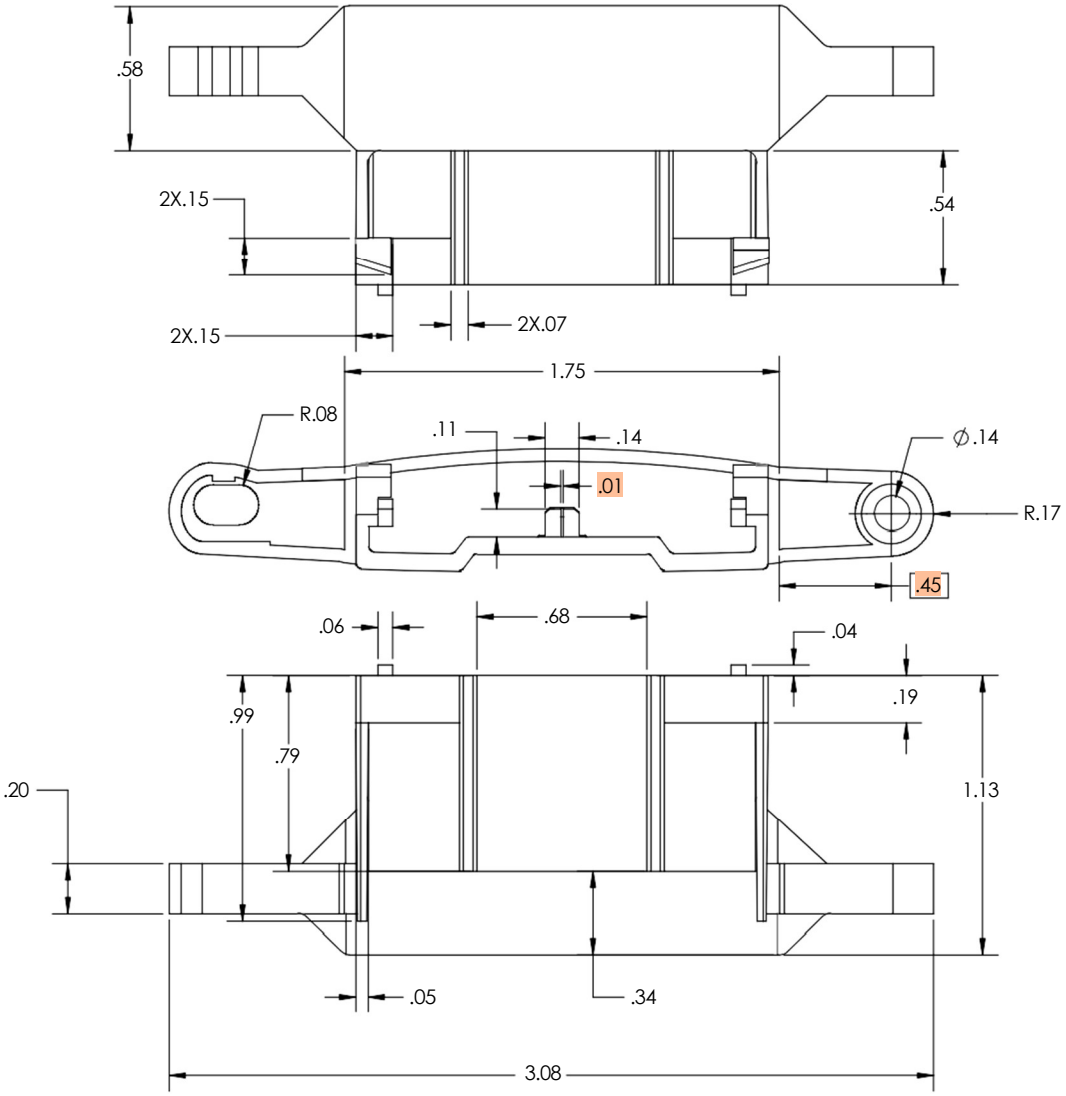


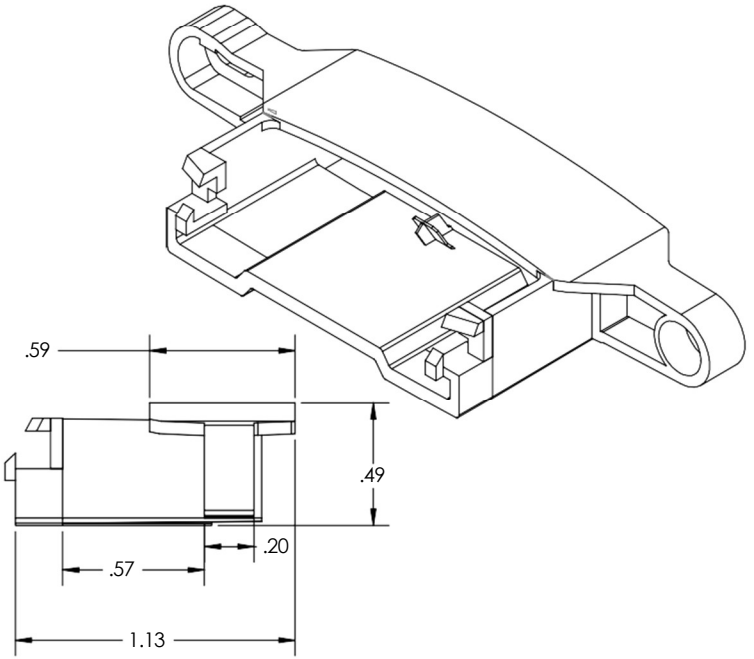
TOLERANCES UNLESS NOTED:	X.X ±0.050 X.XX ±0.010 X.XXX ±0.005	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
UNITS: INCHES	X.X° 0.5°		
MATERIAL: ALUMINUM		DESCRIPTION: BEARING	
FINISH: GLOSSY SILVER		PN: BF008-03	REV: A
			SHEET: 1 of 1

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REV.	DESCRIPTION	DATE	DRAWN BY	APPROVED BY
A	INITIAL RELEASE	10/05/2024	SARAVANA	





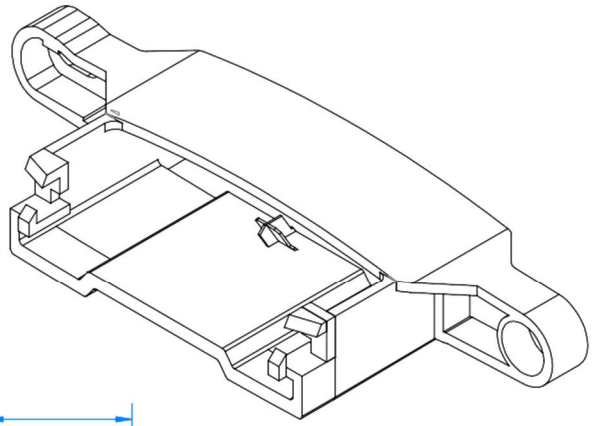
NOTES:
1. PART SHALL BE VISIBLY CLEAN OF ALL DIRT, OIL, AND SOLVENTS.
2. THIS IS A LIMITED DIMENSION DRAWING. UNSPECIFIED DIMENSIONS SHALL BE TAKEN FROM THE ATTACHED COMPUTER-GENERATED SOLID MODEL

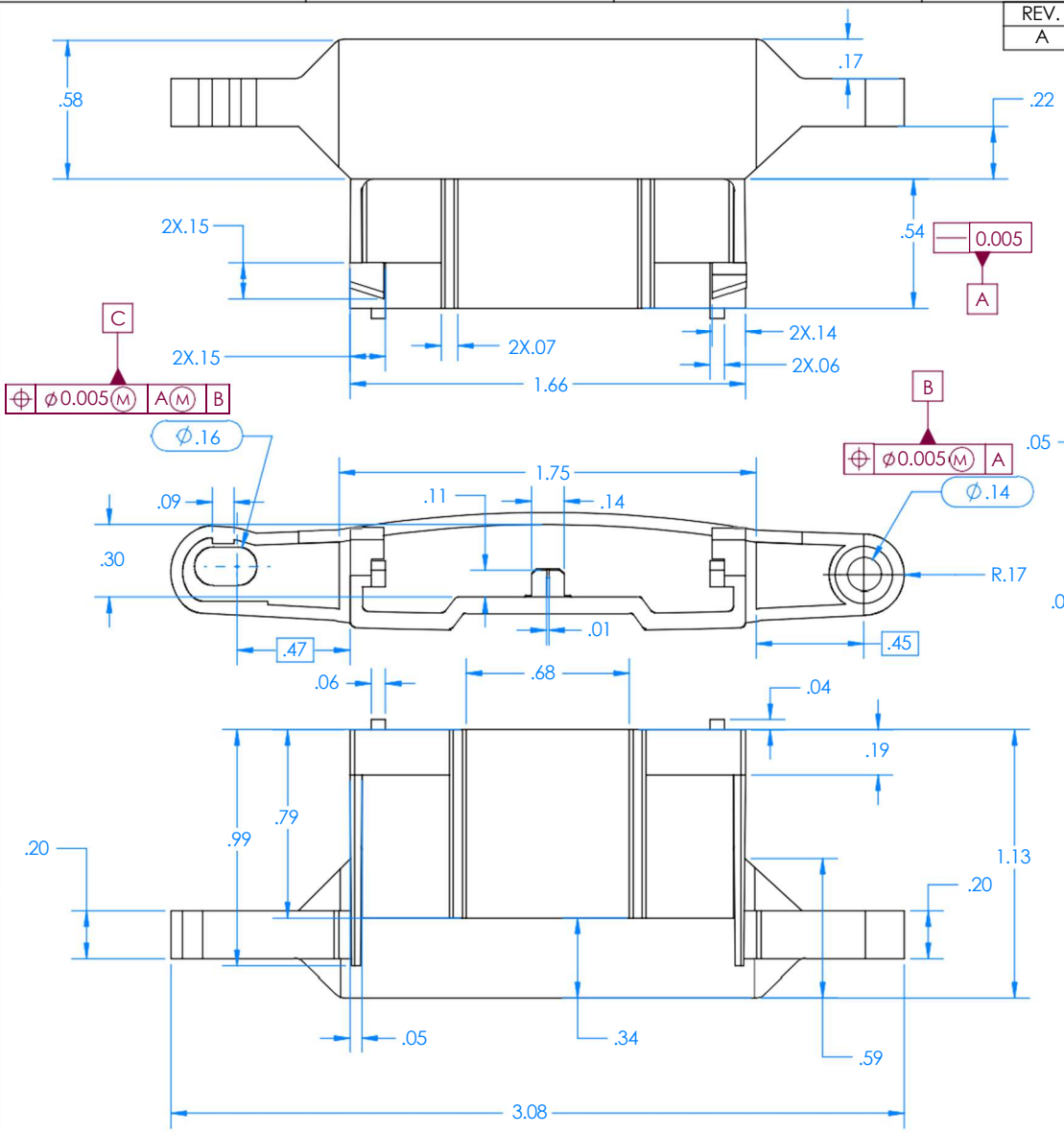
TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ±0.05 X.XX ±0.010 X.XXX ±0.005 X.X ° °1"	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427
MATERIAL: PC/ABS		DESCRIPTION: BASE PLATE SPRING LOCK
FINISH: N/A	SCALE: 2:1	PN: BF008-06 REV: A SHEET: 1 of 1

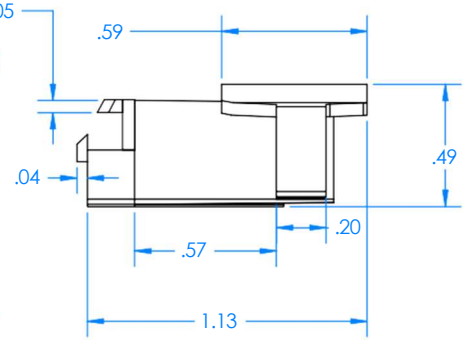
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REV.	DESCRIPTION	DATE	DRAWN BY	APPROVED BY
A	INITIAL RELEASE	10/05/2024	SARAVANA	







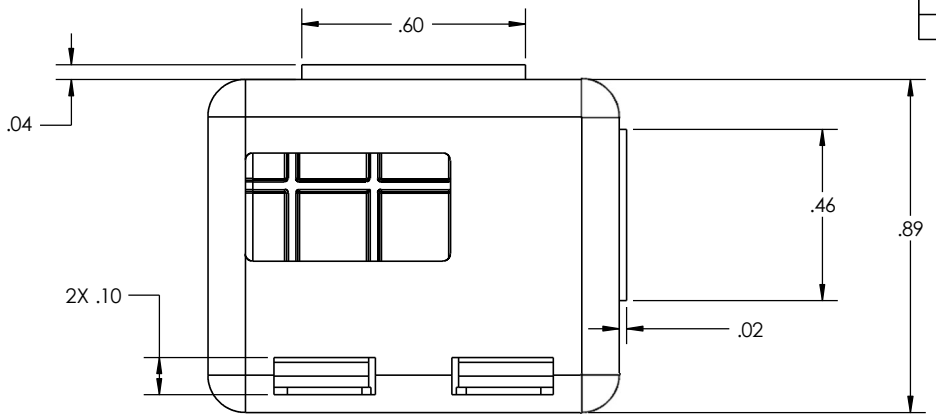
NOTES:
1. PART SHALL BE VISIBLY CLEAN OF ALL DIRT, OIL, AND SOLVENTS.
2. THIS IS A LIMITED DIMENSION DRAWING. UNSPECIFIED DIMENSIONS SHALL BE TAKEN FROM THE ATTACHED COMPUTER-GENERATED SOLID MODEL
3. CRITICAL OR INSPECTION DIMENSIONS ARE ENCLOSED WITH THE SYMBOL

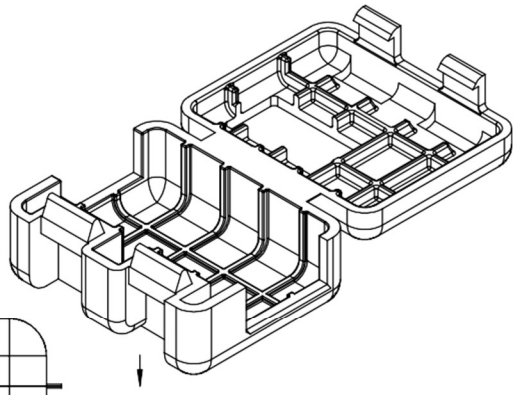
TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ±0.05 X.XX ±0.010 X.XXX ±0.005 X.X ° °1"	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
MATERIAL: PC/ABS		DESCRIPTION: BASE PLATE SPRING LOCK	
FINISH: N/A	SCALE: 2:1	PN: BF008-06	REV: A
			SHEET: 1 of 1

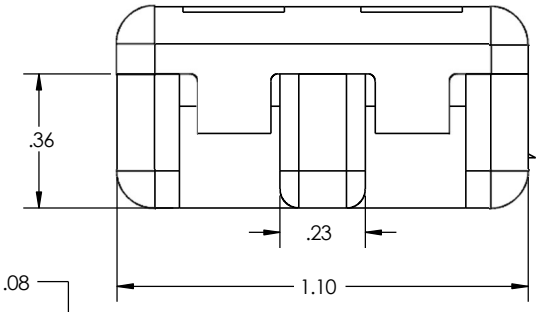
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.

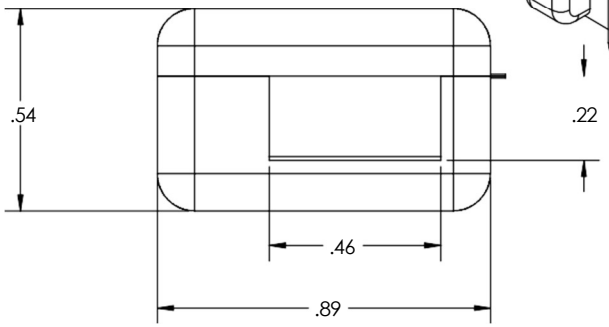
SOLIDWORKS Educational Product. For Instructional Use Only.

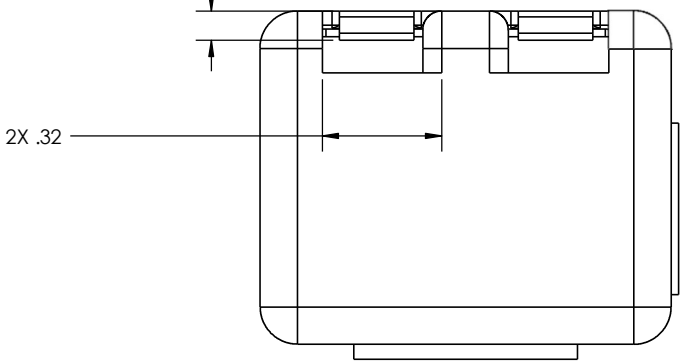
REV.	DESCRIPTION	DATE	DRAWN BY	APPROVED BY
A	INITIAL RELEASE	10/05/2024	SARAVANA	









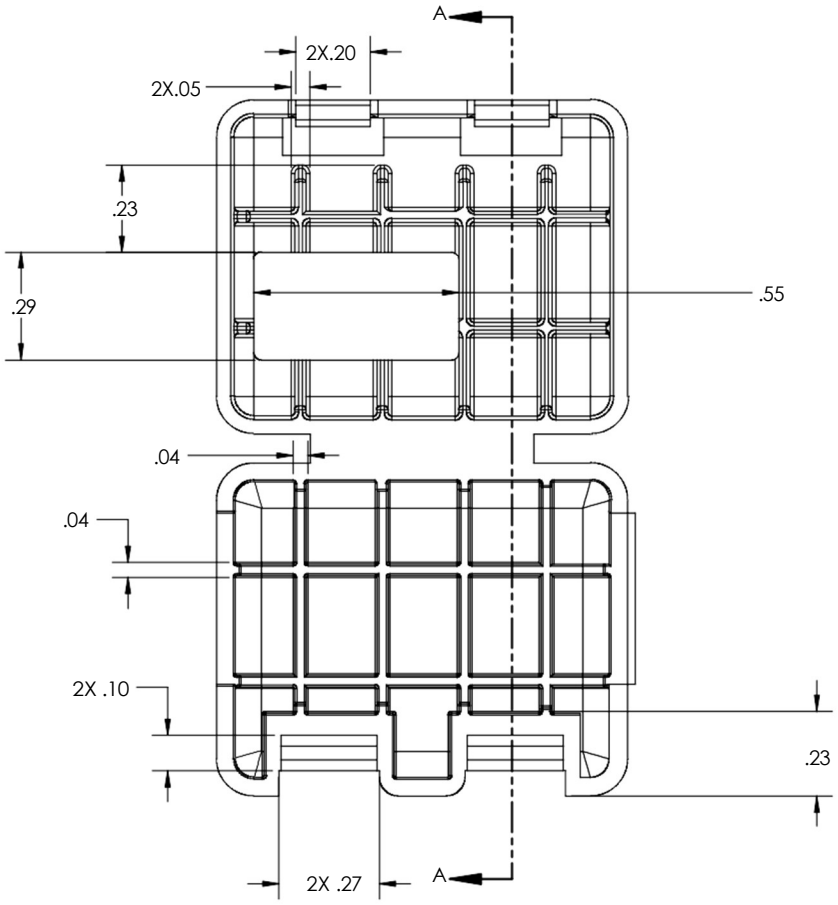


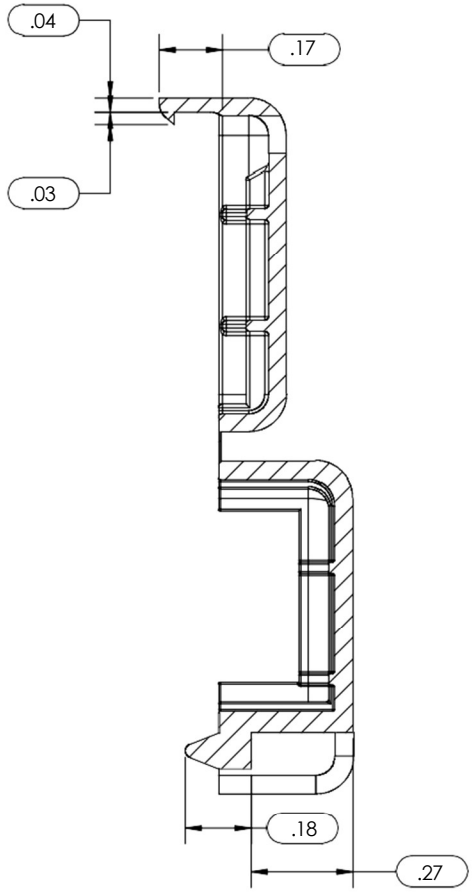
NOTES:
1. PART SHALL BE VISIBLY CLEAN OF ALL DIRT, OIL, AND SOLVENTS.
2. THIS IS A LIMITED DIMENSION DRAWING. UNSPECIFIED DIMENSIONS SHALL BE TAKEN FROM THE ATTACHED COMPUTER-GENERATED SOLID MODEL
3. CRITICAL OR INSPECTION DIMENSIONS ARE ENCLOSED WITH THE SYMBOL ⊖

TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005 X.X ° ° 1"	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427
MATERIAL: PC/ABS		DESCRIPTION: CONNECTOR WIRE
FINISH: N/A	SCALE: 2:1	PN: BF009-09 REV: A SHEET: 1 of 2

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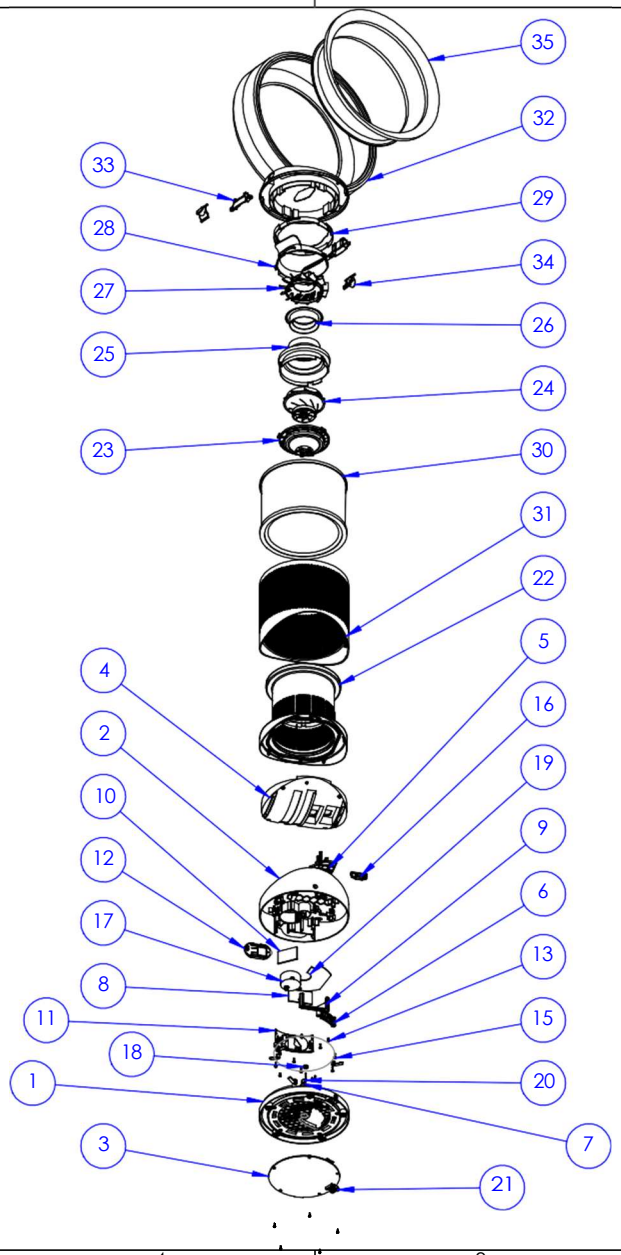
SECTION A-A
SCALE 3 : 1

TOLERANCES UNLESS NOTED:	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
UNITS: INCHES	X.X ° ° 1"		
MATERIAL		DESCRIPTION	
PC/ABS		CONNECTOR WIRE	
FINISH	SCALE	PN / REV	SHEET
N/A	2:1	BF009-09 / A	2 of 2

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REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	10/15/2024



UNIVERSITY OF COLORADO BOULDER

DESCRIPTION
DYSON BLADELESS FAN

PN	REV	SHEET
DP-01	A	1 of 2

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ITEM NO.	PART NUMBER	Description	QTY.
1	BF008-02	Oscillating base	1
2	BF006-01	Lower body housing	1
3	BF008-01	Base plate	1
4	BF005-02	SWIVEL BASE	1
5	BF005-03	Electronics Component Holder	1
6	BF006-04	Base plate lock	3
7	BF006-09	Wire holder 1	6
8	PCB3	PCB for sensor	1
9	BF006-05	LCD & PCB holder	1
10	PCB2	PCB 2	1
11	BF006-08	Sensor hatch housing	1
12	BF006-07	Sensor hatch	1
13	M2.5x06	torx_socket_pan_head_M2.5x06	10
14	M2.5x08	torx_socket_pan_head_M2.5x08	13
15	M2.5x10	torx_socket_pan_head_M2.5x10	6
16	BF006-05	Power Button	1
17	BF007-01	swivel motor	1
18	BF008-03	Bearing	1
19	PCB1	MAIN PCB	1
20	M2.5x20	torx_socket_pan_head_M2.5x20	1
21	BF008-04	Power cord retainer	1
22	BF005-01	Electronic housing	1
23	BF003-02	Impeller top housing	1
24	BF003-06	Stationary plastic blade	1
25	BF003-03	Impeller bottom housing	1
26	BF003-04	Hub base	1
27	BF003-06	Impeller	1
28	BF003-05	Stator housing	1
29	BF002	Motor Bucket retaining ring	1
30	BF004-03	Filter	1
31	BF004-01	Filter Housing	1
32	BF001-02	Amplifier housing	1
33	BF001-03	Release lock Housing	2
34	BF001-04	Release lock	2
35	BF001-01	Amplifier head	1

UNIVERSITY OF COLORADO BOULDER

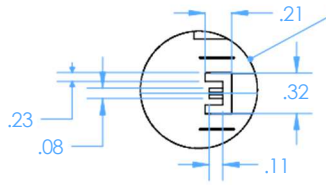
DESCRIPTION
DYSON BLADELESS FAN

PN	REV	SHEET
DP-01	A	2 of 2

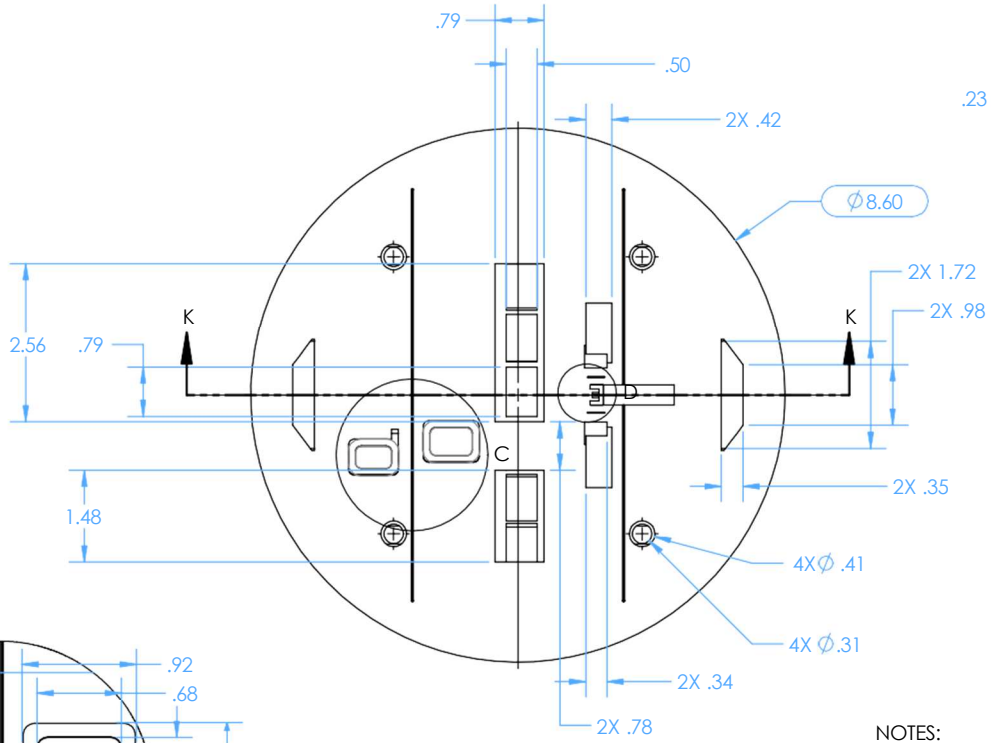
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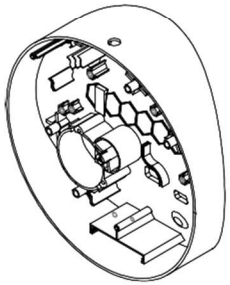
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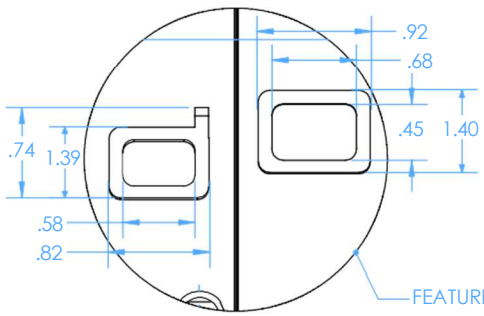
REV.	DESCRIPTION	DATE	DRAWN BY	APPROVED BY
A	INITIAL RELEASE	10/05/2024	SARAVANA	
B	DESIGN CHANGE TO ELIMINATE PARTS	10/13/2024	SARAVANA	



DETAIL D
SCALE 1 : 1







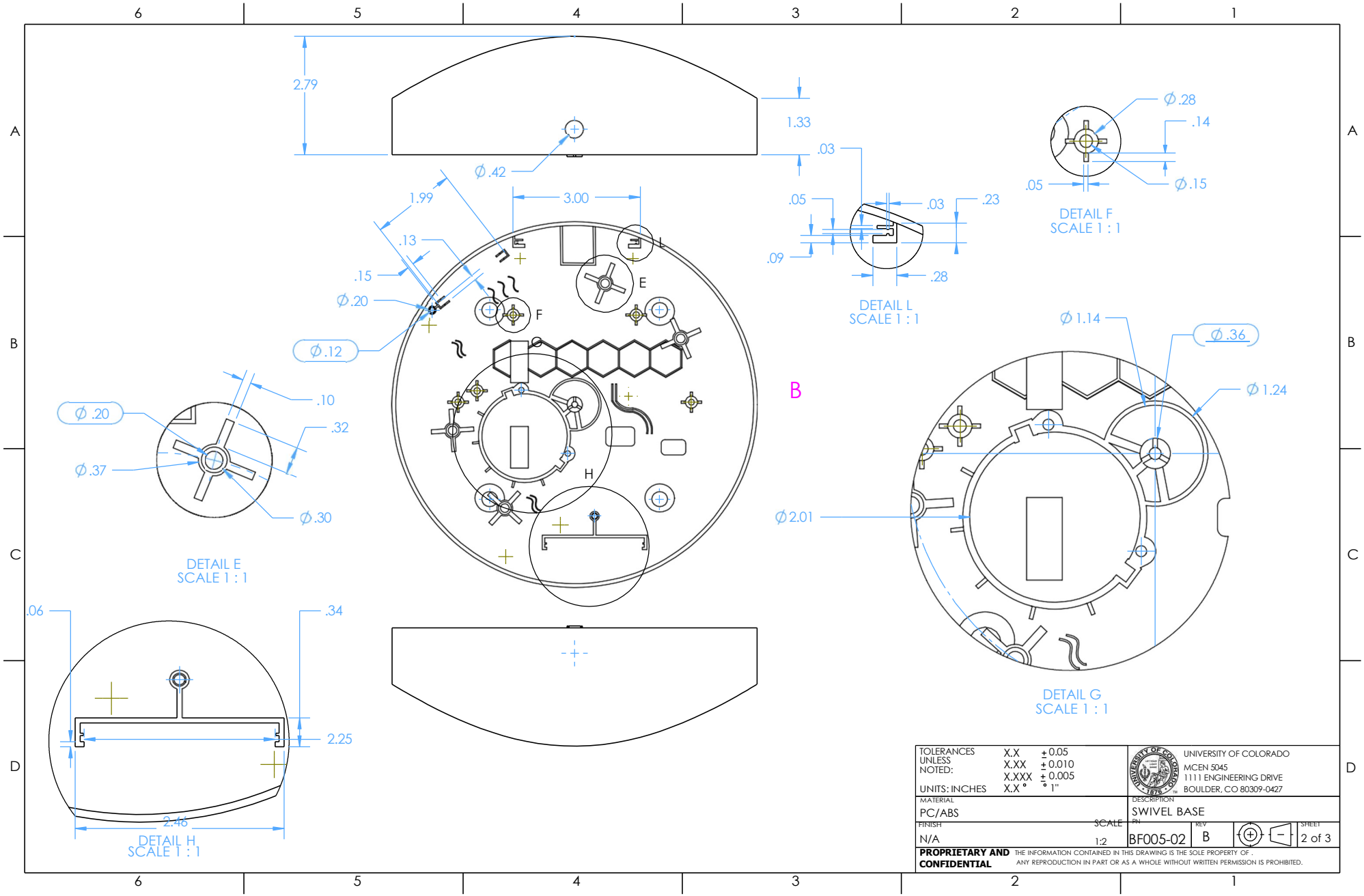
DETAIL C
SCALE 1 : 1

NOTES:
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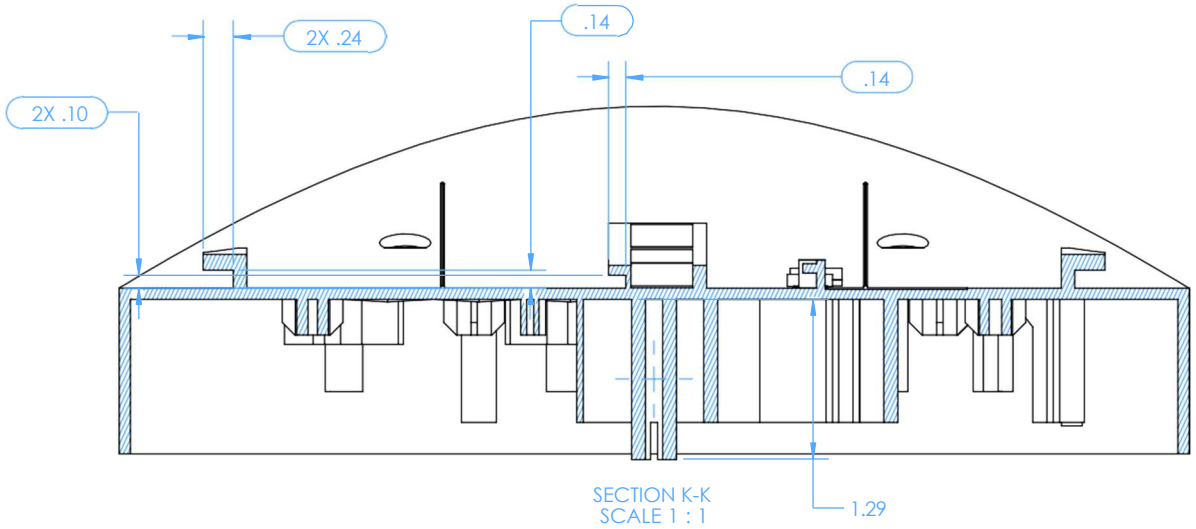
TOLERANCES UNLESS NOTED: UNITS: INCHES	X.X ± 0.05 X.XX ± 0.010 X.XXX ± 0.005 X.X ° ° 1"	UNIVERSITY OF COLORADO MCEN 5045 1111 ENGINEERING DRIVE BOULDER, CO 80309-0427	
MATERIAL: PC/ABS		DESCRIPTION: SWIVEL BASE	
FINISH: N/A	SCALE: 1:2	PN: BF005-02	REV: B / SHEET: 1 of 3

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MATERIAL PC/ABS		DESCRIPTION SWIVEL BASE	
FINISH N/A	SCALE 1:2	PN BF005-02	REV B
			SHEET 3 of 3

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